Quantum World

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Part I

Basics for Quantum Mechanics

Preface

This is a personal review for Quantum Mechanics. While many stuffs would be absent, I shall try to claim the basic thought and hopefully, they arrive at nature.

Chapter 1

Foundation

"Hilbert space is a norm-metric topological vector space which is square-integrable with canonical inner product by conjugate-dual functional space."

1.1 Two State System

It was found experimentally that silver atom with one outer-shell electron would split into two beams after passing through a non-uniform magnetic field vertical to its propagating direction, along each spatial orientation, named as *Stern-Gerlach* experiment. Since by classical analogy

$$\dot{\boldsymbol{p}} = -\nabla U = \boldsymbol{\mu} \cdot \nabla \boldsymbol{B},$$

then one conclude that along each direction, the 'magnetic moment' of the single electron has two values, and they are indeed, related to degrees of freedom of a new physical concept — spin, assigned by *Dirac's ket* $|\alpha, \pm\rangle$, where α denotes to spatial orientation.

What is more intriguing nature shows is after letting quantum spin passing through one SG apparatus with non-uniform magnetic field in one direction, say z, masking the $|z-\rangle$ beam by some board and then measure the remaining $|z+\rangle$ beam using another SG apparatus, the $|z+\rangle$ beam would always remain $|z+\rangle$ if it is still passing SG z, while splitting equally into two beams if the latter apparatus is differently orientated, e.g., if it appears as an x-measurement, we would get 'half' $|x+\rangle$ together with 'half' $|x-\rangle$. Here 'half' is related to the measured result, and in language of ingredient or absolute strength, it gives us directly that

$$\begin{cases} I_1(|z+\rangle) = I_2(|z+\rangle) \\ I_1(|z+\rangle) = I_2(|x+\rangle) + I_2(|x-\rangle) = 2I_2(|x+\rangle) \end{cases},$$

where subscript refers to measuring sequence.

Some may argue that classical physics could also give fairly convincing proof to this phenomenon. Surely it is as one could view spin-x as stochastic variable inside $|z+\rangle$. Then what about this one? Say we put a three-apparatus measuring sequence as SG z - SG x - SG z, each time choosing $|\alpha, +\rangle$ from last to next? Classically, since by first measurement $|z+\rangle$ is certain, no one would expect the third apparatus to give us two beams. However, quantum nature tread that underfoot — one would still see two beams with equal intensity, i.e.,

$$I_2(|x+\rangle) = I_3(|z+\rangle) + I_3(|z-\rangle) = 2I_3(|z+\rangle)!$$

Obviously, under level of classical intensity bounded with probability, one would never reach the mystery quantum nature. Now we start from another point of view, namely, probability amplitude, similar to the amplitude of a wave, to understand the problem. Before that, some pre-knowledge is needed.

1.2 Notations

Dirac ket $|\ \rangle$, as we noted above, is a point in *Hilbert Space* $\mathscr{H}$, which is an L^2 complex linear space where L means it is defined under Lebesgue's measure and its square refers to square integrability. And by linear space we mean that ket addition and number field multiplication is well-defined

$$\exists\, |\gamma\rangle = c\,|\alpha\rangle + d\,|\beta\rangle\,,\ \forall c, d \in \mathbb{C}, \tag{1.1}$$

where $\mathbb{C}$ refers to complex field.

Dirac bra $\langle\ |$ is a point in dual space of $\mathscr{H}$ as $\mathscr{H}^\dagger$, whose existence is guaranteed by Riesz's theorem.

A *quantum state* is a ket $|\alpha\rangle$ with α referring to its *quantum number*. One could define *inner product* and *outer product* between $\mathscr{H}$ and its dual space as

$$\begin{cases} \langle\alpha|\beta\rangle \in \mathbb{C} \\ |\alpha\rangle\langle\beta| \in \mathbb{O} \end{cases}, \tag{1.2}$$

where $\mathbb{O}$ is defined to be the set of *operator*s. In physics, only direction of ket vector is important, so we could build a normalized (if possible) ket by

$$|\alpha\rangle \to |\tilde{\alpha}\rangle = \frac{1}{\langle\alpha|\alpha\rangle}\,|\alpha\rangle\,, \tag{1.3}$$

who contains all information we need except for a trivial phase that could behave non-trivially. And we would use normalized kets without special indication.

An operator lives in Hilbert space and transforms one ket to another by left action

$$\exists\, |\delta\rangle = \hat{A}\,|\alpha\rangle\,,\ \forall\, |\alpha\rangle\,.$$

An *observable* is a self-adjoint operator or roughly speaking, Hermitian operator as

$$\hat{H}^\dagger = \hat{H}, \tag{1.4}$$

where $\dagger$ means Hermitian-conjugate as both transpose and complex-conjugate. As a symmetric operator, an observable $\hat{A}$ always has real eigenvalues as

$$\hat{A}\,|a\rangle = a\,|a\rangle\,,\ a \in \mathbb{R}, \tag{1.5}$$

since

$$a = \langle a|(\hat{A}|a\rangle) = (\langle a|\hat{A}^\dagger)|a\rangle = a^*,$$

where law of combination is used.

Using Hermitian-conjugate, one has

$$\begin{cases} \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^\dagger = \langle\alpha|\beta\rangle^* \\ |\beta\rangle\langle\alpha| = (|\alpha\rangle\langle\beta|)^\dagger \\ (c\,|\alpha\rangle + d\,|\beta\rangle)^\dagger = c^*\,\langle\alpha| + d^*\,\langle\beta| \end{cases}. \tag{1.6}$$

A *complete relation* is also a *closure relation*, which claims that for a certain set of kets, one has

$$\sum_n |a_n\rangle\langle a_n| = \mathbb{I}, \int \mathrm{d}\alpha\, |\alpha\rangle\langle\alpha| = \mathbb{I}, \tag{1.7}$$

where $\mathbb{I}$ is *identity operator* defined for certain space with

$$\mathbb{I}\,|\alpha\rangle = |\alpha\rangle \tag{1.8}$$

and two equations refer to finite and infinite dimensional cases, respectively.

A *commutation relation* between two operators is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{1.9}$$

Notice it that when actually computing this, one needs to examine the action of commutation operator.

The *canonical quantization* is a generalization of classical relations, which claims that

$$[\hat{A}, \hat{B}] = i\hbar\{A, B\}_{classical}, \tag{1.10}$$

where classical $\{\ ,\ \}$ refers to Poisson's bracket, and $\hbar$ is Planck's constant, e.g.

$$[\hat{x}_i, \hat{p}_j] = i\hbar\{x_i, p_j\} = i\hbar\delta_{ij}. \tag{1.11}$$

A *unitary operator* is an operator who satisfies that

$$\langle\hat{U}\alpha|\hat{U}\beta\rangle = \langle\alpha|\beta\rangle\,, \tag{1.12}$$

which would lead to

$$\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \mathbb{I}. \tag{1.13}$$

A similar definition could be applied to get *anti-unitary operator* or *anti-linear operator* as

$$\langle\hat{C}\alpha|\hat{C}\beta\rangle = \langle\alpha|\beta\rangle^*\,, \tag{1.14}$$

which also exhibits $\hat{C}^\dagger\hat{C} = \mathbb{I}$, however, the anti-linearity means that

$$\hat{C}(c\,|\alpha\rangle + d\,|\beta\rangle) = c^*\hat{C}\,|\alpha\rangle + d^*\hat{C}\,|\beta\rangle\,. \tag{1.15}$$

In fact, one could split an anti-unitary operator as

$$\hat{C} = \hat{U}\mathscr{K}, \tag{1.16}$$

where $\mathscr{K}$ is the complex conjugate operator. These two and only two transformations preserving inner product is classified by Wigner's theorem.

A single-parameter unitary operator group $\hat{U}(s)$ with s in a continuous set satisfies that

$$\begin{cases} \hat{U}(0) = \mathbb{I}, & \text{identity} \\ \hat{U}(s_1)\hat{U}(s_2) = \hat{U}(s_1 + s_2), & \text{multiplication} \\ \hat{U}(s_1 + s_2)\hat{U}(s_3) = \hat{U}(s_1)\hat{U}(s_2 + s_3), & \text{associativity} \\ \hat{U}^{-1}(s) = \hat{U}(-s) = \hat{U}^\dagger(s), & \text{inverse} \end{cases}.$$

And it turns out that the multiplication between group elements equals addition of parameter. When this map from s to $\hat{U}$ is smooth, this group becomes a Lie group; and when the set above has finite measure, it is compact. Let us consider an infinitesimal unitary transformation as

$$\hat{U}(\varepsilon) = \mathbb{I} + \left.\frac{\mathrm{d}\hat{U}}{\mathrm{d}s}\right|_{s=0} \varepsilon + O(\varepsilon^2),\ \varepsilon \to 0, \tag{1.17}$$

then by requiring unitality up to first order,

$$\hat{U}^\dagger \hat{U} = \mathbb{I} + \left(\frac{\mathrm{d}\hat{U}}{\mathrm{d}s} + \frac{\mathrm{d}\hat{U}^\dagger}{\mathrm{d}s}\right)\Bigg|_{s=0} \varepsilon + O(\varepsilon^2), \tag{1.18}$$

which means that $\left.\dfrac{\mathrm{d}\hat{U}}{\mathrm{d}s}\right|_{s=0}$ is anti-Hermitian and could be achieved by letting

$$\left.\frac{\mathrm{d}\hat{U}}{\mathrm{d}s}\right|_{s=0} = -i\hat{K},\ \hat{K}^\dagger = \hat{K}, \tag{1.19}$$

where the minus sign is just some convention (annoying not it?). We then name $\hat{K}$ as the generator to group $\hat{U}(\{s\})$, and in this way a finite unitary transformation could be represented as

$$\hat{U}(s) = \lim_{N\to+\infty} \hat{U}^N(\frac{s}{N}) = \lim_{N\to+\infty} \left(\mathbb{I} - i\hat{K}\frac{s}{N}\right)^N = e^{-i\hat{K}s}. \tag{1.20}$$

This method could be generalized to multi-parameters with generators $\hat{\boldsymbol{K}} = (\hat{K}_1, \cdots, \hat{K}_n)$, and

$$\hat{U}(\boldsymbol{s}) = e^{-i\hat{\boldsymbol{K}}\cdot\boldsymbol{s}}, \tag{1.21}$$

and when this forms a *Lie group*, the commutation relations between generators form an algebra as *Lie algebra*, usually,

$$[\hat{K}_i, \hat{K}_j] = i\kappa_{ijk}\hat{K}_k, \tag{1.22}$$

where κ is known as structure constant. Here we emphasise that definition above is purely mathematical, and its relation with physics needs further postulate.

1.3 Matrix Representation

The matrix representation of an operator, which by definition is with finite dimension, specifies which physical situation we are considering, and there should be clearly clarified physical phenomenon standing for that, which is protected by the orthogonal theorem who states eigenkets for a Hermitian operator $\hat{A}$ corresponding to different eigenvalues are orthogonal. The proof follows directly from

$$0 = \langle a|\hat{A} - \hat{A}|a'\rangle = (a - a')\langle a|a'\rangle,$$

since $a \neq a'$, $\langle a|a'\rangle = 0$ with no ambiguity. Clearly, when $a = a'$, one could have $\langle a|a'\rangle \neq 0$ in this subspace, however, this could always be done upon choice or by manual Gram-Schmidt orthogonalization. Then, by using orthonormal kets, i.e.

$$\langle a|a'\rangle = \delta_{aa'}, \tag{1.23}$$

we claim that the eigenkets of operator $\hat{A}$ form complete orthonormal basis as

$$\sum_n |a_n\rangle\langle a_n| = \mathbb{I},\ \langle a_n|a'_n\rangle = \delta_{nn'}, \tag{1.24}$$

and this completeness relation holds as long as we are considering ket inside this space, or physically, state that is measurable by $\hat{A}$.

Now given such a state $|\alpha\rangle$, one has

$$|\alpha\rangle = \mathbb{I}\,|\alpha\rangle = \sum_n |a_n\rangle\,\langle a_n|\alpha\rangle = \sum_n c_n\,|a_n\rangle\,, \tag{1.25}$$

where $c_n = \langle a_n|\alpha\rangle$ is the projection coefficient of an arbitrary *pure state* $|\alpha\rangle$ on $\hat{A}$'s nth eigenstate $|a_n\rangle$, or more precisely, a *pure ensemble* characterized by $|\alpha\rangle$. In matrix form we could write

$$|\alpha\rangle = \begin{pmatrix}|a_1\rangle & \cdots & |a_n\rangle\end{pmatrix}\begin{pmatrix}c_1\\ \vdots\\ c_n\end{pmatrix}$$

so that under basis/representation of A,

$$\alpha^a = \begin{pmatrix}c_1 & \cdots & c_n\end{pmatrix}^T \tag{1.26}$$

as a column vector where T means transpose.

One could also represent an operator in its matrix form, like the ket vector. Given $\hat{O}$,

$$\hat{O} = \mathbb{I}\hat{O}\mathbb{I} = \sum_m |a_m\rangle\langle a_m|\,\hat{O}\sum_n |a_n\rangle\langle a_n| = \sum_{mn}\langle a_m|\hat{O}|a_n\rangle\,|a_m\rangle\langle a_n| = \sum_{mn} O_{mn}\,|a_m\rangle\langle a_n|\,, \tag{1.27}$$

where $O_{mn} = \langle a_m|\hat{O}|a_n\rangle$ is its matrix component in representation of A controlled by

$$\hat{P}_m\hat{O}\hat{P}_n = O_{mn}\,|a_m\rangle\langle a_n|\,. \tag{1.28}$$

Similarly, we could write $\hat{O}$ as a matrix

$$O^a_{mn} = O_{mn}, \tag{1.29}$$

where we emphasise the representation here by superscript since without specifying that the matrix itself is meaningless with infinite physical explanation. It is also clear to see that

$$A^a = \mathrm{diag}\begin{pmatrix}a_1 & \cdots & a_n\end{pmatrix} \tag{1.30}$$

as diagonal matrix from its expansion

$$\hat{A} = \sum_{mn}\langle a_m|\hat{A}|a_n\rangle\,|a_m\rangle\langle a_n| = \sum_{mn} a_n\delta_{mn}\,|a_m\rangle\langle a_n| = \sum_n a_n\,|a_n\rangle\langle a_n|\,. \tag{1.31}$$

1.4 Transformation of Representation

Now consider another observable $\hat{B}$ with $\{|b_n\rangle\}$ who share the same Hilbert space with $\hat{A}$, then we ask, what is the relation between to pictures? In fact, by matrix representation,

we see it that at many times, quantum mechanics is about inserting identities, and so does it work here.

Firstly, one observes that

$$|b_n\rangle = \sum_m |a_m\rangle \langle a_m|b_n\rangle = \sum_m |a_m\rangle U_{mn}, \tag{1.32}$$

and in this way a unitary matrix $U_{mn} = \langle a_m|b_n\rangle$ is introduced under representation of A, explicitly,

$$\begin{pmatrix} |b_1\rangle & \cdots & |b_n\rangle \end{pmatrix} = \begin{pmatrix} |a_1\rangle & \cdots & |a_n\rangle \end{pmatrix} U,$$

which follows the rule like matrix multiplication. Re-write it as an operator,

$$\hat{U} = \sum_{mn} |a_m\rangle U_{mn} \langle a_n| = \sum_m |a_m\rangle\langle a_m| \sum_n |b_n\rangle\langle a_n| = \sum_n |b_n\rangle\langle a_n|, \tag{1.33}$$

whose action is directly on ket of $\hat{A}$ as

$$\hat{U} |a_n\rangle = \sum_n |b_m\rangle \langle a_m|a_n\rangle = \sum_n |b_m\rangle \delta_{mn} = |b_n\rangle. \tag{1.34}$$

The unitality of $\hat{U}$ is easy to verify as

$$\hat{U}^\dagger \hat{U} = \sum_m |a_m\rangle\langle b_m| \sum_n |b_n\rangle\langle a_n| = \sum_{mn} |a_m\rangle\langle a_n| \delta_{mn} = \mathbb{I}. \tag{1.35}$$

As for U, since it is a representation of $\hat{U}$, its unitality is self-evident, but we show it distinctly like

$$(U^\dagger U)_{mn} = \sum_k U^\dagger_{mk} U_{kn} = \langle a_m| \sum_k \hat{U}^\dagger |a_k\rangle\langle a_k| \hat{U} |a_n\rangle = \langle a_m| \hat{U}^\dagger \hat{U} |a_n\rangle = \langle a_m|a_n\rangle = \delta_{mn}, \tag{1.36}$$

as promised.

Now transformation between basis has been achieved, what about represented kets and operators? Now that

$$\begin{aligned} |\alpha\rangle &= \sum_n |a_n\rangle \langle a_n|\alpha\rangle = \sum_n c^a_n |a_n\rangle = \sum_n c^a_n \sum_m |b_m\rangle \langle b_m|a_n\rangle = \sum_m \left(\sum_n c^a_n U^\dagger_{mn} \right) |b_m\rangle \\ &= \sum_m |b_m\rangle \langle b_m|\alpha\rangle = \sum_m c^b_m |b_m\rangle, \end{aligned}$$

one sees immediately that

$$c^b_m = \sum_n U^\dagger_{mn} c^a_n \iff c^b = U^\dagger c^a, \tag{1.37}$$

as if the transformation between vectors of represented ket is the inversed version of basis transformation. For operators, one could similarly make the conclusion, however, we could arrive at it using invariant

$$\langle\alpha|\hat{O}|\alpha\rangle = c^{a\dagger} \langle a|a\rangle O^a \langle a|a\rangle c^a = c^{b\dagger} U^\dagger O^a U c^b = c^{b\dagger} O^b c^b,$$

where

$$\langle a|a\rangle = \begin{pmatrix} \langle a_1| \\ \vdots \\ \langle a_n| \end{pmatrix} \begin{pmatrix} |a_1\rangle & \cdots & |a_n\rangle \end{pmatrix} = \mathbb{I},$$

and one concludes that

$$O^b = U^\dagger O^a U, \tag{1.38}$$

which is once again, seemingly inversed version to transformation between basis and share similar form with vector ket's transformation. And this transformation form, sometime is known as one kind operation of super-operator on operator.

1.5 Measurement

By representation one has that

$$|\alpha\rangle = \sum_n c_n |a_n\rangle\,,\ c_n = \langle a_n|\alpha\rangle\,,\ \hat{A}|a_n\rangle = a_n|a_n\rangle\,.$$

In this way we say that by measuring property A of state $|\alpha\rangle$, which is an observable and should be Hermitian, we are possible to observe result labelled by a_n, which process is governed by the so-called *projection operator*

$$\hat{P}_n = |a_n\rangle\langle a_n|\,, \tag{1.39}$$

with properties

$$\begin{cases} \sum_n \hat{P}_n = \mathbb{I}, & \text{completeness} \\ \hat{P}_i\hat{P}_j = \delta_{ij}\hat{P}_i, & \text{orthogonal} \\ \hat{P}_n^\dagger = \hat{P}_n, & \text{Hermitian} \\ \hat{P}_n^2 = \hat{P}_n, & 0\ \&\ 1 \text{ eigenvalues} \end{cases},$$

and its action is

$$\hat{P}_n|\alpha\rangle = |a_n\rangle\langle a_n|\sum_m c_m|a_m\rangle = c_n|a_n\rangle\,. \tag{1.40}$$

What is more, the probability for this to happen is, by Born's rule or postulate,

$$P_n = |c_n|^2 = |\langle a_n|\alpha\rangle|^2. \tag{1.41}$$

Clearly, by completeness, we have

$$\sum_n |c_n|^2 = \langle\alpha|\sum_n |a_n\rangle\langle a_n|\alpha\rangle = \langle\alpha|\mathbb{I}|\alpha\rangle = 1, \tag{1.42}$$

which also reflects probability conservation as expected. The state after measurement to a_n is certainly $|a\rangle_n$, while it will be enlightening to write in the way that

$$|a_n\rangle = \frac{\hat{P}_n|\alpha\rangle}{\sqrt{P_n}} = \frac{\hat{P}_n|\alpha\rangle}{|\langle a_n|\alpha\rangle|} = \frac{\hat{P}_n|\alpha\rangle}{\|\hat{P}_n|\alpha\rangle\|}, \tag{1.43}$$

where the denominator is just the normalization factor by a Euclidean norm in the Hilbert space.

However, this rule is very strange. Let us introduce *density matrix* operator for a mixed state ensemble $\{p_i, |\alpha_i\rangle\}$

$$\hat{\rho} = \sum_i p_i \, |\alpha_i\rangle\langle\alpha_i| \,, \tag{1.44}$$

with properties

$$\begin{cases} \mathrm{Tr}(\hat{\rho}) = 1, & \text{unit trace} \\ \hat{\rho} = \hat{\rho}^\dagger, & \text{Hermicity} \\ \langle\hat{\rho}\rangle_\alpha \geq 0, \ \forall \, |\alpha\rangle \,, & \text{semi-positive} \end{cases} .$$

Clearly, we could also define density matrix for a pure state as

$$\hat{\psi} = |\alpha\rangle\langle\alpha| = \sum_{m,n} c_m c_n^* \, |a_m\rangle\langle a_n| = \sum_n |c_n|^2 \, |a_n\rangle\langle a_n| + \sum_{m\neq n} c_m c_n^* \, |a_m\rangle\langle a_n| \,. \tag{1.45}$$

And what our postulate tells us is upon measuring, our pure state firstly become a mixed state $\{|c_n|^2, |a_n\rangle\}$ as

$$\hat{\psi} \to \hat{\rho} = \sum_n |c_n|^2 \, |a_n\rangle\langle a_n| \,,$$

which just remains first part. This step eliminates quantum coherence as 'decoherence' and might be explained by open quantum dynamics, but next procedure requires that each time measuring, one $|a_n\rangle$ is picked with probability $|c_n|^2$ and observable character a_n, which is hard to imagine its process. And in my personal point of view, this happens as among (or slightly before) showing phenomenon, strongly correlated interaction happens and tunes the total system into an unstable equilibrium, which would then slide into the above eigenstates upon any tiny perturbation with exact probability defined above. In this picture, the eigen-distribution of measuring apparatus must be exactly compatible and share one-to-one correspondence with the observable so that possible deviation in quantum two-body system is avoided.

Using density matrix defined above, probability of measuring $|\alpha\rangle$ upon $\hat{A}$ to have a_n is

$$P_n = |\, \langle a_n|\alpha\rangle \,|^2 = \langle a_n|\hat{\psi}|a_n\rangle = \mathrm{Tr}\Big(\hat{P}_n\hat{\psi}\Big), \tag{1.46}$$

and given a pure ensemble, one could argue that the expectation measuring $\hat{A}$ is

$$\bar{A}_\psi = \sum_n P_n a_n = \sum_n a_n \, \mathrm{Tr}\Big(\hat{P}_n\hat{\psi}\Big) = \mathrm{Tr}\left(\sum_n a_n \hat{P}_n\hat{\psi}\right) = \mathrm{Tr}\Big(\hat{A}\hat{\psi}\Big), \tag{1.47}$$

or by matrix representation, under any basis $\{b_n\}$,

$$\begin{aligned} \bar{A}_\psi &= \mathrm{Tr}\left(\sum_{mn} A^b_{mn} \, |b_m\rangle\langle b_n| \sum_{ij} c^b_i c^{b*}_j \, |b_i\rangle\langle b_j|\right) = \mathrm{Tr}\left(\sum_{mnj} A^b_{mn} c^b_n c^{b*}_j \, |b_m\rangle\langle b_j|\right) \\ &= \sum_p \langle b_p| \sum_{mnj} A^b_{mn} c^b_n c^{b*}_j \, |b_m\rangle \, \langle b_j|b_p\rangle = \sum_{np} c^{b*}_p A^b_{pn} c^b_n = c^{b\dagger} A^b c^b = \mathrm{Tr}\big(A^b\psi^b\big), \end{aligned} \tag{1.48}$$

where we have introduced representation of density matrix operator of pure ensemble

$$\psi^b = c^b c^{b\dagger}. \tag{1.49}$$

Similarly, the density matrix after measurement reads

$$|a_n\rangle\langle a_n| = \frac{\hat{P}_n\hat{\psi}\hat{P}_n}{\mathrm{Tr}\Big(\hat{P}_n\hat{\psi}\Big)} = \frac{\hat{P}_n\hat{\psi}\hat{P}_n}{P_n}, \tag{1.50}$$

For the measurement of a mixed ensemble, since

$$\hat{\rho} = \sum_i p_i\,|\alpha_i\rangle\langle\alpha_i| = \sum_i p_i\hat{\psi}_i,$$

one has probability of measuring $\hat{\rho}$ upon $\hat{A}$ to have a_n as

$$P_n = \sum_i p_i\,\mathrm{Tr}\Big(\hat{P}_n\hat{\psi}_i\Big) = \mathrm{Tr}\left(\hat{P}_n\sum_i p_i\hat{\psi}_i\right) = \mathrm{Tr}\Big(\hat{P}_n\hat{\rho}\Big), \tag{1.51}$$

together with expectation value

$$\bar{A}_\rho = \mathrm{Tr}\Big(\hat{A}\hat{\rho}\Big) = \mathrm{Tr}\big(A^b\rho^b\big), \tag{1.52}$$

where the latter expression is under certain representation with

$$\rho^b = \sum_i p_i\psi_i^b = \sum_i p_i c_i^b c_i^{b\dagger}. \tag{1.53}$$

1.5.1 Spin-Half Problem

Now we are able to look back on our two-state problem. From experiment, we introduce three observables as quantum spin

$$\hat{\boldsymbol{S}} = (\hat{S}_x, \hat{S}_y, \hat{S}_z), \tag{1.54}$$

with

$$\boldsymbol{S} = \frac{\hbar}{2}\boldsymbol{\sigma}, \tag{1.55}$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is known as Pauli matrix, and we would like derive that from experiment. Note that this is already defined under some representation. $\hbar$ is added here as twice of the eigenvalues and is proved to be Planck's constant as a signal of quantum phenomenon.

We know that

$$\hat{S}_i = \frac{\hbar}{2}\left(|i+\rangle\langle i+| - |i-\rangle\langle i-|\right),$$

and now we stand upon representation of $\hat{S}_z$, which means that, in matrix form,

$$\sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\ |z+\rangle \sim \begin{pmatrix}1\\0\end{pmatrix},\ |z-\rangle \sim \begin{pmatrix}0\\1\end{pmatrix}. \tag{1.56}$$

From two-apparatus experiment, one has

$$|\,\langle j, s'|i, s\rangle\,|^2 = \delta_{ij}\delta_{ss'} + \frac{1}{2}|\varepsilon_{ij}| = \begin{cases}\delta_{ss'}, & i = j\\ 1/2, & i \neq j\end{cases}, \tag{1.57}$$

where $\varepsilon_{\mu_1,\cdots,\mu_n}$ refers to pure-anti-symmetric tensor of nth order, which leads to the assumption

$$|x\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle + e^{i\delta_\pm^x}|z-\rangle),\ |y\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle + e^{i\delta_\pm^y}|z-\rangle). \tag{1.58}$$

Then by using orthogonal theorem together with experiment measurement result one has

$$\langle i+|i-\rangle = 0 \Longrightarrow \delta_-^i = \delta_+^i + \pi,\ |\langle y+|x-\rangle|^2 = \frac{1}{2} \Longrightarrow \delta_+^x = \delta_+^y \pm \frac{\pi}{2},$$

from which we may choose $\delta_+^x = 0$ and $\delta_+^y = \pi/2$, then

$$|x\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle \pm |z-\rangle),\ |y\pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle \pm i|z-\rangle), \tag{1.59}$$

where $\pm$ is also assigned to be bounded with eigenvalue, so

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{1.60}$$

Pseudo-Basis

By construction above we have three Pauli matrices, together with identity matrix

$$\sigma_0 \equiv \mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

we get a pseudo-basis for 2×2 Hermitian matrices — since σ_μ is Hermitian with additivity. They could form basis as they could have one-to-one map with 4 parameters a Hermitian matrix needs in two dimensions. (In fact, one could check it that unitary matrices also own 4 degrees of freedom, DoFs in 2-D, or more generally, Hermitian and unitary matrix both have N^2 DoFs in N-D.) And they are named with 'pseudo' since they do not share common orthogonal basis relation. However, they do own some algebra structure, and it turns out that they are exactly the generators of $SU(2)$ spin-rotation group for spin-half particles or 2-D representation except for quantum constant, as follows

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k,\ [\hat{S}_i, \hat{S}_j] = i\hbar\varepsilon_{ijk}\hat{S}_k, \tag{1.61}$$

which could be examined directly using either matrix form or ket composition. Beyond that, those stuffs possess another set of algebra known as *Clifford algebra*

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij},\ \{\hat{S}_i, \hat{S}_j\} = \frac{1}{2}\hbar^2\delta_{ij}, \tag{1.62}$$

which is rather important in relativistic quantum mechanics or field theory.

Come back to our basis, any 2×2 Hermitian matrix could be written as

$$H = d_0\sigma_0 + \boldsymbol{d}\cdot\boldsymbol{\sigma}, \tag{1.63}$$

as it is easy to verify that $\mathrm{Tr}(\sigma_i) = 0$, one has

$$d_\mu = \frac{1}{2}\mathrm{Tr}(H\sigma_\mu), \tag{1.64}$$

where relation

$$\sigma_i\sigma_j = \delta_{ij} + i\varepsilon_{ijk}\sigma_k \tag{1.65}$$

is used. One could obtain the eigenvalues and eigenkets of this matrix directly

$$\begin{aligned} E_+ &= d_0 + d,\ |n+\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix}, \\ E_- &= d_0 - d,\ |n-\rangle = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2)e^{i\phi} \end{pmatrix}, \end{aligned} \tag{1.66}$$

if we define $d = |\boldsymbol{d}|$ and $\hat{d} = \boldsymbol{d}/d = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. One could see it that vector $\hat{d}$ is defined on surface of a three dimensional sphere known as two-sphere or S^2 if varying (θ, ϕ) is allowed, which is usually called *Bloch sphere*. And when $d_0 = 0$,

$$H = d\,|n+\rangle\langle n+| - d\,|n-\rangle\langle n-|\,, \tag{1.67}$$

just like a quantum spin. In fact, H could be a Hamiltonian describing any two state problem.

If one wants to find the density matrix of a two state system, since it is also Hermitian with unit trace, one may conclude that under certain representation,

$$\rho = \frac{1}{2}(\mathbb{I} + \boldsymbol{r}\cdot\boldsymbol{\sigma}). \tag{1.68}$$

But what is $\boldsymbol{r}$? There should be some limitation on it in common sense. Let us look at pure state first. A pure state must be a quantum spin state in a sense that it is an eigenket of some $\sigma_n = \boldsymbol{\sigma}\cdot\hat{n}$, where $\hat{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, then

$$|\alpha\rangle \sim e^{i\delta}\begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix}, \tag{1.69}$$

where δ is a $U(1)$ gauge phase factor, and

$$\hat{\psi} = |\alpha\rangle\langle\alpha| \sim \begin{pmatrix} \cos^2(\theta/2) & \sin(\theta/2)\cos(\theta/2)e^{-i\phi} \\ \sin(\theta/2)\cos(\theta/2)e^{i\phi} & \cos^2(\theta/2) \end{pmatrix} = \frac{1}{2}(\mathbb{I} + \boldsymbol{\sigma}\cdot\hat{n}) = \psi. \tag{1.70}$$

It is easy to verify that (if such expression is allowed)

$$\sigma_n\,|\alpha\rangle = (2\psi - \mathbb{I})\,|\alpha\rangle = |\alpha\rangle\,, \tag{1.71}$$

which just corresponds our definition to it — $|n+\rangle$.

And now it is time dealing with mixed state, say the simplest case $\hat{\rho} = p_1\hat{\psi}_1 + p_2\hat{\psi}_2$ with $p_1 + p_2 = 1$, then

$$\rho = \frac{1}{2}[\mathbb{I}(p_1 + p_2) + \boldsymbol{\sigma}\cdot(p_1\hat{n_1} + p_2\hat{n_2})] \equiv \frac{1}{2}(\mathbb{I} + \boldsymbol{\sigma}\cdot\boldsymbol{r}), \tag{1.72}$$

where $\boldsymbol{r} = p_1\hat{n_1} + p_2\hat{n_2}$, and clearly, by vector's addition, $r = |\boldsymbol{r}| < 1$, and by argument above for arbitrary two-state system, one has

$$\hat{\rho} = \frac{1+r}{2}\,|r+\rangle\langle r+| + \frac{1-r}{2}\,|r-\rangle\langle r-|\,, \tag{1.73}$$

which deviates from pure state case as 'opposite spin state' appears. And r could be referred to the purity of one system, and $r = 0$ is max-mixed state.

The state space of a two-level system is the Bloch ball, with pure states on the surface (the Bloch sphere) and mixed states inside. The density matrix is a convex combination of pure states, and the extreme points of this convex set are exactly the pure states.

For two pure ensembles, distance between quantum states could be defined as

$$\begin{aligned} |\langle\alpha_1|\alpha_2\rangle|^2 &= \mathrm{Tr}\Big(\hat{\psi_1}\hat{\psi_2}\Big) = \mathrm{Tr}(\psi_1\psi_2) = \frac{1}{4}\,\mathrm{Tr}\left[(\mathbb{I}+\boldsymbol{\sigma}\cdot\hat{n}_1)(\mathbb{I}+\boldsymbol{\sigma}\cdot\hat{n}_2)\right] \\ &= \frac{1}{2}+\frac{1}{4}\,\mathrm{Tr}[(\boldsymbol{\sigma}\cdot\hat{n}_1)(\boldsymbol{\sigma}\cdot\hat{n}_2)] = \frac{1}{2}(1+\hat{n}_1\cdot\hat{n}_2), \end{aligned} \tag{1.74}$$

where relation

$$\begin{aligned} (\boldsymbol{\sigma}\cdot\hat{n}_1)(\boldsymbol{\sigma}\cdot\hat{n}_2) &= \sigma_i n_{1i}\sigma_j n_{2j} = n_{1i}n_{2j}(\delta_{ij}+i\varepsilon_{ijk}\sigma_k) \\ &= \hat{n}_1\cdot\hat{n}_2 + i\boldsymbol{\sigma}\cdot(\hat{n}_1\times\hat{n}_2) \end{aligned} \tag{1.75}$$

is used. Similarly,

$$\mathrm{Tr}(\hat{\rho}_1\hat{\rho}_2) = \frac{1}{2}(1+\boldsymbol{r}_1\cdot\boldsymbol{r}_2). \tag{1.76}$$

This distance is known as Frobenius norm.

If we do measurement along direction $\hat{m}$ to our ensemble, since

$$\hat{P}^m_+ = |m+\rangle\langle m+| = \frac{1}{2}(\mathbb{I}+\boldsymbol{\sigma}\cdot\hat{m}),$$

we find it that,

$$P(m+) = \mathrm{Tr}\Big(\hat{\rho}\hat{P}^m_+\Big) = \frac{1}{2}(1+\boldsymbol{r}\cdot\hat{m}). \tag{1.77}$$

In this way we conclude that quantum spins of silver atoms in SG experiment form a max-mixed state with $r = 0$, so that any first observation gives half-half result along any direction.

Seeing the deep insight of quantum mechanics, however, some may still want to recover the glory of classical physics, and in this time, they argue that they could expand the 'stochastic variable' to a general case as 'hidden-parameter' controlled probability — remember that there should be not concept like ket classically which would lead to the unknowing collapse — know as *Einstein-Podolsky-Rosen paradox* (EPR paradox), which was finally burnt to the ground by examination of Bell's inequality. On the other hand, the stochastic variable may still have some interest as one know from filed theory that disorder average of classical dynamics could emerge lower-one-dimensional quantum mechanics.

** General Solution for Two State Problem*

Here we in fact mean time-independent version. We ignore the σ_0 term and write

$$H = \boldsymbol{h}\cdot\boldsymbol{\sigma} = \begin{pmatrix} h_3 & h_1 - ih_2 \\ h_1 + ih_2 & -h_3 \end{pmatrix}, \tag{1.78}$$

and $\boldsymbol{h} = (h_1, h_2, h_3)$ may depend on extra parameters. By assuming

$$Hu = Eu,\ u = (u_1, u_2)^T, \tag{1.79}$$

it is easy to get

$$\det(H - E\mathbb{I}_2) = E^2 - h_1^2 - h_2^2 - h_3^2 = 0 \Rightarrow E_\pm = \pm h = \pm\sqrt{h_1^2 + h_2^2 + h_3^2}.$$

Then by bring, say $E_- = -h$ back into the equation one would have

$$\begin{pmatrix} h + h_3 & h_1 - ih_2 \\ h_1 + ih_2 & h - h_3 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0,$$

and two different ways to solve this problem appear (physically equivalent).

First way is by observing that we could divide both side by h as long as it is non-zero (gap-closing more professionally), and

$$\begin{pmatrix} 1 + h_3/h & h_1/h - ih_2/h \\ h_1/h + ih_2/h & 1 - h_3/h \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0,$$

which inspires us to define a unit vector $\hat{h} = \boldsymbol{h}/h = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, $\phi = \arctan\dfrac{h_2}{h_1}$, $\theta = \arccos\dfrac{h_3}{h}$, bringing which back would yield

$$\begin{pmatrix} 1 + \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & 1 - \cos\theta \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0,$$

then by focusing on one line one would have (say the first)

$$(1 + \cos\theta)u_1 + \sin\theta e^{-i\phi}u_2 = 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2}u_1 + \sin\frac{\theta}{2}e^{-i\phi}u_2\right) = 0,$$

and one is able to choose

$$u_- = \begin{pmatrix} \sin\frac{\theta}{2}e^{-i\phi} \\ -\cos\frac{\theta}{2} \end{pmatrix} \text{ or } \begin{pmatrix} -\sin\frac{\theta}{2} \\ \cos\frac{\theta}{2}e^{i\phi} \end{pmatrix}$$

depending on the choice. Similarly, for E_+ one would have

$$u_+ = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}e^{i\phi} \end{pmatrix} \text{ or } \begin{pmatrix} \cos\frac{\theta}{2}e^{-i\phi} \\ \sin\frac{\theta}{2} \end{pmatrix}.$$

The other way is to directly simplify the coefficient by observing for the first line

$$u_1 \propto -(h_1 - ih_2),\ u_2 \propto h + h_3,$$

and the normalization condition would give a norm as

$$N^2 = |u_1|^2 + |u_2|^2 = 2h(h + h_3),$$

and one has

$$u_- = \frac{1}{\sqrt{2h(h + h_3)}}\begin{pmatrix} -(h_1 - ih_2) \\ h + h_3 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} -e^{i\phi}\sqrt{1 - \frac{h_3}{h}} \\ \sqrt{1 + \frac{h_3}{h}} \end{pmatrix},\ e^{i\phi} = \frac{h_1 - ih_2}{\sqrt{h_1^2 + h_2^2}}.$$

And similarly for $E_+ = h$,

$$u_1 \propto h_1 - ih_2,\ u_2 \propto h - h_3,\ N^2 = 2h(h - h_3),$$

and one has

$$u_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{i\phi}\sqrt{1 + \frac{h_3}{h}} \\ \sqrt{1 - \frac{h_3}{h}} \end{pmatrix},$$

Completeness of $\mathfrak{su}(2)$ Algebra

Indeed, the pseudo-basis nature of Pauli matrices could be further explained by the completeness of $\mathfrak{su}(2)$ algebra. Consider a general 2×2 Hermitian matrix

$$h = \frac{1}{2}\operatorname{Tr}(h)\sigma_0 + \sum_i \frac{1}{2}\operatorname{Tr}(h\sigma_i)\sigma_i.$$

Then one could let $h_{\alpha\beta} = \delta_{\alpha\delta}\delta_{\beta\gamma}$, which leads to

$$\begin{aligned}\delta_{\alpha\delta}\delta_{\beta\gamma} &= \frac{1}{2}\sum_\nu h_{\nu\nu}\delta_{\alpha\beta} + \sum_i \frac{1}{2}\left(\sum_{\mu\nu} h_{\mu\nu}(\sigma_i)_{\nu\mu}\right)(\sigma_i)_{\alpha\beta} \\ &= \frac{1}{2}\delta_{\alpha\beta}\delta_{\delta\gamma} + \sum_i \frac{1}{2}(\sigma_i)_{\alpha\beta}(\sigma_i)_{\delta\gamma},\end{aligned}$$

which directly leads to

$$\sum_i (\sigma_i)_{\alpha\beta}(\sigma_i)_{\delta\gamma} = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\delta\gamma}. \tag{1.80}$$

1.5.2 General measurement — POVM

More generally in quantum mechanics, a measurement is described by a positive operator-valued measure (POVM), which assigns a set of positive semi-definite operators $\{\hat{E}_i\}$ that satisfy the completeness relation

$$\sum_i \hat{E}_i = \mathbb{I},$$

where especially $i = 1, \cdots, m$ with generally $m \geq N$ for an N-dimensional Hilbert space (note that $\hat{E}_i$s are still represented as $N \times N$ matrices). Each $\hat{E}_i$ corresponds to the outcome i of the measurement, and the probability of obtaining this outcome when measuring a state $\hat{\rho}$ is given by

$$p(i) = \operatorname{Tr}\left(\hat{E}_i\hat{\rho}\right).$$

POVMs are more general than projective measurements, in a sense that it extends the idea that after measurement, the state collapses to an eigenstate of the observable being measured, to that the post-measurement state could be a mixed state.

Then what is the state after PVOM? A first fact is that given semi-positive operator $\hat{E}_i$, there exists a unique semi-positive operator $\hat{M}_i = \sqrt{\hat{E}_i}$ such that

$$\hat{M}_i^2 = \hat{E}_i.$$

This one is easy since $\hat{E}_i$ could be diagonalized as $\hat{E}_i = \sum_n \lambda_n^i \left|e_n^i\right\rangle\!\left\langle e_n^i\right|$ with $\lambda_n^i \geq 0$, then simply let

$$\hat{M}_i = \sum_n \sqrt{\lambda_n^i}\left|e_n^i\right\rangle\!\left\langle e_n^i\right|,$$

and indeed this is a special case of the Kraus operators, as they clearly satisfy

$$\sum_i \hat{M}_i^\dagger \hat{M}_i = \sum_i \hat{E}_i = \mathbb{I}.$$

Now by defining $\hat{\rho}_i$ as the post-measurement state after obtaining outcome i, again with similarity with projective measurement, where we will replace $\hat{P}_n$ by $\hat{M}_i$, we have

$$\hat{\rho}_i = \frac{\sqrt{\hat{E}_i}\hat{\rho}\sqrt{\hat{E}_i}}{\mathrm{Tr}\Big(\hat{E}_i\hat{\rho}\Big)}, \tag{1.81}$$

which is clearly semi-positive and of unit trace.

However, a post-PVOM state does not live 'eigenspace' of $\hat{E}_i$, as opposite to projective measurement. Indeed, we have

$$\mathrm{Tr}\Big(\hat{E}_i\hat{\rho}_i\Big) = \frac{\mathrm{Tr}\Big(\hat{E}_i^2\hat{\rho}\Big)}{\mathrm{Tr}\Big(\hat{E}_i\hat{\rho}\Big)},$$

which is not generally 1 unless $\hat{E}_i$ is a projector, i.e., $\hat{E}_i^2 = \hat{E}_i$.

With the post PVOM state, we can define the quantity called the *fidelity* between two states $\hat{\rho}$ and $\hat{\sigma}$ as

$$F(\hat{\rho},\hat{\sigma}) = \left(\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\hat{\sigma}\sqrt{\hat{\rho}}}\right), \tag{1.82}$$

and by letting $\hat{M} = \sqrt{\hat{\sigma}}\sqrt{\hat{\rho}}$, we find that $F(\hat{\rho},\hat{\sigma}) = |\hat{M}| = \mathrm{Tr}\sqrt{\hat{M}^\dagger\hat{M}}$. This shares the similar concept with overlap between two pure states, as $F(\hat{\psi}_1,\hat{\psi}_2) = |\langle\psi_1|\psi_2\rangle|$.

1.5.3 Uncertainty Relation

Compatible observables are defined as observables with commute relation, say

$$[\hat{A},\hat{B}] = 0.$$

This relation has no transitivity.

By compatible we in fact means that these two observables related to different physical phenomena could be observed and measured simultaneously, and mathematically, this means that two operators share the same set of eigenkets, or the matrices in some representation of the two observables could be diagonalized at the same time, i.e.,

$$\begin{aligned} \hat{A}\,|a_n,b_n\rangle = a_n\,|a_n,b_n\rangle\,,\ \hat{B}\,|a_n,b_n\rangle = b_n\,|a_n,b_n\rangle\,, \\ \hat{A} = \sum_n a_n\,|a_n,b_n\rangle\langle a_n,b_n|\,,\ \hat{B} = \sum_n b_n\,|a_n,b_n\rangle\langle a_n,b_n|\,. \end{aligned} \tag{1.83}$$

Now we could explain 'no transitivity'. In degenerate case, it could happen that $\hat{A}$ and $\hat{B}$ happen to share a set of eigenkets of $\hat{B}$ in some degenerate subspace of $\hat{A}$, while $\hat{A}$ could also share with $\hat{C}$, but with another set of eigenkets different from $\hat{B}$'s, in the same subspace — this could be realized since any unitary transformation of eigenkets in $\hat{A}$'s degenerate subspace would still be its eigenkets — and clearly $\hat{B}$ and $\hat{C}$ do not share the same set of eigenkets and they are *incompatible*. For example, the spin-half operators $\{\hat{S}_i\}$ are not compatible with each other from the commutation relation.

Uncertainty relation claims that for any two observables,

$$\langle(\Delta\hat{A})^2\rangle\,\langle(\Delta\hat{B})^2\rangle \geq \frac{1}{4}|\,\langle[\hat{A},\hat{B}]\rangle\,|^2, \tag{1.84}$$

where $\Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle$. The proof would utilize an analogy with triangle inequality named as Schwarz inequality

$$\langle\alpha|\alpha\rangle\,\langle\beta|\beta\rangle \geq |\,\langle\alpha|\beta\rangle\,|^2, \tag{1.85}$$

since we could generally let two states be normalized states, and utilize

$$|\,\langle\alpha|\beta\rangle\,|^2 = \mathrm{Tr}\Big(\hat{\psi}_\alpha\hat{\psi}_\beta\Big) = \frac{1}{2}(1 + \hat{n}_\alpha \cdot \hat{n}_\beta) \leq 1.$$

Then, by letting $|\alpha\rangle = \Delta\hat{A}\,|\rangle$, $|\beta\rangle = \Delta\hat{B}\,|\rangle$,

$$\begin{aligned}\langle(\Delta\hat{A})^2\rangle\,\langle(\Delta\hat{B})^2\rangle &\geq |\,\langle\Delta\hat{A}\Delta\hat{B}\rangle\,|^2 = \frac{1}{4}\left|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle + \langle\{\Delta\hat{A},\Delta\hat{B}\}\rangle\right|^2 \\ &= \frac{1}{4}\left|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle\right|^2 + \frac{1}{4}\left|\langle\{\Delta\hat{A},\Delta\hat{B}\}\rangle\right|^2 \geq \frac{1}{4}\left|\langle[\Delta\hat{A},\Delta\hat{B}]\rangle\right|^2,\end{aligned}$$

where the property eigenvalues of anti-Hermitian $[A, B]$ is purely imaginary and of Hermitian $\{A, B\}$ is purely real is used.

1.6 Combination of Systems

Clearly in reality, there are no isolated systems, and except for very specific cases like the Stern-Gerlach experiment, measurement requires joint evolution of combined systems. As a first thought, We may try to imagine how two independent systems are put together in language of quantum mechanics. Notice that we are utilizing a matrix formulation, and there is indeed not too many choices to make the combined system again described by a (bigger) matrix, namely, direct sum or tensor product, in similarity to addition and multiplication, respectively. These are both correct for independent or more generally non-interacting systems. Nevertheless, in direct sum, the two space stays back to back with no commutations, making it impossible to describe interaction and entanglement between two systems. Thus, the natural way is to use tensor product (in matrix representation, it becomes Kronecker product), which is also the only way to make the combined system's space to be a Hilbert space again. If one is familiar with differential geometry, one see that tensor product (and its symmetrization) leads to the construction of manifold basis, and so does it here.

Let us firstly focus on states. Giving two systems A and B, with Hilbert spaces $\mathscr{H}_A$ and $\mathscr{H}_B$, the combined system can be described by the tensor product space $\mathscr{H}_{AB} = \mathscr{H}_A \otimes \mathscr{H}_B$. The states of the combined system are then represented by vectors in this larger Hilbert space as

$$|\psi^0_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle\,. \tag{1.86}$$

And by assuming that we have a basis representation (with abbreviation) $|\psi_A\rangle = (\alpha_1, \cdots, \alpha_m)^T$, $|\psi_B\rangle = (\beta_1, \cdots, \beta_n)^T$, we get the product state

$$|\psi^0_{AB}\rangle = \begin{pmatrix}\alpha_1\,|\psi_B\rangle \\ \vdots \\ \alpha_m\,|\psi_B\rangle\end{pmatrix} = \begin{pmatrix}\alpha_1\beta_1 \\ \vdots \\ \alpha_1\beta_n \\ \vdots \\ \alpha_m\beta_1 \\ \vdots \\ \alpha_m\beta_n\end{pmatrix},$$

where $\dim(\mathscr{H}_{AB}) = mn = \dim(\mathscr{H}_A)\dim(\mathscr{H}_B)$. The state $|\psi_{AB}\rangle$ is a totally product state of two systems, while more often in our interest are the entangled states, which cannot be factored into a product of states of the individual systems. Now giving $\{|\psi_A^i\rangle\}_{i=1}^m$ and $\{|\psi_B^j\rangle\}_{j=1}^n$ orthonormal basis for $\mathscr{H}_A$ and $\mathscr{H}_B$, respectively, we can always express a general state in $\mathscr{H}_{AB}$

$$|\psi_{AB}\rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}\,|\psi_A^i\rangle \otimes |\psi_B^j\rangle\,, \tag{1.87}$$

where c_{ij} are complex coefficients satisfying $\sum_{ij}|c_{ij}|^2 = 1$. Thus, as long as there are two (or more) non-zero coefficients c_{ij}, the state $|\psi_{AB}\rangle$ can be entangled. The measurement rule is still a Born's, while at this time we are allowed to measure over one of the systems. Say that we measure B, then

$$p(B=j) = \left\|\langle\psi_B^j|\psi_{AB}\rangle\right\|^2 = \sum_{i=1}^{m}|c_{ij}|^2, \tag{1.88}$$

and we are now left with state

$$|\psi_{A,B=j}\rangle = \frac{1}{\sqrt{p(B=j)}}\sum_{i=1}^{m} c_{ij}\,|\psi_A^i\rangle \otimes |\psi_B^j\rangle\,. \tag{1.89}$$

For instance, we may consider the entangled Bell states

$$|\phi^{\pm}\rangle = \frac{1}{\sqrt{2}}|00\rangle \pm \frac{1}{\sqrt{2}}|11\rangle\,,$$
$$|\psi^{\pm}\rangle = \frac{1}{\sqrt{2}}|01\rangle \pm \frac{1}{\sqrt{2}}|10\rangle\,,$$

over the product of two spin-$\frac{1}{2}$ systems, where we have used abbreviation $|00\rangle = |0\rangle \otimes |0\rangle$ and so on. By considering $|\phi^{\pm}\rangle$, we find we have equal probability to measure 0 or 1 over system B, and if we measure 0, then we are left with $|0\rangle$ for system A, while if we measure 1, then we are left with $|1\rangle$ for system A.

The above combination lies at pure states, while for mixed states, only part of them, namely the separable states, can be expressed as tensor product over density matrices. Firstly, we have product systems, that by denoting $\hat{\rho}_A$ and $\hat{\rho}_B$ as independent density matrices for A and B, we have

$$\hat{\rho}_{AB}^0 = \hat{\rho}_A \otimes \hat{\rho}_B. \tag{1.90}$$

For instance, the above product state $|\psi_{AB}^0\rangle$ gives a pure state example for the case. Sometimes, we just have some product states for the combined system, but with a classical probability to distribute them, and we have

$$\hat{\rho}_{AB} = \sum_k p_k \left|\psi_A^k\right\rangle\!\left\langle\psi_A^k\right| \otimes \left|\psi_B^k\right\rangle\!\left\langle\psi_B^k\right|\,,\ \sum_k p_k = 1, \tag{1.91}$$

which gives rise to a *classically correlated quantum state*. While more generally, the state for each system is also in mixed state, thus giving

$$\hat{\rho}_{AB} = \sum_k p_k \hat{\rho}_A^k \otimes \hat{\rho}_B^k,\ \sum_k p_k = 1, \tag{1.92}$$

which is known as a *separable state* that shares also classical correlations between systems. However, not all density matrices of the combined system can be represented in this way, and when it is not, we call it an *entangled state* with true quantum entanglement between. For instance, the Bell state provides such a minimal example.

Assuming that generally we have got a combined system described by $\hat{\rho}_{AB}$, and we are only interested in system A, then we may introduce the concept of *reduced density matrix* by performing a *partial trace* over the total system that traces out system B:

$$\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}). \tag{1.93}$$

Concretely, assuming we know the basis representation $\{|\psi_B^j\rangle\}_{j=1}^n$ for system B, then

$$\hat{\rho}_A = \sum_{j=1}^{n} (\mathbb{I}_A \otimes \langle\psi_B^j|)\hat{\rho}_{AB}(\mathbb{I}_A \otimes |\psi_B^j\rangle), \tag{1.94}$$

and similarly for B, we have

$$\hat{\rho}_B = \sum_{i=1}^{m} (\langle\psi_A^i| \otimes \mathbb{I}_B)\hat{\rho}_{AB}(|\psi_A^i\rangle \otimes \mathbb{I}_B). \tag{1.95}$$

To represent it explicitly, let $\{|i\rangle\}_{i=1}^m$ and $\{|\alpha\rangle\}_{\alpha=1}^n$ be orthonormal basis for $\mathscr{H}_A$ and $\mathscr{H}_B$, respectively, then $\{|i\rangle \otimes |\alpha\rangle\}_{i,\alpha=1}^{m,n}$ is an orthonormal basis for $\mathscr{H}_{AB}$, and we may write most generally that

$$\hat{\rho}_{AB} = \sum_{i,j=1}^{m}\sum_{\alpha,\beta=1}^{n} \rho_{i\alpha,j\beta}\,|i\rangle\langle j| \otimes |\alpha\rangle\langle\beta| = \sum_{i,j=1}^{m}\sum_{\alpha,\beta=1}^{n} \rho_{i\alpha,j\beta}\,|i,\alpha\rangle\langle j,\beta|\,, \tag{1.96}$$

which is constrained by Hermiticity, normalization and positivity. The first two conditions give $\rho_{i\alpha,j\beta} = \rho^*_{j\beta,i\alpha}$ and $\sum_{i,\alpha}\rho_{i\alpha,i\alpha} = 1$, which is done by

$$\begin{aligned}
\mathrm{Tr}(\hat{\rho}_{AB}) &= \sum_{i,\alpha}\langle i,\alpha|\left(\sum_{j,k=1}^{m}\sum_{\beta,\gamma=1}^{n}\rho_{j\beta,k\gamma}\,|j,\beta\rangle\langle k,\gamma|\right)|i,\alpha\rangle \\
&= \sum_{i,\alpha}\sum_{j,k=1}^{m}\sum_{\beta,\gamma=1}^{n}\rho_{j\beta,k\gamma}\,\langle i,\alpha|j,\beta\rangle\,\langle k,\gamma|i,\alpha\rangle \\
&= \sum_{i,\alpha}\sum_{j,k=1}^{m}\sum_{\beta,\gamma=1}^{n}\rho_{j\beta,k\gamma}\delta_{ij}\delta_{\alpha\beta}\delta_{ki}\delta_{\gamma\alpha} = \sum_{i,\alpha}\rho_{i\alpha,i\alpha} = 1,
\end{aligned}$$

while positivity requires all eigenvalues of $\hat{\rho}_{AB}$ to be non-negative, or in its equivalent form, for any vector $|\psi\rangle = \sum_{k,\gamma} c_{k\gamma}\,|k\gamma\rangle$, the requirement

$$\langle\psi|\hat{\rho}_{AB}|\psi\rangle = \sum_{ij,\alpha\beta} c^*_{i\alpha}\rho_{i\alpha,j\beta}c_{j\beta} = \boldsymbol{c}^\dagger\rho\boldsymbol{c} \geq 0,$$

requires the matrix ρ to be positive semi-definite. Then, by performing partial trace over

B, we have

$$\begin{aligned}\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}) &= \sum_{\gamma=1}^{n}(\mathbb{I}_A \otimes \langle\gamma|)\left(\sum_{i,j=1}^{m}\sum_{\alpha,\beta=1}^{n}\rho_{i\alpha,j\beta}\,|i\rangle\langle j|\otimes|\alpha\rangle\langle\beta|\right)(\mathbb{I}_A\otimes|\gamma\rangle)\\ &= \sum_{i,j=1}^{m}\sum_{\alpha,\beta=1}^{n}\rho_{i\alpha,j\beta}\,|i\rangle\langle j|\left(\sum_{\gamma=1}^{n}\langle\gamma|\alpha\rangle\,\langle\beta|\gamma\rangle\right) = \sum_{i,j=1}^{m}\sum_{\alpha,\beta=1}^{n}\rho_{i\alpha,j\beta}\,|i\rangle\langle j|\,\delta_{\alpha\beta}\\ &= \sum_{i,j=1}^{m}\left(\sum_{\alpha=1}^{n}\rho_{i\alpha,j\alpha}\right)|i\rangle\langle j|\,,\end{aligned} \tag{1.97}$$

and similarly for B, we have

$$\hat{\rho}_B = \sum_{\alpha,\beta=1}^{n}\left(\sum_{i=1}^{m}\rho_{i\alpha,i\beta}\right)|\alpha\rangle\langle\beta|\,. \tag{1.98}$$

It is easy to check that both $\hat{\rho}_A$ and $\hat{\rho}_B$ satisfy Hermiticity, normalization and positivity (if ρ does).

For more about dynamics over such systems, see the Chapter 10, while this should certainly be down after Chapter 2.

1.6.1 Mutual information and Entropy

Let us consider now the entropy. Classically, we have Shannon entropy defined by

$$S_{\text{Shannon}} = -\sum_i p_i \ln p_i, \tag{1.99}$$

while quantum mechanically, we have von Neumann entropy

$$S_{\text{vN}} = -\,\mathrm{Tr}(\hat{\rho}\ln\hat{\rho}), \tag{1.100}$$

and one sees that by spectral decomposition $\hat{\rho} = \sum_i p_i\,|i\rangle\langle i|$, we have $S_{\text{vN}} = S_{\text{Shannon}}$. One important property of von Neumann entropy is its subadditivity, that is, for a combined system AB, we have

$$S_{\text{vN}}(\hat{\rho}_{AB}) \le S_{\text{vN}}(\hat{\rho}_A) + S_{\text{vN}}(\hat{\rho}_B), \tag{1.101}$$

or in another terminology, the *mutual information* is non-negative:

$$I(A:B) = S_{\text{vN}}(\hat{\rho}_A) + S_{\text{vN}}(\hat{\rho}_B) - S_{\text{vN}}(\hat{\rho}_{AB}) \ge 0, \tag{1.102}$$

with the equality holds if and only if $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$. This is proved through the quantum relative entropy (classically known as Kullback-Leibler divergence)

$$S(\hat{\rho}||\hat{\sigma}) = \mathrm{Tr}(\hat{\rho}\ln\hat{\rho}) - \mathrm{Tr}(\hat{\rho}\ln\hat{\sigma}) \ge 0, \tag{1.103}$$

with $\hat{\rho}$ and $\hat{\sigma}$ two density matrices for the same Hilbert space, and the equality holds if and only if $\hat{\rho} = \hat{\sigma}$. It's proof is based on spectral decomposition $\hat{\rho} = \sum_i p_i\,|i\rangle\langle i|$,

$\hat{\sigma} = \sum_\alpha q_\alpha |\alpha\rangle\langle\alpha|$, we have

$$\begin{aligned}
S(\hat{\rho}||\hat{\sigma}) &= \sum_i p_i \ln p_i - \sum_\alpha \langle\alpha|\hat{\rho}|\alpha\rangle \ln q_\alpha = \sum_i p_i \ln p_i - \sum_\alpha \left(\sum_i p_i |\langle\alpha|i\rangle|^2\right) \ln q_\alpha \\
&= \sum_i p_i \left(\ln p_i - \sum_\alpha |\langle\alpha|i\rangle|^2 \ln q_\alpha\right) = \sum_i p_i \ln \frac{p_i}{\prod_\alpha q_\alpha^{|\langle\alpha|i\rangle|^2}} = -\sum_i p_i \ln \frac{\prod_\alpha q_\alpha^{|\langle\alpha|i\rangle|^2}}{p_i} \\
&\geq -\sum_i p_i \left(\frac{\prod_\alpha q_\alpha^{|\langle\alpha|i\rangle|^2}}{p_i} - 1\right) = 1 - \sum_i \prod_\alpha q_\alpha^{|\langle\alpha|i\rangle|^2} = 1 - \sum_i \exp\left(\sum_\alpha |\langle\alpha|i\rangle|^2 \ln q_\alpha\right) \\
&\geq 1 - \sum_i \sum_\alpha |\langle\alpha|i\rangle|^2 q_\alpha = 1 - \sum_\alpha q_\alpha \left(\sum_i |\langle\alpha|i\rangle|^2\right) = 1 - \sum_\alpha q_\alpha = 0,
\end{aligned}$$

where in the third line we used inequality $\ln x \leq x - 1$ for $x > 0$, and in the fourth line we used Jensen's inequality that for convex function f, we have

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2),\ t \in [0,1],$$

and its generalization to $f(\mathbb{E}[X]) \leq \mathbb{E}f(X)$ with X a random variable, and $\mathbb{E}$ the expectation. Note that here $f(x) = e^x$ and $X = q_\alpha$ with probability $|\langle\alpha|i\rangle|^2$. Now by letting $\hat{\rho} = \hat{\rho}_{AB}$, $\hat{\sigma} = \hat{\rho}_A \otimes \hat{\rho}_B$, we have

$$\begin{aligned}
0 \leq S(\hat{\rho}_{AB}||\hat{\rho}_A \otimes \hat{\rho}_B) &= \mathrm{Tr}(\hat{\rho}_{AB} \ln \hat{\rho}_{AB}) - \mathrm{Tr}(\hat{\rho}_{AB} \ln(\hat{\rho}_A \otimes \hat{\rho}_B)) \\
&= -S_{\mathrm{vN}}(\hat{\rho}_{AB}) - \mathrm{Tr}(\hat{\rho}_{AB}(\ln \hat{\rho}_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes \ln \hat{\rho}_B)) \\
&= -S_{\mathrm{vN}}(\hat{\rho}_{AB}) - \mathrm{Tr}((\mathrm{Tr}_B(\hat{\rho}_{AB})) \ln \hat{\rho}_A) - \mathrm{Tr}((\mathrm{Tr}_A(\hat{\rho}_{AB})) \ln \hat{\rho}_B) \\
&= -S_{\mathrm{vN}}(\hat{\rho}_{AB}) + S_{\mathrm{vN}}(\hat{\rho}_A) + S_{\mathrm{vN}}(\hat{\rho}_B),
\end{aligned}$$

and thus the subadditivity is proved. The equality holds if and only if $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$. The proof can rely on Klein's inequality that for convex function f, the operator

$$f(\hat{A}) - f(\hat{B}) - (\hat{A} - \hat{B})f'(\hat{B})$$

is semi-positive definite, and by applying a trace with $f(x) = x \ln x$, we obtain the same result.

1.6.2 Pure entangled states and Schmidt decomposition

The most clean example is a pure entangled state $\hat{\rho}_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|$, where $|\psi_{AB}\rangle = \sum_{i,\alpha} c_{i\alpha} |i,\alpha\rangle$. Then, we have $S_{\mathrm{vN}}(\hat{\rho}_{AB}) = 0$, while the reduced density matrices are generally in mixed states, and thus $S_{\mathrm{vN}}(\hat{\rho}_A), S_{\mathrm{vN}}(\hat{\rho}_B) \geq 0$, with the equality holds if and only if $\hat{\rho}_{AB}$ is a product state. We can further show that $S_{\mathrm{vN}}(\hat{\rho}_A) = S_{\mathrm{vN}}(\hat{\rho}_B)$ by Schmidt decomposition that

$$|\psi_{AB}\rangle = \sum_k \sqrt{\lambda_k} |u_k\rangle \otimes |v_k\rangle,\ \sum_k \lambda_k = 1,$$

and $\hat{\rho}_{AB} = \sum_{ij} \sqrt{\lambda_i \lambda_j} |u_i v_i\rangle\langle u_j v_j|$, $\hat{\rho}_A = \sum_i \lambda_i |u_i\rangle\langle u_i|$, $\hat{\rho}_B = \sum_i \lambda_i |v_i\rangle\langle v_i|$. The Schmidt decomposition is based on the singular-value-decomposition (SVD) of matrix $C = (c_{i\alpha})$, that is, $C = U\Sigma V^\dagger$, with U and V two unitary matrices, and Σ a diagonal matrix

with non-negative (by convention) diagonal elements. Then by letting $|u_i\rangle = \sum_j U_{ji} |j\rangle$, $|v_i\rangle = \sum_\beta V^*_{\beta i} |\beta\rangle$, and $\lambda_i = \Sigma^2_{ii}$, we get the result.

There exists an elegant way to find the Schmidt decomposition, that by spectral decomposition of

$$\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}) = \sum_i \lambda_i |u_i\rangle\langle u_i| ,$$

with certainly $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and $\{|u_i\rangle\}$ forming (part of) orthonormal basis for $\mathscr{H}_A$. Then, we may define

$$|v_i\rangle = \frac{(\langle u_i| \otimes \mathbb{I}_B) |\psi_{AB}\rangle}{\sqrt{\lambda_i}},$$

and we claim that $|\psi_{AB}\rangle = \sum_i \sqrt{\lambda_i} |u_i\rangle \otimes |v_i\rangle$ gives the Schmidt decomposition. To see this, we firstly check

$$\begin{aligned} |\psi_{AB}\rangle &= \sum_i \sqrt{\lambda_i} |u_i\rangle \otimes |v_i\rangle = \sum_i |u_i\rangle \otimes (\langle u_i| \otimes \mathbb{I}_B) |\psi_{AB}\rangle \\ &= \left(\sum_i |u_i\rangle\langle u_i| \otimes \mathbb{I}_B \right) |\psi_{AB}\rangle = (\mathbb{I}_A \otimes \mathbb{I}_B) |\psi_{AB}\rangle = |\psi_{AB}\rangle , \end{aligned}$$

and then utilize

$$\begin{aligned} \hat{\rho}_A &= \mathrm{Tr}_Y(|\psi_{AB}\rangle\langle\psi_{AB}|) = \sum_{ij} \sqrt{\lambda_i \lambda_j} |u_i\rangle\langle u_j| \langle v_j|v_i\rangle \\ &= \sum_i \lambda_i |u_i\rangle\langle u_i| , \end{aligned}$$

thus by calculating

$$\langle u_i|\hat{\rho}_A|u_j\rangle = \sqrt{\lambda_i \lambda_j} \langle v_j|v_i\rangle = \lambda_i \delta_{ij},$$

we have $\langle v_j|v_i\rangle = \delta_{ij}$, and thus $\{|v_i\rangle\}$ also forms (part of) orthonormal basis for $\mathscr{H}_B$.

1.7 Position & Momentum — Infinite Hilbert Space

Classically, the Poisson bracket is basically

$$\{x_i, p_j\} = \sum_n \left(\frac{\partial x_i}{\partial x_n} \frac{\partial p_j}{\partial p_n} - \frac{\partial x_i}{\partial p_x} \frac{\partial p_j}{\partial x_n} \right) = \delta_{ij},$$

over the symplectic phase space $\{x_i, p_i\}$, which gives us

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

quantum mechanically under canonical quantization. And by uncertainty relation,

$$\langle (\Delta\hat{x})^2 \rangle \langle (\Delta\hat{p})^2 \rangle \geq \frac{\hbar^2}{4}, \tag{1.104}$$

or simply, $\Delta x \Delta p \geq \hbar/2$. However, this is not satisfactory enough.

Now since we come to uncountable infinite dimensional Hilbert space, we need to deal with continuous spectrum of an operator, say

$$\hat{\xi} |\xi\rangle = \xi |\xi\rangle , \tag{1.105}$$

whose action and behaviour, are just like the discrete one we discussed using matrix form, but with two things changed. One thing is we replace *Kronecker symbol* by *Dirac's δ function*

$$\delta_{aa'} \longrightarrow \delta(\xi - \xi'), \tag{1.106}$$

and the other is we treat the summation to be integral (generally, this is under Lebesgue's definition)

$$\sum_a \longrightarrow \int \mathrm{d}\xi. \tag{1.107}$$

In this way we also have orthogonal and complete relations

$$\langle \xi | \xi' \rangle = \delta(\xi - \xi'), \ \int \mathrm{d}\xi \, |\xi\rangle\langle\xi| = \mathbb{I}, \tag{1.108}$$

from which we also have projection of a ket

$$|\psi\rangle = \int \mathrm{d}\xi \, |\xi\rangle \, \langle \xi | \psi \rangle = \int \mathrm{d}\xi \ \psi(\xi) \, |\xi\rangle \, , \tag{1.109}$$

where $\psi(\xi)$ is known as the *wavefunction* under representation of $\hat{\xi}$, which could be obtained by projection

$$\hat{P}(\xi) = |\xi\rangle\langle\xi| \, , \ \hat{P}(\xi) \, |\psi\rangle = \psi(\xi) \, |\xi\rangle \, . \tag{1.110}$$

However, by Born's rule, the square of wavefunction is not probability but rather its density since measure $\mathrm{d}\xi$ is not included here. And we may define a probability density function

$$\rho(\xi) = |\psi(\xi)|^2 = |\langle \xi | \psi \rangle|^2 \, , \ \int \mathrm{d}\xi \rho(\xi) = \langle \psi | \left(\int \mathrm{d}\xi \, |\xi\rangle\langle\xi| \right) |\psi\rangle = 1, \tag{1.111}$$

promised that $|\psi\rangle$ is normalizable and normalized. And the probability measuring ξ to be around ξ_0 may be written as

$$P(\xi_0 \pm \Delta\xi) = \int_{\xi_0 - \Delta\xi}^{\xi_0 + \Delta\xi} \mathrm{d}\xi \ \rho(\xi). \tag{1.112}$$

One should notice it that there is no trace form here like in discrete, since trace for 'infinite dimensional matrix' loses its meaning.

Dirac's δ Function

It is defined as a generalized function known as *functional* satisfies that

$$\int_{-\infty}^{+\infty} \mathrm{d}x \ \delta(x) f(x) = f(0). \tag{1.113}$$

And especially,

$$\int_{-\infty}^{+\infty} \mathrm{d}x \ \delta(x) = 1. \tag{1.114}$$

One most important property of δ function may be summarized as

$$\delta[f(x)] = \sum_i \frac{1}{|f'(x)|_{x=x_i}} \delta(x - x_i), \ f(x_i) = 0. \tag{1.115}$$

Another property is

$$f(x)\delta'(x) = -f'(x)\delta(x), \tag{1.116}$$

which could be verified by integration as

$$\int f(x)\delta'(x)\mathrm{d}x = f(x)\delta(x) - \int f'(x)\delta(x)\mathrm{d}x = -\int f'(x)\delta(x)\mathrm{d}x.$$

From its *Fourier transform* one has

$$\int_{-\infty}^{+\infty} \mathrm{d}x\ e^{ikx}\delta(x) = 1 \Longrightarrow \int_{-\infty}^{+\infty} \mathrm{d}k\ e^{-ikx}\cdot 1 = 2\pi\delta(x), \tag{1.117}$$

it is then necessary to give a normalization factor $1/\sqrt{2\pi}$ to each plane wave e^{ikx} if the integral is done on $x-k$. Say $u_k(x) = e^{ikx}/\sqrt{2\pi}$, then

$$\langle u_k|u_{k'}\rangle = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \mathrm{d}x\ e^{-i(k-k')x} = \delta(k-k'). \tag{1.118}$$

Mathematically there could be limitation definition to approach δ function

$$\delta(x) = \lim_{g\to+\infty}\frac{\sin(gx)}{\pi x} = \lim_{\sigma\to 0^+}\frac{1}{\sqrt{\pi}\sigma}e^{-x^2/\sigma^2} = -\frac{1}{\pi}\lim_{\delta\to 0^+}\mathrm{Im}\,\frac{1}{x+i\delta}. \tag{1.119}$$

(*Illustration*) To make sure that the three δ function defined above is proper, i.e., non-zero and approaches infinity only at zero with integral to one, one needs to bring out the integral. The second one, i.e., Gaussian function, is obvious, while the third one employs Dirac's identity following an extension to complex plane with contour integral. Here we focus on the first one by Feynman trick, i.e.,

$$\int_{-\infty}^{+\infty}\mathrm{d}x\frac{\sin(gx)}{\pi x} = \frac{2}{\pi}\int_0^{+\infty}\mathrm{d}x\frac{\sin(x)}{x} = \frac{2}{\pi}\lim_{s\to 0^+} I(s),$$

where in the second step we have made substitution $x\to gx$, while integral

$$I(s) = \int_0^{+\infty}\mathrm{d}x e^{-sx}\frac{\sin(x)}{x}.$$

Now firstly we observe that $I(+\infty) = 0$ by exponential suppressing, then since

$$\frac{\mathrm{d}}{\mathrm{d}s}I(s) = -\int_0^{+\infty}\mathrm{d}x e^{-sx}\sin(x) = \mathrm{Im}\int_0^{+\infty}\mathrm{d}x e^{-(s+i)x} = \mathrm{Im}\,\frac{1}{s+i} = -\frac{1}{s^2+1},$$

we find

$$I(s) = I(0) - \int_0^s \mathrm{d}s\frac{1}{s^2+1} = I(0) - \arctan(s),$$

with gives by letting $s\to+\infty$ that

$$I(0) = \frac{\pi}{2},$$

and the above wanted integral just goes to identity, while clearly

$$\lim_{g\to+\infty}\lim_{x\to 0}\frac{\sin(gx)}{x} = \lim_{g\to+\infty} g\to+\infty.$$

Translation of Position — Physical Guess to Momentum

Position operator is simply defined as

$$\hat{x}\,|x\rangle = x\,|x\rangle\,, \tag{1.120}$$

with $|x\rangle$ living in infinite-dim $\mathscr{H}$ — which means that there would be infinite possible observable positions in real space. Notice that in vector version,

$$\hat{\boldsymbol{x}}\,|\boldsymbol{x}\rangle = \boldsymbol{x}\,|\boldsymbol{x}\rangle\,, \tag{1.121}$$

which means that three component could find eigenkets simultaneously, so

$$[\hat{x}_i, \hat{x}_j] = 0. \tag{1.122}$$

By projection

$$|\psi\rangle = \int \mathrm{d}x\ \psi(x)\,|x\rangle\,, \tag{1.123}$$

where $\psi(x) = \langle x|\psi\rangle$ is wavefunction by position. And for an operator,

$$\hat{O} = \int \mathrm{d}x\mathrm{d}x'\,|x\rangle\langle x|\,\hat{O}\,|x'\rangle\langle x'| = \int \mathrm{d}x\mathrm{d}x' O(x,x')\,|x\rangle\langle x'|\,, \tag{1.124}$$

where $O(x,x') = \langle x|\hat{O}|x'\rangle$ as spatial correlation.

Since position is continuous, we may introduce a parameterized transformation to describe the connection of different position kets, namely, translation, by this we mean that it only acts on DoFs of position

$$\hat{T}(\delta x)\,|x\rangle = |x+\delta x\rangle\,, \tag{1.125}$$

and from the normalization condition $\langle x+\delta x|x+\delta x\rangle = 1$, one sees unitality of $\hat{T}$

$$\hat{T}^\dagger(\delta x)\hat{T}(\delta x) = \mathbb{I}, \tag{1.126}$$

and by examining other properties, it turns out that $\hat{T}(s)$ is just one of single-parameter unitary operator group we defined above, and in fact a Lie group. Anyway, it should own a generator named as $\hat{k}$

$$\hat{T}(\varepsilon) = \mathbb{I} - i\hat{k}\varepsilon, \tag{1.127}$$

and

$$\begin{aligned} &[\hat{x}, \hat{T}(\varepsilon)]\,|x\rangle = \hat{x}\hat{T}(\varepsilon)\,|x\rangle - \hat{T}(\varepsilon)x\,|x\rangle = \varepsilon\hat{T}(\varepsilon)\,|x\rangle \\ \Longrightarrow\ &-i[\hat{x},\hat{k}]\varepsilon\,|x\rangle = \varepsilon\,|x\rangle + O(\varepsilon^2), \end{aligned}$$

from which we conclude that

$$[\hat{x}, \hat{k}] = i, \tag{1.128}$$

which is quite close to our canonical quantization result. In fact, using *de Broglie's relation* as a physical postulate

$$p = \frac{h}{\lambda} = \hbar k, \tag{1.129}$$

where k is the wavevector (in this 1-d case a scalar) describing the propagation of matter wave, and generalizing it to quantum operators

$$\hat{p} = \hbar\hat{k}, \tag{1.130}$$

one has

$$[\hat{x}, \hat{p}] = i\hbar,\ \hat{T}(\delta x) = \exp(-i\hat{p}\delta x/\hbar), \tag{1.131}$$

where $\hat{p}$ is introduced as *momentum operator*. Notice here that we seem to firstly recognize $\hat{k}$ as, in some sense, 'wavevector operator' and then introduce momentum operator using the postulate. This is just an analogy with classical physics, since we would say that momentum generates spatial translation while Hamiltonian generates time evolution classically. And in this way we may say that we first think $\hat{p}$ to be proportional to $\hat{k}$, and then treat $\hat{k}$ as something related to wavevector as two sides of nature, particle and wave.

To see their representation in position space, from the infinitesimal translation

$$\begin{aligned} \langle x|\,\hat{T}(\varepsilon) = \langle x|\,(1 - i\hat{k}\varepsilon) = [\hat{T}^\dagger(\varepsilon)\,|x\rangle]^\dagger = \langle x - \varepsilon| \\ \implies \quad \langle x|\,\hat{k} = \lim_{\varepsilon\to 0} i\frac{\langle x-\varepsilon| - \langle x|}{\varepsilon} = -i\partial_x\,\langle x|\,, \end{aligned}$$

then by right action of $|x'\rangle$,

$$\langle x|\hat{k}|x'\rangle = (-i\partial_x\,\langle x|)\,|x'\rangle = -i\partial_x\,\langle x|x'\rangle = -i\partial_x\delta(x - x'), \tag{1.132}$$

or roughly speaking, $k \sim -i\partial_x$, then,

$$\langle x|\hat{p}|x'\rangle = -i\hbar\partial_x\delta(x - x'),\ p \sim -i\hbar\partial_x.$$

One could verify that $[x, p] = i\hbar$ in position representation.

In fact, momentum operator has its own definition as

$$\hat{p}\,|p\rangle = p\,|p\rangle\,, \tag{1.133}$$

and similarly, in multi-component case,

$$\hat{\boldsymbol{p}}\,|\boldsymbol{p}\rangle = \boldsymbol{p}\,|\boldsymbol{p}\rangle\,,\ [p_i, p_j] = 0, \tag{1.134}$$

and the translation, together with generators and commutation relation would be

$$\hat{\boldsymbol{T}}(\delta\boldsymbol{x})\,|\boldsymbol{x}\rangle = |\boldsymbol{x} + \delta\boldsymbol{x}\rangle\,,\ \hat{\boldsymbol{T}}(\boldsymbol{\varepsilon}) = \mathbb{I} - i\hat{\boldsymbol{k}}\cdot\boldsymbol{\varepsilon},\ \ \hat{\boldsymbol{T}}(\delta\boldsymbol{x}) = \exp(-i\boldsymbol{k}\cdot\delta\boldsymbol{x}); \tag{1.135}$$

$$\hat{\boldsymbol{p}} = \hbar\hat{\boldsymbol{k}}, \qquad [\hat{k}_i, \hat{k_j}] = 0, \qquad \langle\boldsymbol{x}|\hat{\boldsymbol{p}}|\boldsymbol{x}'\rangle = -i\hbar\nabla_x\delta(\boldsymbol{x} - \boldsymbol{x}'). \tag{1.136}$$

Now since

$$\begin{aligned} \langle x|\hat{p}|p\rangle &= \langle x|\,(\hat{p}\,|p\rangle) = p\,\langle x|p\rangle \\ &= (\langle x|\,\hat{p})\,|p\rangle = -i\hbar(\partial_x\,\langle x|)\,|p\rangle = -i\hbar\partial\,\langle x|p\rangle\,, \end{aligned}$$

one solves

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar} \tag{1.137}$$

as a plane wave solution, which is also the wavefunction of eigenket of momentum in position space, and the coefficient is like one in δ function but with an additional $\hbar$ inside since now the normalization is done on $x - p$. Similarly,

$$\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar}. \tag{1.138}$$

Then we could define Fourier transformation of an arbitrary ket in two representations as

$$\begin{aligned}\psi(x) &= \langle x|\psi\rangle = \langle x| \int \mathrm{d}p\, |p\rangle \langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}p\; e^{ipx}\psi(p) \\ \psi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x\; e^{-ipx}\psi(x).\end{aligned} \tag{1.139}$$

And one could verify directly using average by wavefunction

$$\langle\psi|\hat{O}|\psi\rangle = \int \mathrm{d}x\mathrm{d}x'\psi^\dagger(x)O(x,x')\psi(x'), \tag{1.140}$$

where we use † to represent possible multi-component, to evaluate and verify $\Delta x\Delta p \geq \hbar/2$. And at many times, one may need to treat Gaussian package integral like

$$\int_{-\infty}^{+\infty} \mathrm{d}x\; e^{-ax^2} = \sqrt{\frac{\pi}{a}},\ \int \mathrm{d}^n x\; e^{-x^T Ax} = \sqrt{\frac{\pi^n}{\det(A)}}. \tag{1.141}$$

Now we have another relation

$$\langle p|\,\hat{x} = \int \mathrm{d}x\, \langle p|x\rangle \langle x|\,\hat{x} = \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x\, \langle x|\, xe^{-ipx} = \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x\, \langle x|\, (i\partial_p e^{-ipx}),$$

and

$$\langle p|\hat{x}|p'\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x\, \langle x|p'\rangle\, (i\partial_p e^{-ipx}) = i\partial_p \frac{1}{2\pi\hbar} \int \mathrm{d}x e^{-i(p-p')x} = i\partial_p \delta(p-p'). \tag{1.142}$$

Chapter 2

Quantum Dynamics

2.1 Pictures

Like translation in space, we could also define a parameterized unitary group to describe 'translation' — or formally, evolution of a ket in time. Notice that we do not have some time-representation or $|t\rangle$, at least in quantum mechanics, and what we have is a state $|\psi\rangle$ that evolves with time $|\psi(t)\rangle$.

$$\hat{U}(t)\,|\psi(0)\rangle = |\psi(t)\rangle\,. \tag{2.1}$$

We could assign a generator to our time evolution operator

$$\hat{U}(\varepsilon) = \mathbb{I} - i\hat{\Omega}\varepsilon, \tag{2.2}$$

and by another postulate named as *Einstein-Planck relation*

$$E = \hbar\omega, \tag{2.3}$$

we say that

$$\hat{H} = \hbar\hat{\Omega}, \tag{2.4}$$

where $\hat{H}$ is the quantized Hamiltonian that totally governs the quantum world, who is also Hermitian as a requirement of unitary evolution with eigenvalues treated as energy

$$\hat{H}\,|E_n\rangle = E_n\,|E_n\rangle\,. \tag{2.5}$$

It could be proposed in this way, that any stuff, owns at least a trend to travel like a wave, i.e.

$$\exp(ik\cdot x - i\omega t) = \exp(-ik_\mu x^\mu),$$

by physical assumption, we subtract these properties and relate them with quantized quantities that could have meaning classically,

$$\hat{p}_\mu \sim \hbar k_\mu,$$

as shown before and above.

2.1.1 Schrödinger Picture

"We observe how state evolves and leave others along."

To achieve which we need the evolution of time evolution operator

$$\hat{U}(t+\mathrm{d}t) = \hat{U}(\mathrm{d}t)\hat{U}(t) = (\mathbb{I} - i\hat{H}\mathrm{d}t/\hbar)\hat{U}(t) = (1+\mathrm{d}t\partial_t)\hat{U}(t),$$

and

$$i\hbar\frac{\partial}{\partial t}\hat{U}(t) = \hat{H}(t)\hat{U}(t), \tag{2.6}$$

and by adding initial state,

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle, \tag{2.7}$$

and by left acting $\langle x|$ and assume local Hamiltonian,

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = H(x,t)\psi(x,t), \tag{2.8}$$

as *Schrödinger equation*.

For density matrix, $\hat{\rho}(t) = \sum_i p_i \left|\psi_i(t)\right\rangle\!\left\langle\psi_i(t)\right|$, notice that since states follow Schrödingerequation, the probability weight should not change.

$$i\hbar\frac{\partial}{\partial t}\hat{\rho}(t) = \hat{H}(t)\hat{\rho}(t) - \hat{\rho}(t)\hat{H}(t) = [\hat{H}(t), \hat{\rho}(t)], \tag{2.9}$$

which should be distinguished with Heisenberg equation of operators below.

Dyson Series

Generally, assuming that Hamiltonian does not commute at different time,

$$\begin{aligned}
\hat{U}(t) &= \mathbb{I} - \frac{i}{\hbar}\int_0^t \mathrm{d}t_1 \hat{H}(t_1)\hat{U}(t_1) \\
&= \mathbb{I} + \left(-\frac{i}{\hbar}\right)\int_0^t \mathrm{d}t_1 \hat{H}(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \hat{H}(t_1)\hat{H}(t_2)\hat{U}(t_2) = \cdots \\
&= \sum_{n=0}^{+\infty}\left(-\frac{i}{\hbar}\right)^n \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \cdots \int_0^{t_{n-1}} \mathrm{d}t_n \hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n) \longrightarrow \text{Dyson series} \\
&= \sum_{n=0}^{+\infty}\frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \cdots \int_0^t \mathrm{d}t_n \mathscr{T}\left(\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)\right) \\
&\equiv \mathscr{T}\exp\left(-\frac{i}{\hbar}\int_0^t \mathrm{d}t \hat{H}(t)\right),
\end{aligned}$$

where we have added (real-)time-ordered operator $\mathscr{T}$, which obeys 'who acts first if early'. We can further show the combination nature of two sequential time evolution operators,

$\hat{U}(t+T,0)=\hat{U}(t+T,t)\hat{U}(t,0)$, by explicitly expanding the Dyson series as

$$\begin{aligned}
\hat{U}(t+T,0) &= \sum_{n=0}^{+\infty}\frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n\int_0^{t+T}\mathrm{d}t_1\int_0^{t+T}\mathrm{d}t_2\cdots\int_0^{t+T}\mathrm{d}t_n\mathscr{T}\left(\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)\right)\\
&= \sum_{n=0}^{+\infty}\frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n\sum_{m=0}^{n}C_n^m\left[\int_t^{t+T}\mathrm{d}t_1\mathrm{d}t_2\cdots\mathrm{d}t_m\mathscr{T}\left(\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_m)\right)\right]\\
&\qquad\left[\int_0^{t}\mathrm{d}t_{m+1}\mathrm{d}t_{m+2}\cdots\mathrm{d}t_n\mathscr{T}\left(\hat{H}(t_{m+1})\hat{H}(t_{m+2})\cdots\hat{H}(t_n)\right)\right]\\
&= \left[\sum_{n=0}^{+\infty}\frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n\int_t^{t+T}\mathrm{d}t_1\mathrm{d}t_2\cdots\mathrm{d}t_n\mathscr{T}\left(\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)\right)\right]\\
&\qquad\left[\sum_{m=0}^{+\infty}\frac{1}{m!}\left(-\frac{i}{\hbar}\right)^m\int_0^{t}\mathrm{d}t_1\mathrm{d}t_2\cdots\mathrm{d}t_m\mathscr{T}\left(\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_m)\right)\right]\\
&= \hat{U}(t+T,t)\hat{U}(t,0).
\end{aligned}$$

In the second equality, we split each $\int_0^{t+T}=\int_0^t+\int_t^{t+T}$, and further expand the binomial-like expression explicitly. Essentially notice that once we choose m $\int_t^{t+T}$ integrals, the corresponding $\hat{H}$ will automatically move to the left by the time-ordering. In the third equality, we utilize the equivalent re-summation of the binomial series $\sum_{n=0}^{+\infty}\sum_{m=0}^{n}=\sum_{n-m=0}^{+\infty}\sum_{m=0}^{+\infty}$. This result shows that generally the time evolution operator is a combination of sequential time evolution operators, indicating the existence of group property of time translations.

In common problems, we meet case that Hamiltonian commute at different time so that

$$\hat{U}(t)=\exp\left(-\frac{i}{\hbar}\int_0^t\mathrm{d}t\hat{H}(t)\right), \tag{2.10}$$

and when it does not depend on time at all,

$$\hat{U}(t)=\exp\left(-\frac{i}{\hbar}\hat{H}t\right), \tag{2.11}$$

and when it is the case, by letting $c_n=\langle E_n|\psi\rangle$,

$$|\psi(t)\rangle=\exp\left(-\frac{i}{\hbar}\hat{H}t\right)\sum_n c_n|E_n\rangle=\sum_n c_n\mathrm{e}^{-iE_nt/\hbar}|E_n\rangle=\sum_n c_n(t)|E_n\rangle=\sum_n c_n|E_n(t)\rangle, \tag{2.12}$$

and generally, the picture holds by requiring

$$|E_n(t)\rangle=\hat{U}(t)|E_n\rangle. \tag{2.13}$$

Meanwhile, we see it that if $|\psi\rangle$ falls on an eigenket of Hamiltonian say $|E_n\rangle$ at the beginning, the time evolution would just add a trivial phase to it as $|\psi(t)\rangle=\exp(-iE_nt/\hbar)|E_n\rangle$, and at this time,

$$\hat{H}|\psi(t)\rangle=i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle=E_n|\psi(t)\rangle, \tag{2.14}$$

which is just the eigenfunction of Hamiltonian and is known as stationary Schrödinger equation.

Analysis of Wavefunction

Recall the probability density $\rho(\boldsymbol{x},t)=|\psi(\boldsymbol{x},t)|^2$, then in now single component case, we may assume $H(\boldsymbol{x},t)=-\hbar^2\nabla^2/2m+V$ where V is real, by which the Hermicity of Hamiltonian is conserved or we would obtain a non-Hermitian system with dispassion, then

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\rho(\boldsymbol{x},t)&=\left[i\hbar\frac{\partial}{\partial t}\psi^*\right]\psi+\psi^*\left[i\hbar\frac{\partial}{\partial t}\psi\right]=-(H\psi)^*\psi+\psi^*(H\psi)\\&=\frac{\hbar^2}{2m}(\psi^*\nabla^2\psi-\psi\nabla^2\psi^*)+(V-V^*)|\psi|^2\\&=\nabla\cdot\left[\frac{\hbar^2}{2m}(\psi^*\nabla\psi-\psi\nabla\psi^*)\right],\end{aligned}$$

and we conclude that continuity equation

$$\partial_t\rho+\nabla\cdot\boldsymbol{j}=0 \tag{2.15}$$

is conserved, where probability flux is

$$\boldsymbol{j}=-i\frac{\hbar}{2m}(\psi^*\nabla\psi-\psi\nabla\psi^*)=\frac{\hbar}{m}\operatorname{Im}(\psi^*\nabla\psi). \tag{2.16}$$

Now one could let

$$\psi(\boldsymbol{x},t)=\sqrt{\rho(\boldsymbol{x},t)}\mathrm{e}^{iS(\boldsymbol{x},t)/\hbar}, \tag{2.17}$$

and clearly,

$$\boldsymbol{j}=\frac{\hbar}{m}\operatorname{Im}\left[\sqrt{\rho}\mathrm{e}^{-iS/\hbar}\left((\nabla\sqrt{\rho})\mathrm{e}^{iS/\hbar}+\sqrt{\rho}\mathrm{e}^{iS/\hbar}\frac{i}{\hbar}\nabla S\right)\right]=\frac{\rho}{m}\nabla S, \tag{2.18}$$

where the gradient of the seemingly trivial phase determines both strength and direction of the flux, which is amazing with no classical analogy.

2.1.1.1 Spin Precession

This problem shows time evolution of a two state problem and shows how an invariant Hamiltonian could generate non-stationary movement.

We introduce magnetic moment for spin in analogy with classical orbital angular motion of electron as

$$\hat{\boldsymbol{\mu}}=-g\frac{e}{2m}\hat{\boldsymbol{S}}, \tag{2.19}$$

where g is Landé factor and for electron $g_e\approx 2$ in high accuracy, and $\hat{\boldsymbol{S}}$ denotes to spin operator. Also, here $e>0$ is denoted to elementry charge. Then the Hamiltonian is

$$\hat{H}=-\hat{\boldsymbol{\mu}}\cdot\boldsymbol{B}. \tag{2.20}$$

Notice that magnetic field is not quantized and appears as a classical field, and its quantization under first-quantization frame is somehow tricky. Now say we have a magnetic field $\boldsymbol{B}=B\hat{z}$, then under representation of $\hat{S}_z$,

$$H=g_e\frac{e}{2m_e}B\frac{\hbar}{2}\sigma_z=\frac{g_e}{2}\mu_B B\sigma_z\approx\mu_B B\sigma_z=\frac{\hbar}{2}\omega\sigma_z=\omega S_z, \tag{2.21}$$

where $\mu_B=e\hbar/2m_e$ is Bohr magneton and $\omega=eB/m_e$ is precession angular frequency.

Now clearly time evolution operator and its representation is

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\omega\hat{S}_z t\right),\ U(t) = \exp\left(-\frac{i}{\hbar}\omega S_z t\right) = \exp\left(-i\frac{\omega t}{2}\sigma_z\right). \tag{2.22}$$

Then clearly, for a given state at $t = 0$

$$|\alpha\rangle = C_+ |z+\rangle + C_- |z-\rangle\,, \tag{2.23}$$

its evolution would be

$$|\alpha(t)\rangle = \hat{U}(t)\,|\alpha\rangle = C_+ \exp\left(-i\frac{\omega t}{2}\right)|z+\rangle + C_- \exp\left(i\frac{\omega t}{2}\right)|z-\rangle\,. \tag{2.24}$$

And in matrix form, we may write

$$\alpha_z(t) = U(t)\begin{pmatrix} C_+ \\ C_- \end{pmatrix} = \begin{pmatrix} \exp\left(-i\frac{\omega t}{2}\right)C_+ \\ \exp\left(i\frac{\omega t}{2}\right)C_- \end{pmatrix}. \tag{2.25}$$

Firstly one notices that this Hamiltonian form, though very similar to the one in classical, shows its quantum nature by quantized spin operator. Classically, if one wants to describe some precession process, one must correspondingly have an evolving moment appearing in Hamiltonian, and the Hamiltonian form itself would also evolve with time, though some energy conservation mechanism may protect its absolute value. However, now in our Schrödinger picture, Hamiltonian does not change with time but governs time evolution of state. Saying on the other side, operator representating spin always maintain its form whatever direction the quantum state is directing. And this shows fundamental assumption of what is an observable and how does measurement mean in quantum mechanics.

Meanwhile, we see that though time evolution only adds phase factor to each component, it adds differently so that the relative phase would change, which may be measurable. Such state is an unstationary state.

2.1.2 Heisenberg Picture

"Evolution of observables matters."

The relation between Heisenberg picture with Schrödinger's one is similar to the basis transformation we showed in the first chapter, but should be firmly separated from time evolution. The key is, basis transformation keeps expectation value unchanged, while time evolution could violate that. And respecting basis transformation, we require that physical quantity, the expectation value, is invariant under different pictures,

$$\langle\psi(t)_S|\,\hat{O}_S(t)\,|\psi(t)\rangle_S = \langle\psi_H|\,\hat{O}_H(t)\,|\psi\rangle_H\,,$$

where we usually do not require ket in Heisenberg picture to change, we got

$$\hat{O}_H(t) = \hat{U}^\dagger(t)\hat{O}_S(t)\hat{U}(t), \tag{2.26}$$

where subscripts refer to pictures.

And Heisenberg equation of motion for operators would be

$$\begin{aligned} i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\hat{O}_H(t) &= -\hat{U}^\dagger(t)\hat{H}(t)\hat{O}_S(t)\hat{U}(t) + i\hbar\hat{U}^\dagger(t)\left(\frac{\partial}{\partial t}\hat{O}_S(t)\right)\hat{U}(t) + \hat{U}^\dagger(t)\hat{O}_S(t)\hat{H}(t)\hat{U}(t) \\ &= [\hat{O}_H(t), \hat{H}_H(t)] + i\hbar\left(\frac{\partial}{\partial t}\hat{O}_S(t)\right)_H, \end{aligned} \tag{2.27}$$

which could be reduced when Hamiltonian commutes at different time and $\hat{O}_S$ contains no time

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\hat{O}_H(t) = [\hat{O}_H(t), \hat{H}(t)]. \tag{2.28}$$

Based on Heisenberg picture, we may give a hint on why time could not be an operator in non-relativistic quantum mechanics, protected by *Pauli's theorem*.

Proof. Assume that we have

$$[\hat{t}, \hat{H}] = i\hbar,$$

also by Heisenberg EoM, for any $\hat{O} = \hat{O}(\hat{t})$, we have

$$[\hat{O}, \hat{H}] = i\hbar\frac{\mathrm{d}}{\mathrm{d}\hat{t}},$$

then by $\hat{O} = \exp\big(ia\hat{t}\big),\ a \in \mathbb{R}$, and assume $\hat{H}\,|E\rangle = E\,|E\rangle$,

$$\begin{aligned} \hat{H}\hat{O}\,|E\rangle &= ([\hat{H}, \hat{O}] + \hat{O}\hat{H})\,|E\rangle = \left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}\hat{t}}\mathrm{e}^{ia\hat{t}} + \hat{O}\hat{H}\right)|E\rangle \\ &= (E + \hbar a)\hat{O}\,|E\rangle\,, \end{aligned}$$

which gives us another eigenstate with energy shift $\hbar a$. Since $a \in \mathbb{R}$ is continuous, this becomes impossible since it serves us a continuum spectrum, while in general quantum mechanics allows discrete one. □

2.1.3 Interaction Picture

For Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{V}(t),\ \ \hat{H}_0\,|n\rangle = E_n\,|n\rangle\,, \tag{2.29}$$

we have two time evolution operators $\hat{U}(t)$ by whole Hamiltonian and $\hat{U}_0(t)$ by $\hat{H}_0$, then we define

$$|\psi(t)\rangle_I = \hat{U}_0^\dagger(t)\,|\psi(t)\rangle_S = \hat{U}_0^\dagger(t)\hat{U}(t)\,|\psi\rangle \equiv \hat{U}_I(t)\,|\psi\rangle\,, \tag{2.30}$$

where

$$\hat{U}_I(t) = \hat{U}_0^\dagger(t)\hat{U}(t). \tag{2.31}$$

And in spirit of basis transformation, now the operators transform as

$$\hat{O}_I(t) = \hat{U}_0^\dagger(t)\hat{O}\hat{U}_0(t), \tag{2.32}$$

where we assumed simplest case.

Now clearly,

$$i\hbar\frac{\partial}{\partial t}\hat{U}_I(t) = \hat{U}_0^\dagger(t)(-\hat{H}_0 + \hat{H})\hat{U}(t) = \hat{U}_0^\dagger(t)\hat{V}(t)\hat{U}(t) = \hat{V}_I(t)\hat{U}_I(t), \tag{2.33}$$

with $\hat{V}_I(t) = \hat{U}_0^\dagger(t)\hat{V}(t)\hat{U}_0(t)$, thus

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle_I = \hat{V}_I(t)\left|\psi(t)\right\rangle_I. \tag{2.34}$$

From this equation, the state under interaction picture also shares a Dyson series solution, which should be equal to its definition, i.e.,

$$\hat{U}_I(t) = \mathscr{T}\exp\left(-\frac{i}{\hbar}\int_0^t \mathrm{d}t\hat{V}_I(t)\right) = \hat{U}_0^\dagger(t)\hat{U}(t). \tag{2.35}$$

Then similarly, the density matrix operator $\hat{\rho}_I(t) =_I \left|\psi(t)\right\rangle\!\left\langle\psi(t)\right|_I$ changes with time under the Liouville-von Neumann equation

$$i\hbar\frac{\partial}{\partial t}\hat{\rho}_I(t) = [\hat{V}_I(t), \hat{\rho}_I(t)], \tag{2.36}$$

in accordance with the form in Schrödinger picture. In the interaction picture, the operators also evolve with time as

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\hat{O}_I(t) = [\hat{O}_I(t), \hat{H}_0(t)]. \tag{2.37}$$

From evolution of ket we see that by letting $\left|\psi(t)\right\rangle_I = \sum_n c_n(t)\left|n\right\rangle$, we could act on left $\langle n|$

$$i\hbar\frac{\partial}{\partial t}c_n(t) = \langle n|\,\hat{V}_I(t)\left|\psi(t)\right\rangle_I = \sum_m \langle n|\hat{V}_I(t)|m\rangle\left\langle m|\psi(t)\right\rangle_I = \sum_m V_{nm}(t)\mathrm{e}^{i\omega_{nm}t}c_m(t),$$

where $V_{nm}(t) = \langle n|\hat{V}(t)|m\rangle$ and $\hbar\omega_{nm} = E_n - E_m$. Note that this is strict.

2.1.4 A Quantum-Classical Correspondence

Born's rule eliminates possibility of negative probability. Nevertheless, such a negative probability, from the measurement point of view that lies in the heart of physics, is not that unacceptable, as long as the event corresponding to the negative probability is not observable (or no such event exists).

We may generally allow for an over-complete configuration space $\mathscr{C}$ concerning all possible states of Hamiltonian H that we are interested in. For $C \in \mathscr{C}$, we may define its projection operator P_C and we assume $\sum_C P_C = 1$ (may lack proper normalization). We consider the evolution of this projection operator,

$$\partial_t P_C = i[H, P_C] \equiv \sum_{C'} M_{CC'}P_{C'}, \tag{2.38}$$

where $M_{CC'}$ is some matrix with component in configuration space. We have let $\hbar = 1$ here. Due to the completeness of P_C, we further have

$$0 = \partial_t\sum_C P_C = \sum_C\sum_{C'} M_{CC'}P_{C'} = \sum_{C'} P_{C'}\left(\sum_C M_{CC'}\right),$$

and by choosing P_C linear independent, we have that $\sum_C M_{CC'} = 0$, much like a transition matrix.

Now we take care of time-evolution of the probability of measuring C, the above Heisenberg picture leads to

$$\begin{aligned}\partial_t p_C &= \mathrm{Tr}(\rho \partial_t P_C) = \mathrm{Tr}(\rho i[H, P_C]) = \mathrm{Tr}\left(\rho \sum_{C'} M_{CC'} P_{C'}\right) \\ &= \sum_{C'} M_{CC'} p_{C'} = \sum_{C' \neq C} M_{CC'} p_{C'} - M_{C'C} p_C,\end{aligned}$$

which captures feature of a classical Markov process, although the transition matrix $M_{CC'}$ is not necessarily symmetric or positive definite.

If we really want to avoid that, we may introduce always-positive rate

$$M^{\pm}_{CC'} = \pm\Theta(\pm M_{CC'}) M_{CC'},$$

such that $M_{CC'} = M^{+}_{CC'} - M^{-}_{CC'}$ and $p_C = p^{+}_C - p^{-}_C$, with $\pm$ denoting particle and hole part. Then a straightforward calculation shows that

$$\partial_t p^{\pm}_C = \sum_{C' \neq C} \left(M^{+}_{CC'} P^{\pm}_{C'} - M^{+}_{C'C} P^{\pm}_C\right) + \left(M^{-}_{CC'} P^{\mp}_{C'} - M^{-}_{C'C} P^{\pm}_C\right) + 2 \sum_{C' \neq C} M^{-}_{C'C} P^{\pm}_C,$$

where both probability and rate are always positive in the coupled stochastic process.

2.2 Harmonic Oscillator

We use one dimensional harmonic oscillator as the simplest example, and we ignore tip of operator.

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2, \ \omega = \sqrt{\frac{k}{m}}. \tag{2.39}$$

Classically, canonical equations of motion (EoM) tells that

$$\begin{cases} \dot{p} = -\dfrac{\partial H}{\partial x} = -m\omega^2 x \\ \dot{x} = \dfrac{\partial H}{\partial p} = p/m \end{cases} \implies \ddot{x} + \omega^2 x = 0, \tag{2.40}$$

with simple solution

$$x = A\sin(\omega t) + B\cos(\omega t). \tag{2.41}$$

And now, we re-write

$$H = \hbar\omega \left(\frac{p^2}{2m\hbar\omega} + \frac{m\omega x^2}{2\hbar}\right) \equiv \hbar\omega(P^2 + Q^2), \tag{2.42}$$

where dimensionless dynamical variables

$$\begin{cases} P = p/\sqrt{2m\hbar\omega} \\ Q = \sqrt{m\omega/2\hbar}\, x \end{cases} \implies [Q, P] = \frac{1}{2\hbar}[x, p] = \frac{i}{2}. \tag{2.43}$$

Then by observation, if one let

$$\begin{cases} a = Q + iP \\ a^\dagger = Q - iP \end{cases} \implies [a, a^\dagger] = 2i[P, Q] = 1, \tag{2.44}$$

and by inverted transformation, now

$$H = \hbar\omega \left[\left(\frac{a - a^\dagger}{2i} \right)^2 + \left(\frac{a + a^\dagger}{2} \right)^2 \right] = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right). \tag{2.45}$$

Meanwhile,

$$\begin{aligned} [a, H] &= \hbar\omega [a, a^\dagger] a = \hbar\omega a, \\ [a^\dagger, H] &= \hbar\omega a^\dagger [a^\dagger, a] = -\hbar\omega a^\dagger. \end{aligned} \tag{2.46}$$

(This could be done more procedurally by calculating $[Q, H]$ and $[P, H]$ and then find linear combination so that $[Q + i\lambda P, H] = C \cdot (Q + i\lambda P)$ for some λ.)

By defining an operator called number operator

$$N = a^\dagger a,\ [a, N] = a,\ [a^\dagger, N] = -a^\dagger,\ [N, H] = 0, \tag{2.47}$$

we could let

$$N \left| n \right\rangle = n \left| n \right\rangle, \tag{2.48}$$

so that

$$H \left| n \right\rangle = \left(n + \frac{1}{2} \right) \hbar\omega \left| n \right\rangle. \tag{2.49}$$

Now since

$$\begin{aligned} Na \left| n \right\rangle &= (aN + [N, a]) \left| n \right\rangle = (n - 1) a \left| n \right\rangle, \\ Na^\dagger \left| n \right\rangle &= (a^\dagger N + [N, a^\dagger]) = (n + 1) a^\dagger \left| n \right\rangle, \end{aligned}$$

we could say that

$$\begin{cases} a \left| n \right\rangle = C_n^- \left| n - 1 \right\rangle, \\ a^\dagger \left| n \right\rangle = C_n^+ \left| n + 1 \right\rangle, \end{cases} \tag{2.50}$$

as if $a(a^\dagger)$ lowers (lifts) the eigenvalue by one unit. From

$$\begin{aligned} \langle n | a^\dagger a | n \rangle &= \langle n | N | n \rangle = n = (a \left| n \right\rangle)^\dagger (a \left| n \right\rangle) = |C_n^-|^2, \\ \langle n | a a^\dagger | n \rangle &= \langle n | N + 1 | n \rangle = n + 1 = (a^\dagger \left| n \right\rangle)^\dagger (a^\dagger \left| n \right\rangle) = |C_n^+|^2, \end{aligned}$$

one chooses coefficients to be real and has

$$C_n^- = \sqrt{n},\ C_n^+ = \sqrt{n + 1}, \tag{2.51}$$

thus

$$\begin{cases} a \left| n \right\rangle = \sqrt{n} \left| n - 1 \right\rangle \\ a^\dagger \left| n \right\rangle = \sqrt{n + 1} \left| n + 1 \right\rangle \end{cases}, \tag{2.52}$$

and these operators are known as *annihilation operator* and *creation operator*.

Now since $N^\dagger = N$ as Hermitian, we have

$$n = \langle n | N | n \rangle \geq 0; \tag{2.53}$$

on the other hand, our annihilation operator lowers n by exactly one each time it acts on $|n\rangle$, and we claim that n must be non-negative integer with lowest bound as zero, i.e.,

$$a^n \left| n \right\rangle = \sqrt{n!} \left| 0 \right\rangle,\ a \left| 0 \right\rangle = 0, \tag{2.54}$$

where the first $|0\rangle$ denotes *vacuum state* while the last 0 is true zero with nothing. By this we hint that vacuum is not so empty as it has energy $\hbar\omega/2$, together with the potential to generate excitations, whose elementary number quantum is created by $a^\dagger$, and every such quantum carries energy $\hbar\omega$.

Now the normalized state

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n |0\rangle , \tag{2.55}$$

which means that excitations with same quantum could occupy the same oscillator state at the same time, and this is indeed *Bosonic* behavior, shown by the missing commutation relations

$$[a, a] = 0 = [a^\dagger, a^\dagger]. \tag{2.56}$$

And in particle number representation,

$$\langle n'|a|n\rangle = \sqrt{n}\delta_{n',n-1},\ \langle n'|a^\dagger|n\rangle = \sqrt{n+1}\delta_{n',n+1}, \tag{2.57}$$

so none of them is diagonal with infinite extension as matrix form.

Now for real-space form as wavefunction. Firstly for the vacuum state $|0\rangle$,

$$0 = \langle Q|a|0\rangle = \langle Q|Q + iP|0\rangle = (Q + \frac{1}{2}\partial_Q)\psi_0(Q) \Longrightarrow \psi_0(Q) \propto \mathrm{e}^{-Q^2},$$

thus

$$\psi_0(x) = C_0 \mathrm{e}^{-x^2/2x_0^2},\ x_0^2 = \frac{\hbar}{m\omega},$$

and by normalization

$$\int \mathrm{d}x|\psi_0(x)|^2 = 1 \Longrightarrow C_0 = \frac{1}{\pi^{1/4}x_0},$$

thus

$$\psi_0(x) = \frac{1}{\pi^{1/4}x_0}\exp\left(-\frac{x^2}{2x_0^2}\right),\ x_0^2 = \frac{\hbar}{m\omega}. \tag{2.58}$$

For higher excited states,

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\sqrt{n!}\pi^{1/4}x_0}\left(\frac{1}{\sqrt{2}x_0}\right)^n (x - x_0^2\partial_x)^n \exp\left(-\frac{x^2}{2x_0^2}\right). \tag{2.59}$$

What is more, if one considers the average value

$$\langle x\rangle = 0 = \langle p\rangle \tag{2.60}$$

for any state. And for the square, we first consider the vacuum, since

$$x^2 = \frac{2\hbar}{m\omega}Q^2 = \frac{\hbar}{2m\omega}(a^2 + a^{\dagger 2} + aa^\dagger + a^\dagger a),$$

one sees that

$$\langle 0|x^2|0\rangle = \frac{\hbar}{2m\omega}\langle 0|aa^\dagger|0\rangle = \frac{\hbar}{2m\omega}, \tag{2.61}$$

and similarly,

$$\langle 0|p^2|0\rangle = \frac{\hbar m\omega}{2}. \tag{2.62}$$

Then it follows that the kinetic and potential part

$$\langle \frac{p^2}{2m} \rangle = \frac{\hbar\omega}{4} = \frac{\langle H \rangle}{2} = \langle \frac{1}{2} m\omega^2 x^2 \rangle \,, \tag{2.63}$$

as expected from virial theorem. Notice it that this may hold for any state, not only the ground vacuum one. On the other hand, using zero expectation for linear order, one has

$$\langle (\Delta x)^2 \rangle_0 \langle (\Delta p)^2 \rangle_0 = \langle x^2 \rangle_0 \langle p^2 \rangle_0 = \frac{\hbar^2}{4}, \tag{2.64}$$

which satisfies minimal uncertainty relation as expected from the gaussian form of wave-function. And for higher states, one may show that

$$\langle (\Delta x)^2 \rangle_n \langle (\Delta p)^2 \rangle_n = \left(n + \frac{1}{2} \right)^2 \hbar^2. \tag{2.65}$$

Position and Momentum — Heisenberg Picture

From EoM of Heisenberg picture,

$$\begin{cases} i\hbar \dot{Q} = [Q, H] = i\hbar\omega P \\ i\hbar \dot{P} = [P, H] = -i\hbar\omega Q \end{cases} \implies \begin{cases} \ddot{Q} + \omega^2 Q = 0 \\ \ddot{P} + \omega^2 P = 0 \end{cases} , \tag{2.66}$$

which is just similar to the classical canonical EoM. Then given initial $Q(0)$ and $P(0)$, one has

$$\begin{cases} Q(t) = Q(0)\cos(\omega t) + B\sin(\omega t) \\ P(t) = P(0)\cos(\omega t) + D\sin(\omega t) \end{cases} ,$$

and from the coupled first order equation, one sees easily that $B = P(0)$ and $D = -Q(0)$, then

$$\begin{cases} Q(t) = Q(0)\cos(\omega t) + P(0)\sin(\omega t) \\ P(t) = P(0)\cos(\omega t) - Q(0)\sin(\omega t) \end{cases} . \tag{2.67}$$

For x and p, one can bring the form back to the transformation and gets the result easily.

For instance, consider arbitrary initial state to be invariant under Heisenberg picture,

$$|\alpha\rangle = \sum_{n=0}^{+\infty} C_n |n\rangle \,, \quad \sum_n |C_n|^2 = 1, \tag{2.68}$$

the average value of generalized position operator reads

$$\begin{aligned} \langle Q(t) \rangle_\alpha &= \langle \alpha | \frac{a + a^\dagger}{2} \cos(\omega t) + \frac{a - a^\dagger}{2i} \sin(\omega t) |\alpha\rangle \\ &= \frac{1}{2} \sum_{n,m} C_n^* C_m \langle n | \, a e^{-i\omega t} + a^\dagger e^{i\omega t} |m\rangle \\ &= \sum_{n=0}^{+\infty} |C_n C_{n+1}| \cos\big(\omega t + \arg(C_{n+1}^* C_n)\big). \end{aligned} \tag{2.69}$$

Position and Momentum — Schrödinger Picture

Now focusing on time-revolution of state $|\alpha\rangle$, which is easy to follow since H does not depend on t, and we simply write

$$|\alpha(t)\rangle = \sum_n C_n \mathrm{e}^{-i(n+1/2)\omega t} |n\rangle . \tag{2.70}$$

Then

$$\begin{aligned} \langle Q\rangle_{\alpha(t)} &= \langle\alpha(t)| Q |\alpha(t)\rangle = \frac{1}{2}\sum_{n,m} C_n^* C_m \langle n| (a+a^\dagger) e^{i(n-m)\omega t} |m\rangle \\ &= \sum_{n=0}^{+\infty} |C_n C_{n+1}| \cos\big(\omega t + \arg(C_{n+1}^* C_n)\big), \end{aligned} \tag{2.71}$$

which yields the same result as in the Heisenberg picture.

2.2.1 Coherent State

Defined as

$$a |z\rangle = z |z\rangle . \tag{2.72}$$

Always has solution

$$|z\rangle = \exp\left(-\frac{|z|^2}{2}\right) \sum_{n=0}^{+\infty} \frac{z^n}{\sqrt{n!}} |n\rangle . \tag{2.73}$$

One could in fact define a unitary operator to generate the state as

$$|z\rangle = D(z)|0\rangle , \ D(z) = \exp\big(za^\dagger - z^* a\big), \tag{2.74}$$

with help from Baker-Campbell-Hausdorff formula.

This coherent state shares very interesting properties. Firstly, it shares minimal uncertainty relation, with Eq. (2.72), one also sees that $\langle z| a^\dagger = \langle z| \bar{z}$, thus

$$\langle Q\rangle_z = \frac{1}{2}\langle z| a + a^\dagger |z\rangle = \mathrm{Re}\, z, \tag{2.75a}$$

$$\langle P\rangle_z = \frac{1}{2i}\langle z| a - a^\dagger |z\rangle = \mathrm{Im}\, z, \tag{2.75b}$$

$$\langle Q^2\rangle_z = \frac{1}{4}\langle z| (a+\hat{a}^\dagger)^2 |z\rangle = (\mathrm{Re}\, z)^2 + \frac{1}{4}, \tag{2.75c}$$

$$\langle P^2\rangle_z = -\frac{1}{4}\langle z| (a-\hat{a}^\dagger)^2 |z\rangle = (\mathrm{Im}\, z)^2 + \frac{1}{4}, \tag{2.75d}$$

thus we have

$$\langle (\Delta x)^2\rangle_z \langle (\Delta p)^2\rangle_z = \frac{2\hbar}{m\omega}\frac{1}{4}\cdot 2m\hbar\omega\frac{1}{4} = \frac{\hbar^2}{4}, \tag{2.76}$$

which is minimal.

And furthermore, the coherent state is a state that resembles mostly a classical state. Consider that under the Schrödinger picture, the time evolution of coherent state gives

$$\begin{aligned} |z(t)\rangle = U(t)|z\rangle &= e^{-i\omega(a^\dagger a+1/2)t} e^{-|z|^2/2} \sum_n \frac{z^n}{\sqrt{n!}} |n\rangle \\ &= e^{-i\omega t/2} e^{-|z|^2/2} \sum_n \frac{(e^{-i\omega t} z)^n}{\sqrt{n!}} |n\rangle = e^{-i\omega t/2} |e^{-i\omega t} z\rangle , \end{aligned} \tag{2.77}$$

and the time-dependent expectation value of Q and P are, similarly,

$$\langle Q\rangle_z(t) = \mathrm{Re}\, z\cos(\omega t) + \mathrm{Im}\, z\sin(\omega t), \tag{2.78a}$$
$$\langle P\rangle_z(t) = \mathrm{Im}\, z\cos(\omega t) - \mathrm{Re}\, z\sin(\omega t), \tag{2.78b}$$

which are just the classical solution of harmonic oscillator with initial condition fixed by z.

With such properties as minimal wavepacket, we shall see that the coherent states play the extremely important role in re-writing the quantum dynamics into the formalism of path integrals.

2.2.2 A First Index Theorem

The infinitely long energy level list is definitely physically ill defined, which has to be regulated under modern physical language. Such phenomenon or problem often appears in quantum field theory, which is referred to the quantum anomaly. It is rather intriguing that in such a simple system, similar concept also appears, or in other language, standing as the basic model for the quantum field theory, it would be strange that the quantum harmonic oscillator does not contain illness at infinity. The starting point is simple that under energy level representation, the annihilation and creation operators become

$$a = |0\rangle\langle 1| + \sqrt{2}\,|1\rangle\langle 2| + \sqrt{3}\,|2\rangle\langle 3| + \cdots, \tag{2.79a}$$
$$a^\dagger = |1\rangle\langle 0| + \sqrt{2}\,|2\rangle\langle 1| + \sqrt{3}\,|3\rangle\langle 2| + \cdots. \tag{2.79b}$$

And in fact, this is already enough for the index theorem. However, to connect it with another puzzling phenomenon, we may take a leisure first.

Due to the commutation relation, one usually identifies level $|n\rangle$ with n bosonic particles, then encouraged by the action or representation of two operators, one may try to define $a = e^{i\phi}\sqrt{N}$ with N the particle number operator and ϕ the phase operator. Standing as operator, ϕ and N will not commute as a matter of course, and one secure way is to do the (canonical) quantization. On the classical level, we safely let ϕ and N commute, and they give back $P = \sqrt{2N}\cos\phi,\ Q = \sqrt{2N}\sin\phi$ by the classical coordinate transformation copied from the quantum one. Now such a transformation has Jacobian 1 as one can easily verify, then the classical Possion bracket $\{Q, P\} = 1$ naturally induces $\{N, \phi\} = 1$ (again one can do with brute force), which will be $[N, \phi] = i$ under quantization. This gives the famous particle-phase uncertainty relation $\Delta N\Delta\phi \geq \frac{1}{2}$.

Why we stare at this? Because later as we will show, it is the un-unitarity of $e^{i\phi}$ that makes a non-trivial index theorem possible. By definition, we have

$$e^{i\phi} = aN^{-1/2} = (N+1)^{-1/2}a = |0\rangle\langle 1| + |1\rangle\langle 2| + |2\rangle\langle 3| + \cdots, \tag{2.80a}$$
$$(e^{i\phi})^\dagger = |1\rangle\langle 0| + |2\rangle\langle 1| + |3\rangle\langle 2| + \cdots, \tag{2.80b}$$

where we have used an important equality that $af(n) = f(n+1)a$, to prove which we just need to identify that $an = aa^\dagger a = (n+1)a$, and function over n can be treated by Taylor expansion. Similarly, by taking conjugate we have $a^\dagger g(n+1) = g(n)a^\dagger$. Here, one has to be careful since we state nothing on whether ϕ is Hermitian or not, and at most we can write $(e^{i\phi})^\dagger = e^{-i\phi^\dagger}$. What is more, we can easily verify that

$$e^{i\phi}(e^{i\phi})^\dagger = \mathbb{I},\ (e^{i\phi})^\dagger e^{i\phi} = \mathbb{I} - |0\rangle\langle 0|, \tag{2.81}$$

which clearly reveals that $e^{i\phi}$ is not unitary and ϕ is not Hermitian. Be aware that we meet firstly with such kind of strange operator, since usually $AA^\dagger = 1$ means $A^\dagger A = 1$. Such strange behavior is certainly related to the irreversibility of operator, which means that it has 0 eigenvalues, or its rank is lower than its dimension. Counting for such 0s will lead to the index theorem to be discussed below. Sometimes, especially in the context of Bose-Einstein condensation, macroscopically large number of Bosons fall on the same quantum state, which effectively makes the fluctuation of ϕ approaching zero, and we may treat both N and ϕ classical numbers, and ϕ is 'Hermitian' in the sense. This is certainly not the main case.

At first glance, the strange un-unitarity comes from the 0 ground state, while a further thinking clarifies that the cancellation from infinity is also important. This is indeed some fashion of the quantum anomaly: by cancelling the infinite part, a non-trivial theory will always remain something on the ground (low energy observable). And oppositely, if for some reason one wants to remove the ground property which is strange for the observer, one will see the peculiar infinity. You can't have your cake and eat it. For instance, we can fix the strange un-unitarity by a cut-off, from infinite-dimensional level space to $(s+1)$-dim, say

$$e^{i\phi_s} = |0\rangle\langle 1| + |1\rangle\langle 2| + |2\rangle\langle 3| + \cdots + e^{i\theta_s}\,|s\rangle\langle 0|\,, \tag{2.82}$$

with a tail-head coupling gauged by real angle θ_s, and now happily we have $e^{i\phi}(e^{i\phi})^\dagger = (e^{i\phi})^\dagger e^{i\phi} = \mathbb{I}$. However,

$$a_s = e^{i\phi_s}\sqrt{N} = |0\rangle\langle 1| + \sqrt{2}\,|1\rangle\langle 2| + \cdots + \sqrt{s}\,|s-1\rangle\langle s|\,, \tag{2.83a}$$

$$a_s^\dagger = \sqrt{N}e^{-i\phi_s} = |1\rangle\langle 0| + \sqrt{2}\,|2\rangle\langle 1| + \cdots + \sqrt{s}\,|s\rangle\langle s-1|\,, \tag{2.83b}$$

which leads to the ill-defined commutation relation as

$$[a_s, a_s^\dagger] = 1 - (s+1)\,|s\rangle\langle s|\,, \tag{2.84}$$

which is not 1 however large s we choose.

Now we proceed to the index theorem. The original operators, suffering from the un-unitarity of phase operator, has property

$$\dim\ker a - \dim\ker a^\dagger = 1, \tag{2.85}$$

where by kernel, we count on the subspace of the domain where a map f acts on, and maps the subspace to identity element in the range. Basically, for a traditional matrix, it is just the zero-eigenvalue subspace. Here, we have $\ker a = \{|0\rangle\}$, the space spanned by ground state, since $a\,|0\rangle = 0$, and we see $\dim\ker a = 1$. On the other hand, $a^\dagger\,|n\rangle \propto |n+1\rangle$ always has non-empty result, and we recognize $\dim\ker a^\dagger = 0$. Here once again, the non-vanishing dimension comes from the low-energy ground state. The above formula can be written in a more suggestive way as

$$\dim\ker a^\dagger a - \dim\ker aa^\dagger = 1, \tag{2.86}$$

which counts for the identity map dimension of two Hermitian operators now. Such a re-writing holds essentially because once $au = 0$, $a^\dagger au = 0$, and inversely, from $a^\dagger au = 0$, we have $0 = (a^\dagger au, u) = (au, au)$ which yields $au = 0$ thanks to the positively definite inner product. This formula further encourages us to write

$$\mathrm{Tr}\Big(e^{-a^\dagger a}\Big) - \mathrm{Tr}\Big(e^{-aa^\dagger}\Big) = 1, \tag{2.87}$$

where the $-$ sign is assigned to ensure convergence. For this equality to hold, stronger constraint must be imposed so that only zero-eigenvalues are accounted, while the non-zero parts are exactly cancelled. To see this under the circumstance, we notice that for $n \neq 0$, $a^\dagger a u_n = \lambda_n^2 u_n$ (here we can construct $u_n = |n\rangle, \lambda_n = \sqrt{n}$, but we leave it out), then $\tilde{u}_n = a u_n/\lambda_n$ satisfies that $a a^\dagger \tilde{u}_n = \lambda_n^2 \tilde{u}_n$, i.e., the non-zero part of two operators has exact one-one correspondence. Accordingly,

$$\mathrm{Tr}\Big(e^{-a^\dagger a}\Big) - \mathrm{Tr}\Big(e^{-aa^\dagger}\Big) = 1 + \left(\sum_{n=1}^{+\infty} e^{-n}\right) - \left(\sum_{n=1}^{+\infty} e^{-n}\right) = 1. \tag{2.88}$$

As we claimed, the un-cancelled zero-eigenvalue subspace leads to non-vanishing index, together with non-Hermitian phase operator. On the other hand, we also tried to make the phase operator Hermitian, say

$$a_U = U(\phi)\sqrt{N},\ UU^\dagger = U^\dagger U = \mathbb{I}. \tag{2.89}$$

However, at this time, $a_U^\dagger a_U = N$, $a_U a_U^\dagger = UNU^\dagger$, and

$$\mathrm{Tr}\Big(e^{-a_U^\dagger a_U}\Big) - \mathrm{Tr}\Big(e^{-a_U a_U^\dagger}\Big) = \mathrm{Tr}\big(e^{-N}\big) - \mathrm{Tr}\Big(e^{-UNU^\dagger}\Big) = 0, \tag{2.90}$$

which holds since $\mathrm{Tr}\Big(e^{-UNU^\dagger}\Big) = \mathrm{Tr}\big(Ue^{-N}U^\dagger\big) = \mathrm{Tr}\big(e^{-N}\big)$. At this time, the strange ground state property disappears, however, we are left with some ultraviolet incompatibility revealed by the peculiar commutation relation, as derived for the $s+1$ cut-off. Explicitly, one sees

$$\operatorname*{Tr}_{s+1}\left(e^{-a_s^\dagger a_s}\right) - \operatorname*{Tr}_{s+1}\left(e^{-a_s a_s^\dagger}\right) = 0, \tag{2.91}$$

with $s+1$ under trace meaning doing trace in finite space. At this time, certainly the zero-eigenvalue of a_s does not disappear, i.e., $\dim\ker a_s = 1$, however, we also see that $a_s^\dagger |s\rangle = 0$ so that $\dim\ker a_s^\dagger = 1$, which cancels each other.

At above, we see that, if we want the index, we lose unitarity, and if we assign unitarity, we cancel the low-energy index by a high-energy one. Is there any way to fix this? Yes, but also a mind-bending one. Since high-energy gives rise an unwanted additional index, we just give that up, by

$$\operatorname*{Tr}_{s+1}\left(e^{-a_s^\dagger a_s}\right) - \operatorname*{Tr}_{s}\left(e^{-a_s a_s^\dagger}\right) = 1, \tag{2.92}$$

where $\operatorname*{Tr}_s$ is done in first s-dim over $(s+1)$-dim space. In this sense the commutation relation also holds for first s-dim, which makes $[a_s, a_s^\dagger] = 1$ since $|s\rangle$ is not considered here. Now pushing s to infinity we get a theory with Hermitian phase operator, un-vanishing index and healthy commutation relation, while the price paid may be the fuzziness of the procedure.

The actual reason that we find such a strange behavior is that the group generated by a, $a^\dagger$ and another combination, known as the metaplectic group, is a Lie group with no finite-dimensional faithful representation — it is not a matrix group.

2.3 Path Integral

Path integral picture is rather useful in quantum many-body and field theory, and it directly manipulates the space-time.

Define *propagator* or equivalently, *transition amplitude* as

$$K(2;1) = K(\boldsymbol{x}_2, t_2; \boldsymbol{x}_1, t_1) = \langle \boldsymbol{x}_2, t_2 | \boldsymbol{x}_1, t_1 \rangle = \langle 2_S | 1_S \rangle \,, \tag{2.93}$$

where in the first and last step, we write the space-time coordinate in a simpler way, and $|\boldsymbol{x}, t\rangle$ is **eigenket of position operator in Heisenberg picture**, say

$$\hat{\boldsymbol{x}}(t) = \hat{U}^\dagger(t)\hat{\boldsymbol{x}}\hat{U}(t),\ \hat{\boldsymbol{x}}(t)\,|\boldsymbol{x}, t\rangle = \boldsymbol{x}(t)\,|\boldsymbol{x}, t\rangle \,, \tag{2.94}$$

where $\hat{U}$ is generated by system's Hamiltonian. One recognize that they are just Schrödinger states. And in fact, one sees

$$K(2;1) = \langle 2|\,\hat{U}(t_2)\hat{U}^\dagger(t_1)\,|1\rangle = \langle 2|\,\hat{U}(t_2 - t_1)\,|1\rangle \,. \tag{2.95}$$

Bearing its name, propagator stands for basic transition of state,

$$\psi(2) = \langle 2|\psi\rangle = \int \mathrm{d}1\,\langle 2|1\rangle\,\langle 1|\psi\rangle = \int \mathrm{d}\boldsymbol{x}_1 K(2;1)\psi(1),$$

where we emphasise that complete relation is on position, not time. What is more, we say that K is just Green's function as

$$\left[i\hbar\frac{\partial}{\partial t_2} - H(2)\right] K(2;1) = i\hbar\delta(2-1),$$

where right-hand side (RHS) is initial-boundary condition. Notice that to make the equation stand, one has to assign a causality boundary condition as $K(t_2 < t_1) = 0$.

Now our aim is $K(N;0)$. From our writing one knows we would like to insert some identity, and possibly much of them. We ignore difference of bold vector and plain one. Set time interval $\Delta t = t_n - t_{n-1} = (t_N - t_1)/N$.

$$\begin{aligned} K(N;0) &= \langle x_N, t_N | x_1, t_1\rangle = \lim_{N\to+\infty} \langle x_N, t_N|_S \,(\mathrm{e}^{-\frac{i}{\hbar}H\Delta t})^N\,|x_1, t_1\rangle_S \\ &= \cdots \\ &= \int \mathscr{D}[x(t)]\mathrm{e}^{iS[x(t)]/\hbar}, \end{aligned}$$

where $\mathscr{D}[x(t)]$ means we sum over all possible paths, while S is classical action

$$S[x(t)] = \int_{t_1}^{t_N} \mathrm{d}t L(x, \dot{x}, t). \tag{2.96}$$

For evaluation of $\cdots$ above, one may think about one kind of system that is very common and general

$$H = T(p) + V(x),\ T(p) = \frac{p^2}{2m},$$

while inserting identity before every Δt operator as

$$\mathbb{I} = \int \mathrm{d}x_n \mathrm{d}p_n\,|x_n\rangle\,\langle x_n|p_n\rangle\,\langle p_n|\,,$$

with Schrödinger ket, then since

$$\begin{aligned} \langle p_{n+1}|\,\mathrm{e}^{-\frac{i}{\hbar}H\Delta t}\,|x_n\rangle &= \langle p_{n+1}|\,[\mathrm{e}^{-\frac{i}{\hbar}T\Delta t}\mathrm{e}^{-\frac{i}{\hbar}V\Delta t} + O(\Delta t^2)]\,|x_n\rangle \\ &= \exp\left[-\frac{i}{\hbar}\Delta t\left(\frac{p_{n+1}^2}{2m} + V(x_n)\right)\right]\frac{1}{\sqrt{2\pi\hbar}}\exp\left(-\frac{i}{\hbar}p_{n+1}x_n\right), \end{aligned}$$

where the first equal is down by Baker-Campbell-Hausdorff formula (to be ...), and in the second line, all values are just numerical function, and one has

$$\begin{aligned} K(N,0) &= \lim_{N\to+\infty}\int\prod_{n=1}^{N-1}\mathrm{d}x_n\prod_{n=1}^{N}\frac{\mathrm{d}p_n}{2\pi\hbar}\exp\left[-\frac{i}{\hbar}\sum_{n=0}^{N}\Delta t\left(\frac{p_{n+1}^2}{2m}+V(x_n)-p_{n+1}\frac{x_{n+1}-x_n}{\Delta t}\right)\right] \\ &= \int\mathscr{D}x\mathscr{D}p\exp\left[-\frac{i}{\hbar}\int\mathrm{d}t(H-p\dot{x})\right]. \end{aligned}$$

To achieve form above, one needs to carry out integral about p, which reads

$$\begin{aligned} K(N,0) &= \int\mathscr{D}x\exp\left[-\frac{i}{\hbar}\int\mathrm{d}tV\right]\lim_{N\to+\infty}\int\prod_{n=1}^{N}\frac{\mathrm{d}p_n}{2\pi\hbar}\exp\left[-\frac{i}{\hbar}\frac{t}{N}(\frac{p_n^2}{2m}-p_n\dot{x})\right] \\ &= \int\mathscr{D}[x]\exp\left[\frac{i}{\hbar}\int\mathrm{d}t(\frac{1}{2}m\dot{x}^2-V)\right], \end{aligned}$$

where measure

$$\mathscr{D}[x]\equiv\lim_{N\to+\infty}\left(\frac{Nm}{it2\pi\hbar}\right)^{N/2}\prod_{n=1}^{N-1}\mathrm{d}x_n.$$

Here one may think about the similarity between path integral and statistical average $\mathrm{e}^{-\beta H}$ (this seems more likely to be the unitary evolution operator, however, we emphasize here by path integral since it shows a more intuitive picture for multi-channel propagation) and relates β, the (inverse) temperature, to imaginary time (like Wick rotation). By doing so, one may find in certain systems, a zero-temperature quantum system would correspond to classical finite-temperature system in thermodynamic limit. Furthermore, here in single-particle path integral, we choose classical-like phase space to insert identity, and when it comes to field theory, we shall insert identity formed by coherent states, since they are the states mostly close to a classical one, and that is somehow the spirit of path integral.

2.3.1 More about propagators

Propagators bridge the gap between initial and final states, during which the system experience all possible paths. It is related to the measurement observable in the physical way.

Let us firstly consider the classical electrodynamics, where we have Lagrangian density

$$\mathscr{L}=-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}-j_\mu A^\mu, \tag{2.97}$$

where $F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu$, and the equation of motion (EoM) reads

$$\partial^2 A^\mu=j^\mu, \tag{2.98}$$

under Lorentz gauge $\partial_\mu A^\mu=0$. Now we assume the source j^μ is given, and it is zero either when $x_0<x_0'$ or $x_0\to\infty$. Formally, we can write the solution as

$$A^\mu(x)=A_0^\mu(x)+\int\mathrm{d}^4x'G_R(x-x')j^\mu(x'), \tag{2.99}$$

where A_0^μ is the free field solution satisfying $\partial^2 A_0^\mu = 0$, and G_R is retarded Green's function satisfying

$$\partial^2 G_R(x-x') = \delta^{(4)}(x-x'),\ G_R(x-x') = 0 \text{ for } x_0 < x_0'. \tag{2.100}$$

In fact, one can verify that

$$\partial^2 A^\mu(x) = \partial^2 A_0^\mu(x) + \int \mathrm{d}^4x' \partial^2 G_R(x-x') j^\mu(x') = j^\mu(x). \tag{2.101}$$

If we loose the causality condition (retarded, i.e., no $G(x) = 0$ when $x_0 < 0$) and introduce Fourier transformation

$$G(x-x') = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \mathrm{e}^{-ik\cdot(x-x')} \tilde{G}(k),\ \partial^2 G(x-x') = \delta^{(4)}(x-x'), \tag{2.102}$$

we immediately have

$$-k^2 \tilde{G}(k) = 1 \Longrightarrow \tilde{G}(k) = -\frac{1}{k^2}. \tag{2.103}$$

However, notice that $k^2 = 0$, i.e., $k_0^2 - \boldsymbol{k}^2 = 0$ as the on-shell condition, is a singular point on real-energy axis, and we need to assign some prescription to bypass the pole, which actually determines the causality condition.

We may firstly assign retarded Green's function by $G_R(x-x') = \Theta(x_0 - x_0')G(x-x')$, where Θ is Heaviside step function that shares integral representation

$$\Theta(t) = \lim_{\epsilon\to 0^+} \int_{-\infty}^{+\infty} \frac{\mathrm{d}\omega}{2\pi i} \frac{\mathrm{e}^{i\omega t}}{\omega - i\epsilon}, \tag{2.104}$$

which by the Residue theorem gives $\Theta(t) = 1$ for $t > 0$ and 0 for $t < 0$. (If one forgets, recall that we need to integrate over a closed contour in complex plane, and for $t > 0$ ($t < 0$) we close the contour in upper (lower) half plane, where only pole $\omega = i\epsilon$ (no pole) is enclosed.) Inspired by this, we may write

$$G_R(x-x') = -\lim_{\epsilon\to 0^+} \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{\mathrm{e}^{-ik\cdot(x-x')}}{(k_0+i\epsilon)^2 - \boldsymbol{k}^2}, \tag{2.105}$$

where the poles $k_0 = \pm|\boldsymbol{k}| - i\epsilon$ are both located in lower half complex k_0-plane. This leads to $G_R(x-x') = 0$ for $x_0 < x_0'$, since we are forced to choose the contour in upper half plane with no pole enclosed. Also when $x_0 > x_0'$, we close the contour in lower half plane and pick up two poles, leading to

$$\begin{aligned} G_R(x-x') &= -\lim_{\epsilon\to 0^+} \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \int_{-\infty}^{+\infty} \frac{\mathrm{d}k_0}{2\pi} \frac{e^{-ik_0(x_0-x_0')}}{(k_0+i\epsilon)^2 - \boldsymbol{k}^2} \\ &= \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot(\boldsymbol{r}-\boldsymbol{r}')} \frac{\sin(|\boldsymbol{k}|(x_0-x_0'))}{|\boldsymbol{k}|} \\ &= \frac{1}{(2\pi)^2|\boldsymbol{r}-\boldsymbol{r}'|} \int_0^\infty \mathrm{d}k \sin(k(|\boldsymbol{r}-\boldsymbol{r}'|)) \sin(k(x_0-x_0')) \\ &= \frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{x}'|} \delta(x_0 - x_0' - |\boldsymbol{x}-\boldsymbol{x}'|). \end{aligned} \tag{2.106}$$

We thus conclude that

$$G_R(x-x') = \frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{x}'|} \delta(x_0 - x_0' - |\boldsymbol{x}-\boldsymbol{x}'|)\Theta(x_0 - x_0'). \tag{2.107}$$

Similarly, we may define advanced Green's function by $G_A(x-x') = \Theta(x'_0 - x_0)G(x-x')$, which leads to

$$G_A(x-x') = \frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{x}'|}\delta(x'_0 - x_0 - |\boldsymbol{x}-\boldsymbol{x}'|)\Theta(x'_0 - x_0). \tag{2.108}$$

Physically, G_R means the effect propagates from past to future, as when we include the pole $k_0 = -i\epsilon$, it gives a decaying wave propagating factor $\mathrm{e}^{-\epsilon(x_0-x'_0)}$ for $x_0 > x'_0$, while G_A means the effect propagates from future to past.

In quantum mechanics, we may define propagator (retarded Green's function) as

$$G(x',x) = -\frac{i}{\hbar}\Theta(t'-t)\left\langle x'|x\right\rangle, \tag{2.109}$$

with x standing for $(t,\boldsymbol{x})$ (this is just the K we defined above). The propagator obeys the Schrödinger equation

$$\left(i\hbar\frac{\partial}{\partial t'} - H'\right)G(x',x) = \delta^{(4)}(x'-x), \tag{2.110}$$

that basically, the derivative acts on $\Theta(t'-t)$ to give delta function in time, then this constrains $\boldsymbol{x}' = \boldsymbol{x}$.

We separate $H = H_0 + V$, and write down non-interacting propagator G_0 satisfying

$$\left(i\hbar\frac{\partial}{\partial t'} - H'_0\right)G_0(x',x) = \delta^{(4)}(x'-x), \tag{2.111}$$

or equivalently and more intuitively, we write in the matrix form

$$G_0(x',x) = \left\langle x'\right|\frac{1}{i\hbar\frac{\partial}{\partial t} - \hat{H}_0}\left|x\right\rangle, \ \hat{G}_0 = \frac{1}{i\hbar\frac{\partial}{\partial t} - \hat{H}_0}. \tag{2.112}$$

Then the full propagator G satisfies

$$(i\hbar\partial_t - \hat{H}_0 - \hat{V})\hat{G} = \mathbb{I}, \tag{2.113}$$

which leads to the Dyson equation by series expansion,

$$\begin{aligned}\hat{G} &= \frac{1}{i\hbar\partial_t - \hat{H}_0 - \hat{V}} = \frac{1}{\hat{G}_0^{-1} - \hat{V}} = \frac{1}{\hat{G}_0^{-1}(1 - \hat{G}_0\hat{V})} \\ &= \frac{1}{1-\hat{G}_0\hat{V}}\hat{G}_0 = \left(1 + \hat{G}_0\hat{V} + (\hat{G}_0\hat{V})^2 + \cdots\right)\hat{G}_0 \\ &= \hat{G}_0 + \hat{G}_0\hat{V}\hat{G},\end{aligned} \tag{2.114}$$

where one notices that $(\hat{A}\hat{B})^{-1} = \hat{B}^{-1}\hat{A}^{-1}$, or in the integral form in spatial representation,

$$G(x',x) = G_0(x',x) + \int \mathrm{d}^4x''\mathrm{d}^4x'''G_0(x',x'')V(x'',x''')G(x''',x). \tag{2.115}$$

Assuming the **time-independent** non-interacting Hamiltonian contains known spectrum $\hat{H}_0\left|n\boldsymbol{p}\right\rangle = \varepsilon_n(\boldsymbol{p})\left|n\boldsymbol{p}\right\rangle$, with $\left|n\boldsymbol{p}\right\rangle$ being simultaneous eigenket of $\hat{H}_0$ and momentum

operator $\hat{\boldsymbol{p}}$, and n labels different bands as inner degrees of freedom. Then we may write down non-interacting propagator in momentum space by Fourier transformation,

$$G_0(x',x)=\int\frac{\mathrm{d}^3p}{(2\pi\hbar)^3}\int\frac{\mathrm{d}\omega}{2\pi\hbar}\tilde{G}_0(\boldsymbol{p},\omega)\mathrm{e}^{i\boldsymbol{p}\cdot(\boldsymbol{x}'-\boldsymbol{x})/\hbar-i\omega(t'-t)/\hbar}, \tag{2.116}$$

where one notices that in the above, we have utilized momentum rather than wavevector, so that $\boldsymbol{p}=\hbar\boldsymbol{k}$. Notice that by writing only one momentum label, we indeed assumed the system is translational invariant, which is protected by $[\hat{H}_0,\hat{\boldsymbol{p}}]=0$ that leads to momentum conservation during propagation and makes $G_0(x',x)=G_0(x'-x)$. This point is quite interesting as the frequency/energy part is directly assumed to be conserved, as a result of the eigenvalue frame we chose, or because the time-evolution operator satisfies group property $\hat{U}(t_2,t_1)=\hat{U}(t_2-t_1)$ for time-independent Hamiltonian. Then we have

$$(\hbar\omega-H_0(\boldsymbol{p}))\tilde{G}_0(\boldsymbol{p},\omega)=1\Longrightarrow\tilde{G}_0(\boldsymbol{p},\omega)=\frac{1}{\hbar\omega-H_0(\boldsymbol{p})}, \tag{2.117}$$

and by further expanding the momentum-dependent Hamiltonian in eigenbasis, we have

$$\tilde{G}_0(\boldsymbol{p},\omega)=\sum_n\frac{|n\boldsymbol{p}\rangle\langle n\boldsymbol{p}|}{\hbar\omega-\varepsilon_n(\boldsymbol{p})}=\sum_n\tilde{G}_0^n(\boldsymbol{p},\omega)\,|n\boldsymbol{p}\rangle\langle n\boldsymbol{p}|\,, \tag{2.118}$$

where $|n\boldsymbol{p}\rangle$ is better to be understood as spinor in inner space, which is directly some wavefunction (rather than abstract ket).

Let us take the free particle as an example, where $\varepsilon(\boldsymbol{p})=\boldsymbol{p}^2/2m$ with no inner degree of freedom. Then we have

$$\tilde{G}_0(\boldsymbol{p},\omega)=\frac{1}{\hbar\omega-\boldsymbol{p}^2/2m}, \tag{2.119}$$

which has a pole at $\hbar\omega=\boldsymbol{p}^2/2m$ on real axis. This is bad as we cannot define proper contour to do Fourier transformation back to space-time representation. To fix this, we may introduce infinitesimal imaginary part to shift the pole off real axis, i.e.,

$$\tilde{G}_0^R(\boldsymbol{p},\omega)=\frac{1}{\hbar\omega-\boldsymbol{p}^2/2m+i\epsilon},\ \epsilon\to0^+, \tag{2.120}$$

again as retarded Green's function, where the pole is located at $\hbar\omega=\boldsymbol{p}^2/2m-i\epsilon$ in lower half complex ω-plane. One has to notice that the term 'retarded' always means causality in time domain, i.e., no effect before cause, or in language of quasi-particle, no propagation from future to past and it always decays with time. Translate the retarded Green's function back to space-time representation, we have

$$\begin{aligned}G_0^R(x'-x)&=\frac{i}{\hbar}\Theta(t'-t)\int\frac{\mathrm{d}^3p}{(2\pi\hbar)^3}\mathrm{e}^{i\boldsymbol{p}\cdot(\boldsymbol{x}'-\boldsymbol{x})/\hbar-i\frac{\boldsymbol{p}^2}{2m\hbar}(t'-t)}\\&=-i\frac{m}{\hbar^2}\left(\frac{m}{2\pi\hbar(t'-t)}\right)^{3/2}\exp\left[\frac{im|\boldsymbol{x}'-\boldsymbol{x}|^2}{2\hbar(t'-t)}\right]\Theta(t'-t).\end{aligned} \tag{2.121}$$

In the last step, we have utilized Gaussian integral formula. Furthermore, if the wavefunction is known as

$$\psi_{\boldsymbol{p}}(\boldsymbol{x},t)=\frac{1}{(2\pi\hbar)^{3/2}}\exp\left[\frac{i}{\hbar}(\boldsymbol{p}\cdot\boldsymbol{x}-\varepsilon(\boldsymbol{p})t)\right], \tag{2.122}$$

with similar consideration as above, we may conclude that

$$G_0^R(x'-x) = \frac{i}{\hbar}\Theta(t'-t)\int \mathrm{d}^3p\ \psi_{\boldsymbol{p}}(\boldsymbol{x}',t')\psi_{\boldsymbol{p}}^\dagger(\boldsymbol{x},t). \tag{2.123}$$

Note that by lifting $\psi_{\boldsymbol{p}}^*(\boldsymbol{x},t)$ to $\psi_{\boldsymbol{p}}^\dagger(\boldsymbol{x},t)$, we emphasize that the wavefunction could be spinor with inner degrees of freedom. By again writing $\psi_{n\boldsymbol{p}}(\boldsymbol{x},t) = e^{-i\varepsilon_n(\boldsymbol{p})t/\hbar}\psi_{n\boldsymbol{p}}(\boldsymbol{x})$, we can write

$$G_0^R(x'-x) = \lim_{\epsilon\to 0^+}\sum_n \int \mathrm{d}^3p \int \frac{\mathrm{d}\omega}{2\pi}\frac{\psi_{n\boldsymbol{p}}(\boldsymbol{x}')\psi_{n\boldsymbol{p}}^\dagger(\boldsymbol{x})}{\hbar\omega-\varepsilon_n(\boldsymbol{p})+i\epsilon}\mathrm{e}^{-i\omega(t'-t)}, \tag{2.124}$$

or

$$G_0^R(\boldsymbol{x}'-\boldsymbol{x};\omega) = \lim_{\epsilon\to 0^+}\sum_n \int \mathrm{d}^3p \frac{\psi_{n\boldsymbol{p}}(\boldsymbol{x}')\psi_{n\boldsymbol{p}}^\dagger(\boldsymbol{x})}{\hbar\omega-\varepsilon_n(\boldsymbol{p})+i\epsilon}. \tag{2.125}$$

Indeed, the system can have no translational invariance, or in a more general view, we only admit some quantum number n, while the system still maintains time independence, then we may write

$$G_0^R(\boldsymbol{x}',\boldsymbol{x};\omega) = \lim_{\epsilon\to 0^+}\sum_n \frac{\psi_n(\boldsymbol{x}')\psi_n^\dagger(\boldsymbol{x})}{\hbar\omega-\varepsilon_n+i\epsilon} = \lim_{\epsilon\to 0^+}\langle\boldsymbol{x}'|\frac{1}{\hbar\omega-\hat{H}_0+i\epsilon}|\boldsymbol{x}\rangle. \tag{2.126}$$

By

$$\lim_{\epsilon\to 0^+}\frac{1}{x\pm i\epsilon} = \mathscr{P}\frac{1}{x}\mp i\pi\delta(x), \tag{2.127}$$

we have

$$\mathrm{Re}\, G_0^R(\boldsymbol{x}',\boldsymbol{x};\omega) = \sum_n \mathscr{P}\frac{\psi_n(\boldsymbol{x}')\psi_n^\dagger(\boldsymbol{x})}{\hbar\omega-\varepsilon_n}, \tag{2.128a}$$

$$\mathrm{Im}\, G_0^R(\boldsymbol{x}',\boldsymbol{x};\omega) = -\pi\sum_n \psi_n(\boldsymbol{x}')\psi_n^\dagger(\boldsymbol{x})\delta(\hbar\omega-\varepsilon_n). \tag{2.128b}$$

From this, the density of states (DOS) could be defined as

$$\rho(\omega) = -\frac{1}{\pi}\mathrm{Im}\int \mathrm{Tr}\, G_0^R(\boldsymbol{x},\boldsymbol{x};\omega) = \sum_n \delta(\hbar\omega-\varepsilon_n), \tag{2.129}$$

where trace is over both spatial and inner degrees of freedom.

2.4 Gauge Transformation

In electrodynamics, one has Maxwell equations related two fields: electric and magnetic, denoted as $\boldsymbol{E}$ and $\boldsymbol{B}$, which could be generated from four-potential $A^\mu = (\varphi, \boldsymbol{A})$

$$\boldsymbol{E} = -\nabla\varphi - \partial_t\boldsymbol{A},\ \boldsymbol{B} = \nabla\times\boldsymbol{A},\ F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu, \tag{2.130}$$

which are invariant under $U(1)$ gauge transformation by Λ, $\partial^\mu = \frac{\partial}{\partial x_\mu} = (\partial_t, -\nabla)$,

$$A^\mu \to A^\mu - \partial^\mu\Lambda,\ \varphi\to\varphi-\partial_t\Lambda,\ \boldsymbol{A}\to\boldsymbol{A}+\nabla\Lambda. \tag{2.131}$$

Classical Lagrangian for charged particle is

$$L = \frac{1}{2}m\dot{\boldsymbol{x}}^2 - q\varphi(\boldsymbol{x},t) + q\dot{\boldsymbol{x}}\cdot\boldsymbol{A}(\boldsymbol{x},t), \tag{2.132}$$

whose correctness could be verified by EoM,

$$\nabla_x L - \frac{\mathrm{d}}{\mathrm{d}t}(\nabla_{\dot{x}} L) = 0 \Longrightarrow m\ddot{\boldsymbol{x}} = q(\boldsymbol{E} + \dot{\boldsymbol{x}}\times\boldsymbol{B}). \tag{2.133}$$

Then, its canonical momentum is

$$\boldsymbol{p} = \nabla_{\dot{x}} L = m\dot{\boldsymbol{x}} + q\boldsymbol{A}, \tag{2.134}$$

thus Hamiltonian

$$H = \boldsymbol{p}\cdot\boldsymbol{x} - L = \frac{\boldsymbol{\Pi}^2}{2m} + q\varphi, \tag{2.135}$$

where physical momentum

$$\boldsymbol{\Pi} = m\dot{\boldsymbol{x}} = \boldsymbol{p} - q\boldsymbol{A}. \tag{2.136}$$

Notice that by canonical,

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},\ \boldsymbol{p} \sim -i\hbar\nabla. \tag{2.137}$$

At this time, one could examine that requiring probability conservation,

$$\boldsymbol{j} = \frac{\hbar}{m}\operatorname{Im}(\psi^*\nabla\psi) - \frac{q}{m}\boldsymbol{A}|\psi|^2 = \frac{\rho}{m}(\nabla S - q\boldsymbol{A}), \tag{2.138}$$

which could be achieved by the mentioned substitution, also known as minimum coupling

$$\partial_\mu \to D_\mu = \partial_\mu + i\frac{q}{\hbar}A_\mu,\ \partial_t \to \partial_t + i\frac{q}{\hbar}\varphi,\ \nabla \to \nabla - i\frac{q}{\hbar}\boldsymbol{A}. \tag{2.139}$$

One could also examine that by

$$\psi \to \mathrm{e}^{iq\Lambda/\hbar}\psi, \tag{2.140}$$

together with gauge transformation about potential, the Schrödinger equation would stand, and probability flux is also gauge-invariant as a result from covariant derivative D_μ.

2.4.1 Aharonov-Bohm Effect

By classical Lagrangian for charged particle

$$L = \frac{1}{2}m\dot{\boldsymbol{x}}^2 - q\varphi(\boldsymbol{x},t) + q\dot{\boldsymbol{x}}\cdot\boldsymbol{A}(\boldsymbol{x},t) \equiv L_0 + q\dot{\boldsymbol{x}}\cdot\boldsymbol{A}(\boldsymbol{x},t), \tag{2.141}$$

together with the transition amplitude in path integral form,

$$K(x_2,x_1) = \int \mathscr{D}[x(t)]\exp\left(i\frac{S[x(t)]}{\hbar}\right) = \int \mathscr{D}[x(t)]\exp\left(i\frac{S_0[x(t)]}{\hbar}\right)\exp\left(i\frac{e}{\hbar}\int_{x_1}^{x_2}\mathrm{d}\boldsymbol{x}\cdot\boldsymbol{A}\right), \tag{2.142}$$

then given a particle propagating through a zone with vector field, but different paths, say up and down,

$$\begin{aligned} K_{\rm up}(x_2,x_1) &= \int \mathscr{D}[x(t)] \exp\left(i\frac{S_0[x(t)]}{\hbar}\right) \exp\left(i\frac{e}{\hbar}\int_{x_1}^{x_2} \mathrm{d}\boldsymbol{x}_{\rm up}\cdot\boldsymbol{A}\right) \\ &= \int \mathscr{D}[x(t)] \exp\left(i\frac{S_0[x(t)]}{\hbar}\right) \exp\left[i\frac{e}{\hbar}\int_{x_1}^{x_2} \boldsymbol{A}\cdot(\mathrm{d}\boldsymbol{x}_{\rm up}-\mathrm{d}\boldsymbol{x}_{\rm down}+\mathrm{d}\boldsymbol{x}_{\rm down})\right] \\ &= \exp\left(i\frac{e}{\hbar}\oint \mathrm{d}\boldsymbol{x}\cdot\boldsymbol{A}\right) K_{\rm down}(x_2,x_1), \end{aligned}$$

i.e.,

$$\varphi_{\rm up}-\varphi_{\rm down} = \frac{e}{\hbar}\oint \mathrm{d}\boldsymbol{x}\cdot\boldsymbol{A} = \frac{e}{\hbar}\iint \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B} = \frac{e}{\hbar}\phi_B. \tag{2.143}$$

Magnetic Monopole

2.5 Charged Particles in Uniform Magnetic Field

For charged particle in magnetic field, the Hamiltonian of system is (ignore operator tip once again)

$$H = \frac{\boldsymbol{\Pi}^2}{2m} = \frac{(\boldsymbol{p}-q\boldsymbol{A})^2}{2m},\ [x_i,p_j]=i\hbar. \tag{2.144}$$

Now say that, for electron with $q=-e$ ($e>0$) moving in 2-D $x-y$ plane with vertical uniform magnetic field $\boldsymbol{B}=B\hat{z}$ ($B>0$), one has corresponding vector potential $\boldsymbol{A}$ as $\nabla\times\boldsymbol{A}=\boldsymbol{B}$, then immediately,

$$[\Pi_x,\Pi_y] = [p_x+eA_x, p_y+eA_y] = -ie\hbar(\nabla\times\boldsymbol{A})_z = -ie\hbar B, \tag{2.145}$$

thus

$$[\Pi_x - i\Pi_y, \Pi_x+i\Pi_y] = 2i[\Pi_x,\Pi_y] = 2e\hbar B, \tag{2.146}$$

from which one could define Bosonic annihilation and creation operators as

$$a = \frac{1}{\sqrt{2e\hbar B}}(\Pi_x - i\Pi_y),\ a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\Pi_x+i\Pi_y),\ [a,a^\dagger]=1. \tag{2.147}$$

Then one has

$$H = \hbar\omega(a^\dagger a+\frac{1}{2}),\ \omega=\frac{eB}{m}, \tag{2.148}$$

and clearly it will own the same algebra structure like a harmonic oscillator with

$$N = a^\dagger a,\ N|n\rangle = n|n\rangle,\ H|n\rangle = \hbar\omega(n+\frac{1}{2})|n\rangle. \tag{2.149}$$

And the state is constructed by

$$a|0\rangle = 0,\ |n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n|0\rangle, \tag{2.150}$$

with

$$a|n\rangle = \sqrt{n}|n-1\rangle,\ a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \tag{2.151}$$

The above relations hold for any gauge, and the difference lies in the real-space morphology, i.e., the wavefunction.

Landau Gauge

In this gauge, one simply chooses $\boldsymbol{A} = (-By, 0, 0)$, which means that

$$\begin{cases} \Pi_x = p_x - eBy, \ \Pi_y = p_y, \end{cases} \tag{2.152}$$

and one familiar with free-particle solution would yield that now $p_x = \hbar k_x$ with k_x a good quantum number. What is more,

$$a = \frac{1}{\sqrt{2e\hbar B}}(p_x - eBy - ip_y) = \sqrt{\frac{eB}{2\hbar}}(l_B^2 k_x - y - \frac{ip_y}{eB}), \ l_B^2 = \frac{\hbar}{m\omega} = \frac{\hbar}{eB}. \tag{2.153}$$

(Here in the last a is written is position representation, but we ignore the difference.) And in this case, one may write

$$H = \frac{(eB)^2}{2m}\left[(y - l_B^2 k_x)^2 + \frac{p_y^2}{(eB)^2}\right] = \frac{p_y^2}{2m} + \frac{1}{2}m\omega^2(y - l_B^2 k_x)^2, \tag{2.154}$$

which is a harmonic oscillator with central at $y_0 = -l_B^2 k_x$, the energy levels formed here are known as *Landau levels* with high degeneracy, and its ground wavefunction may be written as

$$\psi_{0,k_x}(y) = \frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{ik_x x}\frac{1}{\pi^{1/4}\sqrt{l_B}}\exp\left(-\frac{(y - l_B^2 k_x)^2}{2l_B^2}\right). \tag{2.155}$$

Symmetry Gauge

Here one lets $\boldsymbol{A} = (-By/2, Bx/2, 0)$, clearly, $\boldsymbol{A}^{\mathrm{sym}} = \boldsymbol{A}^{\mathrm{Lan}} + \nabla\Lambda$ with $\Lambda = Bxy/2$. Now, amazingly, some properties would emerge if one defines another set of annihilation/creation operators by introducing

$$\tilde{\boldsymbol{\Pi}} = \boldsymbol{p} + q\boldsymbol{A} = \boldsymbol{p} - e\boldsymbol{A} \tag{2.156}$$

with maybe less physical meaning. However, one could see it that

$$[\tilde{\Pi}_x, \tilde{\Pi}_y] = ie\hbar B, \tag{2.157}$$

and by letting

$$b = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\Pi}_x + i\tilde{\Pi}_y), \ b^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\Pi}_x - i\tilde{\Pi}_y), \ [b, b^\dagger] = 1. \tag{2.158}$$

This set of operators could be defined under any gauge, but only under symmetric gauge we have

$$[\Pi_i, \tilde{\Pi}_j] = [p_i + eA_i, p_j - eA_j] = ie\hbar[\partial_i A_j + \partial_j A_i] = 0 \tag{2.159}$$

so that

$$[a^{(\dagger)}, b^{(\dagger)}] = 0, \tag{2.160}$$

thus one has

$$[N_a, N_b] = 0, \tag{2.161}$$

and one could find simultaneous eigenket as

$$\begin{cases} N_a \left|m, n\right\rangle = n \left|m, n\right\rangle \\ N_b \left|m, n\right\rangle = m \left|m, n\right\rangle \end{cases}, \tag{2.162}$$

with

$$|m,n\rangle = \frac{1}{\sqrt{m!n!}}(a^\dagger)^m(b^\dagger)^n\,|m,n\rangle\,. \tag{2.163}$$

Here one may notice that m describes degeneracy in each Landau level, as Hamiltonian does not contain b and in fact $[H, N_b] = 0$.

What is more, by introducing complex derivative

$$\begin{cases} z = x + iy,\ \partial_z = \dfrac{\partial x}{\partial z}\partial_x + \dfrac{\partial y}{\partial z}\partial_y = \dfrac{1}{2}(\partial_x - i\partial_y) \\ \bar{z} = x - iy,\ \partial_{\bar{z}} = \dfrac{1}{2}(\partial_x + i\partial_y) \end{cases}, \tag{2.164}$$

one finds that

$$\begin{aligned} a &= \frac{1}{\sqrt{2e\hbar B}}\left[p_x + eA_x - i(p_y + eA_y)\right] = \frac{1}{\sqrt{2e\hbar B}}\left[i\frac{eB}{2}(iy + x) - i\hbar(\partial_x - i\partial_y)\right] \\ &= -i\sqrt{2}(l_B\partial_z + \frac{\bar{z}}{4l_B}), \\ b &= -i\sqrt{2}(l_B\partial_{\bar{z}} - \frac{z}{4l_B}), \end{aligned} \tag{2.165}$$

thus, by requiring

$$\langle x,y|a|0,0\rangle = 0 = \langle x,y|b|0,0\rangle\,, \tag{2.166}$$

one has

$$\psi_{0,0}(x,y) \propto \exp\left(-\frac{z\bar{z}}{4l_B^2}\right) = \exp\left(-\frac{x^2+y^2}{4l_B^2}\right), \tag{2.167}$$

while (leading order)

$$\psi_{m,0}(x,y) \propto \left(\frac{z}{l_B}\right)^m \exp\left(-\frac{z\bar{z}}{4l_B^2}\right), \tag{2.168}$$

which serves as a basis for the lowest Landau level.

One may also notice it that upon gauge transformation,

$$\psi_{0,0}(x,y) \to \exp\Big(-i\frac{e}{\hbar}(-\Lambda)\Big)\psi_{0,0}(x,y) \propto \exp\left(-\frac{x^2+y^2}{4l_B^2} + i\frac{xy}{2l_B^2}\right) = \exp\left[-\frac{(x-iy)^2}{4l_B^2} - \frac{y^2}{2l_B^2}\right].$$

And by integrating x out (equivalent to Fourier transformation at $k_x = 0$), one just gets wavefunction under Landau gauge at $k_x = 0$, with a normalization needed.

Chapter 3

Angular Momentum

3.1 Rotational Operator

We define for an arbitrary ket the *rotational operator*

$$|\alpha\rangle \to |\alpha\rangle_R = \hat{D}(R)\,|\alpha\rangle\,, \tag{3.1}$$

with the most natural representation as in 3-D Euclid space as

$$|\boldsymbol{r}\rangle \to |\boldsymbol{r}\rangle_R = \hat{D}(R)\,|\boldsymbol{r}\rangle = |R\boldsymbol{r}\rangle\,,\ \boldsymbol{r}_R = R\boldsymbol{r}, \tag{3.2}$$

where R is the 3×3 matrix satisfying that

$$r^2 = \boldsymbol{r}_R^T\cdot\boldsymbol{r}_R = \boldsymbol{r}^T R^T R\boldsymbol{r} = \boldsymbol{r}^T\boldsymbol{r} \Longrightarrow R^T R = \mathbb{I}, \tag{3.3}$$

and in this way we say that $\{R\}$ form a 3-D representation of $SO(3)$ group, known as *special orthogonal* group. Here, 3 denotes its DoFs. Furthermore, from the operator left-acting form, we realize that generally,

$$\hat{D}^\dagger(R)\hat{D}(R) = \mathbb{I}, \tag{3.4}$$

then they would form *special-unitary* group whose DoF is also three so that it is indeed $SU(2)$ group. In language of group theory, these two groups are very close as *there exists a two-to-one homomorphic mapping of the group SU(2) onto the group SO(3).*

For wavefunctions, we have

$$\psi_{\alpha_R}(\boldsymbol{r}_R) = \langle\boldsymbol{r}|\,\hat{D}^\dagger(R)\hat{D}(R)\,|\alpha\rangle = \langle\boldsymbol{r}|\alpha\rangle = \psi_\alpha(\boldsymbol{r}), \tag{3.5}$$

or

$$\psi_{\alpha_R}(\boldsymbol{r}) = \langle\boldsymbol{r}|\,\hat{D}(R)\,|\alpha\rangle = \langle R^{-1}\boldsymbol{r}|\alpha\rangle = \psi_\alpha(\boldsymbol{r}_{R^{-1}}), \tag{3.6}$$

while LHS is subjective and RHS is passive.

For operators,

$$\hat{O} \to \hat{O}_R = \hat{D}(R)\hat{O}\hat{D}^\dagger(R). \tag{3.7}$$

Euclid Rotation

By group representation we expect that $\{\hat{D}(R)\}$ share the same group properties and possible Lie group algebra with $\{R\}$, then we would like to see more about R. Clearly, in 3-D Euclid space, we have three basic rotations,

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{pmatrix},\ R_y(\phi) = \begin{pmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{pmatrix}, R_z(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3.8}$$

And clearly, $\phi \in [0, 2\pi]$ for each rotation, and we now have a three-parameter generated compact non-Abelian Lie group. To find its generators, we consider infinitesimal rotations as

$$R_x(\varepsilon) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\varepsilon^2/2 & -\varepsilon \\ 0 & \varepsilon & 1-\varepsilon^2/2 \end{pmatrix} = \mathbb{I} - i\varepsilon J_x + O(\varepsilon^2),\ J_x = \begin{pmatrix} 0 & & \\ & & -i \\ & i & \end{pmatrix}. \tag{3.9}$$

Similarly,

$$R_y(\varepsilon) = \mathbb{I} - i\varepsilon J_y + O(\varepsilon^2),\ J_y = \begin{pmatrix} & & i \\ & 0 & \\ -i & & \end{pmatrix}, \tag{3.10}$$

$$R_z(\varepsilon) = \mathbb{I} - i\varepsilon J_z + O(\varepsilon^2),\ J_z = \begin{pmatrix} & -i & \\ i & & \\ & & 0 \end{pmatrix}. \tag{3.11}$$

We have added $-i$ manually to show some unitary nature as mentioned in first chapter and connection to general rotation, however, for the special-orthogonal one could make it simpler by

$$R(\varepsilon) = \mathbb{I} + \varepsilon A + O(\varepsilon^2),\ A + A^T = 0. \tag{3.12}$$

Back to our J, and it is easy to verify that

$$[J_i, J_j] = i\varepsilon_{ijk} J_k, \tag{3.13}$$

similar to the one of spin-half $\boldsymbol{S} = \hbar\boldsymbol{\sigma}/2$, but with no $\hbar$. Using this relation one may show it that

$$[R_i(\varepsilon), R_j(\varepsilon)] = -\varepsilon^2[J_i, J_j] = -i\varepsilon_{ijk}\varepsilon^2 J_k = \varepsilon_{jik} R_k(\varepsilon^2) - \mathbb{I} = O(\varepsilon^2), \tag{3.14}$$

and in this way may say that classical infinitesimal rotations commute with each other. One may also notice it that representation for generators is not unique, and the important thing is their commutation relation. Using generators above, finite angle rotation would be

$$R_i(\phi) = \lim_{N\to+\infty} \left(\mathbb{I} - iJ_i\frac{\phi}{N}\right)^N = e^{-iJ_i\phi}, \tag{3.15}$$

and if it is around some axis $\hat{n}$, one may first observe, say we let $\boldsymbol{\varepsilon}$ to be its infinitesimal case, since infinitesimal rotations commute,

$$R_n(\boldsymbol{\varepsilon}) = R_x(\varepsilon_x)R_y(\varepsilon_y)R_z(\varepsilon_z) + O(\varepsilon^2) = \mathbb{I} - i\boldsymbol{J}\cdot\boldsymbol{\varepsilon} + O(\varepsilon^2), \tag{3.16}$$

which would lead to

$$R_n(\phi) = e^{-i\boldsymbol{J}\cdot\hat{n}\phi} = e^{-i\boldsymbol{J}\cdot\boldsymbol{\phi}},\ \boldsymbol{\phi} = \phi\hat{n}. \tag{3.17}$$

On the other hand, if one writes explicitly,

$$\boldsymbol{V}_R = R_n(\boldsymbol{\varepsilon})\boldsymbol{V} = \begin{pmatrix} 1 & -\varepsilon_z & \varepsilon_y \\ \varepsilon_z & 1 & -\varepsilon_x \\ -\varepsilon_y & \varepsilon_x & 1 \end{pmatrix} \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix} = \boldsymbol{V} + \boldsymbol{\varepsilon} \times \boldsymbol{V}, \tag{3.18}$$

where the latter term describes how rotation proceeds.

Quantum Mechanical Rotation

Now, similarly, we may give generators to $\{\hat{D}(R)\}$ as $\hat{\boldsymbol{J}}$

$$[\hat{J}_i, \hat{J}_j] = i\hbar\varepsilon_{ijk}\hat{J}_k, \tag{3.19}$$

where additional factor $\hbar$ is added to each generator. And similarly, in spirit of group representation,

$$\hat{D}_i(\phi) = \exp\left(-i\frac{\hat{J}_i}{\hbar}\phi\right),\ \hat{D}_n(\phi) = \exp\left(-i\frac{\hat{\boldsymbol{J}}\cdot\hat{n}}{\hbar}\phi\right). \tag{3.20}$$

Here, $\hat{\boldsymbol{J}}$ is explained as *angular momentum* operator in analogy with classical physics.

3.2 Angular Momentum

Orbital

By formally introducing representation of $\hat{D}$ on real-space wavefunction as

$$D_n(\phi)\psi_\alpha(\boldsymbol{r}) \equiv \langle\boldsymbol{r}|\,\hat{D}_n(\phi)\,|\alpha\rangle = \psi_{\alpha_R}(\boldsymbol{r}) = \psi_\alpha(\boldsymbol{r}_{R^{-1}}), \tag{3.21}$$

and for infinitesimal rotation,

$$\begin{aligned} D_n(\boldsymbol{\varepsilon})\psi_\alpha(\boldsymbol{r}) &= \psi_\alpha(R^{-1}\boldsymbol{r}) = \psi_\alpha(\boldsymbol{r} - \boldsymbol{\varepsilon}\times\boldsymbol{r}) = \psi_\alpha(\boldsymbol{r}) - (\boldsymbol{\varepsilon}\times\boldsymbol{r})\cdot\nabla\psi_\alpha(\boldsymbol{r}) \\ &= [1 - \boldsymbol{\varepsilon}\cdot(\boldsymbol{r}\times\nabla)]\psi_\alpha(\boldsymbol{r}) = \left(1 - \frac{i}{\hbar}\boldsymbol{\varepsilon}\cdot\boldsymbol{L}\right)\psi_\alpha(\boldsymbol{r}), \end{aligned} \tag{3.22}$$

where we formally introduce the representation of orbital angular momentum operator

$$\boldsymbol{L} = \boldsymbol{r}\times\boldsymbol{p} = -i\hbar\boldsymbol{r}\times\nabla,\ \ \text{(Euclid)}. \tag{3.23}$$

And back to operator, we could also let

$$\hat{\boldsymbol{L}} = \hat{\boldsymbol{r}}\times\hat{\boldsymbol{p}}, \tag{3.24}$$

with commutation relation

$$[\hat{L}_i, \hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k, \tag{3.25}$$

as Lie algebra of $SO(3)$ group. And spatial rotation operator would be

$$\hat{D}^L_n(\phi) = \exp\left(-\frac{i}{\hbar}\hat{\boldsymbol{L}}\cdot\hat{n}\phi\right). \tag{3.26}$$

Here spin part does not appear as we have chosen somehow explicitly only in real space.

Spin

All spin could be described by some irreducible representation of group $SU(2)$ with half integers: $n/2,\ n = 0, 1, 2, \cdots$. But it seems that until now basic particles prefer $n = 1, 2$, i.e. spin-half *Fermion* and spin-one *Boson*, though some argue we have spinless Higgs Bosons and spin-two gravitons. Notice that this relation between spin and Fermion or Boson is true only under fundamental physics, or simply to some level, relativistic quantum mechanics or field theory. If we turn to condensed matter, there could be exotic emergent phenomenon like spinless Fermions and anyons. Anyway, we just consider our simple case, and the spin-rotational generators as for $SU(2)$ group $\hat{\boldsymbol{S}}$ satisfies that

$$[\hat{S}_i, \hat{S}_j] = i\hbar\varepsilon_{ijk}\hat{S}_k, \tag{3.27}$$

and

$$\hat{D}_n^S(\phi) = \exp\left(-\frac{i}{\hbar}\hat{\boldsymbol{S}}\cdot\hat{n}\phi\right). \tag{3.28}$$

The simplest non-trivial representation is for spin-half, which should be familiar as

$$\boldsymbol{S} = \frac{\hbar}{2}\boldsymbol{\sigma}, \tag{3.29}$$

and from basic multiplication

$$\sigma_i\sigma_j = \delta_{ij}\mathbb{I} + i\varepsilon_{ijk}\sigma_k, \tag{3.30}$$

one could get Lie and Clifford algebra

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k,\ \{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}, \tag{3.31}$$

together with pseudo-scalar multiplication

$$(\boldsymbol{\sigma}\cdot\boldsymbol{m})(\boldsymbol{\sigma}\cdot\boldsymbol{n}) = \boldsymbol{m}\cdot\boldsymbol{n}\mathbb{I} + i\boldsymbol{\sigma}\cdot(\boldsymbol{m}\times\boldsymbol{n}). \tag{3.32}$$

Using generators, we find corresponding rotational matrix in spin-space

$$D_n^{S=1/2}(\phi) = \exp\left(-\frac{i}{\hbar}\boldsymbol{S}\cdot\hat{n}\phi\right) = \exp\left(-i\boldsymbol{\sigma}\cdot\hat{n}\frac{\phi}{2}\right), \tag{3.33}$$

using that

$$(\boldsymbol{\sigma}\cdot\hat{n})^2 = \hat{n}\cdot\hat{n}\mathbb{I} + i\boldsymbol{\sigma}\cdot(\hat{n}\times\hat{n}) = \mathbb{I},$$

$$\begin{aligned}\exp\left(-i\boldsymbol{\sigma}\cdot\hat{n}\frac{\phi}{2}\right) &= \sum_{p=0}^{+\infty}\frac{(-i)^p}{p!}(\boldsymbol{\sigma}\cdot\hat{n})^p\left(\frac{\phi}{2}\right)^p = \sum_{p=0}^{+\infty}\frac{(-i)^{2p}}{(2p)!}\left(\frac{\phi}{2}\right)^{2p}\mathbb{I} - i\sum_{p=0}^{+\infty}\frac{(-i)^{2p}}{(2p+1)!}\left(\frac{\phi}{2}\right)^{2p+1}(\boldsymbol{\sigma}\cdot\hat{n})\\ &= \cos\left(\frac{\phi}{2}\right)\mathbb{I} - i(\boldsymbol{\sigma}\cdot\hat{n})\sin\left(\frac{\phi}{2}\right).\end{aligned}$$

For example, for z-axis rotation,

$$D_z^{S=1/2}(\phi) = \exp\left(-i\sigma_z\frac{\phi}{2}\right) = \cos\left(\frac{\phi}{2}\right)\mathbb{I} - i\sigma_z\sin\left(\frac{\phi}{2}\right), \tag{3.34}$$

from which one sees that

$$D_z^{S=1/2}(2\pi) = -\mathbb{I},$$

as if it only passes half periodicity, and we need $\phi = 4\pi$ to make it back. In this way we could see the two-to-one map.

Meanwhile, by treating $\hat{\boldsymbol{S}}$ as a 'vector', we may have

$$\hat{D}_n^{S=1/2,\dagger}(\phi)\hat{\boldsymbol{S}}\hat{D}_n^{S=1/2}(\phi) = R(\boldsymbol{\phi})\hat{\boldsymbol{S}}, \tag{3.35}$$

where R is just 3-D representation of $SO(3)$ group we start from. This could be verified using matrix representation with infinitesimal rotation

$$\begin{aligned} D_n^{S=1/2,\dagger}(\varepsilon)\sigma_i D_n^{S=1/2}(\varepsilon) &= \left(\cos\left(\frac{\varepsilon}{2}\right)\mathbb{I} + i(\boldsymbol{\sigma}\cdot\hat{n})\sin\left(\frac{\varepsilon}{2}\right)\right)\sigma_i\left(\cos\left(\frac{\varepsilon}{2}\right)\mathbb{I} - i(\boldsymbol{\sigma}\cdot\hat{n})\sin\left(\frac{\varepsilon}{2}\right)\right) \\ &= \cos^2\left(\frac{\varepsilon}{2}\right)\sigma_i + \sin^2\left(\frac{\varepsilon}{2}\right)[2n_i(\boldsymbol{\sigma}\cdot\hat{n}) - \sigma_i] + 2\cos\left(\frac{\varepsilon}{2}\right)\sin\left(\frac{\varepsilon}{2}\right)(\hat{n}\times\boldsymbol{\sigma}) \\ &= \cos\varepsilon\sigma_i + \sin\varepsilon(\hat{n}\times\boldsymbol{\sigma})_i + O(\varepsilon^2) = (\boldsymbol{\sigma} + \boldsymbol{\varepsilon}\times\boldsymbol{\sigma})_i + O(\varepsilon^2), \end{aligned}$$

where

$$\begin{aligned} (\boldsymbol{\sigma}\cdot\hat{n})\sigma_i(\boldsymbol{\sigma}\cdot\hat{n}) &= \sigma_j\sigma_i\sigma_k n_j n_k = (\delta_{ji}\mathbb{I} + i\varepsilon_{jim}\sigma_m)\sigma_k n_j n_k \\ &= \delta_{ij}n_j\sigma_k n_k + \varepsilon_{ijm}\varepsilon_{mkn}\sigma_n n_j n_k = n_i(\boldsymbol{\sigma}\cdot\hat{n}) + n_i(\boldsymbol{\sigma}\cdot\hat{n}) - \sigma_i(\hat{n}\cdot\hat{n}) \\ &= 2n_i(\boldsymbol{\sigma}\cdot\hat{n}) - \sigma_i, \end{aligned}$$

and

$$(\boldsymbol{\sigma}\cdot\hat{n})\sigma_i - \sigma_i(\boldsymbol{\sigma}\cdot\hat{n}) = -[\sigma_i,\sigma_j]n_j = -2i\varepsilon_{ijk}\sigma_k n_j = -2i(\hat{n}\times\boldsymbol{\sigma})_i$$

were used. This relation could be generalized to

$$\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi) = R(\boldsymbol{\phi})\hat{\boldsymbol{J}},\ R(\boldsymbol{\phi}) = \exp(-i\boldsymbol{J}_3\cdot\hat{n}\phi), \tag{3.36}$$

as

$$\begin{aligned} i\hbar\frac{\mathrm{d}}{\mathrm{d}\phi}\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi) &= i\hbar\left(\frac{\mathrm{d}}{\mathrm{d}\phi}\hat{D}_n^{J,\dagger}(\phi)\right)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi) + i\hbar\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\left(\frac{\mathrm{d}}{\mathrm{d}\phi}\hat{D}_n^J(\phi)\right) \\ &= \hat{D}_n^{J,\dagger}(\phi)[\boldsymbol{J},\boldsymbol{J}\cdot\hat{n}]\hat{D}_n^J(\phi) = \hat{D}_n^{J,\dagger}(\phi)i\hbar(\hat{n}\times\boldsymbol{J})\hat{D}_n^J(\phi) \\ &= i\hbar\hat{n}\times\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi) = \hbar\hat{n}\cdot\boldsymbol{J}_3\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi), \end{aligned}$$

then by $\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi)\big|_{\phi=0} = \hat{\boldsymbol{J}}$,

$$\hat{D}_n^{J,\dagger}(\phi)\hat{\boldsymbol{J}}\hat{D}_n^J(\phi) = \exp(-i\boldsymbol{J}_3\cdot\hat{n}\phi)\hat{\boldsymbol{J}},$$

where $\boldsymbol{J}_3$ is generator of $SO(3)$ in 3-D. This formula is related to adjoint representation.

We would also like to mention that rotation to spin-one is much like spatial rotation, but usually they share different generators, one is from the eigenvalue problem below, while the latter is what we talked about early.

Total Angular Momentum

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}}. \tag{3.37}$$

While representation of corresponding operator needs metric involved, i.e., under relativistic discussion, which needs analysis of $SO(1,3)$ group known as *Lorentz group*, which is, locally, a direct summation of two $SU(2)$ groups.

3.3 Eigenvalues and Eigenkets of Angular Momentum

All appeared here are operators and we ignore the ˆ .
Introduce

$$J^2 = J_x^2 + J_y^2 + J_z^2 \tag{3.38}$$

as square of angular momentum and ladder operators

$$J_\pm = J_x \pm iJ_y,\ J_+ = (J_-)^\dagger. \tag{3.39}$$

It is easy to see

$$[J^2, J_i] = 0,\ [J_z, J_\pm] = \pm\hbar J_\pm,\ [J_+, J_-] = 2\hbar J_z,\ [J^2, J_\pm] = 0. \tag{3.40}$$

Now we could choose representation of J_z so that

$$J^2 \left|a, b\right\rangle = a \left|a, b\right\rangle,\ J_z \left|a, b\right\rangle = b \left|a, b\right\rangle \tag{3.41}$$

from commutation relation. Under this representation, $J_{x/y}$ is not diagonalized, however, from ladder operators and their algebra relation,

$$J_z J_\pm \left|a, b\right\rangle = ([J_z, J_\pm] + J_\pm J_z) \left|a, b\right\rangle = (b \pm \hbar) J_\pm \left|a, b\right\rangle, \tag{3.42}$$

which reveals that $J_\pm \left|a, b\right\rangle$ are also eigenkets, by lifting of lowering one quantum unit of eigenvalue

$$J_\pm \left|a, b\right\rangle = c_\pm \left|a, b \pm \hbar\right\rangle. \tag{3.43}$$

On the other hand,

$$J^2 J_\pm \left|a, b\right\rangle = J_\pm J^2 \left|a, b\right\rangle = a J_\pm \left|a, b\right\rangle. \tag{3.44}$$

Meanwhile,

$$\left\langle a, b\right| J_\mp J_\pm \left|a, b\right\rangle = \left\langle a, b\right| (J_\pm)^\dagger J_\pm \left|a, b\right\rangle = |c_\pm|^2. \tag{3.45}$$

At the same time, by observation,

$$J_+^\dagger J_+ + J_-^\dagger J_- = J_+ J_- + J_- J_+ = 2(J_x^2 + J_y^2) = 2(J^2 - J_z^2),$$

then

$$\langle a, b| J^2 - J_z^2 |a, b\rangle = a - b^2 = \frac{|c_+|^2 + |c_-|^2}{2} \geq 0,$$

i.e.,

$$|b| \leq \sqrt{a}. \tag{3.46}$$

This immediately tells us that for a given a, their exist $b_{\max}$ and $b_{\min}$ by which we mean

$$J_+ \left|a, b_{\max}\right\rangle = 0,\ J_- \left|a, b_{\min}\right\rangle = 0. \tag{3.47}$$

Notice that we only have three DoFs, then there might exist a way to express J_+ or $J_+^\dagger J_+$ with J^2 and J_z, whose behaviour is more known by assumption, and we found that

$$J_- J_+ = (J_x - iJ_y)(J_x + iJ_y) = J_x^2 + J_y^2 + i[J_x, J_y] = J^2 - J_z^2 - \hbar J_z, \tag{3.48}$$

then

$$0 = J_- J_+ \left|a, b_{\max}\right\rangle = (J^2 - J_z^2 - \hbar J_z) \left|a, b_{\max}\right\rangle = (a - b_{\max}^2 - \hbar b_{\max}) \left|a, b_{\max}\right\rangle ,$$

which leads to together with $|a, b_{\min}\rangle$

$$a = (b_{\max} + \hbar) b_{\max} = (b_{\min} - \hbar) b_{\min}, \tag{3.49}$$

from which we guess that

$$b_{\max} = -b_{\min} = j\hbar,\ a = j(j+1)\hbar^2. \tag{3.50}$$

Also indicated by the integral up and down, we may let

$$b_{\max} = b_{\min} + n\hbar \Longrightarrow j = \frac{n}{2},\ n = 0, 1, 2, \cdots . \tag{3.51}$$

And here we come to the eigenfunctions by relabel $b/\hbar \to m$, $a \to j$

$$J^2 \left|j, m\right\rangle = j(j+1)\hbar^2 \left|j, m\right\rangle ,\ J_z \left|j, m\right\rangle = m\hbar \left|j, m\right\rangle ,\ m = -j, -j+1, \cdots , j-1, j. \tag{3.52}$$

Also,

$$|c_\pm|^2 = \langle a, b| J_\mp J_\pm \left|a, b\right\rangle = \langle a, b| (J^2 - J_z^2 \mp \hbar J_z) \left|a, b\right\rangle = [j(j+1) - m^2 \mp m]\hbar^2,$$

then choose real part for convenient

$$c_\pm = \sqrt{j(j+1) - m(m \pm 1)}\hbar, \tag{3.53}$$

i.e.,

$$\begin{aligned} J_\pm \left|j, m\right\rangle &= \sqrt{j(j+1) - m(m \pm 1)}\hbar \left|j, m \pm 1\right\rangle \\ &= \sqrt{(j \mp m)(j \pm m + 1)}\hbar \left|j, m \pm 1\right\rangle . \end{aligned} \tag{3.54}$$

Spin One Generator — Another Example of Representation

From above discussion, immediately we see that

$$\langle j, m'| J_\pm \left|j, m\right\rangle = \sqrt{j(j+1) - m(m \pm 1)}\hbar \delta_{m, m' \pm 1}. \tag{3.55}$$

Then for spin-one case, $j = 1$, and one immediately get the matrix form of $J_\pm$ under basis $m = 1, 0, -1$

$$J_+ = \hbar \begin{pmatrix} & \sqrt{2} & \\ & & \sqrt{2} \\ 0 & & \end{pmatrix},\ J_- = \hbar \begin{pmatrix} & & 0 \\ \sqrt{2} & & \\ & \sqrt{2} & \end{pmatrix}, \tag{3.56}$$

which immediately gives us three generators by $J_x = (J_+ + J_-)/2$, $J_y = (J_+ - J_-)/(2i)$

$$J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} & 1 & \\ 1 & & 1 \\ & 1 & \end{pmatrix},\ J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} & -i & \\ i & & -i \\ & i & \end{pmatrix},\ J_z = \hbar \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}. \tag{3.57}$$

I think there needs no explanation for J_z.

One may also use the same methods we used solving Pauli matrices to get spin-one matrices, but that seems tedious.

Representations of Rotational Operators

It is clear that by varying m, $\{|j,m\rangle\}$ form a complete basis for j subspace. Then

$$\hat{D}^j_n(\phi) = \sum_m |j,m\rangle\langle j,m|\, \hat{D}^j_n(\phi) \sum_{m'} |j,m'\rangle\langle j,m'| = \sum_{m,m'} D^j_{n,mm'}(\phi)\, |j,m\rangle\langle j,m'|\,, \tag{3.58}$$

where *Wigner functions*

$$D^j_{n,mm'}(\phi) = \langle j,m|\hat{D}^j_n(\phi)|j,m'\rangle\,, \tag{3.59}$$

is component of $(2j+1)$-D matrix representation $D^j_n(\phi)$, which is also unitary.

3.4 Hydrogen Atom and Sphere Harmonics

The sphere harmonics are defined as

$$Y_{jm}(\theta,\phi) = \sqrt{\frac{(2j+1)(j-m)!}{4\pi(j+m)!}} P^m_j(\cos\theta) e^{im\phi}, \tag{3.60}$$

where P^m_j are associated Legendre polynomials defined as

$$P^m_j(x) = (-1)^m (1-x^2)^{m/2} \frac{\mathrm{d}^m}{\mathrm{d}x^m} P_j(x), \tag{3.61}$$

where $P_j(x)$ are Legendre polynomials defined as

$$P_j(x) = \frac{1}{2^j j!} \frac{\mathrm{d}^j}{\mathrm{d}x^j} (x^2-1)^j. \tag{3.62}$$

A simple way to consider those wavefunctions is

$$Y_{jm}(\theta,\phi) = \langle\theta,\phi|j,m\rangle\,, \tag{3.63}$$

and the orthogonality relation is

$$\int \mathrm{d}\Omega\; Y^*_{j'm'}(\theta,\phi) Y_{jm}(\theta,\phi) = \delta_{jj'}\delta_{mm'}, \tag{3.64}$$

where $\mathrm{d}\Omega = \sin\theta \mathrm{d}\theta \mathrm{d}\phi$ is the solid angle element.

3.5 Addition of Angular Momentum

Considering addition of two angular momentums. Following adjoint formula, angular momentums as generators of rotations could be viewed as 3-D normal vectors from its transformation under rotation. Then immediately their addition follows the triangle rule, i.e., (once again we ignore operator label)

$$\boldsymbol{J} = \boldsymbol{J}_1 + \boldsymbol{J}_2. \tag{3.65}$$

However, these two vectors live in different subspaces and we would show its meaning below. Let

$$|j_1,m_1;j_2,m_2\rangle = |j_1,m_1\rangle \otimes |j_2,m_2\rangle \equiv |m_1,m_2\rangle\,, \tag{3.66}$$

where both sides have $(2j_1+1)(2j_2+1)$ possibilities as multiplication and is a tensor product between representation spaces of certain group element. By infinitesimal rotation

$$\left(\mathbb{I}_1 - \frac{i}{\hbar}\boldsymbol{J}_1\cdot\boldsymbol{\varepsilon}\right)\otimes\left(\mathbb{I}_2 - \frac{i}{\hbar}\boldsymbol{J}_2\cdot\boldsymbol{\varepsilon}\right) = \mathbb{I}_1\otimes\mathbb{I}_2 - \frac{i}{\hbar}(\boldsymbol{J}_1\otimes\mathbb{I}_2 + \mathbb{I}_1\otimes\boldsymbol{J}_2)\cdot\boldsymbol{\varepsilon},$$

and in this way we write

$$\boldsymbol{J} = \boldsymbol{J}_1\otimes\mathbb{I}_2 + \mathbb{I}_1\otimes\boldsymbol{J}_2, \tag{3.67}$$

which is also an angular momentum as

$$[J_i, J_j] = [J_{1i}, J_{1j}]\otimes\mathbb{I}_2 + \mathbb{I}_1\otimes[J_{2i}, J_{2j}] = i\varepsilon_{ijk}\hbar[J_{1k}\otimes\mathbb{I}_2 + \mathbb{I}_1\otimes J_{2k}] = i\hbar\varepsilon_{ijk}J_k. \tag{3.68}$$

Now give eigenket to $\boldsymbol{J}$ as

$$|j,m\rangle \equiv |j_1,j_2;j,m\rangle\,, \tag{3.69}$$

which could be expanded as

$$|j,m\rangle = \sum_{m_1,m_2} |m_1,m_2\rangle\,\langle m_1,m_2|j,m\rangle\,,$$

where $\langle m_1,m_2|j,m\rangle$ is known as *Clebsh-Gordan coefficient*. To calculate that, firstly one notice it that by vector addition,

$$(J_z - J_{1z} - J_{2z})\,|j,m\rangle = 0 \Longrightarrow (m - m_1 - m_2)\,\langle m_1,m_2|j,m\rangle = 0,$$

which forces $m-m_1-m_2=0$ for non-zero CG coefficient. Meanwhile, one could imagine two cases, i.e., top and bottom kets where only one tensor product ket contributes

$$|j,m=j\rangle = |m_1=j_1,m_2=j_2\rangle\,,\ |j,m=-j\rangle = |m_1=-j_1,m_2=-j_2\rangle\,, \tag{3.70}$$

where CG is one. These two states set a bound to j as $j\le j_1+j_2$; on the other hand, j may be lowered by one each time $m_{\max}$ drops $\hbar$, then by

$$\sum_{j=j_{\min}}^{j_1+j_2}(2j+1) = (2j_1+1)(2j_2+1) \Longrightarrow j_{\min} = |j_1-j_2|. \tag{3.71}$$

And the spirit for other coefficients lies in total ladder operators, e.g., when we know how ket $|j,m\rangle$ (and bigger m) is composited, we let

$$\begin{aligned}
J_-\,|j,m\rangle &= \sqrt{j(j+1)-m(m-1)}\hbar\,|j,m-1\rangle \\
&= \sqrt{j(j+1)-m(m-1)}\hbar\sum_{m_1,m_2}|m_1,m_2\rangle\,\langle m_1,m_2|j,m-1\rangle \\
&= \sum_{m_1,m_2}(J_{1-}+J_{2-})\,|m_1,m_2\rangle\,\langle m_1,m_2|j,m\rangle \\
&= \sum_{m_1,m_2}\sqrt{j_1(j_1+1)-m_1(m_1-1)}\hbar\,|m_1-1,m_2\rangle\,\langle m_1,m_2|j,m\rangle \\
&\quad + \sqrt{j_2(j_2+1)-m_2(m_2-1)}\hbar\,|m_1,m_2-1\rangle\,\langle m_1,m_2|j,m\rangle\,,
\end{aligned}$$

then by left action $\langle m_1, m_2|$, one has

$$\sqrt{j(j+1)-m(m-1)}\,\langle m_1, m_2|j, m-1\rangle = \sqrt{j_1(j_1+1)-m_1(m_1+1)}\,\langle m_1+1, m_2|j, m\rangle + \sqrt{j_2(j_2+1)-m_2(m_2+1)}\,\langle m_1, m_2+1|j, m\rangle\,, \tag{3.72}$$

and since in our assumption RHS is all known, we could get the CG coefficient $\langle m_1, m_2|j, m-1\rangle$. And this procedure could only get states with the same j, for those with different j, one may, after getting all CG coefficients for bigger j, assumes

$$|j, m\rangle = \sum_{m_1+m_2=m} c(m_1, m_2)\,|m_1, m_2\rangle\,, \tag{3.73}$$

and uses orthogonal relation

$$\langle j', m'|j, m\rangle = \delta_{jj'}\delta_{mm'} \tag{3.74}$$

to get the CG coefficients.

Spin-half Addition

This might be the simplest case, but it gives spin-singlet and spin-triplet, which are rather useful. As we said before, top and down states

$$|s=1, m=1\rangle = |m_1=1/2, m_2=1/2\rangle\,,\ |s=1, m=-1\rangle = |m_1=-1/2, m_2=-1/2\rangle\,. \tag{3.75}$$

Then,

$$\begin{aligned} J_-\,|s=1, m=1\rangle = \sqrt{2}\hbar\,|s=1, m=0\rangle &= (J_{1-}+J_{2-})\,|m_1=1/2, m_2=1/2\rangle \\ &= |m_1=1/2, m_2=-1/2\rangle + |m_1=-1/2, m_2=1/2\rangle \end{aligned}$$

i.e.

$$|s=1, m=0\rangle = \frac{1}{\sqrt{2}}\left(|m_1=1/2, m_2=-1/2\rangle + |m_1=-1/2, m_2=1/2\rangle\right). \tag{3.76}$$

The above three states are spin-triplet states by which we mean they are symmetric under spin-exchange, for the last state, we assume

$$|s=0, m=0\rangle = c_1\,|m_1=1/2, m_2=-1/2\rangle + c_2\,|m_1=-1/2, m_2=1/2\rangle\,, \tag{3.77}$$

by

$$\langle s=1, m=0|s=0, m=0\rangle = 0 \Longrightarrow c_1+c_2=0,$$

by normalization and make coefficients real, we write

$$|s=0, m=0\rangle = \frac{1}{\sqrt{2}}\left(|m_1=1/2, m_2=-1/2\rangle - |m_1=-1/2, m_2=1/2\rangle\right), \tag{3.78}$$

which is spin-singlet state anti-symmetric to spin-exchange.

3.6 Schwinger's Oscillator Model

3.7 Bell's Inequality

"I, at any rate, am convinced that He (the 'old one') does not throw dice" — said by Albert Einstein.

"Einstein, stop telling God what to do." — Returned by Bohr.

Einstein's argument mainly focuses on *locality* and *reality*. While first may be somehow understood from relativity, the latter is annoying as an old question on what is measurement. Based on *hidden variables*, Bell proposed his famous inequality which was confirmed to be wrong.

The model is simply based on the spin-singlet state

$$|s=0,m=0\rangle = \frac{1}{\sqrt{2}}\left(|m_1=1/2,m_2=-1/2\rangle - |m_1=-1/2,m_2=1/2\rangle\right). \tag{3.79}$$

Upon classical level, whether 'spin' is up and down is certain, and we just do not know it, from which *Bell's inequality* says

$$P(a+;b+) \leq P(a+;c+) + P(b+;c+), \tag{3.80}$$

where a, b, c are arbitrary directions and the probability denotes to simultaneously measurement of two spins with such given directions.

However, quantum mechanically, if 1 is $|a+\rangle$, 2 must be $|a-\rangle$, denoting $\theta_{ab} = \arg(a+,b+)$,

$$P(a+;b+) = |\langle b+|a-\rangle|^2 = |\langle a-|\exp\Big(i\hat{S}_z(\pi-\theta_{ab})/\hbar\Big)|a-\rangle|^2 = \sin^2(\theta_{ab}/2), \tag{3.81}$$

and the inequality means that

$$\sin^2(\theta_{ab}/2) \leq \sin^2(\theta_{ac}/2) + \sin^2(\theta_{bc}/2)$$

for all θ, which is impossible, e.g., let $\theta_{ab} = 2\theta_{ac} = 2\theta_{bc} = 2\theta$, one has

$$\cos^2(\theta/2) \leq \frac{1}{2},\ \forall\theta,$$

which is wrong.

Chapter 4

Symmetry

"das Symmetry Determines Dynamics."

4.1 Noether's Theorem

Classically, continuous symmetry is generated by *Noether's Theorem*, which starts from a Lagrangian $L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$, from which canonical momentum and Euler-Lagrange EoM is

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}},\ \frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}} = 0. \tag{4.1}$$

Then Noether's Theorem claims that under infinitesimal transformation

$$\boldsymbol{q} \to \boldsymbol{q}' = \boldsymbol{q} + \varepsilon\Delta\boldsymbol{q},\ t \to t' + \varepsilon\Delta t, \tag{4.2}$$

if the system bears such symmetry, then one would find corresponding conserved quantity

$$I = \frac{\partial L}{\partial \dot{q}_i}\Delta q_i + (L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i)\Delta t = \frac{\partial L}{\partial \dot{q}_i}\Delta q_i - H\Delta t. \tag{4.3}$$

Proof: under such transformation, $\dot{\boldsymbol{q}} \to \dot{\boldsymbol{q}}' = \dot{\boldsymbol{q}} + \dfrac{\mathrm{d}}{\mathrm{d}t}(\varepsilon\Delta\boldsymbol{q})$, then

$$\begin{aligned} L \to L' = L(\boldsymbol{q}', \dot{\boldsymbol{q}}', t') &= L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) + \frac{\partial L}{\partial q_i}\varepsilon\Delta q_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}(\varepsilon\Delta q_i) + \frac{\partial L}{\partial t}\varepsilon\Delta t \\ &= L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) + \frac{\partial L}{\partial q_i}\varepsilon\Delta q_i + \varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\Delta q_i\right) - \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right)\varepsilon\Delta q_i + \frac{\partial L}{\partial t}\varepsilon\Delta t \\ &= L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) + \varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\Delta q_i\right) + \frac{\partial L}{\partial t}\varepsilon\Delta t, \end{aligned}$$

where EoM is used; on the other hand,

$$\begin{aligned} \frac{\mathrm{d}L}{\mathrm{d}t} &= \frac{\partial L}{\partial q_i}\dot{q}_i + \frac{\partial L}{\partial \dot{q}_i}\ddot{q}_i + \frac{\partial L}{\partial t} = \frac{\partial L}{\partial q_i}\dot{q}_i + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) - \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right)\dot{q}_i + \frac{\partial L}{\partial t} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) + \frac{\partial L}{\partial t}, \end{aligned}$$

where EoM is used again. This term in fact comes from the *Jacobian* introduced by change of integral measure. Bring back and

$$\begin{aligned}\to L' &= L(\boldsymbol{q},\dot{\boldsymbol{q}},t)+\varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{q}_i}\Delta q_i\right)+\varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\left(L-\frac{\partial L}{\partial\dot{q}_i}\dot{q}_i\right)\Delta t\\ &= L+\varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial L}{\partial\dot{q}_i}\Delta q_i+\left(L-\frac{\partial L}{\partial\dot{q}_i}\dot{q}_i\right)\Delta t\right],\end{aligned}$$

requiring invariance of Lagrangian, one gets the conserved quantity. For example,

$$\begin{cases}\Delta t\neq 0\Longrightarrow H \text{ is conserved}\\ \Delta q_i\neq 0\Longrightarrow p_i=\frac{\partial L}{\partial\dot{q}_i} \text{ is conserved}\end{cases}.$$

To see angular momentum conservation, recall infinitesimal 3-D Euclid rotation as

$$\boldsymbol{r}\longrightarrow\boldsymbol{r}+\delta\theta\hat{n}\times\boldsymbol{r},\ \delta\theta\to 0,$$

under which we have

$$\begin{cases}\Delta x_i=\epsilon_{ijk}n_jr_k\\ \Delta t=0\end{cases},$$

when treating $\varepsilon=\delta\theta$, then we have conserved current

$$\frac{\partial L}{\partial\dot{r}_i}\epsilon_{ijk}n_jr_k=\frac{\partial L}{\partial\dot{\boldsymbol{r}}}\cdot(\hat{n}\times\boldsymbol{r})=\hat{n}\cdot(\boldsymbol{r}\times\boldsymbol{p})=\hat{n}\cdot\boldsymbol{L},\tag{4.4}$$

where we have recognized the canonical momentum. And by arbitrary $\hat{n}$ as rotational symmetry requires, we claim that

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{L}=0$$

as angular momentum conservation.

In field theory where $\mathscr{L}=\mathscr{L}(\phi,\partial_\mu\phi)$ with field $\phi=\phi(x)$, $x=x^\mu$, the theory could be generalized as

$$j^\mu=\frac{\partial\mathscr{L}}{\partial(\partial_\mu\phi)}\Delta\phi+(\mathscr{L}\delta^\mu_\nu-\frac{\partial\mathscr{L}}{\partial(\partial_\mu\phi)}\partial_\nu\phi)\Delta x^\nu,\tag{4.5}$$

under

$$\begin{cases}\phi\to\phi'=\phi+\varepsilon\Delta\phi\\ x\to x'=x+\varepsilon\Delta x\end{cases}.$$

To make this clear, we first claim it is $S[\phi]$ that is invariant in field theory, which leads to

$$0=\Delta S=\int\mathrm{d}^nx'\mathscr{L}(\phi',\partial'_\mu\phi')-\int\mathrm{d}^nx\mathscr{L}(\phi,\partial_\mu\phi).$$

Then under our transformation, one has

$$\begin{aligned}&\frac{\partial x'_\mu}{\partial x^\nu}=\delta_{\mu\nu}+\frac{\partial}{\partial x^\nu}(\varepsilon\Delta x_\mu),\ \frac{\partial}{\partial x'^\mu}=\left(\delta_{\mu\nu}-\frac{\partial}{\partial x^\mu}(\varepsilon\Delta x_\nu)\right)\frac{\partial}{\partial x_\nu},\\ &\left|\frac{\partial x'_\mu}{\partial x^\nu}\right|\equiv\det\left(\frac{\partial x'_\mu}{\partial x^\nu}\right)=\exp\left(\mathrm{Tr}\ln\frac{\partial x'_\mu}{\partial x^\nu}\right)\approx\exp\left(\frac{\partial}{\partial x^\mu}(\varepsilon\Delta x_\mu)\right)\approx 1+\frac{\partial}{\partial x^\mu}(\varepsilon\Delta x_\mu),\end{aligned}$$

then,

$$\begin{aligned}\int \mathrm{d}^n x' \mathscr{L}(\phi', \partial'_\mu \phi') &\approx \int \mathrm{d}^n x \left(1 + \frac{\partial}{\partial x^\mu}(\varepsilon \Delta x^\mu)\right) \mathscr{L}\left(\phi + \varepsilon\Delta\phi, \left(\delta^\mu_\nu - \frac{\partial}{\partial x^\mu}(\varepsilon \Delta x_\nu)\right) \frac{\partial}{\partial x_\nu}(\phi + \varepsilon\Delta\phi)\right) \\ &\approx \int \mathrm{d}^n x \left[\mathscr{L}(\phi, \partial_\mu\phi) + \varepsilon \frac{\partial \mathscr{L}}{\partial \phi}\Delta\phi + \frac{\partial \mathscr{L}}{\partial \partial_\mu \phi}\left(\Delta\phi \delta^\mu_\nu \partial^\nu \varepsilon - \Delta x_\nu (\partial_\mu \varepsilon)\partial^\nu \phi\right) + \Delta x^\mu (\partial_\mu \varepsilon)\mathscr{L}\right] \\ &\approx \int \mathrm{d}^n x \left[\mathscr{L}(\phi, \partial_\mu\phi) + \varepsilon \frac{\partial \mathscr{L}}{\partial \phi}\Delta\phi + \Delta\phi\left(\partial^\mu\left(\varepsilon \frac{\partial \mathscr{L}}{\partial \partial_\mu \phi}\right) - \varepsilon \partial^\mu \frac{\partial \mathscr{L}}{\partial \partial_\mu \phi}\right)\right. \\ &\left. - \frac{\partial \mathscr{L}}{\partial \partial_\mu \phi}\Delta x_\nu (\partial_\mu \varepsilon)\partial^\nu \phi + \Delta x^\mu (\partial_\mu \varepsilon)\mathscr{L}\right] \\ &\approx \int \mathrm{d}^n x \mathscr{L}(\phi, \partial_\mu \phi) - \int \mathrm{d}^n x j^\mu \partial_\mu \varepsilon,\end{aligned}$$

which leads to

$$\Delta S = \int \mathrm{d}^n x (\partial_\mu j^\mu)\varepsilon, \tag{4.6}$$

which leads to

$$\begin{cases} \partial_\mu j^\mu = 0 \\ Q \equiv \int \mathrm{d}^{n-1} x j^0 \end{cases}, \tag{4.7}$$

where the first is continuity equation while second is its conserved charge. Notice that

$$\Delta x \equiv \left.\frac{\partial x}{\partial \varepsilon}\right|_{\varepsilon=0}, \ \Delta\phi \equiv \left.\frac{\delta \phi}{\delta \varepsilon}\right|_{\varepsilon=0},$$

where the latter appears as some functional deviation.

4.2 Symmetry in Quantum Mechanics

In quantum mechanics, similar changes like above are generated by unitary transformations like space-time translations and spin-orbital rotations, and by symmetry we mean that the transformation would not influence our system described by Hamiltonian, and in some sense they are 'compatible'. Though unitary transformation operator itself is not an observable, its generators could be, then

$$[\hat{H}, \hat{U}(\hat{\boldsymbol{G}})] = 0 \Longrightarrow \hat{U}^\dagger \hat{H} \hat{U} = \hat{H}, \ [\hat{G}_i, \hat{H}] = 0, \ \frac{\mathrm{d}\hat{G}^H_i}{\mathrm{d}t} = 0. \tag{4.8}$$

For eigenkets,

$$\hat{H}(\hat{U}\left|n\right\rangle) = \hat{U}\hat{H}\left|n\right\rangle = E_n \hat{U}\left|n\right\rangle, \tag{4.9}$$

thus $\hat{U}\left|n\right\rangle$ is also an eigenket with the same energy as $\left|n\right\rangle$, and when they are different, they would become degenerate states. Though we introduce as an analogy with continuous symmetry by Noether's theorem, there could be discrete symmetries, and more amazingly, there could be symmetry described by anti-unitary operator.

4.2.1 Spontaneous Symmetry Breaking

Usually we say that *spontaneous symmetry breaking* happens when Lagrangian or Hamiltonian owns some symmetry while the EoM or ground state does not. And the most familiar example may be the ferromagnetic magnet. Say that in magnet, two nearest spins are coupled by Heisenberg exchange interaction (no $\hbar$ in operator)

$$H = J\hat{\boldsymbol{S}}_1 \cdot \hat{\boldsymbol{S}}_2,\ [\hat{S}_{1i}, \hat{S}_{2j}] = 0. \tag{4.10}$$

Let $s_1 = s_2 = s$, then the coupled state $|S, M\rangle$, with $\hat{\boldsymbol{S}} = \hat{\boldsymbol{S}}_1 + \hat{\boldsymbol{S}}_2$ satisfies

$$H\,|S, M\rangle = \frac{J}{2}[\hat{\boldsymbol{S}}^2 - \hat{\boldsymbol{S}}_1^2 - \hat{\boldsymbol{S}}_2^2]\,|S, M\rangle = \frac{J}{2}[S(S+1) - 2s(s+1)]\,|S, M\rangle\,. \tag{4.11}$$

Since one knows, by angular momentum addition, $0 \leq S \leq 2s$, while for each S, $-S \leq M \leq S$ as $(2S+1)$ fold degeneracy, then one concludes that

$$\begin{cases} |GS\rangle_F = |S = 2s, M\rangle\,,\ E_{GS} = Js^2,\ \text{ferromagnetic} \\ |GS\rangle_{AF} = |S = 0, M = 0\rangle\,,\ E_{GS} = -s(s+1)J,\ \text{anti-ferromagnetic} \end{cases}. \tag{4.12}$$

Clearly, ground state for anti-ferromagnet is non-degenerate, while bears $(4s+1)$ degeneracy in ferromagnetic case. This degeneracy could be reflected as a unitary rotation

$$\hat{U}_n(\phi) = \exp\Big(-i\hat{\boldsymbol{S}} \cdot \hat{n}\phi\Big), \tag{4.13}$$

whose action does not change S, but would braid M altogether. Since this operator is just spin-rotational operator, this is in fact $SU(2)$ symmetry from Hamiltonian, however, in reality the magnet must choose one M as its state, and in this way we say that spontaneous symmetry breaking happens.

Goldstone Mode

A formal expression for the spontaneous symmetry breaking process lies in reducing the global symmetry group G of the Hamiltonian into its subgroup H, with the coset G/H counting the upper-bound number of the Goldstone modes. For instance, the scaler Higgs Boson spontaneously breaks of the $U(1)$ symmetry into the trivial group, with one Goldstone mode further coupled with the gauge field to make the Goldstone mode massive. Formally, let $U = e^{i\alpha Q} \in G$ with Q the generator of the symmetry transformation, then a state $|\psi\rangle$ breaks the symmetry if there exists a locally defined *interpolating field operator* $I(x)$ such that

$$\langle\psi|\,[Q, I(x)]\,|\psi\rangle \neq 0. \tag{4.14}$$

The definition is most easily understood by considering a symmetric state $|\psi\rangle_{Sym}$, which makes the expectation value to vanish automatically, since eigenvalue of $|\psi\rangle_{Sym}$ under a Hermitian Q has to be real. The local order parameter operator can be defined as $\phi(x) = [Q, I(x)]$, which is not necessarily Hermitian. To be complete, let us pay attention to the Goldstone mode as the gapless excitations accompanied with the broken global continuous symmetry. Consider the complete spectrum of Hamiltonian $H\,|n, k\rangle = E_n(k)\,|n, k\rangle$, with orthonormal relation $\langle n, k|n', k'\rangle = (2\pi)^d\delta_{nn'}\delta(k - k')$ and complete relation $\mathbb{I} = \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,|n, k\rangle\langle n, k|$ in $D = d + 1$ spacetime. Being a symmetry, the Noether charge Q corresponds to a local current conservation formula $\partial_\mu j^\mu = 0$, with

$Q(t) = \int \mathrm{d}^d x j^0(x,t)$ satisfying that $\partial_t Q = 0$. Then for the symmetry breaking ground state,

$$\begin{aligned} 0 \neq \langle GS| \left[Q, I(y)\right] |GS\rangle &= \int \mathrm{d}^d x \left(\langle GS| j^0(x,t) I(y) |GS\rangle - \text{c.c.} \right) \\ &= \int_\Omega \mathrm{d}^d x \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d} \left(\langle GS| e^{i(Ht - p\cdot x)} j^0(0) e^{-i(Ht-p\cdot x)} |nk\rangle \langle nk| I(y) |GS\rangle - \text{c.c.} \right) \\ &= \int_\Omega \mathrm{d}^d x \sum_n \int \frac{\mathrm{d}^d k}{(2\pi)^d} e^{-i(\tilde{E}_n(k)t - k\cdot x)} \left(\langle GS| j^0(0) |nk\rangle \langle nk| I(y) |GS\rangle - \text{c.c.} \right) \\ &= \sum_n \int \mathrm{d}^d k \delta_\Omega(k) \left(\langle GS| j^0(0) |nk\rangle \langle nk| I(y) |GS\rangle - \text{c.c.} \right), \end{aligned} \tag{4.15}$$

where in the second line we have replaced the integral over spacetime by the integral over the spacetime region Ω, by assuming locality of the operators; we have also inserted a complete relation and shifted the charge operator. In the third line, $\tilde{E}_n(k) \equiv E_n(k) - E_{GS}$, and we have assumed translational invariance of the ground state and set its momentum to zero. In the last line, a peak function $\delta_\Omega(k)$ near $k \approx 0$ is introduced by integrating x out. As a result of the non-vanishing expectation value, we see that there should be an excitation near the zero momentum such that

$$\begin{cases} I(y) |GS\rangle \sim |n, \delta q\rangle \\ j^0(0) |GS\rangle \sim |n, \delta q\rangle \end{cases}, \tag{4.16}$$

for some n. Notice that in the limit $\Omega \to \infty$, we shall have $\delta q \to 0$, which indicates a zero-wavevector excitation. To further illustrate the gapless nature, we utilize the charge conservation

$$\begin{aligned} 0 &= \partial_t \langle GS| \left[Q, I(y)\right] |GS\rangle \\ &= -i \sum_n \int \mathrm{d}^d k \delta_\Omega(k) \tilde{E}_n(k) \left(\langle GS| j^0(0) |nk\rangle \langle nk| I(y) |GS\rangle - \text{c.c.} \right), \end{aligned} \tag{4.17}$$

and since all terms except $\tilde{E}_n(k)$ contribute to the non-zero expectation value, we conclude that $\tilde{E}_n(\delta q \to 0) \to 0$ indicating the Goldstone mode as a gapless zero-momentum excitation. The counting of the Goldstone modes needs careful treatment, since both the Noether charge and the interpolating field operator can generate the *same* Goldstone mode, and when it is the case that the interpolating field operator is yet another symmetry generator, e.g., $[S_x, S_y] = iS_z$ in the ferromagnetic Heisenberg model, the Goldstone mode is not counted twice. Call this type of Goldstone mode as type-B, and the usual one as type-A, we have

$$\begin{cases} n_A = \dim(G/H) - \mathrm{rank}(M) \\ n_B = \mathrm{rank}(M)/2 \end{cases}, \tag{4.18}$$

where the Watanabe-Brauner matrix $M_{ab} = -i \langle GS| [Q_a, Q_b] |GS\rangle$ is homogeneous in space. It has been shown that in almost all cases type-A modes have linear dispersion while type-B modes have quadratic dispersion, with the latter understood from the coupling of two linear Bosonic modes as

$$\mathscr{L}_{eff} = 2M(\pi_1 \partial_t \pi_2 - \pi_2 \partial_t \pi_1) + \frac{1}{2}(\partial \pi_1)^2 + \frac{1}{2}(\partial \pi_2)^2, \tag{4.19}$$

whose equation of motion $[(\omega^2 - k^2)\sigma_0 - 2M\omega\sigma_y]\pi = 0$ leads to a quadratic excitation $\omega_- = \sqrt{k^2 + M^2} - M \approx \frac{k^2}{2M}$ and a massive mode.

Higher Symmetry in Hydrogen Atom

$$SO(4) = SU(2) \times SU(2). \tag{4.20}$$

Symmetry in a Hint

There are totally $10 + 1$ continuous symmetries when excluding gravity, where $10 = 4+6$ comes from *Poincare group*, a generalized version of Lorentz group, as four space-time translations and $6 = 3 + 3$ as three space rotations and three boosts, where total angular momentum rotations including spin part are combination of both. The additional one is gauge symmetry, from which electro-weak-strong interactions appear. Meanwhile, almost all symmetries above have discrete version, e.g., discrete translations in solid crystal and time crystal. Speaking of crystals, there are point groups and space groups related to certain structures. On the other hand, discrete symmetries with no continuous version exist as permutation, which is purely quantum mechanical, parity, time reversal, charge conjugation (particle-hole) and chiral symmetries. The powerful mathematical tool describing relating categories is named as the 'Group Theory', which I shall put a note link here...

More generally, there are fractal structures in nature like coastline, and in physics it appears as quasi-crystal and chaos. Beyond that, the initial thought for quantum mechanics would be deserted to duality, like wave-particle duality, and this thought has been generalized to some appealing cases like Fermion-Boson duality (super-symmetry) and gravity-gauge duality, aka holographic duality.

4.3 Discrete Symmetry

Parity

A parity transformation (also called parity inversion) is the flip in the sign of one spatial coordinate, which we could assign with a unitary operator $\hat{P}$. Sometimes we would like to assign parity operation as total action on whole space like inversion (point reflection) if space dimension is odd, and mirror reflection if it is even, and we just assume usual 3-D Euclidean space thus no need to try to distinguish in detail. Intuitively,

$$\hat{P}\hat{\boldsymbol{x}}\hat{P}^\dagger = -\hat{\boldsymbol{x}} \Longrightarrow \{\hat{P}, \hat{\boldsymbol{x}}\} = 0, \tag{4.21}$$

where super-operating form is bounded by

$$|\psi\rangle \to \hat{P}\,|\psi\rangle \tag{4.22}$$

and probability conservation.

From

$$\hat{\boldsymbol{x}}\hat{P}\,|\boldsymbol{x}\rangle = -\hat{P}\hat{\boldsymbol{x}}\,|\boldsymbol{x}\rangle = -\boldsymbol{x}\hat{P}\,|\boldsymbol{x}\rangle \Longrightarrow \hat{P}\,|\boldsymbol{x}\rangle = |-\boldsymbol{x}\rangle \tag{4.23}$$

where possible phase is chosen to be one. Then

$$\hat{P}^2\,|\boldsymbol{x}\rangle = |\boldsymbol{x}\rangle \Longrightarrow \hat{P}^2 = \mathbb{I}, \tag{4.24}$$

with eigenvalues $+$ and $-$ as even or odd parity. Using unitary condition,

$$\hat{P}^\dagger = \hat{P}^{-1} = \hat{P}. \tag{4.25}$$

Furthermore, it is also Intuitive

$$\hat{P}\hat{\boldsymbol{p}}\hat{P}^\dagger = -\hat{\boldsymbol{p}},\ \hat{P}\hat{\boldsymbol{J}}\hat{P}^\dagger = \hat{\boldsymbol{J}}, \tag{4.26}$$

where the latter is positive since $L = r \times p$. More strictly,

$$\hat{P}\hat{T}(\delta x)\left|x\right\rangle = \hat{P}\left|x+\delta x\right\rangle = \left|-x-\delta x\right\rangle = \hat{T}(-\delta x)\hat{P}\left|x\right\rangle \Longrightarrow \hat{P}\hat{T}(\delta x)\hat{P}^\dagger = \hat{T}(-\delta x),$$

which means that

$$\hat{P}\hat{\boldsymbol{p}}\hat{P}^\dagger = -\hat{\boldsymbol{p}},$$

as $\delta\boldsymbol{x}$ is just number. Meanwhile, for rotations generated by angular momentum, in 3-D Euclid representation

$$[R, P] = 0 \Longrightarrow P = \mathrm{diag}(-1,-1,-1), \tag{4.27}$$

where the minus sign reflects action on position.

For wavefunction in position space, assume that ket

$$\hat{P}\left|\psi\right\rangle = \pm\left|\psi\right\rangle, \tag{4.28}$$

then

$$\begin{aligned} \langle x|\hat{P}|\psi\rangle &= \langle x|\pm|\psi\rangle = \pm\psi(x) \\ &= \langle -x|\psi\rangle = \psi(-x) \\ \Longrightarrow \psi(x) &= \pm\psi(-x). \end{aligned} \tag{4.29}$$

For Hamiltonian satisfying $[\hat{H}, \hat{P}] = 0$, i.e.,

$$H(x) = \langle x|\hat{H}|x\rangle = \langle x|\hat{P}^\dagger\hat{P}\hat{H}\hat{P}^\dagger\hat{P}|x\rangle = H(-x), \tag{4.30}$$

then if $H(x)\psi(x) = E\psi(x)$, one must have $H(x)\psi(-x) = H(-x)\psi(-x) = E\psi(-x)$, i.e., $\psi(-x)$ is also a solution with the same energy, and one may construct manually

$$\psi_E(x) \sim \psi(x) + \psi(-x),\ \psi_O(x) \sim \psi(x) - \psi(-x). \tag{4.31}$$

However, if $|\psi\rangle$, i.e. $\psi(x)$ itself is even, then $\psi_O = 0$ and loses meaning, vice versa.

By even and odd parity we may let *polar vector* (operator) to be those odd under parity like position and momentum; *axial/pseudo vector* to be even like angular momentum and magnetic field; and *pseudo scalar* to be scalar (operator) odd under parity like $\hat{\boldsymbol{S}}\cdot\hat{\boldsymbol{x}}$.

Parity Selection

For two states being eigenkets of parity operator

$$\hat{P}\left|\alpha\right\rangle = s_\alpha\left|\alpha\right\rangle,\ \hat{P}\left|\beta\right\rangle = s_\beta\left|\beta\right\rangle, \tag{4.32}$$

the scattering process $\langle\beta|\hat{O}|\alpha\rangle$ with intermediate operator

$$\hat{P}\hat{O}\hat{P}^\dagger = s_o\hat{O} \tag{4.33}$$

is non-zero unless $s_\alpha s_\beta s_o = +$, since

$$\langle\beta|\hat{O}|\alpha\rangle = \langle\beta|\hat{P}^\dagger\hat{P}\hat{O}\hat{P}^\dagger\hat{P}|\alpha\rangle = s_\alpha s_\beta s_o\left\langle\beta|\hat{O}|\alpha\right\rangle. \tag{4.34}$$

Lattice — Discrete Spatial Translation

As we introduced in Chapter 1, spatial translation operator acts like

$$\hat{T}(\delta x)\,|x\rangle = |x+\delta x\rangle\,,$$

which is a unitary operator and could be viewed as a faithful unitary Lie group representation of translation with

$$\hat{T}(\delta x) = \exp(-i\hat{p}\delta x/\hbar).$$

Then its action on ket leads to change of wavefunction

$$\psi'(x) \equiv \langle x|\,\hat{T}(\delta x)\,|\psi\rangle = (\hat{T}^\dagger(\delta x)\,|x\rangle)^\dagger\,|\psi\rangle = \langle x-\delta x|\,|\psi\rangle = \psi(x-\delta x). \tag{4.35}$$

Now for single-particle local Hamiltonian on a lattice, which assume works in non-relativistic limit with spatial translational symmetry on each direction it lives, we have

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{U},$$

whose real-space representation is

$$H(\boldsymbol{r}) = -\frac{\hbar^2}{2m}\nabla_r^2 + U(\boldsymbol{r}),\ U(\boldsymbol{r}+\boldsymbol{a}_i) = U(\boldsymbol{r}), \tag{4.36}$$

where local means U only depends on the coordinate of single particle we want to figure out, and it is periodic on each direction i ($\boldsymbol{a}_i$ as unit lattice vector). Clearly to get this Hamiltonian we have already made some approximations, say Born-Oppenheimer, mean-field and periodic field approximations as proposed.

Theorem 4.3.1 (Bloch's Theorem)**.** *Admittedly we have such a Hamiltonian as written above, its solution must be like*

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u_{n,\boldsymbol{k}}(\boldsymbol{r}),\ u_{n,\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{a}_i) = u_{n,\boldsymbol{k}}(\boldsymbol{r}) \tag{4.37}$$

known as Bloch state.

Proof. Due to existence of lattice, continuous spatial translational symmetry is spontaneously broken into a discrete one, and we define discrete lattice translation operator as

$$\hat{T}_i\,|\boldsymbol{r}\rangle = |\boldsymbol{r}+\boldsymbol{a}_i\rangle\,,$$

with

$$T_i\psi(\boldsymbol{r}) \equiv \psi'(\boldsymbol{r}) = \psi(\boldsymbol{r}-\boldsymbol{a}_i),\ T_iO(\boldsymbol{r})T_i^\dagger \equiv O(\boldsymbol{r}-\boldsymbol{a}_i)$$

for consistence. Due to the addition form of position, we have

$$[T_i, T_j] = 0.$$

Then from $H(\boldsymbol{r}) = H(\boldsymbol{r}+\boldsymbol{a}_i)$, we clearly have

$$[T_i, H] = 0,\ \forall i. \tag{4.38}$$

From this we immediately see there exist simultaneous eigenstates as

$$\begin{cases} H\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}) \\ T_i\psi(\boldsymbol{r}) = \lambda_i\psi(\boldsymbol{r}) \end{cases}.$$

Then by periodic boundary condition, one has

$$\psi(\boldsymbol{r}) = \psi(\boldsymbol{r} + N_i \boldsymbol{a}_i) = T_i^{-N_i}\psi(\boldsymbol{r}) = \lambda_i^{-N_i}\psi(\boldsymbol{r})$$

with N_i lattice site number along i. Then we have

$$\lambda_i^{N_i} = 1 = \exp(i2\pi n_i) \Longrightarrow \lambda_i = \exp\left(i\frac{2\pi n_i}{N_i}\right), \tag{4.39}$$

and by introducing quasi-momentum vector

$$\boldsymbol{k} = \sum_i \frac{n_i}{N_i}\boldsymbol{b}_i,\ \ \boldsymbol{b}_i \cdot \boldsymbol{a}_j = 2\pi\delta_{ij},$$

we have

$$\lambda_i = \exp(i\boldsymbol{k} \cdot \boldsymbol{a}_i). \tag{4.40}$$

Now given arbitrary lattice vector

$$\boldsymbol{R}_l \equiv \sum_i l_i \boldsymbol{a}_i,\ \ l_i \in \mathbb{Z},$$

we have

$$\begin{aligned}\psi(\boldsymbol{r} + \boldsymbol{R}_l) = \psi(\boldsymbol{r} + \sum_i l_i\boldsymbol{a}_i) &= \prod_i \lambda_i^{l_i}\psi(\boldsymbol{r}) \\ &= \exp\left(i\boldsymbol{k} \cdot \sum_i l_i \boldsymbol{a}_i\right) = e^{i\boldsymbol{k}\cdot\boldsymbol{R}_l}\psi(\boldsymbol{r}),\end{aligned}$$

then every ψ is decorated with a quantum number $\boldsymbol{k}$; and by letting

$$u_{\boldsymbol{k}}(\boldsymbol{r}) = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{\boldsymbol{k}}(\boldsymbol{r}),$$

we have

$$\begin{aligned}u_{\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}_l) &= e^{-i\boldsymbol{k}\cdot(\boldsymbol{r}+\boldsymbol{R}_l)}\psi_{\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}_l) \\ &= e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_l}e^{i\boldsymbol{k}\cdot\boldsymbol{R}_l}\psi(\boldsymbol{r}) = e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{\boldsymbol{k}}(\boldsymbol{r}) = u_{\boldsymbol{k}}(\boldsymbol{r}),\end{aligned}$$

as a periodic function on lattice. And we have

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u_{\boldsymbol{k}}(\boldsymbol{r}),\ \ u_{\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{a}_i) = u_{\boldsymbol{k}}(\boldsymbol{r}),$$

with n omitted as band index. □

Notice that our proof utilizes wavefunction language, but one could also work that out using purely operator relation, especially

$$\hat{T}_i\hat{U}\hat{T}_i^\dagger = \hat{U},$$

and it will turn out

$$|\psi_{n,\boldsymbol{k}}\rangle = \exp(i\boldsymbol{k} \cdot \hat{\boldsymbol{r}})\,|u_{n,\boldsymbol{k}}\rangle\,, \tag{4.41}$$

where r on exponential should be recognized as position operator while k keeps as quantum number composing reciprocal space and Brillouin zone.

One may stay and take a glance on k, which has limited (though large) choices. Take orthogonal basis a_i as example, but the conclusion should be general. Now we have

$$b_i = \frac{2\pi}{a_i},\ k_i = \frac{2\pi n_i}{N_i a_i} = \frac{2\pi}{L_i} n_i,\ n_i = 1, 2, \cdots, N_i,$$

that is, we have N_i quantum numbers along each direction with $2\pi/L_i$ gap between neighbours. We could define the zone in k-space composed by unit vector b_i where $\exp(ik_i a_i)$ remains single-valued as Brillouin zone. Then with continuous real space bounded by lattice system and discrete Brillouin zone, we could define Fourier transformation between as

$$\phi_k = \frac{1}{\sqrt{V}} \int \mathrm{d}^n r \phi(r) e^{-ik\cdot r},\ \phi(r) = \frac{1}{\sqrt{V}} \sum_k \phi_k e^{ik\cdot r},\ V = \prod_i^n L_i. \tag{4.42}$$

This makes sense by considering initial definition with 2π factor for one-dimensional continuous Fourier transformation:

$$1 = \int \mathrm{d}r \delta(r) e^{-ik\cdot r},\ \delta(r) = \frac{1}{2\pi} \int \mathrm{d}k 1 \cdot e^{ik\cdot r}.$$

Then we start from continuous version and assume

$$\begin{aligned}
\phi(r) &= C \int_{\mathrm{BZ}} \mathrm{d}^n k \phi_k e^{ik\cdot r} = \frac{C}{\sqrt{V}} \int_{\mathrm{BZ}} \mathrm{d}^n k \int \mathrm{d}^n r' \phi(r') e^{-ik\cdot(r'-r)} \\
&= \frac{C}{\sqrt{V}} \int \mathrm{d}^n r' \phi(r') \int_{\mathrm{BZ}} \mathrm{d}^n k e^{-ik\cdot(r'-r)} \\
&= \frac{C}{\sqrt{V}} \int \mathrm{d}^n r' \phi(r') (2\pi)^n \delta(r'-r) = C \frac{(2\pi)^n}{\sqrt{V}} \phi(r),
\end{aligned}$$

which gives us $C = \dfrac{\sqrt{V}}{(2\pi)^n}$, and

$$\phi(r) = \frac{\sqrt{V}}{(2\pi)^n} \int \mathrm{d}^n k \phi_k e^{ik\cdot r}.$$

Then by replacing integral with summation where $\mathrm{d}k$ is recognized with multi-gap $(2\pi/L)^n = (2\pi)^n/V$ (density over 1 in fact), we have

$$\phi(r) = \frac{\sqrt{V}}{(2\pi)^n} \sum_k \frac{(2\pi)^n}{V} \phi_k e^{ik\cdot r} = \frac{1}{\sqrt{V}} \sum_k \phi_k e^{ik\cdot r}.$$

On the other hand, by

$$1 = \sum_{k'} \delta_{k,k'} = \frac{V}{(2\pi)^n} \int_{\mathrm{BZ}} \mathrm{d}^n k' C' \delta(k-k'),$$

we recognize

$$\delta_{k,k'} = \frac{(2\pi)^n}{V} \delta(k-k').$$

Time Reversal

Lorentz force is written as

$$m\ddot{\boldsymbol{r}} = q(\boldsymbol{E} + \dot{\boldsymbol{r}} \times \boldsymbol{B}).$$

Intuitively, under time reversal $t \to -t$ so that only $\dot{\boldsymbol{r}}$ changes sign, then we see that *external* magnetic field would break time reversal invariance.

Now given time reversal operator $\hat{T}$, intuitively,

$$\hat{T}\hat{x}\hat{T}^{-1} = \hat{x},\ \hat{T}\hat{p}\hat{T}^{-1} = -\hat{p},\ \hat{T}\hat{J}\hat{T}^{-1} = -\hat{J}, \tag{4.43}$$

which serves as a definition for time reversal operation and corresponding operator. And one may also admit if we have 'time operator'

$$\hat{T}\hat{t}\hat{T}^{-1} = -\hat{t}.$$

Then, similar to the parity case one has

$$\begin{aligned} & \hat{x}\hat{T}\left|x\right\rangle = \hat{T}\hat{x}\left|x\right\rangle = x\hat{T}\left|x\right\rangle,\ \hat{p}\hat{T}\left|p\right\rangle = -p\hat{T}\left|p\right\rangle,\ \hat{J}\hat{T}\left|J\right\rangle = -J\hat{T}\left|J\right\rangle, \\ \Longrightarrow \quad & \hat{T}\left|x\right\rangle = \left|x\right\rangle, \qquad \hat{T}\left|p\right\rangle = \left|-p\right\rangle, \qquad \hat{T}\left|J\right\rangle = \left|-J\right\rangle, \end{aligned}$$

where a possible phase factor is chosen to be identity. And when it acts on some general ket, also intuitively we have

$$\hat{T}\left|\psi(\delta t)\right\rangle = \hat{T}\left(\mathbb{I} - \frac{i}{\hbar}\hat{H}\delta t\right)\left|\psi(0)\right\rangle \sim \left|\psi(-\delta t)\right\rangle = \left(\mathbb{I} + \frac{i}{\hbar}\hat{H}\delta t\right)\left|\psi(0)\right\rangle,$$

then confusingly we have $\{\hat{T}, \hat{H}\} = 0$? This is not the symmetry we want. Then we have to assume that $\hat{T} = \hat{U}\mathscr{K}$ is anti-unitary with

$$\mathscr{K}i = -i\mathscr{K}, \tag{4.44}$$

so that

$$[\hat{T}, \hat{H}] = 0. \tag{4.45}$$

And our mission reduces to find $\hat{U}$.

Before that, some general properties need discussion. First of all, from the definition one could see it that this symmetry operation is related to certain representation, and the anti-unitarity further leads to more sensitive choice for a representation. Say, given state $\left|\psi\right\rangle$ and representation $\hat{A}$, one would have

$$\begin{aligned} \left|T\psi\right\rangle = \hat{T}\left|\psi\right\rangle &= \hat{U}\mathscr{K}\sum_n \left|a_n\right\rangle\left\langle a_n|\psi\right\rangle = \sum_n \hat{U}\mathscr{K}\left\langle a_n|\psi\right\rangle\left|a_n\right\rangle \\ &= \sum_n \left\langle a_n|\psi\right\rangle^* \hat{U}\mathscr{K}\left|a_n\right\rangle = \sum_n \left\langle\psi|a_n\right\rangle\hat{U}\left|a_n\right\rangle, \end{aligned} \tag{4.46}$$

where in the last step we have used the fact that we are under representation of $\hat{A}$ so that action of $\mathscr{K}$ on its eigenket $\left|a_n\right\rangle$ just changes nothing. One needs to notice it that $\mathscr{K}$ is an operator so it could not just pass a ket and says nothing, however, on the other hand, it only acts on complex field, i.e., the expansion coefficient of a ket. One suitable example here is spin-half case. Say now we are under representation of $\hat{S}_z$, then

$$\mathscr{K}\left|z\pm\right\rangle = \left|z\pm\right\rangle,$$

as the expansion coefficient

$$\langle z \pm | z \pm \rangle = 1,\ \langle z \pm | z \mp \rangle = 0,$$

and there is nothing to change. However, consider that

$$|y \pm\rangle = \frac{1}{\sqrt{2}}(|z+\rangle \pm i\,|z-\rangle),$$

and the action of complex conjugate would lead to

$$\mathscr{K}\,|y \pm\rangle = \frac{1}{\sqrt{2}}(\mathscr{K}\,|z+\rangle \pm \mathscr{K} i\,|z-\rangle) = \frac{1}{\sqrt{2}}(\mathscr{K}\,|z+\rangle \mp i\mathscr{K}\,|z-\rangle) = |y \mp\rangle\,.$$

For the action of time reversal on a bra, we need to utilize the dual correspondence as Hermitian conjugate

$$\langle T\psi| = (|T\psi\rangle)^\dagger = \sum_n \langle a_n|\,\hat{U}^\dagger\,\langle a_n|\psi\rangle\,. \tag{4.47}$$

One may formally introduce the complex conjugate of a ket as

$$|\psi\rangle^* = \mathscr{K}\,|\psi\rangle = \sum_n \langle a_n|\psi\rangle^*\,|a_n\rangle\,, \tag{4.48}$$

and one may formally write

$$\begin{cases} |T\psi\rangle = \hat{U}\,|\psi\rangle^* \\ \langle T\psi| = \langle\psi|^*\,\hat{U}^\dagger \end{cases}. \tag{4.49}$$

However, we would better avoid writing in this way to get away from the right action of anti-unitary operator.

Using expressions above, one finds

$$\begin{aligned} \langle T\phi|T\psi\rangle &= \sum_{mn} \langle a_m|\,\hat{U}^\dagger\,\langle a_m|\phi\rangle\,\langle\psi|a_n\rangle\,\hat{U}\,|a_n\rangle \\ &= \sum_{mn} \langle\psi|a_n\rangle\,\delta_{mn}\,\langle a_m|\phi\rangle = \langle\psi|\phi\rangle \\ &= \langle\phi|\psi\rangle^*\,, \end{aligned}$$

which just conforms the definition of anti-unitary operator. What is more,

$$\begin{aligned} \langle\phi|\hat{A}|\psi\rangle = \langle\phi A^\dagger|\psi\rangle = \langle T\psi|TA^\dagger\phi\rangle = \langle T\psi|\,\hat{T}\hat{A}^\dagger\,|\phi\rangle \\ = \langle T\psi|\,\hat{T}\hat{A}^\dagger\hat{T}^{-1}\,|T\phi\rangle\,. \end{aligned} \tag{4.50}$$

A Glance of Spin-half particle

By representation under $\hat{\sigma}_z$, we may let $T = U\mathscr{K}$, and from $T\boldsymbol{\sigma}T^{-1} = -\boldsymbol{\sigma}$ as time reversal operation on spin-angular momentum, one find

$$U\sigma_{x/z}U^{-1} = -\sigma_{x/z},\ U\sigma_y U^{-1} = \sigma_y, \tag{4.51}$$

then by construction one could let

$$U = i\sigma_y,\ T = i\sigma_y\mathscr{K} = \exp\Big(i\sigma_y\frac{\pi}{2}\Big)\mathscr{K},\ \hat{T} = \exp\left(\frac{i}{\hbar}\hat{S}_y\pi\right)\mathscr{K}, \tag{4.52}$$

which could be generalized to common angular momentum

$$\hat{T} = \exp\left(\frac{i}{\hbar}\hat{J}_y\pi\right)\mathscr{K}. \tag{4.53}$$

Now since

$$\begin{aligned}\hat{T}^2\left|\psi\right\rangle &= \exp\left(\frac{i}{\hbar}\hat{J}_y 2\pi\right)\mathscr{K}^2\sum_m \left|j,m\right\rangle\left\langle j,m|\psi\right\rangle \\ &= (-1)^{2j}\sum_m \left|j,m\right\rangle\left\langle j,m|\psi\right\rangle = (-1)^{2j}\left|\psi\right\rangle,\end{aligned} \tag{4.54}$$

we have

$$\hat{T}^2 = \begin{cases}+\mathbb{I}, \text{ integer spin} \\ -\mathbb{I}, \text{ half-integer spin}\end{cases}. \tag{4.55}$$

Notice that we have used rotational property $R_{\hat{n}}(2\pi) = (-1)^{2j}$ regardless of rotation axis, as easy to be generalized from spin-$\frac{1}{2}$ and spin-1 cases, since the property independent of concrete representation. On the other hand, one may argue from

$$\exp\left(\frac{i}{\hbar}\hat{J}_z 2\pi\right)\left|j,m\right\rangle = (-1)^{2m}\left|j,m\right\rangle = (-1)^{2j}\left|j,m\right\rangle,$$

since $2m$ has the same odevity with $2j$, and J_z can be again chosen to be arbitrary axis.

Kramers Degeneracy

For system with time reversal invariance,

$$\hat{H}\hat{T}\left|n\right\rangle = \hat{T}\hat{H}\left|n\right\rangle = E_n\hat{T}\left|n\right\rangle. \tag{4.56}$$

If assuming $\hat{T}\left|n\right\rangle = e^{i\theta}\left|n\right\rangle$,

$$\hat{T}^2\left|n\right\rangle = \hat{T}(e^{i\theta}\left|n\right\rangle) = e^{-i\theta}\hat{T}\left|n\right\rangle = \left|n\right\rangle, \tag{4.57}$$

which holds only when $\hat{T}^2 = \mathbb{I}$. And in this way we say that states electronic system with time reversal invariance must be at least doubly degenerated as $\hat{T}^2 = -\mathbb{I}$.

Chapter 5

Perturbation Theory

5.1 Time Independent Perturbation Theory

"That perturbation related to series expansion is about two projections: to subspace energy got corrected, and to others state got corrected."

Now say we have Hamiltonian (again ignore operator tip)

$$H = H_0 + \lambda V,\ H_0 \left|n^{(0)}\right\rangle = E_n^{(0)} \left|n^{(0)}\right\rangle, \tag{5.1}$$

where λ is small and H_0 is all-known.

5.1.1 Non-degenerate Case

By degenerate or non-degenerate we refer to the unperturbed subspace of H_0. For n-th state we want to know, define projection operator and its complementary operator using *unperturbed information*

$$P_n = \left|n^{(0)}\right\rangle\!\left\langle n^{(0)}\right|,\ Q_n = \mathbb{I} - P_n = \sum_{m\neq n} \left|m^{(0)}\right\rangle\!\left\langle m^{(0)}\right|, \tag{5.2}$$

with

$$\langle n^{(0)}|m^{(0)}\rangle = \delta_{nm}. \tag{5.3}$$

In spirit of series expansion, we let *exact solution* to be

$$\begin{cases} |n\rangle = \sum_{i=p}^{+\infty} \lambda^p \left|n^{(p)}\right\rangle = \left|n^{(0)}\right\rangle + \lambda \left|n^{(1)}\right\rangle + \lambda \left|n^{(2)}\right\rangle + \cdots \\ E_n = \sum_{i=p}^{+\infty} \lambda^p E_n^{(p)} = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots \end{cases}. \tag{5.4}$$

Bring which back to eigenfunction gives

$$(H_0 + \lambda V)\left(\sum_{i=p}^{+\infty} \lambda^p \left|n^{(p)}\right\rangle\right) = \left(\sum_{i=p}^{+\infty} \lambda^p E_n^{(p)}\right)\left(\sum_{i=p}^{+\infty} \lambda^p \left|n^{(p)}\right\rangle\right), \tag{5.5}$$

which readily leads to iterate series of equations, to p-th order of λ

$$H_0 \left|n^{(p)}\right\rangle + V \left|n^{(p-1)}\right\rangle = \sum_{i+j=p} E_n^{(i)} \left|n^{(j)}\right\rangle, \tag{5.6}$$

or

$$(H_0 - E_n^{(0)})\,|n^{(p)}\rangle = (E_n^{(1)} - V)\,|n^{(p-1)}\rangle + E_n^{(2)}\,|n^{(p-2)}\rangle + \cdots + E_n^{(p)}\,|n^{(0)}\rangle\,, \tag{5.7}$$

written explicitly,

$$\begin{aligned}
(H_0 - E_n^{(0)})\,|n^{(0)}\rangle &= 0 \\
(H_0 - E_n^{(0)})\,|n^{(1)}\rangle &= (E_n^{(1)} - V)\,|n^{(0)}\rangle \\
(H_0 - E_n^{(0)})\,|n^{(2)}\rangle &= (E_n^{(1)} - V)\,|n^{(1)}\rangle + E_n^{(2)}\,|n^{(0)}\rangle \\
&\cdots .
\end{aligned}$$

Assume that $(p-1)$-th and lower order information is known, then for p-th, by left action of P_n,

$$0 = -\langle n^0|\,V\,|n^{(p-1)}\rangle + E_n^{(p)} \Longrightarrow E_n^{(p)} = \langle n^{(0)}|\,V\,|n^{(p-1)}\rangle\,, \tag{5.8}$$

where we have assumed that direct inner product between $|n^{(0)}\rangle$ and $|n^{(p)}\rangle\,(p \geq 1)$ vanishes; and by left action of Q_n,

$$Q_n(H_0 - E_n^{(0)})\,|n^{(p)}\rangle = Q_n\left[(E_n^{(1)} - V)\,|n^{(p-1)}\rangle + E_n^{(2)}\,|n^{(p-2)}\rangle + \cdots + E_n^{(p-1)}\,|n^{(1)}\rangle\right], \tag{5.9}$$

since LHS equals $(H_0 - E_n^{(0)})\,|n^{(p)}\rangle$, and

$$T_n \equiv \frac{1}{E_n^{(0)} - H_0} Q_n = \sum_{m \neq n} \frac{1}{E_n^{(0)} - E_m^{(0)}}\,\big|m^{(0)}\big\rangle\big\langle m^{(0)}\big| \tag{5.10}$$

is well-defined, we may write

$$|n^{(p)}\rangle = \frac{1}{H_0 - E_n^{(0)}} Q_n\left[(E_n^{(1)} - V)\,|n^{(p-1)}\rangle + E_n^{(2)}\,|n^{(p-2)}\rangle + \cdots + E_n^{(p-1)}\,|n^{(1)}\rangle\right]. \tag{5.11}$$

And from this equation, we may say that our inner product assumption is right, since clearly by Q_n, or T_n, $|n^{(p)}\rangle\,(p \geq 1)$ is always contributed from $|m^{(0)}\rangle$ with $m \neq n$.

Let us see first several examples. For $p = 1$,

$$E_n^{(1)} = \langle n^{(0)}|\,V\,|n^{(0)}\rangle\,,\ |n^{(1)}\rangle = \frac{1}{H_0 - E_n^{(0)}} Q_n(E_n^{(1)} - V)\,|n^{(0)}\rangle = \sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}\,|m^{(0)}\rangle\,, \tag{5.12}$$

where

$$V_{mn} = \langle m^{(0)}|\,V\,|n^{(0)}\rangle\,. \tag{5.13}$$

For $p = 2$,

$$\begin{aligned}
E_n^{(2)} &= \langle n^{(0)}|\,V\,|n^{(1)}\rangle = \langle n^{(0)}|\,V \frac{1}{E_n^{(0)} - H_0} Q_n V\,|n^{(0)}\rangle = \sum_{m \neq n} \frac{|V_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}, \\
|n^{(2)}\rangle &= \frac{1}{H_0 - E_n^{(0)}} Q_n\left[(E_n^{(1)} - V)\,|n^{(1)}\rangle + E_n^{(2)}\,|n^{(0)}\rangle\right] \\
&= \frac{1}{E_n^{(0)} - H_0} Q_n V \frac{1}{E_n^{(0)} - H_0} Q_n V\,|n^{(0)}\rangle - \frac{\langle n^{(0)}|\,V\,|n^{(0)}\rangle}{E_n^{(0)} - H_0} Q_n \frac{1}{E_n^{(0)} - H_0} Q_n V\,|n^{(0)}\rangle\,,
\end{aligned} \tag{5.14}$$

and one thing interesting is that for ground state $E_m^{(0)} > E_n^{(0)}$ so that 2-nd order perturbation energy correction is also minus.

This procedure could be explored more enlighteningly. Let $\Delta_n = E_n - E_n^{(0)}$, then

$$(H_0 + \lambda V)\left|n\right\rangle = (E_n^{(0)} + \Delta_n)\left|n\right\rangle \Longrightarrow (E_n^{(0)} - H_0)\left|n\right\rangle = (\lambda V - \Delta_n)\left|n\right\rangle, \tag{5.15}$$

by left acting P_n one finds

$$\left\langle n^{(0)}\right|(\lambda V - \Delta_n)\left|n\right\rangle = 0 \Longrightarrow \Delta_n = \lambda\left\langle n^{(0)}\right|V\left|n\right\rangle, \tag{5.16}$$

by left acting Q_n one has

$$(E_n^{(0)} - H_0)\left|n\right\rangle = Q_n(E_n^{(0)} - H_0)\left|n\right\rangle = Q_n(\lambda V - \Delta_n)\left|n\right\rangle,$$

from the well-defined operator one may write

$$\left|n\right\rangle = c_n(\lambda)\left|n^{(0)}\right\rangle + \frac{1}{E_n^{(0)} - H_0} Q_n(\lambda V - \Delta_n)\left|n\right\rangle, \tag{5.17}$$

where the first term makes sure that the perturbated state would come back as $\lambda \to 0$, and by convenience we let

$$c_n(\lambda) = \langle n|n^{(0)}\rangle = 1, \tag{5.18}$$

which may make $\left|n\right\rangle$ unnormalized. Now

$$\left|n\right\rangle = \left|n^{(0)}\right\rangle + T_n(\lambda V - \Delta_n)\left|n\right\rangle \tag{5.19}$$

as iterate equation, and one could also expand this order by order.

5.1.2 Degenerate Case

In this case, if one directly applies theory above to one of degenerate states of certain subspace, the T_n would not be well-defined anymore.

However, one may have imagined a method to solve that, i.e., by splitting the degenerate energy to several non-degenerate levels so that our theory still applies. And clearly this work would be done by $\{\left|n^{(0)}\right\rangle\}$ and V.

At this time, we would also like to define projection operator and its complementary one, but to the *degenerate subspace*

$$P_D = \sum_{n\in D}\left|n^{(0)}\right\rangle\!\left\langle n^{(0)}\right|, \; Q_D = \mathbb{I} - P_D = \sum_{m\notin D}\left|m^{(0)}\right\rangle\!\left\langle m^{(0)}\right|. \tag{5.20}$$

Since any unitary transformation in this subspace would leave the transformed state a new eigenket of H_0, we may generally expect it that the splitting is not done directly on $\{\left|n^{(0)}\right\rangle\}$, but their linear combinations, say $\{\left|l^{(0)}\right\rangle\}$, as unitary transformation, which maintains P_D and Q_D invariant. And we may expect that exact solutions labelled as $\{\left|l\right\rangle\}$.

Then a natural question emerges as how to choose the unitary transformation, or more directly, $\{\left|l^{(0)}\right\rangle\}$? Generally speaking, all we have for the subspace is P_D and V, then in spirit of projection, this may lead to a new all-known Hamiltonian

$$H_0 \to H_0' = H_0 + \lambda P_D V P_D, \; H_0'\left|l^{(0)}\right\rangle = E_l^{(0)'}\left|l^{(0)}\right\rangle, \tag{5.21}$$

i.e., we assume that the transformed kets $\{|l^{(0)}\rangle\}$ should be simultaneous eigenkets of H_0 and projected perturbation, since

$$H_0' \left|l^{(0)}\right\rangle = (E_D^{(0)} + \lambda P_D V P_D) \left|l^{(0)}\right\rangle = E_l^{(0)'} \left|l^{(0)}\right\rangle,$$

which is clearly an eigenfunction problem with $\{|l^{(0)}\rangle\}$ and $\{E_l^{(0)'}\}$ to be determined. Then under representation of $\{|n^{(0)}\rangle\}$, the matrix representation form would lead to a secular equation

$$\det\left(E_D^{(0)} + \lambda P_D V P_D - E^{(0)'}\right) = 0, \tag{5.22}$$

whose roots give $\{E_l^{(0)'}\}$ that differ from $E_D^{(0)}$ up to order λ; bringing which back gives linear combination coefficients of $\{|n^{(0)}\rangle\}$ to $\{|l^{(0)}\rangle\}$, and one may find it that

$$E_l^{(0)'} - E_D^{(0)} = \lambda \left\langle l^{(0)}\right| V \left|l^{(0)}\right\rangle. \tag{5.23}$$

Suppose that this practically splits energy levels, and now our problem is transferred to a non-degenerate problem

$$H = H_0' + \lambda V',\ H_0' \left|l^{(0)}\right\rangle = E_l^{(0)'} \left|l^{(0)}\right\rangle,\ V' = V - P_D V P_D. \tag{5.24}$$

Clearly this is upon a first order manipulation, and what if this does not work? Say $E_l^{(0)'} \equiv E_D^{(0)}$, $\forall l \in D$? This could happen only when $P_D V P_D = 0$, which in turn means that V has no influence up to first order. And in this case, $V_{mn} \equiv 0$ in the subspace so that we need not to be worried about T_n as the summation would avoid the subspace. However, even in this case, it turns out that the non-degenerate perturbation theory still does not work, and as we shall see, we indeed need the effective Hamiltonian approach to solve this problem.

As an illustrate example, we may consider Hamiltonian $H = H_0 + V$ with $H_0 \left|n\right\rangle = E_n \left|n\right\rangle$ fully known and V small quantity. Now we focus on two levels, and without losing generality, we choose $n = 1, 2$, which are not necessarily degenerate. In this subspace defined with projection operator $P = |1\rangle\langle 1| + |2\rangle\langle 2|$, a reduced Hamiltonian (which is indeed the first-order effective Hamiltonian) can be written as

$$\begin{aligned} H_{\text{eff}} = PHP &= \begin{pmatrix} E_1 + V_{11} & V_{12} \\ V_{21} & E_2 + V_{22} \end{pmatrix} \\ &= \overline{E+V} + \delta(E+V)\sigma_z + \operatorname{Re} V_{12}\sigma_x + \operatorname{Im} V_{12}\sigma_y, \end{aligned} \tag{5.25}$$

with basis ignored and $V_{ij} = \langle i| V |j\rangle$, and $\overline{E+V} = (E_1 + V_{11} + E_2 + V_{22})/2$, $\delta(E+V) = (E_1 + V_{11} - E_2 - V_{22})/2$, and $\sigma_{x,y,z}$ the Pauli matrices. The eigenvalues of H_{eff} can be written as

$$E_\pm = \overline{E+V} \pm \sqrt{\delta(E+V)^2 + |V_{12}|^2}, \tag{5.26}$$

which contains the key information. Firstly, we consider the non-degenerate case, where $E_1 \neq E_2$ thus generally $E_2 - E_1 \gg |V_{12}|$, and the eigenvalues can be expanded as

$$E_\pm^{\text{non-degenerate}} = \overline{E+V} \pm \delta(E+V) \pm \frac{1}{2}\frac{|V_{12}|^2}{\delta(E+V)} + O(|V_{12}|^3), \tag{5.27}$$

with $\overline{E+V} + \delta(E+V) = E_1 + V_{11}$ and $\overline{E+V} - \delta(E+V) = E_2 + V_{22}$ the standard first-order perturbation results. Essentially, V_{12} enters only at the second order, and we

do not need to consider it, or equivalently, the projected Hamiltonian in the first order. However, when it is the case that $E_1 = E_2$, we find that

$$E_{\pm}^{\text{degenerate}} = \overline{E+V} \pm \sqrt{(V_{11}-V_{22})^2 + |V_{12}|^2}, \tag{5.28}$$

and clearly, both V_{11}, V_{22} and V_{12} will enter at the first order, thus the off-diagonal term V_{12} will be crucial in this case.

To find the general case, we utilize mathematical deduction. The proposition is that

Claim 1. *For $n \times n$ Hamiltonian $H = H_0 + V$ with H_0 fully known, with eigenvalues E_i and eigenstates $|i\rangle_{i=1,\cdots,n}$, the first-order corrections to the eigenvalue E_i do not contain $V_{ij} = \langle i|V|j\rangle$ terms with $i \neq j$ if and only if $|E_i - E_j| \gg |V_{ik}|,\ \forall j \neq i,\ \forall k$.*

Proof. We use mathematical deduction to prove this claim. For $n = 1$ the claim is trivial, while for $n = 2$ we have already shown the result. Now we assume that the claim is true for $n-1$, then for n, we consider that under $|i\rangle$ basis, the Hamiltonian can be written as

$$H = \begin{pmatrix} D_{n-1} + V_{n-1} & \nu \\ \nu^\dagger & E_n + V_{nn} \end{pmatrix},$$

where $D_{n-1} = \text{diag}(E_1, \cdots, E_{n-1})$ and $\nu = (V_{1,n}, V_{2,n}, \cdots, V_{n-1,n})^T$. Consider unitary transformation $U = \begin{pmatrix} U_{n-1} & 0 \\ 0 & 1 \end{pmatrix}$, where U_{n-1} is the unitary transformation that fully diagonalizes $D_{n-1} + V_{n-1}$, i.e., $U_{n-1}^\dagger (D_{n-1} + V_{n-1}) U_{n-1} = \tilde{D}_{n-1} = \text{diag}(\tilde{E}_1, \cdots, \tilde{E}_{n-1})$, then the transformed Hamiltonian reads

$$\tilde{H} = U^\dagger H U = \begin{pmatrix} \tilde{D}_{n-1} & \tilde{\nu} \\ \tilde{\nu}^\dagger & E_n + V_{nn} \end{pmatrix},$$

where $\tilde{\nu} = U_{n-1}^\dagger \nu$. To find the n-th eigenvalue of $\tilde{H}$, we ask help from Section A.3, and write

$$\begin{aligned} \det\left(\tilde{H} - \lambda_n\right) &= \det\left(\tilde{D}_{n-1} - \lambda_n\right) \det\left(E_n + V_{nn} - \lambda_n - \tilde{\nu}^\dagger \frac{1}{\tilde{D}_{n-1} - \lambda_n} \tilde{\nu}\right) \\ &= \prod_{i=1}^{n-1} (\tilde{E}_i - \lambda_n) \left(E_n + V_{nn} - \lambda_n - \sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{\tilde{E}_i - \lambda_n}\right), \end{aligned}$$

from which we see that λ_n is determined self-consistently by

$$\lambda_n = E_n + V_{nn} - \sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{\tilde{E}_i - \lambda_n}.$$

Notice that this decomposition does not fit for the first $n-1$ eigenvalues, since $\tilde{D}_{n-1} - \lambda_i$ can be non-invertible. Also notice that the equation can also be obtained from the perturbation theory we discussed above.

Now generally we can assume that $\lambda_n = E_n + V_{nn} + \alpha O(V_{ij}) + O(V_{ij}^2)$, with $\alpha = 0$ or 1. Under the assumption that the claim stands for $n-1$, we have $\tilde{E}_i = E_i + V_{ii} + O(V_{ij}^2)$ with no V_{ij} terms in the first order, and the equations becomes

$$\alpha O(V_{ij}) + O(V_{ij}^2) = \sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{E_n - E_i + V_{nn} - V_{ii} + \alpha O(V_{ij}) + O(V_{ij}^2)},$$

and since $|E_i - E_j| \gg |V_{ik}|$, $\forall j \neq i$, $\forall k$, the RHS becomes $\sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{E_n - E_i + O(V)} = \sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{E_n - E_i} + O(V^3)$, which starts from the second order, and thus $\alpha = 0$.

On the other hand, when there exists degeneracy, we can, without losing generality, assuming n-fold degeneracy to continue using the equation. And since now $\tilde{E}_i = E_i + V_{ii} + O(V_{ij})$ with V_{ij} appearing in the first order, the equation becomes

$$\alpha O(V_{ij}) + O(V_{ij}^2) = \sum_{i=1}^{n-1} \frac{|\tilde{\nu}|^2}{V_{nn} - V_{ii} + O(V_{ij})},$$

which continues to allow V_{ij} terms in the first order, and thus $\alpha = 1$. □

5.1.3 The Effective Hamiltonian

The above approach, often referred to as the *Rayleigh-Schrödinger perturbation theory*, is a direct way to solve the perturbation problem. However, it is not always the most efficient or the most physical way to do so. There exists another way to attack the perturbation problem, possibly developed by Bloch and followers, by constructing an effective Hamiltonian that acts only on the desired subspace (like the ground state). In this section, we shall write the Hamiltonian as $H = H_0 + V$ with V small, with $H_0 |i/I\rangle = \varepsilon_{i/I} |i/I\rangle$ known, and $P = \sum_{i=1}^{d} |i\rangle\langle i|$ the desired subspace projector, $Q = \mathbb{I} - P = \sum_{I=d+1}^{D} |I\rangle\langle I|$ its complementary projector.

Let us consider the full solution assumed as $H |\psi_\alpha\rangle = E_\alpha |\psi_\alpha\rangle$, and we denote

$$\begin{cases} |\phi_\alpha\rangle = P |\psi_\alpha\rangle \\ |\chi_\alpha\rangle = Q |\psi_\alpha\rangle \end{cases},$$

such that $|\psi_\alpha\rangle = |\phi_\alpha\rangle + |\chi_\alpha\rangle$ serves as a decomposition, and the desired effective Hamiltonian can be defined through an eigen-problem as

$$H_{\text{eff}} |\phi_\alpha\rangle = E_\alpha |\phi_\alpha\rangle . \tag{5.29}$$

To solve this, we notice that by left multiplying P/Q over the full solution, we have (using $P |\phi_\alpha\rangle = |\phi_\alpha\rangle$, $Q |\chi_\alpha\rangle = |\chi_\alpha\rangle$)

$$\begin{pmatrix} PHP & PHQ \\ QHP & QHQ \end{pmatrix} \begin{pmatrix} |\phi_\alpha\rangle \\ |\chi_\alpha\rangle \end{pmatrix} = E_\alpha \begin{pmatrix} |\phi_\alpha\rangle \\ |\chi_\alpha\rangle \end{pmatrix}, \tag{5.30}$$

which leads to the following equations (perturbative regime without level crossings)

$$\begin{cases} |\chi_\alpha\rangle = \dfrac{1}{E_\alpha - QHQ} QVP |\phi_\alpha\rangle \\ |\psi_\alpha\rangle = \left(P + \dfrac{1}{E_\alpha - QHQ} QVP \right) |\phi_\alpha\rangle \\ H_{\text{eff}} = PHP + PVQ \dfrac{1}{E_\alpha - QHQ} QVP \end{cases}, \tag{5.31}$$

where we have utilized that $PHQ = PVQ$. At this time, we seem to have obtained the effective Hamiltonian, which, however, directly depends on the eigenvalue E_α and makes the eigen-problem a non-linear one. Such a Hamiltonian is known as the Bloch-Horowitz

Hamiltonian. For a degenerate subspace, we can expect that E_α is closed to $\varepsilon_i \equiv \varepsilon$, and can thus write down the effective Hamiltonian as

$$H_{\text{eff}} = PHP + PVQ\frac{1}{\varepsilon - QHQ}QVP, \tag{5.32}$$

which is indeed correct for our purpose, up to the second order accuracy. For instance, we may consider a half-filled two-site Hubbard model

$$H_2 = t\sum_{\sigma=\uparrow\downarrow}\left(c_{1\sigma}^\dagger c_{2\sigma} + \text{h.c.}\right) + U\left(n_{1\uparrow}n_{1\downarrow} + n_{2\uparrow}n_{2\downarrow}\right),$$

where $c_{i\sigma}$ denotes the electron annihilation operator at site i with spin σ, and $n_{i\sigma}$ is the corresponding number operator. The basis to write down this Hamiltonian can be chosen as $|n_{1\uparrow}n_{1\downarrow}n_{2\uparrow}n_{2\downarrow}\rangle$, and for our half-filled case with constraint $n_{1\uparrow} + n_{1\downarrow} + n_{2\uparrow} + n_{2\downarrow} = 2$, we choose the corresponding basis in sequence as

$$\begin{cases} |1\rangle = |1010\rangle\,, & |2\rangle = |0101\rangle \\ |3\rangle = |1001\rangle\,, & |4\rangle = |0110\rangle \\ |5\rangle = |1100\rangle\,, & |6\rangle = |0011\rangle \end{cases},$$

under which the matrix-form Hamiltonian can be written as

$$H = \left(\begin{array}{cc:cccc} 0 & 0 & & & & \\ 0 & 0 & & & & \\ \hdashline & & 0 & 0 & t & t \\ & & 0 & 0 & -t & -t \\ & & t & -t & U & \\ & & t & -t & & U \end{array}\right), \tag{5.33}$$

as two separable sub-blocks, one as the (classical) ferromagnetic subspace and the other as the anti-ferromagnetic subspace. Notice that one has to be cautious about the signs of t in the Hamiltonian induced by Fermionic exchange, and for instance,

$$\begin{cases} c_{1\uparrow}^\dagger c_{2\uparrow}\,|0011\rangle = c_{1\uparrow}^\dagger c_{2\uparrow}c_{2\uparrow}^\dagger c_{2\downarrow}^\dagger\,|0\rangle = c_{1\uparrow}^\dagger c_{2\downarrow}^\dagger\,|0\rangle = |1001\rangle \\ c_{1\downarrow}^\dagger c_{2\downarrow}\,|0011\rangle = c_{1\downarrow}^\dagger c_{2\downarrow}c_{2\uparrow}^\dagger c_{2\downarrow}^\dagger\,|0\rangle = c_{2\uparrow}^\dagger c_{1\downarrow}^\dagger\,|0\rangle = -\,|0110\rangle \end{cases}.$$

Consider the right-down 4×4 matrix as the nontrivial block, with $\text{diag}(0,0,U,U)$ to be the large and unperturbed part (H_0), we have four states of it

$$\begin{cases} |3\rangle = (1,0,0,0)^T, & |4\rangle = (0,1,0,0)^T, & \varepsilon = 0, \\ |5\rangle = (0,0,1,0)^T, & |6\rangle = (0,0,0,1)^T, & \varepsilon' = U, \end{cases}$$

with $P = |3\rangle\langle 3| + |4\rangle\langle 4|$, $\varepsilon = 0$, and Q, V are defined correspondingly. Then the effective Hamiltonian can be written as

$$H_{\text{eff}} = -\frac{2t^2}{U}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \tag{5.34}$$

whose lower eigenvalue reads $-4t^2/U$, which is indeed the second order perturbation result as one can verify by a direct diagonalization. One can see that the corresponding

wavefunction is $(|3\rangle - |4\rangle)/\sqrt{2}$, as an anti-ferromagnetic state, while the state with higher energy corresponds to a spin-z zero ferromagnetic (triplet) state. For Bosonic system, the sign of t should be positive, while the ground state is unchanged. Also notice that to preserve the $SU(2)$ symmetry, we need to consider also the (classical) ferromagnetic subspace, which has been abandoned since it decouples.

The above procedure suggests that once we find a way to deal with the non-linear E_α part, we obtain the desired effective Hamiltonian. Notice that this is always achievable as we can represent $E_\alpha - \varepsilon$ order by order with known quantities, which, however, may be to tedious to be written down. In this sense, we need some clever way, and the key may lies in the relation between $|\phi_\alpha\rangle$ and $|\psi_\alpha\rangle$, which are related in the non-linear way as

$$\begin{cases} |\phi_\alpha\rangle = P\,|\psi_\alpha\rangle \\ |\psi_\alpha\rangle = \Omega^\alpha(E_\alpha)\,|\phi_\alpha\rangle \end{cases}, \ \Omega^\alpha(E_\alpha) = P + \frac{1}{E_\alpha - QHQ} QVP,$$

from which we recognize Ω^α as some 'section' operator that shows some inverting property to P. Especially, we have $P\Omega^\alpha = P$, $\Omega^\alpha P = \Omega^\alpha$, and the effective Hamiltonian can be written as

$$H_{\text{eff}} = PHP + PVQ\frac{1}{E_\alpha - QHQ}QVP = PH_0P + PV\Omega^\alpha,$$

as a projected H_0 plus additional terms.

Based on the observation, we may make an assumption that our effective Hamiltonian can be written as

$$\begin{cases} H_{\text{eff}} = PH_0P + v \\ v = PV\Omega \\ \Omega P = \Omega,\ P\Omega = P \end{cases},$$

with v an effective interaction. Here, Ω is also a section operator which satisfies that $|\psi_\alpha\rangle = \Omega\,|\phi_\alpha\rangle$, which stands for all $\alpha = 1, \cdots, d$ in the aim subspace. Importantly, Ω does not contain the unknown eigenvalues, and here we do not require the unperturbed basis $|i = 1, \cdots, d\rangle$ to be all degenerate. The requirements by combining P and Ω may be argued by considering their actions over the target subspace, as

$$\begin{aligned} &P\,|\phi_\alpha\rangle = |\phi_\alpha\rangle = P\,|\psi_\alpha\rangle = P\Omega\,|\phi_\alpha\rangle\,, \\ &\Omega\,|\phi_\alpha\rangle = \Omega P\,|\phi_\alpha\rangle\,, \end{aligned}$$

while a strict derivation comes as follows: Expand the P, Ω definitions over unperturbed states as

$$\begin{aligned} |\psi_\alpha\rangle &= \sum_{\mu=i,I} |\mu\rangle\,\langle\mu|\psi_\alpha\rangle \\ &= \Omega P\,|\psi_\alpha\rangle = \sum_i \Omega\,|i\rangle\,\langle i|\psi_\alpha\rangle\,, \end{aligned}$$

which becomes, by left acting $\langle\nu|$, as

$$\sum_i \langle\nu|\Omega|i\rangle\,\langle i|\psi_\alpha\rangle = \langle\nu|\psi_\alpha\rangle\,.$$

This equation can be written by defining $d \times d$ square matrix $\Psi_{i\alpha} = \langle i|\psi_\alpha\rangle$, together with its extension $\bar{\Psi}_{\nu\alpha} = \langle\nu|\psi_\alpha\rangle$ as

$$\Omega\Psi = \bar{\Psi}. \tag{5.35}$$

Notice that in spirit of perturbation, Ψ should be close to identity and is thus invertible, and we can then multiply the equation by Ψ^{-1} to obtain, in the schematic form

$$\Omega = \bar{\Psi}\Psi^{-1} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ \hline * & \cdots & * \\ \vdots & & \vdots \end{pmatrix}_{D\times d} . \tag{5.36}$$

If we try to extend it as some $D \times D$ matrix, the upper-left $d \times d$ block should be identity, which is just the represented P, i.e., the full representation of Ω reads

$$\Omega = \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline * & \cdots & * & \\ \vdots & & \vdots & 0 \end{array}\right)_{D\times D} . \tag{5.37}$$

It then follows quickly that

$$\Omega P = \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline * & \cdots & * & \\ \vdots & & \vdots & 0 \end{array}\right) \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline 0 & & & 0 \end{array}\right) = \Omega,$$

$$P\Omega = \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline 0 & & & 0 \end{array}\right) \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline * & \cdots & * & \\ \vdots & & \vdots & 0 \end{array}\right) = P,$$

$$\Omega^2 = \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline * & \cdots & * & \\ \vdots & & \vdots & 0 \end{array}\right) \left(\begin{array}{ccc:c} 1 & & & 0 \\ & \ddots & & \\ & & 1 & \\ \hline * & \cdots & * & \\ \vdots & & \vdots & 0 \end{array}\right) = \Omega,$$

and one sees that $\{P, Q\}$ forms a ring. Indeed, the relation between v and Ω can be obtained by left-acting the full eigen-equation by P, as

$$\begin{aligned} E_\alpha |\phi_\alpha\rangle &= PE_\alpha |\psi_\alpha\rangle = PH\Omega |\phi_\alpha\rangle = P(H_0\Omega + V\Omega) |\phi_\alpha\rangle \\ &= (PH_0P\Omega + PV\Omega) |\phi_\alpha\rangle = (PH_0P + PV\Omega) |\phi_\alpha\rangle , \end{aligned}$$

where $P^2 = P, [P, H_0] = 0$ have been used. On the other hand, if we consider a left action by Ω, we obtain

$$\begin{aligned} E_\alpha |\psi_\alpha\rangle &= \Omega E_\alpha |\psi_\alpha\rangle = \Omega(H_0 + V) |\psi_\alpha\rangle = \Omega PH_0 |\psi_\alpha\rangle + \Omega V\Omega |\phi_\alpha\rangle \\ &= (\Omega H_0 + \Omega V\Omega) |\phi_\alpha\rangle , \end{aligned}$$

while a direct expansion of the full eigen-equation gives

$$(H_0\Omega + V\Omega)\,|\phi_\alpha\rangle = E_\alpha\,|\psi_\alpha\rangle\,,$$

and comparison of the two equations gives

$$[\Omega, H_0] = V\Omega - \Omega V\Omega, \tag{5.38}$$

which is referred to as the (generalized) Bloch equation, solving which we shall get the desired Ω. Notice that this equation is strict disregarding whether V is perturbative or not, while it is only for our interest sometimes to deal it with perturbative V. At this stage, we expand $\Omega = \Omega^{(0)} + \Omega^{(1)} + \cdots$, and the Bloch equation can be written order-by-order as

$$[\Omega^{(n)}, H_0] = V\Omega^{(n-1)} - \sum_{m=1}^{n-1} \Omega^{(m)} V \Omega^{(n-1-m)},\ n = 1, 2, \cdots, \tag{5.39}$$

with $\Omega^{(0)} = P$ similar to that in Ω^α, which satisfies the equation by $[\Omega^{(0)}, H_0] = [P, H_0] = 0$. To solve this, we introduce the notation

$$\tilde{A}_{Ij} = \frac{1}{\varepsilon_j - \varepsilon_I} A_{Ij}, \tag{5.40}$$

and obtain that

$$\begin{aligned}\langle I|[\Omega^{(n)}, H_0]|j\rangle &= (\varepsilon_j - \varepsilon_I)\Omega^{(n)}_{Ij} \\ &= \left[V\Omega^{(n-1)}\right]_{Ij} - \sum_{m=1}^{n-1} (\Omega^{(m)} V \Omega^{(n-1-m)})_{Ij},\end{aligned}$$

which gives

$$\Omega^{(n)} = \widetilde{V\Omega^{(n-1)}} - \sum_{m=1}^{n-1} \Omega^{(m)} \widetilde{V\Omega^{(n-1-m)}}. \tag{5.41}$$

Notice that from the structure of Ω, we have only $\Omega^{(n>0)}_{Ij} \neq 0$. For instance, we may consider the first and second order corrections, which read

$$\Omega^{(1)}_{Ij} = \frac{1}{(\varepsilon_j - \varepsilon_I)} V_{Ij}, \tag{5.42a}$$

$$\begin{aligned}\Omega^{(2)}_{Ij} &= \frac{1}{(\varepsilon_j - \varepsilon_I)} \left(\sum_J V_{IJ}\Omega^{(1)}_{Jj} - \sum_k \Omega^{(1)}_{Ik} V_{kj} \right) \\ &= \frac{1}{(\varepsilon_j - \varepsilon_I)} \left(\sum_J V_{IJ} \frac{1}{(\varepsilon_j - \varepsilon_J)} V_{Jj} - \sum_k \frac{1}{(\varepsilon_k - \varepsilon_I)} V_{Ik} V_{kj} \right),\end{aligned} \tag{5.42b}$$

and the effective Hamiltonian follows. Such a series hints a diagrammatic expansion, and one may find that the effective Hamiltonian is a sum of all possible diagrams with V as the vertex and Ω as the propagator.

When the target subspace to be a degenerate one with $\varepsilon_i \equiv \varepsilon$, we can introduce the operator $T = \dfrac{1}{\varepsilon - H_0} Q$ to modify the iteration equation as

$$\Omega^{(n)} = TV\Omega^{(n-1)} - \sum_{m=1}^{n-1} T\Omega^{(m)} V \Omega^{(n-1-m)}, \tag{5.43}$$

which gives

$$\Omega^{(1)} = TVP, \tag{5.44a}$$
$$\Omega^{(2)} = (TV)^2P - T^2VPV, \tag{5.44b}$$
$$\Omega^{(3)} = (TV)^3P - T^2VPVTVP - T(TV)^2PV + T^3VPV^2P, \tag{5.44c}$$

and so on. Up to the second order, the effective Hamiltonian over a degenerate subspace can be written as

$$\begin{aligned} H_{\text{eff}} &= PH_0P + PV(\Omega^{(0)} + \Omega^{(1)})P + O(V^3) \\ &= PHP + PVTVP + O(V^3), \end{aligned} \tag{5.45}$$

which is indeed the same one as we replace E_α with ε before.

Now let us define $u = V\Omega - \Omega V\Omega$, with a name 'effective transition potential', whose meaning will be understood later. Since $Pu = P(V\Omega - \Omega V\Omega) = PV\Omega - PV\Omega = 0$, we have

$$QuP = (1-P)uP = uP = u,$$

and $QuQ = Qu(1-P) = 0$ follows. Now by $[\Omega, H_0] = u$, we have $\Omega_{Ij} = \tilde{u}_{Ij}$, or $Q\Omega P = \tilde{u}$, which can be written as

$$\tilde{u} = (1-P)\Omega P = \Omega - P,$$

or

$$\Omega = P + \tilde{u}, \tag{5.46}$$

which is indeed the same to the iteration formula above and leads to

$$|\psi_\alpha\rangle = \Omega\,|\phi_\alpha\rangle = |\phi_\alpha\rangle + \tilde{u}\,|\phi_\alpha\rangle\,, \tag{5.47}$$

which will become, for the degenerate case,

$$|\psi_\alpha\rangle = |\phi_\alpha\rangle + Tu\,|\phi_\alpha\rangle\,, \tag{5.48}$$

which is the Lippmann-Schwinger equation with replacement $V \to u$ and brings u the name effective transition potential, although here the perturbed state generally contains a different energy. We may bring this relation back to the definition of u as

$$\begin{aligned} u &= V\Omega - \Omega V\Omega = (1-\Omega)V\Omega = (Q-\tilde{u})V(P+\tilde{u}) \\ &= QVP + QV\tilde{u} - \tilde{u}VP - \tilde{u}V\tilde{u} \\ &= Q(V + V\tilde{u} - \tilde{u}V - \tilde{u}V\tilde{u})P, \end{aligned}$$

such that

$$u = V + V\tilde{u} - \tilde{u}V - \tilde{u}V\tilde{u}, \tag{5.49}$$

inside which we have utilized that

$$\begin{cases} P\Omega = P \\ \Omega P = \Omega \end{cases} \implies \begin{cases} P\tilde{u} = 0 \Rightarrow \tilde{u} = Q\tilde{u} \\ \tilde{u}P = \tilde{u} \end{cases}.$$

To read the equation, we may introduce $(A) = \tilde{A}$ such that

$$(u) = (V) + (V(u)) - ((u)V) - ((u)V(u)), \tag{5.50}$$

which is suitable for a diagrammatic expansion with each $)V$ contributing a minus sign. A series expansion of $u = u^{(1)} + u^{(2)} + \cdots$, which starts from $u^{(1)} = V$, gives

$$(u^{(n)}) = (V(u^{(n-1)})) - ((u^{(n-1)})V) - \sum_{m=1}^{n-2}((u^{(m)})V(u^{(n-2-m)})), \tag{5.51}$$

with first lowest orders as

$$(u^{(1)}) = (V), \tag{5.52a}$$
$$(u^{(2)}) = (V(V)) - ((V)V), \tag{5.52b}$$
$$(u^{(3)}) = (V(V(V))) - (V((V)V)) - ((V(V))V) + (((V)V)V) - ((V)V(V)), \tag{5.52c}$$

and so on. One may observe that only $(V())$, $(()V)$ and $(()V())$ appear, and for a diagrammatic expansion, a reading from inner to outer, while with fixed outermost bracket, is suitable.

Revisit of the Hubbard model

Now we may derive the effective Hamiltonian for the general half-filled Hubbard model in the large U limit, which reads

$$\begin{aligned} H &= t\sum_{\langle rr'\rangle\sigma}\left(c_\sigma^\dagger(r)c_\sigma(r') + \text{h.c.}\right) + U\sum_r n_\uparrow(r)n_\downarrow(r) \\ &= t\sum_{\langle rr'\rangle\sigma}\left(c_\sigma^\dagger(r)c_\sigma(r') + \text{h.c.}\right) - \frac{2U}{3}\sum_r(\boldsymbol{S}(r))^2 + \frac{NU}{2}, \end{aligned} \tag{5.53}$$

with a nearest-neighbor hopping and a on-site repulsion. To arrive at the second line, we have utilized the completeness of $\mathfrak{su}(2)$ algebra (Eq. (1.80))

$$\sum_a \sigma^a_{\alpha\beta}\sigma^a_{\gamma\delta} = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta},$$

and the second-quantized spin operator ($\hbar = 1$) (Eq. (7.22))

$$\boldsymbol{S} = \frac{1}{2}\sum_{\sigma,\sigma'} c_\sigma^\dagger \boldsymbol{\sigma}_{\sigma\sigma'} c_{\sigma'}.$$

Still, we recognize $V = t\sum_{\langle rr'\rangle\sigma}\left(c_\sigma^\dagger(r)c_\sigma(r') + \text{h.c.}\right)$ to be the small perturbation, and H_0 the U term. We are still interested in the ground state, then we focus on the degenerate subspace of H_0 as $\{|\alpha\rangle\} = \{|\cdots n_\uparrow(r)n_\downarrow(r)\cdots\rangle \,|n_\uparrow(r) + n_\downarrow(r) = 1\}$ with zero unperturbed energy, then up to second order, the effective Hamiltonian can be written as

$$\begin{aligned} H_{\text{eff}} &= PHP + PVTVP \\ &= -PV\frac{Q}{H_0}VP = -\sum_{\langle rr'\rangle} PV_{rr'}Q\frac{1}{QH_0Q}QV_{r'r}P \\ &= -\frac{2}{U}\sum_{\langle rr'\rangle} PV_{rr'}QV_{r'r}P, \end{aligned}$$

where in the second equality we utilize the zero-energy property of the target subspace together with $\langle\alpha'|V|\alpha\rangle \equiv 0$ by half-filling; and to the third equality we notice that by

nearest-neighbor hopping,to come back to the subspace we have to let the two steps induced by V to form a loop, i.e., $r \to r' \to r$, and denote corresponding hopping as $V_{rr'}$ and $V_{r'r}$; and the last equality is obtained by noticing that a single action of V lifts the system's energy by $2(\boldsymbol{S}^2)/3\,|1/2\rangle = 1/2\,|1/2\rangle$ ($\hbar = 1$).

From the above deduction, we may further infer that in the perturbative regime, the effective Hamiltonian of the half-filled Hubbard model is composed of all possible loops of hopping terms, some a single loop while others may contain multiple connected loops. For instance, the third order effective Hamiltonian on a square lattice can not exist, since the nearest-neighbor hopping forbids the formation of triangle loops, while it is definitely possible for a Kagome lattice. As we shall show later, the second order effective will give rise to a Heisenberg model with nearest-neighbor spin-exchange interaction, and thus the higher order terms will contain multiple spin interactions on a loop, like that in Kitaev's honeycomb model. Such an expansion reflects the strongly-correlated nature of electron-electron interactions.

Let us come back to the second-order. Since V is quadratic in electron operators, we can safely restrict our attention to the subspace of r, r' without sign problems. Utilizing the basis $|\cdots n_{r\uparrow} n_{r\downarrow} n_{r'\uparrow} n_{r'\downarrow} \cdots\rangle$, we similarly have the effective Hamiltonian

$$-\frac{2t^2}{U}\left(|\cdots 1001\cdots\rangle, |\cdots 0110\cdots\rangle\right)\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \langle\cdots 1001\cdots| \\ \langle\cdots 0110\cdots| \end{pmatrix}, \tag{5.54}$$

which leads to the operator form of the effective antiferromagnetic Heisenberg Hamiltonian as

$$\begin{aligned} H_{\text{eff}} &= -\frac{1}{2}\frac{2t^2}{U}\sum_{\langle rr'\rangle}\left(c^\dagger_{r\uparrow}c^\dagger_{r'\downarrow}c_{r'\uparrow}c_{r\downarrow} - c^\dagger_{r\uparrow}c^\dagger_{r'\downarrow}c_{r'\uparrow}c_{r\downarrow} - c^\dagger_{r\downarrow}c^\dagger_{r'\uparrow}c_{r'\downarrow}c_{r\uparrow} + c^\dagger_{r\downarrow}c^\dagger_{r'\uparrow}c_{r'\uparrow}c_{r\downarrow}\right) \\ &= \frac{2t^2}{U}\sum_{\langle rr'\rangle}\left(\boldsymbol{S}_r\cdot\boldsymbol{S}_{r'} - \frac{1}{4}n_r n_{r'}\right), \end{aligned}$$

with the $1/2$ coefficient cancelling the double counting of neighbors and the utilization of

$$\begin{cases} \boldsymbol{S}_r\cdot\boldsymbol{S}_{r'} = \dfrac{1}{2}\left(S_{r+}S_{r'-} + S_{r-}S_{r'+}\right) + S_{rz}S_{r'z} \\ S_+ = c^\dagger_\uparrow c_\downarrow,\ S_- = c^\dagger_\downarrow c_\uparrow,\ S_z = \dfrac{1}{2}(n_\uparrow - n_\downarrow) \end{cases}.$$

Notice that the second term is a constant and can be dropped.

Under the large U limit, if the system is slightly away from half-filling, we see that the first order term PVP will allow for a constrained moving between $n_r = 1$ and $n_{r'} = 0$ with $\langle rr'\rangle$ the nearest neighbors. If we denote $J = 2t^2/U$, the effective Hamiltonian reads the $t - J$ model

$$H_{\text{eff}} = t\sum_{\langle rr'\rangle\sigma}\left(c^\dagger_\sigma(r)c_\sigma(r') + \text{h.c.}\right) + J\sum_{\langle rr'\rangle}\boldsymbol{S}_r\cdot\boldsymbol{S}_{r'}, \tag{5.55}$$

with constraint $n_r = 0, 1$.

5.2 Variational Method

Variational method is used for ground state. By assuming a parameterized normalized trial ket $|\psi(\{\lambda_i\})\rangle$ for ground state, whose energy is approximated to be

$$E_\psi = E(\{\lambda_i\}) = \langle\psi(\{\lambda_i\})|\,\hat{H}\,|\psi(\{\lambda_i\})\rangle, \tag{5.56}$$

further assuming $\hat{H}\left|n\right\rangle = E_n\left|n\right\rangle$,

$$\begin{aligned} E_\psi &= \langle\psi(\{\lambda_i\})|\sum_n |n\rangle\langle n|\,\hat{H}\sum_m |m\rangle\langle m|\psi(\{\lambda_i\})\rangle = \sum_n |\langle\psi(\{\lambda_i\})|n\rangle|^2 E_n \\ &\geq E_0\langle\psi(\{\lambda_i\})|(\sum_n |n\rangle\langle n|)|\psi(\{\lambda_i\})\rangle = E_0, \end{aligned}$$

then, what we do is try to minimize E_ψ varying $\{\lambda_i\}$, while the choice of initial form of $|\psi(\{\lambda_i\})\rangle$ directly determines how good the approximation is.

5.3 Time Dependent Perturbation Theory

From interaction picture, for Hamiltonian

$$\hat{H} = \hat{H}_0 + \lambda\hat{V}(t),\ \hat{H}_0\left|n\right\rangle = E_n\left|n\right\rangle, \tag{5.57}$$

one has

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle_I = \lambda\hat{V}_I(t)\left|\psi(t)\right\rangle_I, \tag{5.58}$$

with

$$\begin{aligned} \left|\psi(t)\right\rangle_I &= \hat{U}_I(t)\left|\psi\right\rangle_H = \hat{U}_0^\dagger(t)\left|\psi(t)\right\rangle_S,\ \hat{U}_I(t) = \hat{U}_0^\dagger(t)\hat{U}(t), \\ \hat{O(t)}_I &= \hat{U}_I(t)\hat{O(t)}_H\hat{U}_I^\dagger(t) = \hat{U}_0^\dagger(t)\hat{O(t)}_S\hat{U}_0(t). \end{aligned} \tag{5.59}$$

By letting $\left|\psi(t)\right\rangle_I = \sum_n c_n(t)\left|n\right\rangle$, we could act on left $\langle n|$

$$i\hbar\frac{\partial}{\partial t}c_n(t) = \lambda\langle n|\hat{V}_I(t)\left|\psi(t)\right\rangle_I = \lambda\sum_m\langle n|\hat{V}_I(t)|m\rangle\langle m|\psi(t)\rangle_I = \lambda\sum_m V_{nm}(t)e^{i\omega_{nm}t}c_m(t), \tag{5.60}$$

where $V_{nm}(t) = \langle n|\hat{V}(t)|m\rangle$ and $\hbar\omega_{nm} = E_n - E_m$.

Then in spirit of series expansion, let

$$c_n = \sum_{p=0}^{+\infty}\lambda^p c_n^{(p)} = c_n^{(0)} + \lambda c_n^{(1)} + \lambda^2 c_n^{(2)} + \cdots, \tag{5.61}$$

then

$$i\hbar\frac{\partial}{\partial t}\left(\sum_{p=0}^{+\infty}\lambda^p c_n^{(p)}(t)\right) = \lambda\sum_m V_{nm}(t)e^{i\omega_{nm}t}\left(\sum_{q=0}^{+\infty}\lambda^q c_m^{(q)}(t)\right),$$

which leads to p-th order

$$i\hbar\frac{\partial}{\partial t}c_n^{(p)}(t) = \sum_m V_{nm}(t)e^{i\omega_{nm}t}c_m^{(p-1)}(t). \tag{5.62}$$

By assuming that information lower than $(p-1)$-th order is known, and perturbation is added from t_0,

$$c_n^{(p)}(t) = -\frac{i}{\hbar}\sum_m\int_{t_0}^t \mathrm{d}t'\ V_{nm}(t')e^{i\omega_{nm}t'}c_m^{(p-1)}(t'). \tag{5.63}$$

And now since we expand our state to a complete basis, and in spirit of perturbation, we say that our system is more likely to transit between states of $|n\rangle$, rather than with certain energy at t — which is also forbidden by energy-time duality.

For lowest order, assume that initially we have a state of $|m\rangle$, i.e., $c_n^{(0)} = \delta_{nm}$, then for zeroth order,

$$i\hbar\frac{\partial}{\partial t}c_n^{(0)}(t) \equiv 0; \tag{5.64}$$

while for $p = 1$,

$$c_n^{(1)}(t) = -\frac{i}{\hbar}\int_{t_0}^{t} \mathrm{d}t' \; V_{nm}(t')e^{i\omega_{nm}t'}, \tag{5.65}$$

while especially for $n = m$,

$$c_m^{(1)}(t) = -\frac{i}{\hbar}\int_{t_0}^{t} \mathrm{d}t' \; V_{mm}(t'), \tag{5.66}$$

which may give

$$c_m(t) = c_m^{(0)}(t) + \lambda c_m^{(1)}(t) + O(\lambda^2) \approx \exp\left(-\frac{i}{\hbar}\int_{t_0}^{t} \mathrm{d}t' \; \langle m|V(t')|m\rangle\,(t')\right), \tag{5.67}$$

and

$$\begin{aligned} |\psi(t)\rangle_I &= |m\rangle + \lambda\sum_n c_n^{(1)}(t)\,|n\rangle + O(\lambda^2) \\ &\approx \exp\left(-\frac{i}{\hbar}\int_{t_0}^{t} \mathrm{d}t' \; \langle m|V(t')|m\rangle\,(t')\right)|m\rangle - \frac{i\lambda}{\hbar}\sum_{n\neq m}\int_{t_0}^{t} \mathrm{d}t' \; V_{nm}(t')e^{i\omega_{nm}t'}\,|n\rangle\,. \end{aligned} \tag{5.68}$$

And $|\psi(t)\rangle_I$ has probability to transit as

$$P(m \to n)(t) = |c_n^{(1)}(t)|^2 = \frac{\lambda^2}{\hbar^2}\left|\sum_{n\neq m}\int_{t_0}^{t} \mathrm{d}t' \; V_{nm}(t')e^{i\omega_{nm}t'}\right|^2. \tag{5.69}$$

5.3.1 Adiabatic Approximation

Here Hamiltonian is also time dependent, however, the varying is no longer expressed explicitly by V but hidden in the Hamiltonian itself. Still, we ignore operator tip, and we assume that the time dependence is described by some time-dependent parameter λ as

$$H = H(\lambda(t)). \tag{5.70}$$

And we require now, instead of λ, but $\dot{\lambda}$ is small, and the dependence of Hamiltonian on λ is smooth, then we could say Hamiltonian varies slowly with time. Meanwhile, we assume that *instantaneous eigenstates* are well-known as

$$H(\lambda)\,|n(\lambda)\rangle = E_n(\lambda)\,|n(\lambda)\rangle\,. \tag{5.71}$$

Furthermore, we require that no degeneracy is presented here, which is referred to the natural *Abelian* case (though still an approximation).

Consider now an initial state $|\psi(t_0 = 0)\rangle$, by Schrödinger equation one has

$$i\hbar\frac{\partial}{\partial t}\,|\psi(t)\rangle = H(\lambda(t))\,|\psi(t)\rangle\,. \tag{5.72}$$

It is also clear that the state also depends on λ as $|\psi(\lambda(t))\rangle$, then

$$H(\lambda(t))\,|\psi(\lambda(t))\rangle = i\hbar\frac{\partial}{\partial t}\,|\psi(\lambda(t))\rangle = i\hbar\dot{\lambda}\frac{\partial}{\partial\lambda}\,|\psi(\lambda(t))\rangle\,, \tag{5.73}$$

and now, by assuming that

$$|\psi(\lambda(t))\rangle = \sum_n c_n(\lambda(t))e^{i\theta_n(\lambda(t))}\,|n(\lambda(t))\rangle\,, \tag{5.74}$$

where

$$i\theta_n(\lambda(t)) = -\frac{i}{\hbar}\int_{\lambda(0)}^{\lambda(t)}\frac{E_n(\lambda(t))}{\dot{\lambda}}\mathrm{d}\lambda = -\frac{i}{\hbar}\int_0^t E_n(\lambda(t))\mathrm{d}t \tag{5.75}$$

as usual dynamical phase. Bring the assumption back to equation and one has

$$\begin{aligned}
H(\lambda(t))\,|\psi(\lambda(t))\rangle &= \sum_n c_n(\lambda(t))e^{i\theta_n(\lambda(t))}E_n(\lambda(t))\,|n(\lambda(t))\rangle = i\hbar\dot{\lambda}\frac{\partial}{\partial\lambda}\,|\psi(\lambda(t))\rangle \\
&= i\hbar\dot{\lambda}\sum_n\left[[\partial_\lambda c_n(\lambda(t))]e^{i\theta_n(\lambda(t))}\,|n(\lambda(t))\rangle + c_n(\lambda(t))e^{i\theta_n(\lambda(t))}[\partial_\lambda\,|n(\lambda(t))\rangle]\right] \\
&\quad + \sum_n c_n(\lambda(t))E_n(\lambda(t))e^{i\theta_n(\lambda(t))}\,|n(\lambda(t))\rangle\,,
\end{aligned}$$

which leads to

$$\sum_n e^{i\theta_n(\lambda)}\left\{[\partial_\lambda c_n(\lambda)]\,|n(\lambda)\rangle + c_n(\lambda)[\partial_\lambda\,|n(\lambda)\rangle]\right\} = 0. \tag{5.76}$$

Now similarly, we could define projection operator

$$P_n = |n(\lambda)\rangle\langle n(\lambda)|\,,\ Q_n = \mathbb{I} - P_n. \tag{5.77}$$

With left action of P_n,

$$\partial_\lambda c_n(\lambda) = i\sum_m e^{i[\theta_m(\lambda)-\theta_n(\lambda)]}c_m(\lambda)A_{nm}(\lambda), \tag{5.78}$$

where we have introduced *non-Abelian berry connection*

$$A_{nm}(\lambda) = i\,\langle n(\lambda)|\,\frac{\partial}{\partial\lambda}\,|m(\lambda)\rangle\,, \tag{5.79}$$

and essentially, if λ has multicomponent,

$$A_{\mu,nm}(\lambda) = i\,\langle n(\lambda)|\,\frac{\partial}{\partial\lambda^\mu}\,|m(\lambda)\rangle\,. \tag{5.80}$$

This formula is strict in a way that no approximation is introduced, and the only guess on form of solution is based on complete-basis expansion, which is also rigorous. However, now we just want to focus on simplest case as single-component berry connection and ignore hopping between different bands, then only

$$A(\lambda) = A_{nn}(\lambda) \tag{5.81}$$

is conserved, aka single *Abelian berry connection*. And in this way we may call it *adiabatic*, and it is a lowest-order approximation. Now

$$\partial_\lambda c_n(\lambda) = ic_n(\lambda)A(\lambda) \Longrightarrow c_n(\lambda) = \exp\left(i\int_0^t A(\lambda)\mathrm{d}t\right) = \exp(i\gamma(t)). \tag{5.82}$$

Also, from

$$0 = \partial_\lambda \langle n|n\rangle = -i(A - A^*)$$

we see that A is purely real. In sum,

$$|\psi(\lambda(t))\rangle = \sum_n e^{i\gamma(t)} e^{i\theta_n(\lambda(t))} |n(\lambda(t))\rangle \tag{5.83}$$

under our adiabatic approximation.

5.4 Linear Response Theory

This part of discussion usually needs the introduction of Green's function, here, however, we may also directly arrive at that just use our perturbation theory, since they are just some equivalent stuff to some extent expressed in different ways.

A more inspiring way to start is to consider an eigenket, and then follow its revolution with changing parameter (in general space, say Brillouin zone). Say now we have (in operator form, which I think one is able to distinguish by now)

$$H(\lambda) |n(\lambda)\rangle = E(\lambda) |n(\lambda)\rangle ,$$

then by action of $\partial_\lambda = \mathrm{d}/\mathrm{d}\lambda$ on both sides,

$$(E_n - H) |\partial_\lambda n\rangle = \partial_\lambda (H - E_n) |n\rangle , \tag{5.84}$$

with dependence on λ omitted. One may notice that this is a more general form than $H = H_0 + \lambda V$ where $\partial_\lambda H = V$ as a single perturbation. Now by parameter-varying projection operator $P_n = |n(\lambda)\rangle\langle n(\lambda)|$ acting from left, one has

$$\partial_\lambda E_n = \langle n|\partial_\lambda H|n\rangle , \tag{5.85}$$

and then the λ evolution equation becomes

$$\begin{aligned} (E_n - H) |\partial_\lambda n\rangle &= \partial_\lambda H |n\rangle - |n\rangle \langle n|\partial_\lambda H|n\rangle = (1 - |n\rangle\langle n|)\partial_\lambda H |n\rangle \\ &= (1 - P_n)(\partial_\lambda H) |n\rangle = Q_n (\partial_\lambda H) |n\rangle , \end{aligned} \tag{5.86}$$

known as *Sternheimer approach* to linear response, with Q_n as complementary operator of projection. Since it shares a similar structure as in time-independent perturbation theory, but with a parameter variation on ket, we may also guess since $\dfrac{1}{E_n - H} Q_n$ is well defined

$$|\partial_\lambda n\rangle = c_n(\lambda) |n\rangle + \frac{1}{E_n - H} Q_n (\partial_\lambda H) |n\rangle = c_n(\lambda) |n\rangle + \sum_{m \neq n} \frac{|m\rangle\langle m|}{E_n - E_m} (\partial_\lambda H) |n\rangle ,$$

and by P_n we have

$$c_n(\lambda) = \langle n|\partial_\lambda n\rangle = -iA_n(\lambda),$$

where $A_n(\lambda)$ is just the one component Abelian Berry connection of nth energy level, and we have the solution as

$$|\partial_\lambda n\rangle = -iA_n(\lambda) |n\rangle + \sum_{m \neq n} \frac{|m\rangle\langle m|}{E_n - E_m} (\partial_\lambda H) |n\rangle = -iA_n(\lambda) |n\rangle + T_n(\partial_\lambda H) |n\rangle . \tag{5.87}$$

One may easily verify that $Q_n T_n = T_n = T_n Q_n$ so that

$$Q_n \left|\partial_\lambda n\right\rangle = T_n(\partial_\lambda H) \left|n\right\rangle . \tag{5.88}$$

The physical result people care from linear response is the change (or more exactly, the changing rate as some leading order stuff) of some observable by varying parameter(s) that we are able to control. For an observable O (that does not depend on λ), we have

$$\begin{aligned}\partial_\lambda \left\langle O\right\rangle_n &= \partial_\lambda \left\langle n|O|n\right\rangle = \left\langle \partial_\lambda n|O|n\right\rangle + \left\langle n|O|\partial_\lambda n\right\rangle = 2\,\mathrm{Re}\left\langle n|O|\partial_\lambda n\right\rangle \\ &= 2\,\mathrm{Re}\left\langle n\right| OT_n(\partial_\lambda H)\left|n\right\rangle + 2\,\mathrm{Re}(-iA_n(\lambda)) = 2\,\mathrm{Re}\left\langle n\right| OT_n(\partial_\lambda H)\left|n\right\rangle \\ &= 2\,\mathrm{Re}\left\langle n|OQ_n|\partial_\lambda n\right\rangle .\end{aligned}$$

Now in a system with energy levels or 'bands', like a multi-atom molecule, a crystal system, etc., there exist some bands that are occupied while others are not, then we change our projection operator to project onto all such occupied bands (manifold) as

$$P = \sum_n^{\mathrm{occ}} \left|n\right\rangle\!\left\langle n\right| , \tag{5.89}$$

together with its complementary operator as

$$Q = \mathbb{I} - P = \mathbb{I} - \sum_n^{\mathrm{occ}} \left|n\right\rangle\!\left\langle n\right| = \sum_m^{\mathrm{unocc}} \left|m\right\rangle\!\left\langle m\right| . \tag{5.90}$$

Then it comes with the conclusion that counting all change/response from occupied bands one has

$$\partial_\lambda \left\langle O\right\rangle = 2\,\mathrm{Re}\sum_n^{\mathrm{occ}} \left\langle n|OQ|\partial_\lambda n\right\rangle , \tag{5.91}$$

where we simply replace Q_n with Q besides the summation, and one may ask for the correctness of this procedure, which we argue by showing that such transition-like response between occupied bands complete vanishes as

$$\begin{aligned}&2\,\mathrm{Re}\sum_n^{\mathrm{occ}} \left\langle n|O(Q_n - Q)|\partial_\lambda n\right\rangle = 2\,\mathrm{Re}\sum_{n\neq n'}^{\mathrm{occ}} \left\langle n|O|n'\right\rangle \left\langle n'|\partial_\lambda n\right\rangle \\ =&\sum_{n\neq n'}^{\mathrm{occ}} \left(\left\langle n|O|n'\right\rangle \left\langle n'|\partial_\lambda n\right\rangle + \left\langle n'|O|n\right\rangle \left\langle \partial_\lambda n|n'\right\rangle\right),\ \partial_\lambda \left\langle n|n'\right\rangle = 0, \\ =&\sum_{n\neq n'}^{\mathrm{occ}} \left(\left\langle n|O|n'\right\rangle \left\langle n'|\partial_\lambda n\right\rangle - \left\langle n'|O|n\right\rangle \left\langle n|\partial_\lambda n'\right\rangle\right),\ n\leftrightarrow n', \\ =&0,\end{aligned}$$

and this is indeed because transition between occupied bands is a unitary transformation, which may be viewed some gauge variation and thus does not influence our physical observation. Now as for new T_n operator, it could still be defined considering both one occupied band n and all occupied bands P so that

$$T_n = \frac{1}{E_n - H} Q = \sum_m^{\mathrm{unocc}} \frac{\left|m\right\rangle\!\left\langle m\right|}{E_n - E_m}, \tag{5.92}$$

and still one has

$$Q\left|\partial_\lambda n\right\rangle = T_n(\partial_\lambda H)\left|n\right\rangle, \tag{5.93}$$

as from the strict solution, where one needs to calculate

$$Q\sum_{m\neq n}\frac{|m\rangle\langle m|}{E_n-E_m}=\sum_{m}^{\text{unocc}}\frac{|m\rangle\langle m|}{E_n-E_m}=T_n.$$

Then one may yield

$$\partial_\lambda\left\langle O\right\rangle = 2\,\mathrm{Re}\sum_{n}^{\text{occ}}\left\langle n\right|OT_n(\partial_\lambda H)\left|n\right\rangle. \tag{5.94}$$

We may consider an example when

$$H=H_0+\lambda_A A+\lambda_B B,\ \frac{\partial H}{\partial\lambda_A}=A,\ \frac{\partial H}{\partial\lambda_B}=B,$$

and then one may have the off-diagonal linear response as

$$\begin{aligned}\frac{\partial^2 E}{\partial\lambda_A\partial\lambda_B}&=\frac{\partial\left\langle A\right\rangle}{\partial\lambda_B}=\frac{\partial\left\langle B\right\rangle}{\partial\lambda_A}=2\,\mathrm{Re}\sum_{n}^{\text{occ}}\left\langle n\right|AT_nB\left|n\right\rangle\\&=\sum_{n}^{\text{occ}}\sum_{m}^{\text{unocc}}\frac{\langle n|A|m\rangle\langle m|B|n\rangle}{E_n-E_m}+\text{c.c},\end{aligned}$$

as the familiar response form, where one may notice that when $A=B$ this is just standard second-order perturbation for energy change. And one should not be confused that how our strict theory generates a perturbation, since in that case we in fact calculated the second order derivative of energy with respect to λ, using non-perturbed energy as some $E^{(0)}$ stuff, and in order to get the exact energy change, we need all orders of expansion, which is equivalent to calculating all order of derivatives in our response theory using 0th order information.

5.4.1 Hall Conductance

Hall effect is a classical electro-magnetic effect as follows: say apply I_x along longitudinal and B_z along perpendicular direction, then one is able to measure the transverse voltage V_y to obtain the Hall effect (and longitudinal voltage V_x for magnetoresistance). This result is just some linear response, and its physical core is balance of Lorentz force,

$$0=\boldsymbol{F}=q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B})\Longrightarrow E_y=v_xB_z,$$

then for transverse conductance

$$\sigma_{xy}=\frac{j_x}{E_y}=\frac{nqv_x}{E_y}=\frac{nq}{B_z},$$

which varies linear to $1/B_z$ and changes sign for electrons and holes. However, from 1980s, people find this quantity quantized over a universal unit (for electrons)

$$\sigma_{xy}=\nu\frac{e^2}{h},\ \nu=1,2,\cdots, \tag{5.95}$$

where $h = 2\pi\hbar$. And this effect is known as Integer Quantum Hall Effect (IQHE), and the corresponding (insulating) system holding the phenomenon is called Chern insulator. By insulating we in fact mean that longitudinal conductance vanishes as $\rho_{xx} = 0$, and using the conductance matrix

$$\begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}^{-1},$$

one immediately finds that

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2} = 0 = \rho_{yy},\ \rho_{xy} = -\rho_{yx} = -\frac{\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2} = -\frac{e^2}{\nu h}.$$

(*Streda Formula*) With differential form, we conclude easily that

$$\sigma_{xy} = q\frac{\delta n}{\delta B}, \tag{5.96}$$

which, however, needs further examination. Say consider definition $j_x = \sigma_{xy}E_y$, where we have set zone to insulating phase so that longitudinal conductance equals zero, then to relate this formula with charge density and magnetic field, one notices that by continuity equation and gauge potential form of electro-magnetic field,

$$\begin{cases} \partial_t \rho + \partial_x j_x = 0 \\ E_y = -\partial_t A_y \\ B_z = (\nabla \times \boldsymbol{A})_z = \partial_x A_y \end{cases},$$

one obtains

$$\partial_t \rho = -\partial_x j_x = \sigma_{xy}\partial_x \partial_t A_y = \sigma_{xy}\partial_t B_z,$$

i.e., Streda formula we write above.

Quantization from Landau Levels

This phenomenon could firstly be explained by Landau levels of the electron, where one finds the key is to figure out the electron density n for each Landau level (LL) occupied, if one believes that each fully filled LL could contribute to the conductance in some way — extended states and edge states are all ok. One could try to view from symmetric gauge where one finds the wavefunction

$$\psi_{m,0}(x,y) \propto \exp\left(-\frac{z\bar{z}}{4l_B^2}\right) = \exp\left(-\frac{x^2+y^2}{4l_B^2}\right),\ z = x + iy,$$

while the second quantum number is lifted by $b^\dagger = -i\sqrt{2}(l_B\partial_z - \bar{z}/4l_B)$ as

$$\psi_{m,n}(x,y) = (b^\dagger)^n \psi_{m,0}(x,y) \propto \left(\frac{z}{l_B}\right)^n \exp\left(-\frac{z\bar{z}}{4l_B^2}\right);$$

then by angular momentum operator

$$J = J_z = (r \times p)_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = \hbar(z\partial_z - \bar{z}\partial_{\bar{z}}),$$

one finds that the wavefunction under symmetric gauge is eigenstate of angular momentum

$$J\psi_{m,n}(x,y) = m\hbar\psi_{m,n}(x,y).$$

Then one may say that each state may occupy an area with radius $r_n = \sqrt{2m}l_B$, where the seemingly extra 2 factor comes from choice of physical momentum, say now roughly speaking,

$$rp = r(mv - qA) = m\hbar,$$

and using $qvB = mv^2/r$ to have $mv = qrB$, substituting which and integrating over the circle (unit one) one has

$$2q\pi r^2 B - q(2\pi rA) = 2\pi m\hbar \Longrightarrow 2q\Phi - q\Phi = q\Phi = mh,$$

which is just semi-classical angular momentum quantization. Now,

$$r_m = \sqrt{\frac{\Phi}{\pi B}} = \sqrt{\frac{mh}{q\pi B}} = \sqrt{2m}l_B.$$

On the contrary, if one just use mv as momentum for quantization, one would find the factor 2 lost. Now given an area A with radius R, density of state, i.e., number of state it could hold per unit area (when $r_m = R$) averagely is

$$\Omega = \frac{m}{A}\bigg|_{r_n=R} = \frac{1}{A}\frac{R^2}{2l_B^2} = \frac{1}{2\pi l_B^2} = \frac{Bq}{h}, \tag{5.97}$$

and if one looks for total degeneracy, it becomes $A\Omega = \Phi/\Phi_0$, where $\Phi_0 = h/e$ denotes flux quantum. And one yields

$$\sigma_{xy} = \frac{\Omega q}{B} = \frac{q^2}{h},$$

as one quantum unit conductance. Notice that we did not consider inner DoFs for more degeneracy and this is only for one Landau level, considering all would give

$$\sigma_{xy} = gN\frac{q^2}{h},\ N = 1, 2, \cdots, \tag{5.98}$$

where g describes inner DoFs like spin and N counts for occupied LLs.

On the other hand, one could obtain exactly the same degeneracy for each LL using Landau gauge. Clearly, the degeneracy now comes from varying of center of one state, then

$$\Omega = \frac{1}{S}\frac{L_y}{\Delta y} = \frac{1}{L_xL_y}\frac{L_y}{l_B^2\Delta k_x} = \frac{1}{L_xL_y}\frac{L_yqB}{\hbar 2\pi/L_x} = \frac{qB}{h},$$

where a rectangular area is exploited and periodic boundary condition (PBC) is used which forces that

$$k_x \cdot L_x = 2\pi \Longrightarrow k_x = n\frac{2\pi}{L_x},$$

respecting the periodicity of plane wave.

Quantization from Berry Terminology

A quick conclusion is, using wave-packet dynamics,

$$\dot{\boldsymbol{r}}_c = \boldsymbol{v}_n(\boldsymbol{k}_c) - \dot{\boldsymbol{k}}_c \times \boldsymbol{\Omega}_n(\boldsymbol{k}_c), \tag{5.99}$$

where c for centre and n for band index, and Berry curvature, mathematically equivalent to Gaussian curvature, is generally defined as a 2nd-order tensor

$$\Omega_n^{\mu\nu}(k) \equiv \partial^\mu A_n^\nu(k) - \partial^\nu A_n^\mu(k), \tag{5.100}$$

from which one immediately yields A^μ and $F^{\mu\nu}$ are just $U(1)$ vector field and electromagnetic tensor, but in k-space. Then, by considering pure-magnetic part, we could transform 3×3 anti-symmetric tensor into vector magnetic field, and transform

$$\boldsymbol{k} \cdot \Omega \to \boldsymbol{k} \times \boldsymbol{\Omega}$$

correspondingly.

Then, current is defined through

$$\boldsymbol{J} = -e \sum \dot{\boldsymbol{r}}_c = -e \sum_{n\boldsymbol{k}} \left(\boldsymbol{v}_n(\boldsymbol{k}_c) - \dot{\boldsymbol{k}}_c \times \boldsymbol{\Omega}_n(\boldsymbol{k}_c) \right), \tag{5.101}$$

and for Hall conductance given to an 2-D insulator, we have

$$\begin{aligned} J_y &= -e \sum_{n\boldsymbol{k}} (\dot{\boldsymbol{k}}_c)_x \boldsymbol{\Omega}_n(\boldsymbol{k}_c)_z = -e \sum_{n\boldsymbol{k}} \dot{k}_x \Omega_n^{xy}(\boldsymbol{k}),\ \hbar \dot{k}_x = -eE_x \\ &= \frac{e^2}{\hbar} \sum_n \frac{S}{(2\pi)^2} \int_{\mathrm{BZ}} \mathrm{d}^2 k \Omega_n^{xy}(k) E_x, \end{aligned}$$

where one needs to notice that summation over k is in Brillouin zone and S is area of 2-D system. Then one sees

$$\sigma_{yx} = \frac{j_y}{E_x} = \frac{J_y}{S E_x} = \frac{e^2}{h} c_1 = -\sigma_{xy}, \tag{5.102}$$

where

$$c_1 = \frac{1}{2\pi} \sum_n \int_{\mathrm{BZ}} \mathrm{d}^2 k \Omega_n^{xy}(k) = \frac{1}{2\pi} \sum_n \int_{\mathrm{BZ}} \mathrm{d}^2 k \Omega_n^z(k) \tag{5.103}$$

is known as first Chern number. Be aware that there seems to exist a minus sign difference with Chern number mathematically.

Kubo Formula

Given a perturbative Hamiltonian

$$H = H_0 + V,$$

we could express quantum dynamics in interaction picture as

$$V_I(t) = U_0^\dagger(t) V_S U_0(t),\ \ |\psi_I(t)\rangle = U_0^\dagger(t) \left|\psi_S(t)\right\rangle,\ U_0(t) = e^{-iH_0 t/\hbar},$$

where S means Schrödinger picture. Essentially the LHS of second equation is equivalent to

$$|\psi_I(t)\rangle = U_I(t) \left|\psi_I(0)\right\rangle,$$

where

$$U_I(t) = \mathscr{T} \exp\left(-\frac{i}{\hbar}\int_0^t \mathrm{d}t' V_I(t')\right) \approx \mathrm{I} - \frac{i}{\hbar}\int_0^t \mathrm{d}t_1 V_I(t_1)$$

up to first linear order. Now given arbitrary observable, its expectation value could be approximated up to linear order by

$$\begin{aligned}\langle O(t)\rangle &= \langle\psi_I(t)|O_I(t)|\psi_I(t)\rangle = \langle\psi_I(0)|\, U_I^{-1}(t) O_I(t) U_I(t)\,|\psi_I(0)\rangle \\ &\approx \langle\psi_I(0)|\left(\mathrm{I} + \frac{i}{\hbar}\int_0^t \mathrm{d}t_1 V_I(t_1)\right) O_I(t) \left(\mathrm{I} - \frac{i}{\hbar}\int_0^t \mathrm{d}t_1 V_I(t_1)\right)|\psi_I(0)\rangle \\ &\approx \langle\psi_I(0)|\, O_I(t) + \frac{i}{\hbar}\int_0^t \mathrm{d}t' [V_I(t'), O_I(t)]\,|\psi_I(0)\rangle\,.\end{aligned}$$

Then, if we push initial time 0 to t_0 where we recognize $|\psi_I(0)\rangle$ as eigenstate of H_0 in equilibrium, and replace single-state average with statistical equilibrium average, we get Kubo formula as linear response

$$\langle O(t)\rangle = \langle O_I(t)\rangle_0 + \frac{i}{\hbar}\int_{t_0}^t \mathrm{d}t' \,\langle [V_I(t'), O_I(t)]\rangle_0\,. \tag{5.104}$$

Now we come back to our conductance. From single-particle Lagrangian term $q\dot{x}\cdot A$ which generates basically Lorentz force with vector potential, we see that when written in Hamiltonian, its generalized form as current-potential interaction would become

$$H = H_0 - J\cdot A,\ H_0\,|n\rangle = E_n\,|n\rangle\,. \tag{5.105}$$

And we want its linear response from Kubo formula. Consider an electrical field added, by letting $A \propto e^{-i\omega t}$ as an alternating current (AC) potential, we get a AC field by

$$E = -\partial_t A = i\omega A \Rightarrow A = \frac{E}{i\omega}e^{-i\omega t}.$$

Later on, we will let ω go to zero so the normal direct current (DC) in experiment is restored. This may sound tedious but could save us from infrared divergence problem. Now by pushing $t_0 \to -\infty$ so that A is viewed as added adiabatically, and also use single state average as an example which we would restore afterwards, we have, for the response part (i.e. V included part),

$$\begin{aligned}\langle J_\alpha(t)\rangle &= \frac{i}{\hbar}\int_{-\infty}^t \mathrm{d}t' \,\langle [-J_I(t')\cdot A(t'), J_{I,\alpha}(t)]\rangle_0 \\ &= -\frac{i}{\hbar}\int_{-\infty}^t \mathrm{d}t' \,\langle 0|\,[J_{I,\beta}(t'), J_{I,\alpha}(t)]\,|0\rangle \,\frac{E_\beta}{i\omega}e^{-i\omega t'},\ t'' = t - t' \\ &= -\frac{1}{\hbar\omega}\left(\int_0^{+\infty} \mathrm{d}t'' e^{i\omega t''}\,\langle 0|\,[J_{I,\beta}(0), J_{I,\alpha}(t'')]\,|0\rangle\right) E_\beta e^{-i\omega t}.\end{aligned}$$

Now, the averaged part is basically some correlation which attach time translational invariance to so we could minus $t - t''$ for each part; then, t-dependent part is just embedded in the last AC oscillating part sharing the same frequency with external A,

which reflects linear response. Since J is current, we have current density defined as it divided by general volume, then we have frequency-dependent conductance in 2-D case:

$$\begin{aligned}\sigma_{\alpha\beta}(\omega) &\equiv \frac{\langle j_\alpha(t)\rangle}{E_\beta e^{-i\omega t}} = \frac{1}{S}\frac{\langle J_\alpha(t)\rangle}{E_\beta e^{-i\omega t}} \\ &= -\frac{1}{S\hbar\omega}\int_0^{+\infty} \mathrm{d}t e^{i\omega t}\left\langle 0\right|\left[J_{I,\beta}(0), J_{I,\alpha}(t)\right]\left|0\right\rangle \\ &= -\frac{S}{\hbar\omega}\int_0^{+\infty} \mathrm{d}t e^{i\omega t}\left\langle 0\right| j_\beta e^{iH_0 t} j_\alpha e^{-iH_0 t} - e^{iH_0 t} j_\alpha e^{-iH_0 t} j_\beta \left|0\right\rangle \\ &= -\frac{S}{\hbar\omega}\sum_{n\neq 0}\int_0^{+\infty} \mathrm{d}t e^{i\omega t}\left[\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle e^{i(E_n-E_0)t/\hbar} - \langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle e^{i(E_0-E_n)t/\hbar}\right],\end{aligned}$$

where we have again replaced current with density and inserted identity, $n \neq 0$ is attributed to general ground state property as with no net current. Then by doing integral upon t where retarded nature should be introduced as $\omega \to \omega + i\epsilon$ for convergence, as

$$\begin{aligned}\int_0^{+\infty} \mathrm{d}t e^{i(\omega+i\epsilon)t} e^{\pm i(E_n-E_0)t/\hbar} &= \frac{1}{i(\omega+i\epsilon)\pm i(E_n-E_0)/\hbar} e^{-\epsilon t+\cdots}\Big|_0^{+\infty} \\ &= \frac{i\hbar}{\hbar\omega \pm (E_n-E_0)},\end{aligned}$$

we have

$$\sigma_{\alpha\beta}(\omega) = -\frac{iS}{\omega}\sum_{n\neq 0}\left[\frac{\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle}{\hbar\omega + (E_n-E_0)} - \frac{\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle}{\hbar\omega - (E_n-E_0)}\right],$$

then by letting ω goes to zero, we see

$$\begin{aligned}\lim_{\omega\to 0}\frac{1}{\omega}\frac{1}{\hbar\omega \pm (E_n-E_0)} &= \pm\frac{1}{E_n-E_0}\lim_{\omega\to 0}\frac{1}{\omega}\frac{1}{1\pm\hbar\omega/(E_n-E_0)} \\ &= \pm\frac{1}{E_n-E_0}\frac{1}{\omega}\left(1 \mp \frac{\hbar\omega}{E_n-E_0} + O(\omega^2)\right) \\ &\overset{\text{P.V.}}{=} -\frac{\hbar}{(E_n-E_0)^2},\end{aligned}$$

where higher order together with infrared divergent terms are abandoned. Indeed, by noticing that $\mathrm{Im}\,\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle = -\mathrm{Im}\,\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle$ and requiring $\sigma_{\alpha\beta}$ to be real, we see that the infrared divergence is exactly canceled out. Now we get

$$\begin{aligned}\sigma_{\alpha\beta} &= iS\hbar\sum_{n\neq 0}\left[\frac{\langle 0|j_\beta|n\rangle\langle n|j_\alpha|0\rangle - \langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle}{(E_n-E_0)^2}\right] \\ &= 2S\hbar\sum_{n\neq 0}\frac{\mathrm{Im}(\langle 0|j_\alpha|n\rangle\langle n|j_\beta|0\rangle)}{(E_n-E_0)^2}\end{aligned}$$

Generically, we have multi-states occupied as total ground state to replace single $|0\rangle$, then it is easy to write

$$\sigma_{\alpha\beta} = -iS\hbar\sum_{E_m<\mu<E_n}\frac{\langle m|j_\alpha|n\rangle\langle n|j_\beta|m\rangle - \langle m|j_\beta|n\rangle\langle n|j_\alpha|m\rangle}{(E_n-E_m)^2},$$

where we have introduced chemical potential μ. And when temperature enters, the relation $E_m < \mu < E_n$ should be replaced by general Fermi-Dirac distribution counting as

$$\sigma_{\alpha\beta} = -iS\hbar \sum_{m,n} [f_z(E_m) - f_z(E_n)] \frac{\langle m|j_\alpha|n\rangle \langle n|j_\beta|m\rangle}{(E_n - E_m)^2},$$

where

$$f_z(\varepsilon) = \frac{1}{z^{-1}e^{\beta\varepsilon} + 1},\ z = e^{\beta\mu}.$$

Notice it that this formula is still upon single-particle response, as

$$J = -ev,\ e > 0,$$

with canonical velocity

$$v = \dot{x} = \frac{1}{i\hbar}[x, H] = \frac{1}{\hbar}\nabla_k H,$$

which leads further to single particle response

$$\sigma_{\alpha\beta} = -i\frac{e^2}{S\hbar} \sum_{E_m<\mu<E_n} \frac{\langle m|\partial_{k_\alpha} H|n\rangle \langle n|\partial_{k_\beta} H|m\rangle - \langle m|\partial_{k_\beta} H|n\rangle \langle n|\partial_{k_\alpha} H|m\rangle}{(E_n - E_m)^2},$$

where we have gone back to zero-temperature to see topological feature for ground state.

To account for many-body state, one needs to sum over contribution from all particles, which is equivalent to summing over k thanks to its good-quantum-number property as a (quasi-)particle label. This leads to

$$\begin{aligned}
\sigma_{\alpha\beta} &= -i\frac{e^2}{S\hbar} \sum_{E_m<\mu<E_n} \sum_k \frac{\langle m|\partial_{k_\alpha} H|n\rangle \langle n|\partial_{k_\beta} H|m\rangle - \langle m|\partial_{k_\beta} H|n\rangle \langle n|\partial_{k_\alpha} H|m\rangle}{(E_n - E_m)^2} \\
&= -i\frac{e^2}{S\hbar} \sum_{E_m<\mu<E_n} \frac{S}{(2\pi)^2} \int \mathrm{d}^2k \frac{\langle m|\partial_{k_\alpha} H|n\rangle \langle n|\partial_{k_\beta} H|m\rangle - \langle m|\partial_{k_\beta} H|n\rangle \langle n|\partial_{k_\alpha} H|m\rangle}{(E_n - E_m)^2} \\
&= \frac{e^2}{h} \left[-\frac{i}{2\pi} \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k \frac{\langle m|\partial_{k_\alpha} H|n\rangle \langle n|\partial_{k_\beta} H|m\rangle - \langle m|\partial_{k_\beta} H|n\rangle \langle n|\partial_{k_\alpha} H|m\rangle}{(E_n - E_m)^2} \right].
\end{aligned}$$

Now we want to show that this formula is equivalent to that we give above using Berry terminology, and to achieve which we need to utilize an equation parallel to Feynman-Hellman theorem, which we have also used when deriving linear response before. By taking derivative upon eigen-equation,

$$(\partial_k H)\,|n\rangle + H\,|\partial_k n\rangle = (\partial_k E_n)\,|n\rangle + E_n\,|\partial_k n\rangle\,,$$

then by left-multiplying $\langle n|$, we get F-H theorem, but if we multiply $\langle m \neq n|$ similar to the case when we do series expansion perturbation, we have

$$\langle m|\,\partial_k H\,|n\rangle + E_m\,\langle m|\partial_k n\rangle = 0 + E_n\,\langle m|\partial_k n\rangle\,,$$

which leads to

$$\langle m|\,\partial_k H\,|n\rangle = (E_n - E_m)\,\langle m|\partial_k n\rangle\,,$$

bring which back to conductance would leads to

$$\sigma_{\alpha\beta} = \frac{e^2}{h}\left[\frac{i}{2\pi}\int \mathrm{d}^2k \sum_{E_m<\mu<E_n} \left(\langle m|\partial_{k_\alpha} n\rangle \langle n|\partial_{k_\beta} m\rangle - \langle m|\partial_{k_\beta} n\rangle \langle n|\partial_{k_\alpha} m\rangle\right)\right]. \tag{5.106}$$

In fact, RHS could already be viewed as some special summation form with non-Abelian Berry curvature, which, however, is much tedious for this simple case. By noticing that $\langle n|m\rangle = 0$ and re-arranging partial-derivative, we obtain

$$\sigma_{\alpha\beta} = \frac{e^2}{h}\sum_m^{\text{occ}}\left[-\frac{i}{2\pi}\int \mathrm{d}^2k \sum_n^{\text{un-o}} \left(\langle \partial_{k_\alpha} m|n\rangle \langle n|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|n\rangle \langle n|\partial_{k_\alpha} m\rangle\right)\right],$$

then by introducing projection operator on occupied states as

$$P_{\text{occ}} = \sum_m^{\text{occ}} |m\rangle\langle m|\,,$$

we have

$$\begin{aligned}
&\sum_n^{\text{un-o}}\left(\langle \partial_{k_\alpha} m|n\rangle \langle n|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|n\rangle \langle n|\partial_{k_\alpha} m\rangle\right)\\
&\quad= \langle \partial_{k_\alpha} m|\left(\mathbb{I} - P_{\text{occ}}\right)|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|\left(\mathbb{I} - P_{\text{occ}}\right)|\partial_{k_\alpha} m\rangle\\
&\quad= \langle \partial_{k_\alpha} m|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|\partial_{k_\alpha} m\rangle - \sum_{m'}^{\text{occ}}\left(\langle \partial_{k_\alpha} m|m'\rangle \langle m'|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|m'\rangle \langle m'|\partial_{k_\alpha} m\rangle\right)\\
&\quad= \langle \partial_{k_\alpha} m|\partial_{k_\beta} m\rangle - \langle \partial_{k_\beta} m|\partial_{k_\alpha} m\rangle = \Omega_m^{\alpha\beta}/i,
\end{aligned}$$

where the last summation disappears because $\langle \partial_{k_\alpha} m|m'\rangle$ is purely imaginary like shown for Berry phase. And finally, we come back to the so-called Thouless-Kohmoto-Nightingale-den Nijs (TKNN) formula:

$$\sigma_{\alpha\beta} = -\frac{e^2}{h}\sum_n^{\text{occ}}\frac{1}{2\pi}\int \mathrm{d}^2k \Omega_n^{\alpha\beta}. \tag{5.107}$$

And especially in 3-D (or lower dimension), we could map the 2-tensor $\Omega^{\alpha\beta}$ to a vector $\boldsymbol{\Omega}$, viewed as k-space magnetic field by

$$\boldsymbol{\Omega} = \nabla_k \times \boldsymbol{A},\ A_n^\alpha = i\langle n|\partial_{k_\alpha} n\rangle\,. \tag{5.108}$$

Two-Band System

General two-band system as discussed above could be written as

$$H = \boldsymbol{d}\cdot\boldsymbol{\sigma},\ \hat{H} = \hat{\boldsymbol{d}}\cdot\boldsymbol{\sigma},$$

where we have also introduced reduced Hamiltonian with $\hat{\boldsymbol{d}} \equiv \boldsymbol{d}/d$, where $d = |\boldsymbol{d}|$ as absolute eigenvalue of original Hamiltonian, whose eigenvalue now is ± 1 (we tend to not include the tricky 0 energy as gapless case, which might be considered using limit).

On the other hand, we may offer a formula with more geometric meaning. Notice that for our two band system,

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{\pi}\int_{\mathrm{BZ}} \mathrm{d}^2k \frac{\mathrm{Im}(\langle +|\partial_{k_x} H|-\rangle \langle -|\partial_{k_y} H|+\rangle)}{(E_+ - E_-)^2},$$

with $H|\pm\rangle = E_\pm|\pm\rangle$. Then if we continuously deform $H = d\cdot\sigma$ to a unit form $\hat{H} = \hat{d}\cdot\sigma$ as shown above, topological property should be conserved, while giving convenience that

$$\hat{H}|\pm\rangle = \pm|\pm\rangle,$$

where $|\pm\rangle$ would also be modified accordingly (but due to normalization there should actually no difference). This procedure provides that

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}} \mathrm{d}^2k\, \mathrm{Im}(\langle +|\partial_{k_x} \hat{H}|-\rangle \langle -|\partial_{k_y} \hat{H}|+\rangle), \tag{5.109}$$

which could be further modified since

$$\langle +|\partial_{k_x}\hat{H}|-\rangle = \partial_{k_x}\hat{d}_i \langle +|\sigma_i|-\rangle,$$

and

$$\langle +|\partial_{k_x}\hat{H}|-\rangle \langle -|\partial_{k_y}\hat{H}|+\rangle = \partial_{k_x}\hat{d}_i\partial_{k_y}\hat{d}_j \langle +|\sigma_i|-\rangle \langle -|\sigma_j|+\rangle.$$

Now, using complete relation (or from point view of projection operator) by $P_- = \mathbb{I} - P_+$, we have

$$\begin{aligned}\langle +|\sigma_i|-\rangle \langle -|\sigma_j|+\rangle &= \langle +|\sigma_i\sigma_j|+\rangle - \langle +|\sigma_i|+\rangle \langle +|\sigma_j|+\rangle \\ &= i\epsilon_{ijk}\langle +|\sigma_k|+\rangle + (\langle +|\delta_{ij}|+\rangle - \langle +|\sigma_i|+\rangle \langle +|\sigma_j|+\rangle),\end{aligned}$$

where the second term does not contribute by operation of Im( ) since σ_i as Pauli matrix is Hermitian thus owns real expectation value; and the first term could be further modified using eigenfunction with $\hat{E}_+ = +1$ as

$$\begin{aligned}\langle +|\sigma_k|+\rangle &= \langle +|\sigma_k(\hat{d}\cdot\sigma)|+\rangle = \hat{d}_m \langle +|\left(\delta_{km} + i\epsilon_{kmn}\sigma_n\right)|+\rangle \\ &= \hat{d}_k + i\epsilon_{kmn}\hat{d}_m \langle +|\sigma_n|+\rangle,\end{aligned}$$

and the second term again does not contribute since we have extra factor $i\epsilon_{ijk}$ ahead and Im( ) operation. Then totally we find

$$\mathrm{Im}(\langle +|\partial_{k_x}\hat{H}|-\rangle \langle -|\partial_{k_y}\hat{H}|+\rangle) = \epsilon_{ijk}\partial_{k_x}\hat{d}_i\partial_{k_y}\hat{d}_j\hat{d}_k,$$

which gives us final result

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}} \mathrm{d}^2k(\partial_{k_x}\hat{\boldsymbol{d}}\times\partial_{k_y}\hat{\boldsymbol{d}})\cdot\hat{\boldsymbol{d}} \tag{5.110}$$

written in a compact form. Equivalently, we can write

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int_{\mathrm{BZ}} \mathrm{d}^2k\frac{(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d}}{d^3}, \tag{5.111}$$

by direct expansion of $\hat{\boldsymbol{d}}$, since $(\cdots\times\boldsymbol{d})\cdot\boldsymbol{d}$ terms vanish. Notice that the integral form just serves as solid angle, whose geometric meaning is most straightforward when considering integral on sphere. This offers with some 'winding' like formula, whose 1-D partner is written as

$$w_\alpha = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}k[(\partial_k\hat{\boldsymbol{d}})\times\hat{\boldsymbol{d}}]_\alpha, \tag{5.112}$$

as usual winding number for 1-D system (there seems to be several winding numbers but only when calculated with chiral symmetry makes it invariant and robust), and may be generalized into arbitrary dimension

$$I = \frac{1}{S_n}\int \mathrm{d}^n k\epsilon_{\alpha_1\alpha_2\cdots\alpha_{n+1}}(\partial_{k_1}\hat{d}_{\alpha_1})(\partial_{k_2}\hat{d}_{\alpha_2})\cdots(\partial_{k_n}\hat{d}_{\alpha_n})\hat{d}_{\alpha_{n+1}},$$

with S_n area of S^n, super-sphere surface of dim-n (n-sphere). In fact, the n-th winding number is equivalent to n-th Chern number defined in $(2n+1)$D space-time, i.e., even spatial dimension. And what makes the definition of winding number possible instead of a more general Chern number lies in the special structure of Hamiltonian, constructed upon Clifford algebra, which effectively maps curvature integral on T^n into winding angle on S^n.

Anyway, in the continuum model case for the two-band model when

$$\boldsymbol{d} = (\lambda k_x, \lambda k_y, m(\boldsymbol{k})),$$

where we require dependence of $m(\boldsymbol{k})$ over k is quadratic, i.e., depending on k^2. Now performing calculation upon Berry curvature using winding form leads to

$$(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d} = -\lambda^2(k_x\partial_{k_x}+k_y\partial_{k_y})m + \lambda^2 m,$$

where by noticing that

$$\frac{\partial}{\partial k_x} = \frac{\partial}{\partial k_x^2}\frac{\partial k_x^2}{\partial k_x} = 2k_x\frac{\partial}{\partial k_x^2},$$

and m depends only on k^2 so that $\partial_{k_x^2}m = \partial_{k^2}m$, we have

$$(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d} = -2\lambda^2k^2\partial_{k^2}m + \lambda^2 m.$$

Now the integral form for the Hall conductance becomes (continuum)

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int \mathrm{d}^2k\frac{-2\lambda^2k^2\partial_{k^2}m+\lambda^2m}{(\lambda^2k^2+m^2)^{3/2}} = -\frac{e^2}{h}\frac{1}{4}\int_{k_F}^{+\infty}\mathrm{d}k^2\frac{-2\lambda^2k^2\partial_{k^2}m+\lambda^2m}{(\lambda^2k^2+m^2)^{3/2}},$$

where we have manually set an integral limit by possible metallic Fermi surface k_F, and it would be suitable to just $k_F = 0$ for insulator. By observation, one finds that if we define

$$\cos\theta = \frac{m}{(\lambda^2k^2+m^2)^{1/2}},$$

we shall obtain

$$\frac{\partial\cos\theta}{\partial k^2} = \frac{1}{2}\frac{2\lambda^2k^2\partial_{k^2}m-\lambda^2m}{(\lambda^2k^2+m^2)^{3/2}},$$

which leads to

$$\sigma_{xy}/(e^2/h) = \frac{1}{2}\int_{k_F}^{+\infty}\mathrm{d}k^2\frac{\partial}{\partial k^2}\cos\theta = \frac{1}{2}[\cos\theta(+\infty)-\cos\theta(k_F)]. \tag{5.113}$$

Especially for a Chern insulator with m written in some standard regulated form $m(k) = m_0 - Bk^2$, we have

$$C = -\frac{\mathrm{sgn}(B) + \mathrm{sgn}(m_0)}{2} = -\mathrm{sgn}(B)\Theta(m_0 B),$$

which is requiring a kind soliton solution in 1-D to achieve non-trivial property.

The Sum Rule

There is a famous sum rule in the context of Hall conductance, which we start from the general zero-temperature frequency-dependent conductance formula

$$\begin{aligned}\sigma_{\alpha\beta}(\omega) &= \frac{e^2}{h^2} i \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k \frac{1}{\omega}\left[\frac{\langle m|\partial_\alpha H|n\rangle\langle n|\partial_\beta|m\rangle}{\hbar\omega + E_n - E_m} - \frac{\langle m|\partial_\beta H|n\rangle\langle n|\partial_\alpha|m\rangle}{\hbar\omega - (E_n - E_m)}\right] \\ &= \frac{e^2}{h^2}(-i) \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k \frac{\omega_{nm}^2}{\omega}\left[\frac{\langle m|\partial_\alpha n\rangle\langle n|\partial_\beta m\rangle}{\hbar\omega + \omega_{nm}} - \frac{\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle}{\hbar\omega - \omega_{nm}}\right],\end{aligned}$$

where we have introduced $\omega_{nm} = E_n - E_m$, and utilize a derived formula $\langle m|\,\partial_k H\,|n\rangle = (E_n - E_m)\langle m|\partial_k n\rangle$. Now by considering the retarded nature of the virtual process, we let $\omega \to \omega + i\epsilon$ with $\epsilon > 0$ a small positive quantity counting for the finite life-time of the quasi-excitation, and then by Dirac identity $1/(x + i\epsilon) = \mathrm{P.V.}1/x - i\pi\delta(x)$, we take the dissipation part represented by the delta-function as virtual excitation as

$$\begin{aligned}\sigma_{\mathrm{dis},\alpha\beta}(\omega) &= \frac{e^2}{h^2}\pi \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k \frac{\omega_{nm}^2}{\omega}\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\,\delta(\hbar\omega - \omega_{nm}) \\ &= \frac{e^2}{2h} \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k\,\omega_{nm}\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\,\delta(\hbar\omega - \omega_{nm}),\end{aligned}$$

where we have abandoned a $\delta(\hbar\omega + \omega_{nm})$ term since $\omega > 0$ is required for the frequency. In the second line we have brought into a $\omega_{nm} = \hbar\omega$. It is also easy to check that $\sigma_{\mathrm{dis},\alpha\beta}(\omega) = \sigma^*_{\mathrm{dis},\beta\alpha}(\omega)$, which indicates that the considered part of conductance matrix is Hermitian thus dissipative.

Now we introduce the sum rule integral as

$$\begin{aligned}\int_0^{+\infty} \mathrm{d}\omega \frac{\sigma_{\mathrm{dis},\alpha\beta}(\omega)}{\omega} &= \frac{e^2}{2h} \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k\,\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle \int_0^{+\infty} \mathrm{d}\omega \frac{\omega_{nm}}{\omega}\delta(\hbar\omega - \omega_{nm}) \\ &= \frac{e^2}{2h} \sum_{E_m<\mu<E_n} \int \mathrm{d}^2k\,\langle m|\partial_\beta n\rangle\langle n|\partial_\alpha m\rangle\,,\end{aligned}$$

and immediately, one finds that a sum rule is satisfied as (refer to Eq. (5.106))

$$\mathrm{Im}\int_0^{+\infty} \mathrm{d}\omega \frac{\sigma_{\mathrm{dis},\alpha\beta}(\omega)}{\omega} = -\frac{\pi}{2}\sigma_{\alpha\beta}(\omega = 0). \tag{5.114}$$

The rule reveals that weighted dissipative transverse conductance is topologically determined by the static Hall conductance.

Equivalent Form of Kubo Formula — Kubo-Streda-Bastin Approach

Restoring static ($\omega = 0$) conductance matrix component with arbitrary temperature with counting band and k-index leads to

$$\sigma_{\alpha\beta} = -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}[f_z(E_m) - f_z(E_n)]\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - E_n)^2}, \tag{5.115}$$

with $\boldsymbol{v} = \frac{1}{i\hbar}[\boldsymbol{x}, H] = \frac{1}{\hbar}\nabla_k H$ denoting the single-particle velocity operator in momentum representation, and notice that generally we do not have correspondence $x \leftrightarrow i\partial_k$, and the sum over $\boldsymbol{k}, m, n$ should be replaced by general quantum index. Also notice that we have indeed abandoned a small converging factor, which could be restored as

$$\sigma_{\alpha\beta} = -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}[f_z(E_m) - f_z(E_n)]\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - E_n + i\delta)^2},$$

with $\delta > 0$ with a physical meaning usually referred to as band broadening by disorder. Now observation leads to the conclusion that we could manually insert Dirac-δ function integral over certain energy variable as

$$\begin{aligned}
\sigma_{\alpha\beta} &= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k},m,n}\int \mathrm{d}\varepsilon\left[f_z(\varepsilon)\delta(\varepsilon - E_m)\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(\varepsilon - E_n + i\delta)^2} - f_z(\varepsilon)\delta(\varepsilon - E_n)\frac{\langle m|v_\alpha|n\rangle\langle n|v_\beta|m\rangle}{(E_m - \varepsilon + i\delta)^2}\right] \\
&= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\sum_n \mathrm{Tr}\left(\frac{\delta(\varepsilon - H)v_\alpha\,|n\rangle\langle n|\,v_\beta}{(\varepsilon - E_n + i\delta)^2}\right) - \sum_m \mathrm{Tr}\left(\frac{\delta(\varepsilon - H)v_\beta\,|m\rangle\langle m|\,v_\alpha}{(E_m - \varepsilon + i\delta)^2}\right)\right] \\
&= -i\frac{e^2\hbar}{S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\left(\delta(\varepsilon - H)v_\alpha\frac{1}{(\varepsilon - H + i\delta)^2}v_\beta\right) - \mathrm{Tr}\left(\delta(\varepsilon - H)v_\beta\frac{1}{(H - \varepsilon + i\delta)^2}v_\alpha\right)\right] \\
&= -i\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\big(\mathrm{Im}\,G^R v_\alpha \partial_\varepsilon G^R v_\beta\big) - \mathrm{Tr}\big(\mathrm{Im}\,G^R v_\beta \partial_\varepsilon G^A v_\alpha\big)\right] \\
&= -i\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon f_z(\varepsilon)\left[\mathrm{Tr}\big(v_\alpha \partial_\varepsilon G^R v_\beta\,\mathrm{Im}\,G^R\big) - Tr(v_\alpha\,\mathrm{Im}\,G^R v_\beta \partial_\varepsilon G^A)\right],
\end{aligned}$$

which is the plausible Kubo-Streda-Bastin formula. Notice that sequence inside trace is important, since all quantities inside are operators, and in order to formally simplify the expression, we have introduced the retarded/advanced Green function as

$$G^{R/A} = \frac{1}{\varepsilon - H \pm i\delta},\ \delta \to 0^+,$$

and

$$\mathrm{Im}\,G^R = -\frac{\delta}{(\varepsilon - H)^2 + \delta^2} = -\pi\delta(\varepsilon - H)$$

is identified with Dirac identity

$$\frac{1}{x + i0^+} = \mathrm{P.V}\frac{1}{x} - i\pi\delta(x).$$

To show how this formula works, we may try to attempt at one famous phenomenon as minimal conductivity of one single Dirac cone, whose Hamiltonian and eigen-quantities are

$$H = \lambda(k_x\sigma_x + k_y\sigma_y),\ E_\pm = \pm\lambda k,\ |\pm\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ \pm e^{i\phi}\end{pmatrix},\ \tan\phi = \frac{k_y}{k_x},$$

and our purpose is to obtain the DC longitudinal conductivity of this band at $\mu = 0$ ($z = 1$). With formula above, the calculation of this problem is quite direct as

$$\begin{aligned}
\sigma_{xx} &= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_k \int n_F(\epsilon-\mu)\mathrm{d}\epsilon \mathrm{Tr}\left[v_x\partial_\epsilon G^R v_x \,\mathrm{Im}\, G^R - v_x \,\mathrm{Im}\, G^R v_x \partial_\epsilon G^A\right] \\
&= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int n_F(\epsilon-\mu)\mathrm{d}\epsilon \left[\langle v_x\rangle_{mn}\partial_\epsilon G^R_n \langle v_x\rangle_{nm}\,\mathrm{Im}\, G^R_m - \langle v_x\rangle_{mn}\,\mathrm{Im}\, G^R_n\langle v_x\rangle_{nm}\partial_\epsilon G^A_m\right] \\
&\overset{\text{I.B.P}}{=} \frac{1}{2}\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int -\partial_\epsilon n_F(\epsilon-\mu)\mathrm{d}\epsilon \left[\langle v_x\rangle_{mn}\,\mathrm{Im}\, G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\, G^R_m - \langle v_x\rangle_{mn}\,\mathrm{Im}\, G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\, G^A_m\right] \\
&\overset{T=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn} \int \delta(\epsilon-\mu) d\epsilon \langle v_x\rangle_{mn}\,\mathrm{Im}\, G^R_n\langle v_x\rangle_{nm}\,\mathrm{Im}\, G^R_m \\
&\overset{\mu=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn} \frac{\delta^2|\langle v_x\rangle_{mn}|^2}{(\epsilon_n^2+\delta^2)(\epsilon_m^2+\delta^2)} \\
&= 2\frac{e^2\hbar}{\pi S}\frac{S}{(2\pi)^2}\int d^2k\left[\frac{\delta^2|\langle v_x\rangle_{++}|^2}{(\lambda^2k^2+\delta^2)^2} + \frac{\delta^2|\langle v_x\rangle_{+-}|^2}{(\lambda^2k^2+\delta^2)^2}\right] \\
&= \frac{e^2}{2\pi^3\hbar}\int d^2k \frac{\lambda^2\delta^2}{(\lambda^2k^2+\delta^2)^2} \\
&\overset{t=(\lambda k/\delta)^2}{=} \frac{e^2}{h}\frac{1}{\pi}\int_0^{+\infty} dt \frac{1}{(t+1)^2} = \frac{e^2}{h}\frac{1}{\pi}.
\end{aligned}$$

Note that we have used $n_F(\varepsilon-\mu)$ to re-write Fermi-Dirac distribution with explicit chemical potential dependence. In the first step, Im in the formula is equivalent to multiplying $-i$ in the front. In the third line for integration by part, the factor $1/2$ comes from the fact that we have two Green functions in each part, as

$$n_F(\partial G)G = \partial(n_F GG) - (\partial n_F)GG - n_F G(\partial G),$$

where on RHS, the first term is integrated out to be zero, and the minus third term is exactly the same as the original, with only indices symmetrically permuted, and that accounts for a one-half factor. After taking zero temperature and neutral limit, the sum m, n runs over all two states and gives double counting in the sixth line. Direct calculation leads to $|\langle v_x\rangle_{++}| = (\lambda\cos\phi/\hbar)^2$, $|\langle v_x\rangle_{+-}| = (\lambda\sin\phi/\hbar)^2$, bringing which inside integral formula gives the final result presented, as $1/\pi$ fraction of quantum unit conductance. Take caution that δ as disorder broadening is not thrown in the whole calculation while the final answer is independent of it, which gives a hint on the existence of 'intrinsic disorder'; also be careful on that our calculation is based on limit sequence $\lim_{\delta\to 0}\lim_{\omega\to 0}$, i.e., firstly a DC limit with clean limit afterwards, while inverse limit sequence may cause a different result, and experiment compare may have to depend on whether system is clean enough.

Chapter 6

Scattering Theory

6.1 Lippmann-Schwinger Equation

Assuming known Hamiltonian H_0 with un-perturbed eigenstate,

$$H_0 \left|\phi\right\rangle = E \left|\phi\right\rangle ,$$

when this system is perturbed by a elastic-scattering potential V with scattered eigenstate $|\Psi\rangle$, one has

$$(H_0 + V) \left|\psi\right\rangle = E \left|\psi\right\rangle ,$$

where the elastic nature is reflected by the energy conservation. Putting two equations together and one sees that

$$V \left|\psi\right\rangle = (E - H_0)(|\psi\rangle - |\phi\rangle),$$

which gives us the dynamical change between before-scattering and scattered states, and leads to that

$$|\psi\rangle = |\phi\rangle + (|\psi\rangle - |\phi\rangle) = |\phi\rangle + \frac{1}{E - H_0} V \left|\psi\right\rangle ,$$

and to account for causality, one writes down the Lippmann-Schwinger equation

$$|\psi^{\pm}\rangle = |\phi\rangle + G_0^{R/A} V \left|\psi^{\pm}\right\rangle , \tag{6.1}$$

where

$$G_0^{R/A}(\varepsilon) = \frac{1}{\varepsilon - H_0 \pm i\delta} \tag{6.2}$$

with $\delta \to 0^+$ is the unperturbed retarded/advanced Green function. The Lippmann-Schwinger equation is exactly equivalent to original stationary eigenvalue equation, with a formal iterate solution

$$|\psi^{\pm}\rangle = \left[1 + (G_0^{R/A} V) + (G_0^{R/A} V)^2 + \cdots\right] |\phi\rangle = \frac{1}{1 - G_0^{R/A} V} \left|\phi\right\rangle . \tag{6.3}$$

(Or one just solves it formally directly.) Such a series is known as the Born series, and if we keep only the first order, then the equation is called the Born approximation.

Inserting complete relation of free-of-scattering states leads to

$$|\psi_\alpha^{\pm}\rangle = |\phi_\alpha\rangle + \sum_\beta G_0^{R/A} \left|\phi_\beta\right\rangle\langle\phi_\beta| V \left|\psi_\alpha^{\pm}\right\rangle = |\phi_\alpha\rangle + \sum_\beta \frac{T_{\beta\alpha}^{\pm}}{E_\alpha - E_\beta \pm i\delta} \left|\phi_\beta\right\rangle , \tag{6.4}$$

with the scatter T-matrix defined as

$$T^{\pm}_{\beta\alpha} = \langle\phi_\beta|V|\psi^{\pm}_\alpha\rangle\,, \tag{6.5}$$

which turns the equation into a summation/integral form for discrete/continuous spectrum.

Noting that the T-matrix element is defined under representation of $|\phi\rangle$, then the T-matrix (operator) itself is written as

$$\begin{aligned}
T &= \sum_{\beta,\alpha} |\phi_\beta\rangle\, T_{\beta\alpha}\, \langle\phi_\alpha| = \sum_\beta |\phi_\beta\rangle\langle\phi_\beta| \sum_\alpha V\, |\psi_\alpha\rangle\langle\phi_\alpha| \\
&= V \sum_\alpha \left[|\phi_\alpha\rangle + \sum_\beta G_\beta(E_\alpha) T_{\beta\alpha}\, |\phi_\beta\rangle \right] \langle\phi_\alpha| \\
&= V + V \sum_{\alpha,\beta} G_\beta(E_\alpha)\, \langle\phi_\beta|V|\psi_\alpha\rangle\, |\phi_\beta\rangle\langle\phi_\alpha| \\
&= V + V \sum_\alpha G(E_\alpha) \sum_\beta |\phi_\beta\rangle\langle\phi_\beta|\, V\, |\psi_\alpha\rangle\langle\phi_\alpha| \\
&= V + VGT,
\end{aligned}$$

where the formal equation in the last should be understood as

$$T\,|\phi_\alpha\rangle = V\,|\phi_\alpha\rangle + VG(E_\alpha)T\,|\phi_\alpha\rangle\,, \tag{6.6}$$

and this state dependent form could be substituted by changing from energy representation to time representation.

Chapter 7

Identical Particles and Second Quantization

The indistinguishability of basic particles like photons and electrons gives rises to a special subspace of total Hilbert space, which describes our physical world.

7.1 Identical Particles

The difference between Fermion and Boson lies in Pauli's exclusion principle, which states that Fermions could not occupy the same quantum state simultaneously while Bosons are unlimited. Combining which with the identical property immediately tells us that wavefunction of Fermion/Bosons is anti-symmetric/symmetric. Given two particles as an example,

$$\psi_F(x_1, x_2) = \frac{1}{\sqrt{2}}(\psi_1(x_1)\psi_2(x_2) - \psi_2(x_1)\psi_1(x_2))$$
$$\psi_B(x_1, x_2) = \frac{1}{\sqrt{2}}(\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)),$$

and the state, using λ as general quantum number, is

$$|\lambda_1, \lambda_2\rangle_{F/B} = \frac{1}{\sqrt{2}}(|\lambda_1\rangle \otimes |\lambda_2\rangle + \epsilon\, |\lambda_2\rangle \otimes |\lambda_1\rangle),$$

where $\epsilon = -1$ is for Fermion and $\epsilon = 1$ for Boson. Now 'symmetric' in fact means sign change under pair permutation. Generalize this to arbitrary number of particles, we have

$$|\lambda_1, \lambda_2, \cdots, \lambda_N\rangle = \frac{1}{\sqrt{N! \prod_{n_\lambda} n_\lambda!}} \sum_P \epsilon^{\mathrm{sgn}P} |\lambda_{P1}\rangle \otimes |\lambda_{P2}\rangle \otimes \cdots \otimes |\lambda_{PN}\rangle\,, \tag{7.1}$$

where P denotes permutation operator, and $\mathrm{sgn}P = \pm 1$ upon even/odd permutation to $(1, 2, \cdots, N)$. N is total particle number and n_λ counts number of particle occupying λ state. Clearly, $n_\lambda = 0, 1$ for Fermion while is only required non-negative integer for Boson. In fact, using P we could already classify Boson or Fermion with its irreducible one-dimensional representation, which contains two kinds as identity for Boson and alternation for Fermion. Think with group representation is useful especially when generalizing permutation to braiding group, which provides us representation related to anyon in 2-D.

7.2 Second Quantization

Discussion above relies on N-particle Hilbert space

$$\mathscr{H}^N = \mathscr{H} \otimes \mathscr{H} \otimes \cdots \otimes \mathscr{H},$$

where $\otimes$ denotes tensor product between vector spaces. And by symmetrization, we reduce to symmetric Hilbert space as physical space. This, however, still limits our understanding not only technically but also physically. Physical system at many times contains multi-particles which makes description above troublesome; also, we want to figure out system with particle-number fluctuation sometimes like in grand canonical ensemble; more importantly, we need some physical picture, i.e., simple physical objects instead of tedious function to understand what happens.

To our understanding, physical process could almost always be described by some 'particles', real or imagined, where the latter usually refers to collective quasi-particle or wave-packet stuff. Then it would be convenient if we could build some particle-number Hilbert space so we may have some probability to virtualize physical process. Such a space is known as Fock space,

$$\mathscr{F} = \bigoplus_{N=0}^{\infty} \mathscr{F}^N, \tag{7.2}$$

where $\mathscr{F}^N$ is somehow equal to symmetrized subspace of $\mathscr{H}^N$ as $\mathscr{H}_S^N$, but may under different representation. Essentially this direct sum starts from $N = 0$ as the space with only vacuum state $|0\rangle$ as an actual physical quantum state, and should be distinguished from 0. This process opens the door to second quantization and many-body physics, together with one way to field theory. To my own understanding, the name second quantization still matches in some sense, as it provides one way to quantize classical fields by studying algebra of operators that we shall introduce.

Completeness Relation

Within $\mathscr{H}_S^N$, we have subspace completeness relation inside $\mathscr{H}$ as

$$\frac{\prod_{n_\lambda} n_\lambda!}{N!} \int_{\mathscr{H}_S^N} \mathrm{d}\lambda_1 \mathrm{d}\lambda_2 \cdots \mathrm{d}\lambda_N \, |\lambda_1, \lambda_2, \cdots, \lambda_N\rangle\langle\lambda_1, \lambda_2, \cdots, \lambda_N| = \mathbb{I}_N, \tag{7.3}$$

where $\mathbb{I}_N$ denotes N-particle symmetric identity, which is reasonable since $\mathscr{H}_S^N$ itself is a close vector space. And one could replace integral by summation in discrete case. Notice the pre-factor, which reveals that though the vector form defined above is normalized in some sense especially for discrete case, it could not serve as completeness relation, to achieve which one needs to re-define the normalization relation by defining

$$\begin{aligned} |\lambda_1, \lambda_2, \cdots, \lambda_N\rangle_N &= \sqrt{\frac{\prod_{n_\lambda} n_\lambda!}{N!}} \, |\lambda_1, \lambda_2, \cdots, \lambda_N\rangle \\ &= \frac{1}{N!} \sum_P \epsilon^{\mathrm{sgn}P} \, |\lambda_{P1}\rangle \otimes |\lambda_{P2}\rangle \otimes \cdots \otimes |\lambda_{PN}\rangle \,, \end{aligned} \tag{7.4}$$

which then gives us inner product as

$${}_N\langle \alpha_1, \cdots, \alpha_N | \lambda_1, \cdots, \lambda_N \rangle_N = \frac{1}{(N!)^2} \sum_{P,P'} \epsilon^{\mathrm{sgn}P + \mathrm{sgn}P'} \langle \alpha_{P_1'} | \lambda_{P1} \rangle \cdots \langle \alpha_{P_N'} | \lambda_{P_N} \rangle$$

$$= \frac{1}{N!}\sum_{P_\lambda} \epsilon^{\mathrm{sgn}P_\lambda} P_\lambda\{\delta(\alpha_1-\lambda_1)\cdots\delta(\alpha_N-\lambda_N)\},$$

where P_λ permutes $(\lambda_1,\cdots,\lambda_N)$ to $(\lambda_{P_1},\cdots,\lambda_{P_N})$, and we used face that there are $N!$ permutations in for each P. After this we claim that

$$\int_{\mathscr{H}_S^N} \mathrm{d}\lambda_1\mathrm{d}\lambda_2\cdots\mathrm{d}\lambda_N \left|\lambda_1,\lambda_2,\cdots,\lambda_N\right\rangle_N {}_N\left\langle\lambda_1,\lambda_2,\cdots,\lambda_N\right| = \mathbb{I}_N \tag{7.5}$$

is indeed a complete relation, since

$$\begin{aligned}
&\int_{\mathscr{H}_S^N} \mathrm{d}\lambda_1\mathrm{d}\lambda_2\cdots\mathrm{d}\lambda_N \left|\lambda_1,\lambda_2,\cdots,\lambda_N\right\rangle_N {}_N\left\langle\lambda_1,\lambda_2,\cdots,\lambda_N|\alpha_1,\cdots,\alpha_N\right\rangle_N \\
=&\int_{\mathscr{H}_S^N} \mathrm{d}\lambda_1\cdots\mathrm{d}\lambda_N \frac{1}{N!}\sum_{P_\alpha}\epsilon^{\mathrm{sgn}P_\alpha}P_\alpha\{\delta(\alpha_1-\lambda_1)\cdots\delta(\alpha_N-\lambda_N)\}\left|\lambda_1,\cdots,\lambda_N\right\rangle_N \\
=&\frac{1}{N!}\sum_{P_\alpha}\epsilon^{\mathrm{sgn}P_\alpha}P_\alpha\left|\alpha_1,\cdots,\alpha_N\right\rangle_N = \left|\alpha_1,\cdots,\alpha_N\right\rangle_N,
\end{aligned}$$

where $\left|\alpha_1,\cdots,\alpha_N\right\rangle_N$ is already symmetrized and we have

$$P_\alpha\left|\alpha_1,\cdots,\alpha_N\right\rangle_N = \epsilon^{\mathrm{sgn}P_\alpha}\left|\alpha_1,\cdots,\alpha_N\right\rangle_N.$$

Now, by assuming that we have already arranged particles in original $\left|\lambda_1,\lambda_2,\cdots,\lambda_N\right\rangle$ to certain known sequence according to their quantum number, say $\lambda_1=\lambda_2=\cdots=\lambda_{n_1}$ as n_1 particles occupying first n_1 states with the same quantum number, and so on, we could rewrite such state merely using

$$\left|n_1,n_2,\cdots\right\rangle,\ \sum_{i=1} n_i = N,$$

by which we have successfully transformed into so-called particle-number representation, although nothing has been really changed, since we still have λ underneath. Notice that for continuous representation, we are not able to write all quantum numbers in a single state since its uncountable; however, as long as we have a finite N, we could do this by only write down quantum number with non-zero occupation. Of course, such description using particle numbers is just equivalent to the one using quantum numbers, and we still have

$$\sum_{\substack{n_1,n_2,\cdots\\ \sum_i n_i=N}} \left|n_1,n_2,\cdots\right\rangle\!\left\langle n_1,n_2,\cdots\right| = \mathbb{I}_N, \tag{7.6}$$

without additional factor correction. Notice that $\mathbb{I}_N$ for $\mathscr{F}^N$ is different from that for $\mathscr{H}_S^N$ since they are under different representation, although we claimed nothing has been changed. However, unfortunately, this summation only makes sense for discrete λ, since we require n_i to run over all possible states for completeness. But still, we may pretend that this works for continuous case as an intuitive picture, and remember that true completeness relation is give by integral above. To show completeness relation, let us consider

$$\left\langle n_1',\cdots\right|\left.n_1,\cdots\right\rangle = \left\langle\lambda_1',\cdots,\lambda_N'\right|\left.\lambda_1,\cdots,\lambda_N\right\rangle$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{\prod_{n'_\lambda} n'_\lambda! \prod_{n_\lambda} n_\lambda!}} \sum_{P_\lambda} \epsilon^{\mathrm{sgn} P_\lambda} P_\lambda \{\delta_{\lambda'_1,\lambda_1} \cdots \delta_{\lambda'_N,\lambda_N}\} \\
&= \delta_{n'_1,n_1} \cdots ,
\end{aligned}
$$

which I think is able to work out. The first step holds since we have claimed they only differ by permutation and sign if possible. Notice that this inner product is more precise, and it becomes straightforward to see that

$$
\begin{aligned}
\sum_{\substack{n_1,n_2,\cdots \\ \sum_i n_i = N}} |n_1, n_2, \cdots\rangle \langle n_1, n_2, \cdots | n'_1, n'_2, \cdots\rangle &= \sum_{\substack{n_1,\cdots \\ \sum_i n_i = N}} |n_1, \cdots\rangle \delta_{n_1,n'_1} \cdots \\
&= |n'_1, n'_2, \cdots\rangle ,
\end{aligned}
$$

which confirms our result.

Let us come back, generally a state in $\mathscr{F}^N$ is then written as

$$
|\Psi\rangle_N = \sum_{\substack{n_1,n_2,\cdots \\ \sum_i n_i = N}} c_{n_1,n_2,\cdots} |n_1, n_2, \cdots\rangle ,
$$

where $c_{n_1,n_2,\cdots}$ may be recognized as discrete wavefunction. And by throwing away particle-number constraint, we get general state in $\mathscr{F}$ as

$$
|\Psi\rangle = \sum_{n_1,n_2,\cdots} c_{n_1,n_2,\cdots} |n_1, n_2, \cdots\rangle , \tag{7.7}
$$

which, again, hints completeness relation

$$
\sum_{n_1,n_2,\cdots} |n_1, n_2, \cdots\rangle\langle n_1, n_2, \cdots| = \mathbb{I} \tag{7.8}
$$

for total Fock space $\mathscr{F}$.

Creation and Annihilation

With discussion above in mind, we are able to define transformation inside $\mathscr{F}$, but between $\mathscr{F}^N$ and $\mathscr{F}^{N+1}$ as creation operator (known as ladder operator sometimes, say simple harmonic oscillator and angular momentum) $a_i^\dagger : \mathscr{F} \to \mathscr{F}$, and its action on basis vector is defined as

$$
a_i^\dagger |n_1, \cdots, n_i, \cdots\rangle = \sqrt{n_i + 1} \epsilon^{s_i} |n_1, \cdots, n_i + 1, \cdots\rangle , \tag{7.9}
$$

where

$$
s_i \equiv \sum_{j=i}^{i-1} n_j
$$

as permutation time between $a_i^\dagger$ and $a_j^\dagger$ needed. Notice that due to exclusion principle, n_i for Fermion should be understood as mod 2. Then we could construct a general vector by

$$
|n_1, n_2, \cdots\rangle = \prod_i \frac{1}{\sqrt{n_i!}} (a_i^\dagger)^{n_i} |0\rangle , \tag{7.10}
$$

where $|0\rangle$ is quantum vacuum state and the acting order should be understood as from last to first, to eliminate order problem. Under such definition, we could again see our commutation relation since

$$[a_i^\dagger a_j^\dagger - \epsilon a_j^\dagger a_i^\dagger] \left|\cdots, n_i, \cdots, n_j, \cdots\right\rangle,$$

which is easy to verify and gives us

$$[a_i^\dagger, a_j^\dagger]_\epsilon \equiv a_i^\dagger a_j^\dagger - \epsilon a_j^\dagger a_i^\dagger = 0, \tag{7.11}$$

as Fermionic/Bosonic commutation relation for $\epsilon = -1/+1$, and we may come back to

$$\{a_i^\dagger, a_j^\dagger\} = 0,\ [a_i^\dagger, a_j^\dagger] = 0, \tag{7.12}$$

for Fermions and Bosons, respectively.

Operator on Hilbert space is promised to have adjoint operator, and usually, a Hermit conjugate one as $a_i : \mathscr{F} \to \mathscr{F}$ known as annihilation operator. By

$$\begin{aligned}
\langle n_1', \cdots, n_i', \cdots| a_i \left|n_1, \cdots, n_i, \cdots\right\rangle &= \langle n_1, \cdots, n_i, \cdots| a_i^\dagger \left|n_1', \cdots, n_i', \cdots\right\rangle^* \\
&= \langle n_1, \cdots, n_i, \cdots| \sqrt{n_i'+1}\epsilon^{s_i'} \left|n_1', \cdots, n_i'+1, \cdots\right\rangle^* \\
&= (\sqrt{n_i'+1}\epsilon^{s_i'}\delta_{n_1,n_1'}\cdots\delta_{n_i,n_i'+1}\cdots)^* \\
&= \sqrt{n_i}\epsilon^{s_i}\delta_{n_1,n_1'}\cdots\delta_{n_i,n_i'+1}\cdots,
\end{aligned}$$

which offers us

$$a_i \left|n_1, \cdots, n_i, \cdots\right\rangle = \sqrt{n_i}\epsilon^{s_i} \left|n_1, \cdots, n_i - 1, \cdots\right\rangle, \tag{7.13}$$

and especially,

$$a_i \left|n_i = 1\right\rangle = \left|0\right\rangle,\ a_i \left|0\right\rangle = 0. \tag{7.14}$$

And similarly, we have

$$[a_i, a_j]_\epsilon \equiv a_i a_j - \epsilon a_j a_i = 0. \tag{7.15}$$

What is more, it is also likely to obtain

$$[a_i, a_j^\dagger]_\epsilon = a_i a_j^\dagger - \epsilon a_j^\dagger a_i = \delta_{ij}. \tag{7.16}$$

One may also work out the action of $a_i^{(\dagger)}$ on $|\lambda_1, \lambda_2, \cdots, \lambda_N\rangle_{(N)}$, which is simple for creation as

$$\begin{aligned}
a_i^\dagger \left|\lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle &= \sqrt{n_i+1} \left|\lambda_i, \lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle, \\
a_i^\dagger \left|\lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle_N &= \sqrt{\frac{\prod_{n_\lambda} n_\lambda!}{N!}}\sqrt{n_i+1} \left|\lambda_i, \lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle \\
&= \sqrt{N+1} \left|\lambda_i, \lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle.
\end{aligned} \tag{7.17}$$

However, the story becomes tedious for annihilation operator, since quantum number inside $|\lambda_1, \lambda_2, \cdots, \lambda_N\rangle$ has no well-defined order. We only simply add one more in the beginning for creation case, but to annihilate some particle, we have to search over all λs. Basically we write

$$\begin{aligned}
a_i \left|\lambda_1, \lambda_2, \cdots, \lambda_N\right\rangle = {} & \delta(\lambda_i - \lambda_1)\sqrt{n_{\lambda_1}} \left|\lambda_2, \cdots, \lambda_N\right\rangle \\
& + \epsilon\delta(\lambda_i - \lambda_2)\sqrt{n_{\lambda_2}} \left|\lambda_1, \lambda_3, \cdots, \lambda_N\right\rangle + \cdots
\end{aligned}$$

$$+\epsilon^{N-1}\delta(\lambda_i-\lambda_N)\sqrt{n_{\lambda_N}}\,|\lambda_1,\lambda_2,\cdots,\lambda_{N-1}\rangle\,,$$

and it is similar with

$$\begin{aligned}a_i\,|\lambda_1,\lambda_2,\cdots,\lambda_N\rangle_N=\frac{1}{\sqrt{N}}\{&\delta(\lambda_i-\lambda_1)\,|\lambda_2,\cdots,\lambda_N\rangle_N\\&+\epsilon\delta(\lambda_i-\lambda_2)\,|\lambda_1,\lambda_3,\cdots,\lambda_N\rangle_N+\cdots\\&+\epsilon^{N-1}\delta(\lambda_i-\lambda_N)\,|\lambda_1,\lambda_2,\cdots,\lambda_{N-1}\rangle_N\}\,.\end{aligned}$$

All those expressions make it unclear and annoying to work in λ representation. Then we shall focus below the n representation, which basically works for discrete case. However, by basis transformation, especially using Fourier transformation, we are able to restore the continuous stuff.

Representation of Operators

Introduce particle number operator as $\hat{n}_i=a_i^\dagger a_i$ with property

$$[\hat{n}_i,a_j]=-a_i\delta_{ij},\ [\hat{n}_i,a_j^\dagger]=a_i^\dagger\delta_{ij},$$

and essentially,

$$[\hat{n}_i,(a_i^\dagger)^n]=n_i(a_i^\dagger)^n,$$

then it is readily to see

$$\hat{n}_i\,|n_1,\cdots,n_i,\cdots\rangle=n_i\,|n_1,\cdots,n_i,\cdots\rangle\,;\tag{7.18}$$

and by defining total particle number operator $\hat{N}=\sum_i\hat{n}_i$, we have

$$\hat{N}\,|n_1,\cdots,n_i,\cdots\rangle=\sum_i n_i\,|n_1,\cdots,n_i,\cdots\rangle\,.\tag{7.19}$$

Now, for a general one-body operator

$$\hat{O}_1=\sum_{n=1}^N\hat{o}_n,$$

where $\hat{o}_n$ acts on nth particle only, we may choose λ to be its quantum number and

$$\hat{o}\,|\lambda_i\rangle=o_i\,|\lambda_i\rangle\,,\ \hat{o}=\sum_i o_i\,|\lambda_i\rangle\langle\lambda_i|\,,\ o_i\equiv\langle\lambda_i|\hat{o}|\lambda_i\rangle\,.$$

Then, to expand it under particle number representation, we have matrix element

$$\begin{aligned}\langle n_1',n_2',\cdots|\hat{O}_1|n_1,n_2,\cdots\rangle&=\sum_{n=1}^N\langle n_1',n_2',\cdots|\hat{o}_n|n_1,n_2,\cdots\rangle\\&=\sum_i o_in_i\,\langle n_1',n_2',\cdots|n_1,n_2,\cdots\rangle\\&=\langle n_1',n_2',\cdots|\sum_i o_i\hat{n}_i\,|n_1,n_2,\cdots\rangle\,,\end{aligned}$$

where the second comes from

$$\begin{aligned}
\sum_{n=1}^{N}\hat{o}_n\left|n_1,n_2,\cdots\right\rangle &= \sum_{n=1}^{N}\hat{o}_n\frac{1}{\sqrt{N!\prod_j n_j!}}\sum_P \epsilon^{\mathrm{sgn}P}P\left\{\underbrace{|\lambda_1\rangle\cdots|\lambda_1\rangle}_{n_1}\cdots\underbrace{|\lambda_i\rangle\cdots|\lambda_i\rangle}_{n_i}\cdots\right\}\\
&= \frac{1}{\sqrt{N!\prod_j n_j!}}\sum_P \epsilon^{\mathrm{sgn}P}P\sum_{n=1}^{N}\hat{o}_n\left\{\underbrace{|\lambda_1\rangle\cdots|\lambda_1\rangle}_{n_1}\cdots\underbrace{|\lambda_i\rangle\cdots|\lambda_i\rangle}_{n_i}\cdots\right\}\\
&= \frac{1}{\sqrt{N!\prod_j n_j!}}\sum_P \epsilon^{\mathrm{sgn}P}P\sum_i o_i n_i\left\{\underbrace{|\lambda_1\rangle\cdots|\lambda_1\rangle}_{n_1}\cdots\underbrace{|\lambda_i\rangle\cdots|\lambda_i\rangle}_{n_i}\cdots\right\}\\
&= \sum_i o_i n_i\frac{1}{\sqrt{N!\prod_j n_j!}}\sum_P \epsilon^{\mathrm{sgn}P}P\left\{\underbrace{|\lambda_1\rangle\cdots|\lambda_1\rangle}_{n_1}\cdots\underbrace{|\lambda_i\rangle\cdots|\lambda_i\rangle}_{n_i}\cdots\right\}\\
&= \sum_i o_i n_i\left|n_1,n_2,\cdots\right\rangle.
\end{aligned}$$

Thus, using single-particle representation λ for many-body basis element, we get

$$\hat{O}_1 = \sum_i o_i\hat{n}_i = \sum_i \langle\lambda_i|\hat{o}|\lambda_i\rangle\, a_i^\dagger a_i \equiv \sum_\lambda \langle\lambda|\hat{o}|\lambda\rangle\, a_\lambda^\dagger a_\lambda. \tag{7.20}$$

Clearly, the result would be just the same under $|\lambda_1,\cdots,\lambda_N\rangle_N$, where we use renormalized one for completeness. For example, kinetic energy $\hat{T}=\sum_n \hat{p}_n^2/2m$, could be quantized under wavevector representation:

$$\hat{T} = \sum_k \langle k|\frac{\hat{p}^2}{2m}|k\rangle\, a_k^\dagger a_k = \sum_k \frac{\hbar^2k^2}{2m}a_k^\dagger a_k.$$

Another example might be spin-half operator, for which we have

$$\hat{\boldsymbol{S}} = \sum_{\lambda,\sigma,\sigma'} a_{\lambda\sigma}^\dagger \boldsymbol{S}_{\sigma\sigma'}a_{\lambda\sigma'},$$

where λ for quantum number by representation and we have chosen $\boldsymbol{S}_{\sigma\sigma'}=\hbar\boldsymbol{\sigma}_{\sigma\sigma'}/2$ with $\boldsymbol{\sigma}$ Pauli matrices. To derive this, we need to calculate matrix projection. Using $|k,\sigma\rangle$ as projecting basis where σ denotes spin number ($\uparrow,\downarrow$ or $\pm$) for z while k for other DoFs, then

$$\begin{aligned}
\langle k,\sigma|\hat{S}_z|k',\sigma'\rangle &= (\langle k|\otimes\langle\sigma|)\hat{S}_z(|k'\rangle\otimes|\sigma'\rangle)\\
&= \langle k|k'\rangle\,\langle\sigma|\hat{S}_z|\sigma'\rangle = \frac{\hbar}{2}\delta_{k,k'}\sigma'\delta_{\sigma,\sigma'},
\end{aligned}$$

and we say that

$$\hat{S}_z = \frac{\hbar}{2}\sum_k (a_{k\uparrow}^\dagger a_{k\uparrow} - a_{k\downarrow}^\dagger a_{k\downarrow}) \tag{7.21}$$

Also clearly, utilizing physical explanation on ladder operator related to angular momentum, we have (which could also be carried out explicitly like above, but not necessary)

$$\hat{S}_+ = \hat{S}_x + i\hat{S}_y = \hbar\sum_k a_{k\uparrow}^\dagger a_{k\downarrow},$$

$$\hat{S}_- = \hat{S}_x - i\hat{S}_y = \hbar \sum_k a^\dagger_{k\downarrow} a_{k\uparrow},$$

and finally, we write that

$$\begin{aligned}\hat{\boldsymbol{S}} &= (\hat{S}_x, \hat{S}_y, \hat{S}_z) \\ &= \frac{\hbar}{2} \sum_k (a^\dagger_{k\uparrow} a_{k\downarrow} + a^\dagger_{k\downarrow} a_{k\uparrow}, -i(a^\dagger_{k\uparrow} a_{k\downarrow} - a^\dagger_{k\downarrow} a_{k\uparrow}), a^\dagger_{k\uparrow} a_{k\uparrow} - a^\dagger_{k\downarrow} a_{k\downarrow}) \\ &= \frac{\hbar}{2} \sum_{k,\sigma,\sigma'} a^\dagger_{k\sigma} \boldsymbol{\sigma}_{\sigma\sigma'} a_{k\sigma'}.\end{aligned} \tag{7.22}$$

Two-body operator

$$\hat{V}_2 = \frac{1}{2} \sum_{n\neq m}^{N} \hat{v}_{nm}$$

is defined by its action

$$\hat{v}_{nm}[\cdots \otimes \underbrace{|\lambda_i\rangle}_{n} \otimes \cdots \otimes \underbrace{|\lambda_j\rangle}_{m} \otimes \cdots] = V(i,j)[\cdots \otimes \underbrace{|\lambda_i\rangle}_{n} \otimes \cdots \otimes \underbrace{|\lambda_j\rangle}_{m} \otimes \cdots], \tag{7.23}$$

where formally

$$V(i,j) = \langle \lambda_i | \otimes \langle \lambda_j | \, \hat{v} \, | \lambda_i \rangle \otimes | \lambda_j \rangle .$$

Then, similarly as one could also expand and verify, we have

$$\begin{aligned}\langle n_1', n_2', \cdots | \hat{V}_2 | n_1, n_2, \cdots \rangle &= \frac{1}{2} \sum_{n\neq m}^{N} \langle n_1', n_2', \cdots | \hat{v}_{nm} | n_1, n_2, \cdots \rangle \\ &= \frac{1}{2} \left(\sum_i V(i,i) n_i (n_i - 1) + \sum_{i\neq j} V(i,j) n_i n_j \right) \langle n_1', n_2', \cdots | n_1, n_2, \cdots \rangle \\ &= \frac{1}{2} \sum_{i,j} \langle n_1', n_2', \cdots | \, a_i^\dagger a_j^\dagger V(i,j) a_j a_i \, | n_1, n_2, \cdots \rangle ,\end{aligned}$$

the reason why instead of density-density $\hat{n}_i \hat{n}_j$ interaction (known as Hartree approximation), we obtain the order-changed form lies in the essence of classical and quantum difference. Then we write

$$\hat{V}_2 = \frac{1}{2} \sum_{i,j} a_i^\dagger a_j^\dagger V(i,j) a_j a_i = \frac{1}{2} \sum_{\lambda,\lambda'} \langle \lambda | \langle \lambda' | \, \hat{v} \, | \lambda' \rangle \, | \lambda \rangle \, a^\dagger_\lambda a^\dagger_{\lambda'} a_{\lambda'} a_\lambda. \tag{7.24}$$

For example, Coulomb interaction, we have

$$\hat{V}_e = \frac{1}{2} \int \mathrm{d}^n r \mathrm{d}^n r' a^\dagger(r) a^\dagger(r') V(r,r') a(r') a(r),$$

where we have used continuous representation, but we still claim that this works, and one just need to re-define

$$V(r,r') = {}_N \langle r, r' | \, \hat{v}_e \, | r, r' \rangle_N ,$$

which is in fact, as one may check, equivalent to the one we defined above.

Change of Representation

Given another complete single-particle basis $\sum_\mu |\mu\rangle\langle\mu| = \mathbb{I}$ and $|\mu\rangle = a_\mu^\dagger |0\rangle$, by

$$|\lambda\rangle = \mathbb{I}\,|\lambda\rangle = \sum_\mu \langle\mu|\lambda\rangle\,|\mu\rangle\,,\ \ |\lambda\rangle = a_\lambda^\dagger\,|0\rangle\,,$$

we immediately see

$$a_\lambda^\dagger = \sum_\mu \langle\mu|\lambda\rangle\, a_\mu^\dagger,\ a_\lambda = \sum_\mu \langle\lambda|\mu\rangle\, a_\mu. \tag{7.25}$$

Under such transformation, operator may lose sense of 'diagonal', and generally we have

$$\begin{aligned}\hat{O}_1 &= \sum_{\mu,\nu} \langle\mu|\hat{o}|\nu\rangle\, a_\mu^\dagger a_\nu,\\ \hat{V}_2 &= \frac{1}{2}\sum_{\mu\mu'\nu\nu'} \langle\mu'|\,\langle\nu'|\,\hat{v}\,|\nu\rangle\,|\mu\rangle\, a_{\mu'}^\dagger a_{\nu'}^\dagger a_\nu a_\mu.\end{aligned} \tag{7.26}$$

One case we often meet is the dual between real space and wavevector (momentum) space, one is continuous while the other is discrete (when confined). And we have

$$\begin{aligned}a_k &= \int \mathrm{d}^n r\,\langle k|r\rangle\,\psi(r) = \frac{1}{\sqrt{V}}\int \mathrm{d}r e^{-ik\cdot r}\psi(r),\\ \psi(r) &= \sum_k \langle r|k\rangle\, a_k = \frac{1}{\sqrt{V}}\sum_k e^{ik\cdot r} a_k,\end{aligned} \tag{7.27}$$

where $\psi(r)$ is known as field operator since it is easy to check

$$\psi^\dagger(r)\,|0\rangle = \frac{1}{\sqrt{V}}\sum_k e^{-ik\cdot r} a_k^\dagger\,|0\rangle = \frac{1}{\sqrt{V}}\sum_k \langle k|r\rangle\,|k\rangle = |r\rangle\,.$$

Notice that we have used 'free' wave and particle coordinates above. However, in many cases especially in condensed matter physics (CMP), one mainly deals with lattice model where due to Bloch's theorem, Bloch states emerge as

$$|\psi_{n,k}\rangle = \exp(i\boldsymbol{k}\cdot\hat{\boldsymbol{r}})\,|u_{n,k}\rangle\,,\ \ \psi_{n,k}(\boldsymbol{r}) = \exp(i\boldsymbol{k}\cdot\boldsymbol{r})u_{n,k}(\boldsymbol{r}),$$

with discrete translational symmetry as

$$\hat{T}_i\,|u_{n,k}\rangle = |u_{n,k}\rangle\,,\ \ u_{n,k}(\boldsymbol{r}+\boldsymbol{a}_i) = u_{n,k}(\boldsymbol{r}).$$

The plane-wave factor hints k as somehow good quantum number, and motivates us to similarly define its dual state that takes r as 'good quantum number', which turns out to be Wannier state also defined by Fourier transformation

$$|\phi_{n,i}\rangle = \frac{1}{\sqrt{V}}\sum_k^{\mathrm{BZ}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_i}\,|\psi_{n,k}\rangle\,,\ \ |\psi_{n,k}\rangle = \frac{1}{\sqrt{V}}\sum_i e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i}\,|\phi_{n,i}\rangle\,. \tag{7.28}$$

My comment on $\boldsymbol{R}_i$ here up to now is since it represents certain lattice coordinates, it is just fine by either taking it to be real discrete vector as number multiplication defined on

Hilbert space, or recognizing it as site operator as if it is some transformation. And it might be more clear in Wannier function form

$$\phi_{n,i}(\boldsymbol{r}) = \frac{1}{\sqrt{V}} \sum_{k}^{\mathrm{BZ}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_i} \psi_{n,k}(\boldsymbol{r}),\ \ \phi_{n,i}(\boldsymbol{r}) \equiv \langle \boldsymbol{r}|\phi_{n,i}\rangle\,. \tag{7.29}$$

Since clearly both Fourier transformation and exponential group transformation are unitary, we say that Wannier states compose also complete basis. By defining

$$|\psi_{n,k}\rangle = a^\dagger_{n,k}\,|0\rangle\,,\ \ |\phi_{n,i}\rangle = a^\dagger_{n,i}\,|0\rangle\,, \tag{7.30}$$

we similarly have transformation between creation operators,

$$a^\dagger_{n,i} = \frac{1}{\sqrt{V}} \sum_{k}^{\mathrm{BZ}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_i} a^\dagger_{n,k},\ \ a^\dagger_{n,k} = \frac{1}{\sqrt{V}} \sum_{i} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} a^\dagger_{n,i}.$$

Now for a Bloch Hamiltonian, since

$$\hat{H}\,|\psi_{n,k}\rangle = \varepsilon_{n,k}\,|\psi_{n,k}\rangle\,,$$

we have

$$\begin{aligned} \hat{H} &= \sum_{n,k} \varepsilon_{n,k} a^\dagger_{n,k} a_{n,k} = \frac{1}{V} \sum_{n,k} \sum_{i,j} \varepsilon_{n,k} e^{i\boldsymbol{k}\cdot(\boldsymbol{R}_i - \boldsymbol{R}_j)} a^\dagger_{n,i} a_{n,j} \\ &= \sum_{n,i,j} t^n_{ij} a^\dagger_{n,i} a_{n,j}, \end{aligned} \tag{7.31}$$

where hopping integral is defined as

$$t^n_{ij} \equiv \frac{1}{V} \sum_{k} \varepsilon_{n,k} e^{i\boldsymbol{k}\cdot(\boldsymbol{R}_i - \boldsymbol{R}_j)}. \tag{7.32}$$

This gives the familiar form Hamiltonian in CMP, which also hint that Wannier state is not eigenstate. Here, one needs to be careful to distinguish the Wannier state operator and the field operator defined above. The field operator is a true basic electron operator, while the Wannier operator, which is close to an eigenstate in the non-interacting case upon a unitary diagonalization and is obtained from Fourier transformation of Bloch state operator, is an operator for a quasi-electron, in the sense of Landau-Fermi liquid and the analogy. The quasi-electron is a superposition of true electron, while the latter is a specie of fundamental particle, the former reflects the collective behavior of the latter in crystalline system, and is of more importance in condensed matter. Such insight means the difference between the two is a unitary basis transformation, and the transformation 'matrix' element is Wannier wavefunction,

$$a^\dagger(r)\,|0\rangle = |r\rangle = \sum_{ni} |\phi_{ni}\rangle\,\langle\phi_{ni}|r\rangle = \sum_{ni} \phi_{ni}(r) a^\dagger_{ni}\,|0\rangle\,, \tag{7.33}$$

where completeness relation of Wannier states has been used. Similar transformation can be down between field operators and Bloch operators with Bloch wavefunction as transformation matrix element.

Chapter 8

Relativistic Quantum Mechanics

"Quantum mechanics is fundamentally a many-body theory."

8.1 Klein-Gordon Equation

Choose natural units so that $c = \hbar = 1$, which means that

$$m = s = \mathrm{kg}^{-1}. \tag{8.1}$$

Also introduce covariant signals as

$$x^\mu = (t, \boldsymbol{x}),\ x_\mu = (t, -\boldsymbol{x}),\ \partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_t, \nabla). \tag{8.2}$$

By this we mean that metric

$$g^{\mu\nu} = g_{\mu\nu} = \mathrm{diag}(1, -\mathbf{1}), \tag{8.3}$$

and four-inner product

$$x^\mu g_{\mu\nu} x^\nu = x^\mu x_\mu = \sum_\mu x^\mu x_\mu = x_0^2 - \boldsymbol{x} \cdot \boldsymbol{x}. \tag{8.4}$$

From Einstein's energy-mass relation in special relativity,

$$E^2 = p^2 + m^2, \tag{8.5}$$

and if we relate $E \to H \to i\partial_t$, $\boldsymbol{p} \to -i\nabla$, then in real space,

$$(-E^2 + p^2 + m^2)\psi(x) = (\partial^\mu \partial_\mu + m^2)\psi(x) = 0, \tag{8.6}$$

known as *Klein-Gordon equation*.

This equation is extremely simple and harmonic but with problems. The simplest solution would be a plane wave

$$\psi(x) = Ce^{-ip^\mu x_\mu} = C\exp(i\boldsymbol{p} \cdot \boldsymbol{x} - iEt), \tag{8.7}$$

however, both $E = \pm|\boldsymbol{p}|$ would make this satisfy K-G equation, as it is an energy square form. And minus energy could not be explained in this frame and no positive energy

solution is stable — as from basic mechanics or thermodynamically one prefers to stay at low energy state.

Another even more serious problem would be the negative probability. If we still require flux to be

$$\boldsymbol{j} = -i\frac{(\hbar)}{2m}(\psi^*\nabla\psi - \psi\nabla\psi^*), \tag{8.8}$$

then in spirit of consistent covariant expression, we would have

$$\rho = j_0 = i\frac{1}{2m}(\psi^*\partial_t\psi - \psi\partial_t\psi^*) \tag{8.9}$$

so that

$$j^\mu = i\frac{1}{2m}(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*), \tag{8.10}$$

and indeed, this satisfies

$$\partial_\mu j^\mu = \partial_t\rho + \nabla\cdot\boldsymbol{j} = 0. \tag{8.11}$$

However, now ρ has changed its form and is always minus than zero, which is unacceptable.

8.2 Dirac Equation

The negative energy appeared above is seemingly possibly coming from the quadratic form of energy, which is transferred to second order derivative of time. And if one re-think for a moment the Schrödinger equation, it is first order in fact. Then, one may try to find also a relativistic quantum mechanic equation with first order derivative of time, together with first order derivative of space in spirit of equality of space-time. Or simply speaking, we extract 'positive' square root of Klein-Gordon equation, as *Dirac equation*.

By first order we may write a Hamiltonian as

$$H = \boldsymbol{\alpha}\cdot\boldsymbol{p} + m\beta, \tag{8.12}$$

where $\{\beta, \boldsymbol{\alpha}\}$ are 'numbers' we defined. To be consistent with classical energy interpretation we let

$$\begin{aligned}\boldsymbol{p}^2 + m^2 = H^2 &= (\alpha_i p_i + m\beta)(\alpha_j p_j + m\beta)\\ &= \sum_i \alpha_i^2 p_i^2 + \sum_{i\neq j}\{\alpha_i,\alpha_j\}p_i p_j + \sum_i\{\alpha_i,\beta\}mp_i + m^2\beta^2,\end{aligned}$$

from which we see that

$$\alpha_i^2 = \beta^2 = \mathbb{I},\ \{\alpha_i,\beta\} = \{\alpha_i,\alpha_j\}_{i\neq j} = 0. \tag{8.13}$$

And we have Dirac equation

$$i\partial_t\psi = (\boldsymbol{\alpha}\cdot\boldsymbol{p} + m\beta)\psi. \tag{8.14}$$

This form is not compact enough. By left multiplying β one has

$$(i\beta\partial_t + i\beta\boldsymbol{\alpha}\cdot\nabla - m)\psi = 0,$$

or with covariant form, let

$$\gamma^0 = \beta,\ \gamma^i = \beta\alpha^i, \tag{8.15}$$

one has

$$(i\gamma^\mu\partial_\mu - m)\psi = 0, \tag{8.16}$$

as the more compact form. Clearly, we have Clifford algebra for γ as

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{I}. \tag{8.17}$$

And the representation could be matrices. Different from 2×2 case we now have four in (3+1)-D, then one finds we at least need 4×4 matrices as 2^2. Using Dirac's representation

$$\gamma^0 = \begin{pmatrix} \mathbb{I} & \\ & -\mathbb{I} \end{pmatrix},\ \gamma^i = \begin{pmatrix} & \sigma_i \\ -\sigma_i & \end{pmatrix}, \tag{8.18}$$

then

$$\beta = \gamma^0 = \tau_z \otimes \sigma_0,\ \alpha^i = \gamma^0\gamma^i = \begin{pmatrix} & \sigma_i \\ \sigma_i & \end{pmatrix} = \tau_x \otimes \sigma_i, \tag{8.19}$$

where τ and σ are all Pauli matrices.

In language of field theory, one may introduce $\psi(x)$ as field and $\bar{\psi}(x) = \psi^\dagger(x)\gamma^0$, then the Lagrangian density

$$\mathscr{L}(\psi, \partial_\mu\psi) = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi, \tag{8.20}$$

and its EoM is just Dirac equation. And its conjugate form is

$$\frac{\partial\mathscr{L}}{\partial\psi} - \partial_\mu\frac{\partial\mathscr{L}}{\partial(\partial_\mu\psi)} = -(i\partial_\mu\bar{\psi}\gamma^\mu + m\bar{\psi}) = 0. \tag{8.21}$$

Now we could find similar form for probability flux as

$$j^\mu = \bar{\psi}\gamma^\mu\psi,\ \rho = \psi^\dagger\psi,\ \boldsymbol{j} = \psi^\dagger\boldsymbol{\alpha}\psi. \tag{8.22}$$

Write Dirac equation more explicitly as

$$i\partial_t\psi = \begin{pmatrix} m & \boldsymbol{\sigma}\cdot\boldsymbol{p} \\ \boldsymbol{\sigma}\cdot\boldsymbol{p} & -m \end{pmatrix}\psi. \tag{8.23}$$

Symmetry

Define

$$\boldsymbol{L} = \boldsymbol{r}\times\boldsymbol{p}, \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & \\ & \boldsymbol{\sigma} \end{pmatrix}, \tag{8.24}$$

one finds

$$[\boldsymbol{L}, H] = i\boldsymbol{\alpha}\times\boldsymbol{p},\ [\Sigma, H] = -2i\boldsymbol{\alpha}\times\boldsymbol{p}, \tag{8.25}$$

then total angular momentum $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{\Sigma}/2$ satisfies

$$[\boldsymbol{J}, H] = 0. \tag{8.26}$$

Parity. In such representation, both matrix and operator exists, then define

$$\hat{\pi} = \beta\hat{P}. \tag{8.27}$$

Time reversal.

$$T = i\sigma_y\mathscr{K}, \tag{8.28}$$

as Dirac equation describes electron.

Charge conjugation.

$$C\psi = i\gamma^2\gamma^0(\bar{\psi})^T. \tag{8.29}$$

8.2.1 Zeeman and Spin-Orbital Coupling

By low energy expansion,

$$H_{NR} = \frac{1}{2m}(p - qA)^2 + V + \frac{e(\hbar)}{2m}\sigma \cdot B + \frac{(\hbar)}{4m^2(c^2)}\sigma \cdot (\nabla V \times (p - qA)) + \cdots , \quad (8.30)$$

where the third term is known as Zeeman splitting effect and the fourth one is spin-orbit coupling (SOC) term.

Part II

Advanced Topics in Quantum Mechanics

Chapter 9

Path Integrals for Quantum Fields

"It is there, upon your perspective."

The core of path integral lies in finding a representation for the time-evolution operator. To do that, we need two states under the same representation, which constrains the initial and final points of the path. Furthermore, we insert complete relations through the path to count for experience over different world lines — the superposition principle. With this recognition or definition, we realize that we are actually calculating some matrix element of the time-evolution operator with intermediate steps. Similar ideas can be easily applied to stochastic processes and statistical mechanics, by treating possible configurations as the world sections to be summed over.

We have introduced the path integral for quantum mechanics in the previous chapter with phase space as the representation of states. Here for quantum fields, it makes indeed no difference about the steps, but we need to construct an appropriate Hilbert space to represent the states instead.

9.1 Bosonic Field Path Integral

9.1.1 Coherent State Path Integral for Harmonic Oscillator

The coherent state path integral for harmonic oscillator is a good start to understand the path integral for Bosons and Bosonic fields. By introducing the so-called Weyl-Heisenberg algebra,

$$[a, a^\dagger] = 1,\ [a, a] = 0,\ [a^\dagger, a^\dagger] = 0, \tag{9.1}$$

we can define the coherent state $|\alpha\rangle$ as

$$|z\rangle = e^{za^\dagger}\,|0\rangle = \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}}\,|n\rangle\,, \tag{9.2}$$

with Fock space $|n\rangle$ as complete orthonormal basis, as

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}\,|0\rangle\,,\ \ \langle m|n\rangle = \delta_{mn},\ \ \sum_{n=0}^{\infty} |n\rangle\langle n| = 1. \tag{9.3}$$

The dual state $\langle z|$ is defined as

$$\langle z| = \langle 0|\, e^{\bar{z}a}, \tag{9.4}$$

with $\bar{z}$ as the complex conjugate of z. We did not utilize directly z^* since we will use $^-$ to represent a different degree of freedom in the fermionic case. It is easy to check that

$$a\left|z\right\rangle = z\left|z\right\rangle, \tag{9.5a}$$

$$a^\dagger\left|z\right\rangle = \frac{\partial}{\partial z}\left|z\right\rangle, \tag{9.5b}$$

$$\left\langle z\right| a^\dagger = \bar{z}\left\langle z\right|, \tag{9.5c}$$

$$\left\langle z\right| a = \frac{\partial}{\partial \bar{z}}\left\langle z\right|, \tag{9.5d}$$

by writing down the coherent state in the Fock space basis, with one formula $[a,(a^\dagger)^n] = n(a^\dagger)^{n-1}$ needed as a derivative. In this case we see that the coherent can be defined as the right eigenket of the annihilation operator. Essentially notice that the commutation relation of the Bosonic operators is reflected by

$$\begin{aligned}[a,a^\dagger]\left|z\right\rangle &= (aa^\dagger - a^\dagger a)\left|z\right\rangle = a\partial_z\left|z\right\rangle - a^\dagger z\left|z\right\rangle \\ &= \partial_z a\left|z\right\rangle - za^\dagger\left|z\right\rangle = \partial_z z\left|z\right\rangle - z\partial_z\left|z\right\rangle = \left|z\right\rangle,\end{aligned}$$

i.e., $[a,a^\dagger]=1$ is reflected by $[\partial_z, z]=1$.

Over Completeness and Resolution of Identity

For $[A,B]\propto\mathbb{I}$, the Baker-Campbell-Hausdorff formula is

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\frac{1}{12}[A,[A,B]]+\frac{1}{12}[B,[B,A]]+\cdots} = e^{A+B+\frac{1}{2}[A,B]} = e^{[A,B]}e^B e^A. \tag{9.6}$$

Applying this to the inner product of coherent state, we find

$$\left\langle z|z'\right\rangle = \left\langle 0\right| e^{\bar{z}a}e^{z'a^\dagger}\left|0\right\rangle = e^{\bar{z}z'}\left\langle 0|e^{z'a^\dagger}e^{\bar{z}a}|0\right\rangle = e^{\bar{z}z'}, \tag{9.7}$$

i.e., the coherent state is neither orthogonal nor normalized.

With this inner product, we can find the wavefunction of an arbitrary state $|\psi\rangle$ onto coherent state. By letting $\psi_n = \langle n|\psi\rangle$ as a discrete wavefunction over Fock states, we have

$$\left\langle z|\psi\right\rangle = \sum_{n=0}^{\infty}\frac{\bar{z}^n}{\sqrt{n!}}\left\langle n|\psi\right\rangle = \sum_{n=0}^{\infty}\frac{z^n}{\sqrt{n!}}\psi_n \equiv \psi(\bar{z}), \tag{9.8}$$

i.e., the projection of a state onto coherent state is an anti-holomorphic function. Similarly, we have $\langle\psi|z\rangle = \bar{\psi}(z)$ as a holomorphic function, and clearly this function is complex conjugate to the one above.

We can further decompose normal ordered operator onto coherent states. By defining $A = \sum_{nm} A_{nm}(a^\dagger)^n a^m$, we have

$$\left\langle z|A|z'\right\rangle = \sum_{nm} A_{nm}\left\langle z|(a^\dagger)^n a^m|z'\right\rangle = \sum_{nm} A_{nm}\bar{z}^n(z')^m e^{\bar{z}z'} \equiv A(\bar{z},z')e^{\bar{z}z'}. \tag{9.9}$$

(By normal ordered we mean that creation operators are always on the left of annihilation operators.)

Now we claim that the following expression serves as a resolution of identity,

$$\mathbb{I} = \int \frac{\mathrm{d}z\mathrm{d}\bar{z}}{2\pi i} e^{-\bar{z}z}\left|z\right\rangle\!\left\langle z\right|. \tag{9.10}$$

Proof. Notice that $\frac{\mathrm{d}z\mathrm{d}\bar{z}}{2\pi i} = \frac{1}{\pi}\mathrm{d}\,\mathrm{Re}\,z\mathrm{d}\,\mathrm{Im}\,z$, we have

$$\begin{aligned}\langle n|\mathbb{I}|m\rangle &= \int \frac{\mathrm{d}z\mathrm{d}\bar{z}}{2\pi i} e^{-\bar{z}z}\langle n|z\rangle\langle z|m\rangle = \int \frac{\mathrm{d}z\mathrm{d}\bar{z}}{2\pi i} e^{-\bar{z}z}\frac{z^n\bar{z}^m}{\sqrt{n!m!}} \\ &\overset{\text{polar}}{=} \int_0^{+\infty}\rho\mathrm{d}\rho\int_0^{2\pi}\frac{\mathrm{d}\varphi}{\pi}e^{-\rho^2}\frac{\rho^{n+m}}{\sqrt{n!m!}}e^{i(n-m)\varphi} \\ &\overset{x=\rho^2}{=} 2\delta_{nm}\frac{1}{2n!}\int_0^{+\infty}\mathrm{d}x x^n e^{-x} = \delta_{nm}\frac{\Gamma(n+1)}{n!} = \delta_{nm}.\end{aligned}$$

□

Path Integral

By path integral, we mean that we are focusing on the transition amplitude $\langle f|U(T)|i\rangle$, with initial state $|i\rangle$ and final state $|f\rangle$ connected by the time evolution operator $U(T) = e^{-iHT/\hbar}$, where $T = t_f - t_i$ and $H = H(a^\dagger, a)$ is normal ordered so that $H|0\rangle = 0$.

Now we perform the standard two steps to derive the path integral, i.e., breaking the path into pieces and inserting complete relations:

$$\begin{aligned}\langle f|U(T)|i\rangle &= \langle f|e^{-iHT/\hbar}|i\rangle = \lim_{N\to\infty}\langle f|\left(1 - i\frac{\varepsilon}{\hbar}H(a^\dagger,a)\right)^N|i\rangle\,,\ \varepsilon = \frac{T}{N} \\ &= \lim_{N\to\infty}\left[\prod_{k=0}^{N}\int\frac{\mathrm{d}z_k\mathrm{d}\bar{z_k}}{2\pi i}e^{-|z_k|^2}\right]\left[\prod_{j=0}^{N-1}\langle z_{j+1}|\,1 - i\frac{\varepsilon}{\hbar}H(a^\dagger,a)\,|z_j\rangle\right]\langle f|z_N\rangle\langle z_0|i\rangle\,,\end{aligned} \tag{9.11}$$

where abbreviation $|z_j\rangle = |z(t_j)\rangle$ is utilized and we have defined $j = 0$ to be initial and $j = N$ to be final. For the intermidiate steps, we have

$$\langle z_{j+1}|\,1 - i\frac{\varepsilon}{\hbar}H(a^\dagger,a)\,|z_j\rangle = e^{\bar{z}_{j+1}z_j}\left(1 - i\frac{\varepsilon}{\hbar}H(\bar{z}_{j+1},z_j)\right) \cong e^{\bar{z}_{j+1}z_j}e^{-i\varepsilon H(\bar{z}_{j+1},z_j)/\hbar}, \tag{9.12}$$

and thus

$$\begin{aligned}&\langle f|U(T)|i\rangle \\ &= \lim_{N\to\infty}\left[\prod_{k=0}^{N}\int\frac{\mathrm{d}z_k\mathrm{d}\bar{z_k}}{2\pi i}e^{-|z_k|^2}\right]\left[\prod_{j=0}^{N-1}e^{\bar{z}_{j+1}z_j}e^{-i\varepsilon H(\bar{z}_{j+1},z_j)/\hbar}\right]\bar{\psi}_f(z_f)\psi_i(\bar{z}_i) \\ &= \lim_{N\to\infty}\left[\prod_{k=0}^{N}\int\frac{\mathrm{d}z_k\mathrm{d}\bar{z_k}}{2\pi i}\bar{\psi}_f(z_f)\psi_i(\bar{z}_i)e^{-(|z_f|^2+|z_i|^2)/2}\right]\left[\prod_{j=0}^{N-1}e^{-\frac{1}{2}\bar{z}_{l+1}z_{l+1}+\bar{z}_{l+1}z_l-\frac{1}{2}\bar{z}_l z_l}e^{-i\varepsilon H(\bar{z}_{j+1},z_j)/\hbar}\right] \\ &= \int \mathrm{D}z\mathrm{D}\bar{z}\exp\left(\frac{i}{\hbar}\int_{t_i}^{t_f}\mathrm{d}t L(\bar{z},z)\right)\bar{\psi}_f(z_f)\psi_i(\bar{z}_i)e^{-(|z_f|^2+|z_i|^2)/2},\end{aligned} \tag{9.13}$$

where

$$\mathrm{D}z\mathrm{D}\bar{z} = \lim_{N\to\infty}\prod_{k=0}^{N}\frac{\mathrm{d}z_k\mathrm{d}\bar{z_k}}{2\pi i}, \tag{9.14a}$$

$$L(\bar{z},z) = \frac{\hbar}{2i}(z\partial_t\bar{z} - \bar{z}\partial_t z) - H(\bar{z},z) \overset{\text{IBP}}{=} i\hbar\bar{z}\partial_t z - H(\bar{z},z). \tag{9.14b}$$

Partition Function

Partition function, defined as $Z = \operatorname{Tr} e^{-\beta H}$ with $\beta = 1/k_B T$, can be viewed as an imaginary time path integral, with final state equivalent to the initial state and to be summed. To be specific, we define imaginary time as τ to range from $\tau = 0$ to $\tau = \beta$, and thus the partition function is, after a similar procedure like path integral,

$$Z = \int \frac{\mathrm{d}z_i \mathrm{d}\bar{z}_i}{2\pi i} e^{-|z_k|^2} \langle z_i | e^{-\beta H} | z_i \rangle = \int \mathrm{D}z \mathrm{D}\bar{z} \exp(-S_E), \tag{9.15}$$

where the Euclidean action is defined as

$$S_E = \int_0^\beta \mathrm{d}\tau \left[\bar{z} \partial_\tau z + H(\bar{z}, z) \right]. \tag{9.16}$$

Notice that the periodic boundary condition of τ is reflected by $z(\tau + \beta) = z(\tau)$. Also, the boundary term appeared in path integral vanishes due to the periodic boundary condition and one-more integral.

To find the correspondence between the two, we can firstly perform a Wick rotation $\tau = it$, and then a scaling $t \to t' = t\hbar$ which leads to

$$\begin{aligned} Z &= \int \mathrm{D}z \mathrm{D}\bar{z} \exp\left(i \int_0^{-i\beta} \mathrm{d}t \left[i\bar{z} \partial_t z - H(\bar{z}, z) \right] \right) \\ &= \int \mathrm{D}z \mathrm{D}\bar{z} \exp\left(\frac{i}{\hbar} \int_0^T \mathrm{d}t' \left[i\hbar \bar{z} \partial_{t'} z - H(\bar{z}, z) \right] \right), \end{aligned} \tag{9.17}$$

where a time is defined as $T = -i\beta\hbar$.

Notice that in the last two sections, we did not in fact specify about the actual form of the Hamiltonian, with the only requirements that it can be represented by a set of Bosonic operators and is normal ordered.

9.1.2 Path Integral for Bosonic Fields

Now we consider the path integral for Bosonic fields. As we saw previously, the most important information we need to know is a set of complete states together the resolution of identity by them. For Bosonic fields, we can similarly define the coherent state

$$|\{\phi(\boldsymbol{x})\}\rangle = \exp\left(\int \mathrm{d}^d \boldsymbol{x} \phi(\boldsymbol{x}) \hat{\phi}^\dagger(\boldsymbol{x}) \right) |0\rangle, \tag{9.18}$$

in $D = d+1$ spacetime, where $|0\rangle$ is the empty ket and $\hat{\phi}^\dagger(\boldsymbol{x})$ is the field creation operator satisfying the equal-time commutation relation

$$[\hat{\phi}(\boldsymbol{x}), \hat{\phi}^\dagger(\boldsymbol{y})] = \delta^d(\boldsymbol{x} - \boldsymbol{y}), \ \hat{\phi}(\boldsymbol{x}) |0\rangle = 0. \tag{9.19}$$

Similar to single particle case, we have

$$\hat{\phi}(\boldsymbol{x}) |\{\phi(\boldsymbol{x})\}\rangle = \phi(\boldsymbol{x}) |\{\phi(\boldsymbol{x})\}\rangle, \tag{9.20}$$

and one needs essentially to distinguish the field operator and field value at a point. The state $|\{\phi(\boldsymbol{x})\}\rangle$ in the sense of coherent state, may represent a state for the most classical

field configuration over the space. Considering a complex field generally, the resolution of identity is

$$\mathbb{I} = \int \mathrm{D}\phi\mathrm{D}\bar{\phi} e^{-\int \mathrm{d}^d\boldsymbol{x}\bar{\phi}(\boldsymbol{x})\phi(\boldsymbol{x})} |\{\phi(\boldsymbol{x})\}\rangle\langle\{\phi(\boldsymbol{x})\}|, \tag{9.21}$$

where the measure $\mathrm{D}\phi\mathrm{D}\bar{\phi}$ is defined as

$$\mathrm{D}\phi\mathrm{D}\bar{\phi} = \lim_{N\to\infty}\prod_{i=1}^{N}\frac{\mathrm{d}\phi(\boldsymbol{x}_i)\mathrm{d}\bar{\phi}(\boldsymbol{x}_i)}{2\pi i} = \prod_{a=x,y,z}\lim_{N_a\to\infty}\prod_{i=1}^{N}\frac{\mathrm{d}\phi((x^a)_i)\mathrm{d}\bar{\phi}((x^a)_i)}{2\pi i}. \tag{9.22}$$

Path Integral

Similarly, given Hamiltonian $H(\hat{\phi}^\dagger, \hat{\phi})$, we have

$$\langle f|U(T)|i\rangle = \int \mathrm{D}\phi\mathrm{D}\bar{\phi}\exp\left(\frac{i}{\hbar}\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}t\mathscr{L}(\bar{\phi},\phi)\right)\bar{\Psi}_f(\phi(\boldsymbol{x},t_f))\Psi_i(\bar{\phi}(\boldsymbol{x},t_i))e^{-\frac{1}{2}\int \mathrm{d}^d\boldsymbol{x}(|\phi_f|^2+|\phi_i|^2)}, \tag{9.23}$$

where

$$\mathscr{L}(\bar{\phi},\phi) = i\hbar\bar{\phi}(x)\partial_t\phi(x) - H(\bar{\phi},\phi), \tag{9.24a}$$

$$\mathrm{D}\phi\mathrm{D}\bar{\phi} = \lim_{N\to\infty}\prod_{i=1}^{N}\frac{\mathrm{d}\phi(x_i)\mathrm{d}\bar{\phi}(x_i)}{2\pi i}, \tag{9.24b}$$

where the additional pieces along time direction should be recognized. We have also defined

$$\langle f|\{\phi(\boldsymbol{x},t_f)\}\rangle = \bar{\Psi}_f(\phi(\boldsymbol{x},t_f)),\ \ \langle\{\phi(\boldsymbol{x},t_i)\}|i\rangle = \Psi_i(\bar{\phi}(\boldsymbol{x},t_i)). \tag{9.25}$$

For instance, given non-relativistic Bose gas

$$\hat{H} = \int \mathrm{d}^d\boldsymbol{x}\hat{\phi}^\dagger(\boldsymbol{x})\left(-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{x})\right)\hat{\phi}(\boldsymbol{x}) + \frac{1}{2}\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}^d\boldsymbol{y}\hat{\phi}^\dagger(\boldsymbol{x})\hat{\phi}^\dagger(\boldsymbol{y})V(\boldsymbol{x}-\boldsymbol{y})\hat{\phi}(\boldsymbol{y})\hat{\phi}(\boldsymbol{x}), \tag{9.26}$$

which is explicitly normal ordered, we have its action in path integral as

$$\begin{aligned} S = \int \mathrm{d}^D x\bar{\phi}(x)\left(i\hbar\partial_t + \frac{\hbar^2}{2m}\nabla^2 + \mu - V(x)\right)\phi(x) \\ -\frac{1}{2}\int \mathrm{d}^D x\mathrm{d}^D y V(x-y)|\phi(x)|^2|\phi(y)|^2\Big], \end{aligned} \tag{9.27}$$

with $V(x-y) = V(\boldsymbol{x}-\boldsymbol{y})\delta(x_0-y_0)$ recognized.

Partition Function

Similarly, the partition function for Bosonic fields becomes

$$\begin{aligned} Z &= \mathrm{Tr}\, e^{-\beta\hat{H}} = \int \mathrm{D}\phi\mathrm{D}\bar{\phi}\exp(-S_E), \\ S_E &= \int_0^\beta \mathrm{d}\tau\int \mathrm{d}^d\boldsymbol{x}\bar{\phi}(x)\left(\hbar\partial_\tau - \frac{\hbar^2}{2m}\nabla^2 - \mu + V(x)\right)\phi(x) \\ &\quad + \frac{1}{2}\int_0^\beta \mathrm{d}\tau\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}^d\boldsymbol{y}V(x-y)|\phi(x)|^2|\phi(y)|^2\Big], \end{aligned} \tag{9.28}$$

with Bosonic complex field satisfying the periodic boundary condition $\phi(\boldsymbol{x}, \tau + \beta) = \phi(\boldsymbol{x}, \tau)$. This fact suggests us to perform a discrete Fourier transformation over the time direction,

$$\phi(\boldsymbol{x}, \tau) = \sum_n \phi_n(\boldsymbol{x}, \omega_n) e^{-i\omega_n \tau},\ \omega_n = \frac{2\pi n}{\beta},\ n \in \mathbb{Z}, \tag{9.29}$$

where ω_n are known as the Bosonic Matsubara frequencies.

9.2 Fermionic Field Path Integral

Unlike the bosonic case, where we map the Bosonic (field) operators to a set of complex numbers which are naturally embedded in the Hilbert space, in the fermionic case we have to extend the Hilbert space to include another field that is suitable to describe the Fermionic operators.

9.2.1 Fermion Coherent State

Grassmann Numbers

The field to be included turns to be the Grassmann number field. Given fermions $\{c_i\}\big|_{i=1}^N$ satisfying

$$\{c_i, c_j^\dagger\} = \delta_{ij},\ \{c_i, c_j\} = 0,\ \{c_i^\dagger, c_j^\dagger\} = 0,\ c_i^2 = 0 = (c_i^\dagger)^2, \tag{9.30}$$

we introduce two sets of corresponding Grassmann numbers $\{\xi_i\}$, $\{\bar{\xi}_i\}$ satisfying

$$\{\xi_i, \xi_j\} = \{\xi_i, \bar{\xi}_j\} = \{\bar{\xi}_i, \bar{\xi}_j\} = 0,\ \xi_i^2 = 0 = \bar{\xi}_i^2, \tag{9.31}$$

i.e., they obey purely anticommutative relations, or more precisely, they obey exterior algebra. Notice that Grassmann number is not a number in the sense of complex number, and the defined $\bar{\xi}$ is thus not the complex conjugate of ξ but instead an independent variable.

For further convenience, we introduce the analytic function of Grassmann numbers

$$\psi(\xi) = \psi_0 + \psi_1\xi + \psi_2\xi^2 + \cdots = \psi_0 + \psi_1\xi, \tag{9.32a}$$
$$\bar{\psi}(\bar{\xi}) = \bar{\psi}_0 + \bar{\psi}_1\bar{\xi} + \bar{\psi}_2\bar{\xi}^2 + \cdots = \bar{\psi}_0 + \bar{\psi}_1\bar{\xi}, \tag{9.32b}$$

where ψ_i and $\bar{\psi}_i$ are conjugate complex numbers. We can also define functions over a pair of Grassmann numbers as

$$A(\bar{\xi}, \xi) = a_0 + a_1\xi + \bar{a}_1\bar{\xi} + a_{12}\bar{\xi}\xi, \tag{9.33}$$

where $\bar{a}_1$ may not be complex conjugate of a_1.

Differentiation View by eye we have

$$\partial_\xi \psi(\xi) = \psi_1, \tag{9.34a}$$
$$\partial_{\bar{\xi}} \bar{\psi}(\bar{\xi}) = \bar{\psi}_1, \tag{9.34b}$$
$$\partial_\xi(\bar{\xi}\xi) = -\partial_\xi(\xi\bar{\xi}) = -\bar{\xi}, \tag{9.34c}$$

$$\partial_\xi A(\bar{\xi},\xi) = a_1 - a_{12}\bar{\xi}, \tag{9.34d}$$
$$\partial_{\bar{\xi}} A(\bar{\xi},\xi) = \bar{a}_1 + a_{12}\xi. \tag{9.34e}$$

We also assign anticommutation relations to the differentiation operators, i.e.,

$$\partial_{\bar{\xi}}\partial_\xi A(\bar{\xi},\xi) = -a_{12} = -\partial_\xi\partial_{\bar{\xi}} A(\bar{\xi},\xi), \tag{9.35}$$

or more concisely,

$$\{\partial_\xi, \partial_{\bar{\xi}}\} = 0,\ \partial_\xi^2 = 0 = \partial_{\bar{\xi}}^2. \tag{9.36}$$

Integration We assign exact differentiation to Grassmann numbers as

$$0 = \int \mathrm{d}\xi \partial_\xi \xi = \int \mathrm{d}\xi \cdot 1. \tag{9.37}$$

We also assign

$$\int \mathrm{d}\xi \xi = 1, \tag{9.38}$$

so that the integration of Grassmann numbers is essentially equivalent to the differentiation. Notice that the anticommutation relation manifests as

$$\{\mathrm{d}\xi, \mathrm{d}\bar{\xi}\} = 0,\ \mathrm{d}\xi^2 = 0 = \mathrm{d}\bar{\xi}^2, \tag{9.39}$$

and so on. The basic rules above suggests that

$$\int \mathrm{d}\xi \psi(\xi) = \psi_1, \tag{9.40a}$$
$$\int \mathrm{d}\xi A(\bar{\xi},\xi) = a_1 - a_{12}\bar{\xi}, \tag{9.40b}$$
$$\int \mathrm{d}\bar{\xi} A(\bar{\xi},\xi) = \bar{a}_1 + a_{12}\xi, \tag{9.40c}$$
$$\int \mathrm{d}\xi \mathrm{d}\bar{\xi} A(\bar{\xi},\xi) = -a_{12} = \int \mathrm{d}\bar{\xi}\mathrm{d}\xi A(\bar{\xi},\xi). \tag{9.40d}$$

Fermionic Coherent State

We now follow the similar procedure to define the Fermionic coherent state

$$|\xi\rangle = e^{-\xi c^\dagger}|0\rangle = (1 - \xi c^\dagger)|0\rangle, \tag{9.41a}$$
$$\langle\xi| = \langle 0| e^{\bar{\xi}c} = \langle 0|(1 + \bar{\xi}c). \tag{9.41b}$$

Notice that to make the expression meaningful we need to assign commutation relation between Grassmann numbers and Fermionic operators, i.e.,

$$\{\xi_i, c_j\} = \{\bar{\xi}_i, c_j\} = \{\xi_i, c_j^\dagger\} = \{\bar{\xi}_i, c_j^\dagger\} = 0, \tag{9.42}$$

thus the Grassmann numbers contain almost no other algebra than anticommutation relation. The inner product of Fermionic coherent states is

$$\langle\xi|\eta\rangle = \langle 0|(1 + \bar{\xi}c)(1 - \eta c^\dagger)|0\rangle = 1 + \bar{\xi}\eta = e^{\bar{\xi}\eta}, \tag{9.43}$$

which reproduces the exponential function of Grassmann numbers like usual (Bosonic) coherent states. Nevertheless, the simplicity is certain since the Hilbert space is only two dimensional if we choose Fock space as the basis. Also notice that since $\xi c^\dagger$ or $\bar{\xi}c$ forms a bilinear about anticommute variables, the exponential function is easy to be generalized for multi-fermion system as we only need to add then together on the exponential. Once again, one can verify that

$$c\left|\xi\right\rangle = \xi\left|\xi\right\rangle, \tag{9.44a}$$
$$c^\dagger\left|\xi\right\rangle = -\partial_\xi\left|\xi\right\rangle, \tag{9.44b}$$
$$\left\langle\xi\right| c = \partial_{\bar{\xi}}\bar{\xi}, \tag{9.44c}$$
$$\left\langle\xi\right| c^\dagger = \left\langle\xi\right|\bar{\xi}, \tag{9.44d}$$

by expansion. This holds also for the multi-fermion case by commutation relation between $c^{(\dagger)}$ and a bilinear. We can also define the resolution of identity in the fermionic case as

$$\mathbb{I} = \int \mathrm{d}\bar{\xi}\mathrm{d}\xi e^{-\bar{\xi}\xi}\left|\xi\right\rangle\!\left\langle\xi\right|, \tag{9.45}$$

which is easily verified by

$$\begin{aligned}\left\langle n|\mathbb{I}|m\right\rangle &= \int \mathrm{d}\bar{\xi}\mathrm{d}\xi e^{-\bar{\xi}\xi}\left\langle n|1-\xi c^\dagger|0\right\rangle\left\langle 0|1+\bar{\xi}c|m\right\rangle \\ &= \int \mathrm{d}\bar{\xi}\mathrm{d}\xi(1-\bar{\xi}\xi)(\delta_{n,0}+\xi\delta_{n,1})(\delta_{m,0}+\bar{\xi}\delta_{m,1}) \\ &= \int \mathrm{d}\bar{\xi}\mathrm{d}\xi(\delta_{n,1}\delta_{m,1}+\delta_{n,0}\delta_{m,0})\xi\bar{\xi} = \delta_{n,1}\delta_{m,1}+\delta_{n,0}\delta_{m,0} = \delta_{nm},\end{aligned}$$

where one needs to be careful about $\left\langle n|\xi c^\dagger|0\right\rangle = -\left\langle n|c^\dagger|0\right\rangle\xi$. Thus by similarly defining

$$\left\langle\xi|f\right\rangle = f(\bar{\xi}), \tag{9.46a}$$
$$\left\langle f|\xi\right\rangle = \bar{f}(\xi), \tag{9.46b}$$

with f a complex function and $\bar{f}$ is complex conjugate, we have

$$\left\langle\xi|c|f\right\rangle = \partial_{\bar{\xi}}f(\bar{\xi}), \tag{9.47a}$$
$$\left\langle\xi|c^\dagger|f\right\rangle = \bar{\xi}f(\bar{\xi}), \tag{9.47b}$$
$$\left\langle\xi|\hat{A}(c^\dagger,c)|\xi'\right\rangle = e^{\bar{\xi}\xi'}A(\bar{\xi},\xi), \tag{9.47c}$$
$$\left\langle f|g\right\rangle = \int \mathrm{d}\bar{\xi}\mathrm{d}\xi e^{-\bar{\xi}\xi}\bar{f}(\xi)g(\bar{\xi}) = \bar{f}_0 g_0 + \bar{f}_1 g_1, \tag{9.47d}$$

as is easy to carry out.

Gaussian Integral for Grassmann Numbers

Consider multi Grassmann numbers $\{\xi_i\}$ and $\{\bar{\xi}_i\}$ with $i = 1,\cdots,N$, we define the Gaussian integral as

$$Z[\bar{\eta},\eta] = \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}M\xi+\bar{\xi}\eta+\bar{\eta}\xi} = \det(M)e^{\bar{\eta}M^{-1}\eta}. \tag{9.48}$$

Here, $\mathrm{D}\xi = \prod_i \mathrm{d}\xi_i$ and is straightforward to push to a true path integral measure; M is a Hermitian matrix and terms on the exponents should be understood within usual vector and matrix product.

Proof. The proof is straightforward by expanding the exponential function and then performing the integration. We can firstly show that

$$\begin{aligned} Z[0,0] &= \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}M\xi} = \int \prod_i \mathrm{d}\bar{\xi}_i \mathrm{d}\xi_i \prod_{ij}(1-\bar{\xi}_i M_{ij}\xi_j) \\ &= (-1)^N M_{11}\cdots M_{NN}\int \prod_i \mathrm{d}\bar{\xi}_i \mathrm{d}\xi_i \ \bar{\xi}_1\xi_1\cdots\bar{\xi}_N\xi_N + (\text{permutations}) \\ &= \sum_P \mathrm{sgn}(P) M_{1,P(1)}\cdots M_{N,P(N)} = \det(M), \end{aligned} \tag{9.49}$$

by a straightforward calculation. Or on the other hand, we decompose $M = U^{-1}DU$ with D diagonal and U unitary, and perform variant transformation $\xi' = U\xi$, $\bar{\xi}' = \bar{\xi}U^{-1}$, which brings Jacobian $\mathrm{D}\xi' = |\det(U)|\mathrm{D}\xi$ through a direct expansion and so on. Then we have

$$Z[0,0] = \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}U^{-1}DU\xi} = \int \mathrm{D}\bar{\xi}'\mathrm{D}\xi' e^{-\bar{\xi}'D\xi'} = \det(D) = \det(M). \tag{9.50}$$

We then have, by also defining that $\eta = U^{-1}\eta'$, $\bar{\eta} = \bar{\eta}'U$,

$$\begin{aligned} Z[\bar{\eta},\eta] &= \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}M\xi+\bar{\xi}\eta+\bar{\eta}\xi} = \int \mathrm{D}\bar{\xi}'\mathrm{D}\xi' e^{-\bar{\xi}'D\xi'+\bar{\xi}'\eta'+\bar{\eta}'\xi'} \\ &= \prod_i \int \mathrm{d}\bar{\xi}_i'\mathrm{d}\xi_i' e^{-\bar{\xi}_i' D_{ii}\xi_i' + \bar{\xi}_i'\eta_i' + \bar{\eta}_i'\xi_i'} \\ &= \prod_i \int \mathrm{d}\bar{\xi}_i'\mathrm{d}\xi_i' \left(1 - D_{ii}\bar{\xi}_i'\xi_i' + \cancel{\bar{\xi}_i'\eta_i'} + \cancel{\bar{\eta}_i'\xi_i'} + \bar{\xi}_i'\eta_i'\bar{\eta}_i'\xi_i'\right) \\ &= \prod_i (D_{ii} + \bar{\eta}_i'\eta_i') = \prod_i D_{ii}\prod_i (1 + \bar{\eta}_i' D_{ii}^{-1}\eta_i') = \det(M)\left(1 + \bar{\eta}U^{-1}D^{-1}U\eta\right) \\ &= \det(M) e^{\bar{\eta}M^{-1}\eta}. \end{aligned} \tag{9.51}$$

□

The Wick's theorem is then obtained, by noticing that

$$\begin{aligned} \langle \xi_i\bar{\xi}_i\rangle &= \frac{1}{Z[0,0]}\int \mathrm{D}\bar{\xi}\mathrm{D}\xi \ \xi_i\bar{\xi}_i e^{-\bar{\xi}M\xi} \\ &= \frac{1}{Z[0,0]}\left(\prod_i \int \mathrm{d}\bar{\xi}_i\mathrm{d}\xi_i\right)\left(\prod_{(j,k)\neq(i,i)} e^{-\bar{\xi}_j D_{jk}\xi_k}\right)\cdot \partial_{\eta_i}\partial_{\bar{\eta}_i} e^{-\bar{\xi}_i D_{ii}\xi_i + \bar{\xi}_i\eta_i + \bar{\eta}_i\xi_i} \\ &= \partial_{\eta_i}\partial_{\bar{\eta}_i}\frac{1}{Z[0,0]}\int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}M\xi + \bar{\xi}_i\eta_i + \bar{\eta}_i\xi_i} \\ &= \partial_{\eta_i}\partial_{\bar{\eta}_i} e^{\bar{\eta}_i (M^{-1})_{ii}\eta_i} = (M^{-1})_{ii}, \end{aligned}$$

and by generalization, we have

$$\begin{aligned} \langle \xi_1\cdots\xi_n\bar{\xi}_n\cdots\bar{\xi}_1\rangle &= \partial_{\eta_1}\cdots\partial_{\eta_n}\partial_{\bar{\eta}_n}\cdots\partial_{\bar{\eta}_1} e^{\bar{\eta}M^{-1}\eta} \\ &= \sum_P \mathrm{sgn}(P)(M^{-1})_{1,P(1)}\cdots(M^{-1})_{n,P(n)}. \end{aligned} \tag{9.52}$$

Notice that we have used abbreviation $\bar{\eta}M^{-1}\eta = \sum_{i,j=1}^n \bar{\eta}_i (M^{-1})_{ij}\eta_j$.

As a comment, different from the Bosonic (usual complex) Gaussian integral where we get $\det(M)^{-1/2}$, here $\det(M)$ appears in the Fermionic case, and this is fully due to the exterior algebra bear by Grassmann variables which again lies in the fundamental anticommutative property of Fermions.

9.2.2 Fermionic Path Integral

Path Integral for Multi Fermions

Since the inner product of Fermionic coherent states just contains a similar form with the Boson case, we are legitimate to directly write down the path integral for Fermionic fields as

$$\begin{aligned}\langle f|U(T)|i\rangle &= \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}\xi}\exp\left(\frac{i}{\hbar}\int \mathrm{d}t\mathscr{L}(\bar{\xi},\xi)\right)\langle f|\xi(t_f)\rangle\langle\xi(t_i)|i\rangle\, e^{-\frac{1}{2}\int \mathrm{d}t(\bar{\xi}_f\xi_f+\bar{\xi}_i\xi_i)} \\ &= \int \mathrm{D}\bar{\xi}\mathrm{D}\xi e^{-\bar{\xi}\xi}\exp\left(\frac{i}{\hbar}\int \mathrm{d}t\mathscr{L}(\bar{\xi},\xi)\right)\bar{\Psi}_f(\xi(t_f))\Psi_i(\bar{\xi}(t_i))e^{-\frac{1}{2}\int \mathrm{d}t(\bar{\xi}_f\xi_f+\bar{\xi}_i\xi_i)},\end{aligned} \tag{9.53}$$

where the Lagrangian is defined as

$$\mathscr{L}(\bar{\xi},\xi) = i\hbar\bar{\xi}\partial_t\xi - H(\bar{\xi},\xi), \tag{9.54}$$

with the Hamiltonian obtained by replacing $c\to\xi$, $c^\dagger\to\bar{\xi}$ in the normal ordered Hamiltonian $H(c^\dagger,c)$. Notice that $\mathrm{D}\xi$ is an abbreviation for $\lim_{N\to\infty}\prod_{n=1}^{N}(\prod_i \mathrm{d}\xi_i(t_n))$ with i for Fock space label and n for time label.

Path Integral for Fermion Field

For Fermionic field, we can define the Fermionic coherent state as

$$|\{\psi(\boldsymbol{x},t)\}\rangle = \exp\left(-\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}t\psi(\boldsymbol{x},t)\hat{\psi}^\dagger(\boldsymbol{x},t)\right)|0\rangle\,, \tag{9.55}$$

where $\hat{\psi}^\dagger(\boldsymbol{x},t)$ is the Fermionic field creation operator satisfying the equal-time anticommutation relation

$$\{\hat{\psi}(\boldsymbol{x},t),\hat{\psi}^\dagger(\boldsymbol{y},t)\} = \delta^d(\boldsymbol{x}-\boldsymbol{y}),\ \{\hat{\psi}(\boldsymbol{x},t),\hat{\psi}(\boldsymbol{y},t)\} = 0,\ \{\hat{\psi}^\dagger(\boldsymbol{x},t),\hat{\psi}^\dagger(\boldsymbol{y},t)\} = 0. \tag{9.56}$$

Such a definition follows completely parallel with single Fermion case and the field integral form is similar to the Bosonic case. As a coherent state, we have

$$\hat{\psi}(\boldsymbol{x},t)\,|\{\psi(\boldsymbol{x},t)\}\rangle = \psi(\boldsymbol{x},t)\,|\{\psi(\boldsymbol{x},t)\}\rangle\,, \tag{9.57}$$

and the resolution of identity is

$$\mathbb{I} = \int \mathrm{D}\psi\mathrm{D}\bar{\psi}e^{-\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}t\bar{\psi}(\boldsymbol{x},t)\psi(\boldsymbol{x},t)}\,|\{\psi(\boldsymbol{x},t)\}\rangle\langle\{\psi(\boldsymbol{x},t)\}|\,, \tag{9.58}$$

where the measure $\mathrm{D}\psi\mathrm{D}\bar{\psi}$ is defined as

$$\mathrm{D}\psi\mathrm{D}\bar{\psi} = \lim_{N\to\infty}\prod_{n=1}^{N}\mathrm{d}\psi_i(\boldsymbol{x}_n)\mathrm{d}\bar{\psi}_i(\boldsymbol{x}_n). \tag{9.59}$$

The path integral for Fermionic field is then

$$\langle f|U(T)|i\rangle = \int \mathrm{D}\psi\mathrm{D}\bar{\psi}\exp\left(\frac{i}{\hbar}\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}t\mathscr{L}(\bar{\psi},\psi)\right)\bar{\Psi}_f(\psi(\boldsymbol{x},t_f))\Psi_i(\bar{\psi}(\boldsymbol{x},t_i))e^{-\frac{1}{2}\int \mathrm{d}^d\boldsymbol{x}\mathrm{d}t(|\psi_f|^2+|\psi_i|^2)}, \tag{9.60}$$

where the Lagrangian density is defined as

$$\mathscr{L}(\bar{\psi},\psi) = i\hbar\bar{\psi}(\boldsymbol{x},t)\partial_t\psi(\boldsymbol{x},t) - H(\bar{\psi},\psi), \tag{9.61}$$

with field operator replaced by Grassmann numbers in the Hamiltonian.

Partition Function

The partition function for Fermionic field is then, by a similar treatment with Bosoinc field case

$$\begin{aligned} Z &= \mathrm{Tr}\, e^{-\beta\hat{H}} = \int \mathrm{D}\psi\mathrm{D}\bar{\psi}\exp(-S_E), \\ S_E &= \int_0^\beta \mathrm{d}\tau\int \mathrm{d}^d\boldsymbol{x}\bar{\psi}(\boldsymbol{x},\tau)\left(\partial_\tau + H(\bar{\psi},\psi)-\mu\right)\psi(\boldsymbol{x},\tau). \end{aligned} \tag{9.62}$$

However, difference lies in that, instead of periodic boundary condition, the Grassmann numbers should satisfy antiperiodic boundary condition, i.e.,

$$\psi(\boldsymbol{x},\tau+\beta) = -\psi(\boldsymbol{x},\tau),\ \ \bar{\psi}(\boldsymbol{x},\tau+\beta) = -\bar{\psi}(\boldsymbol{x},\tau). \tag{9.63}$$

To see this, let us consider the partition function as a sum over imaginary time transition amplitude on a closed loop, and we treat the relevant part as the final state to be close the initial one, i.e., we let $A(\bar{\psi}_i,\psi_f) = A_{00} + A_{11}\bar{\psi}_i\psi_f$ and consider

$$\mathrm{tr}_i = \int d\bar{\psi}_i d\psi_i e^{-\bar{\psi}_i\psi_i}\lim_{f\to i} e^{-A(\bar{\psi}_i,\psi_f)}, \tag{9.64}$$

where for simplicity we utilize single particle formula to illustrate, which should be easy to be generalize. Now if ψ obeys antiperiodic boundary condition, i.e., $\psi_f = -\psi_i = \psi$, we have

$$\mathrm{tr}_i = \int d\bar{\psi}d\psi(1-\bar{\psi}\psi)(A_{00} - A_{11}\bar{\psi}\psi) = A_{00} + A_{11}, \tag{9.65}$$

which correctly reproduces a trace as the partition needs. On the other hand, if we persist periodic boundary condition, we have $\tilde{\mathrm{tr}}_i = A_{00} - A_{11}$ as a supertrace and should be abandoned. And since for partition function, ψ_i is chosen arbitrarily, we obtain the general antiperiodic boundary condition for Fermionic field. Correspondingly, the Matsubara frequencies for Fermionic field are given by

$$\omega_n = \frac{(2n+1)\pi}{\beta},\ \ n\in\mathbb{Z}. \tag{9.66}$$

9.2.3 Path Integral for Dirac Field

Consider free Dirac field, we have

$$\mathscr{L} = \bar{\psi}(i\not{\partial} - m)\psi, \tag{9.67}$$

where by Feynman slash notation $\not{\partial} = \gamma^\mu \partial_\mu$. In the operator form, $\bar{\psi} = \psi^\dagger \gamma^0$, which satisfies equal-time anticommutation relation in components as

$$\{\psi_\alpha(\boldsymbol{x}), \psi^\dagger_\beta(\boldsymbol{y})\} = \delta_{\alpha\beta}\delta^d(\boldsymbol{x} - \boldsymbol{y}). \tag{9.68}$$

Putting now ψ as Grassmann field, we have the partition function for Dirac field as

$$Z = \int \mathrm{D}\bar{\psi}\mathrm{D}\psi e^{iS} = \det(i\not{\partial} - m), \tag{9.69}$$

with $S = \int \mathrm{d}^4x \mathscr{L}$ recognized. Essentially by charge conjugation induced by $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, which satisfies $(\gamma^5)^2 = \mathbb{I}$ and $\{\gamma^5, \gamma^\mu\} = 0$, so that

$$\gamma^5(i\not{\partial} - m)\gamma^5 = -(i\not{\partial} + m), \tag{9.70}$$

which leads to

$$\begin{aligned} \det(i\not{\partial} - m) = \det(i\not{\partial} + m) &= \sqrt{\det(i\not{\partial} - m)(i\not{\partial} + m)} \\ &= \sqrt{\det(-\partial^2 - m^2) \cdot \mathbb{I}_{4\times 4}} = [\det\big(\partial^2 + m^2\big)]^2, \end{aligned} \tag{9.71}$$

i.e., square of determinant of Klein-Gordon field. Now since Klein-Gordon field is Bosonic, we have $Z_{\text{K-G}} = [\det(\partial^2 + m^2)]^{-1/2}$, and by thermodynamic relation $E = -\partial_\beta \ln Z$, we have

$$E_{\text{Dirac}} = -4E_{\text{K-G}}, \tag{9.72}$$

for the ground state/vacuum energy of free fields.

Considering the so-called Majorana fermion, which is self-conjugate so can be viewed as real fermion reduces its degrees of freedom by half compared with a usual complex one, we have

$$Z_{\text{Majorana}} = \mathrm{Pf}(i\not{\partial} - m) = \sqrt{\det(\partial^2 + m^2)}, \tag{9.73}$$

where Pf is an abbreviation for Pfaffian, which is defined for an antisymmetric (skew-symmetric) matrix A as product of its block-diagonalized off-diagonalized elements. And one sees

$$E_{\text{Majorana}} = -2E_{\text{K-G}}, \tag{9.74}$$

Similarly for a Weyl fermion, which is chiral and can be viewed as a half of Dirac fermion, we have

$$E_{\text{Weyl}} = -2E_{\text{K-G}}. \tag{9.75}$$

And if we combine and form a Majorana-Weyl fermion, we have

$$E_{\text{Majorana-Weyl}} = -E_{\text{K-G}}. \tag{9.76}$$

9.3 Spinor Path Integral

When Feynman introduced the path integral formalism, he did not know yet how to generalize it to Fermionic and spinor fields. The way to deal with fermionic path integrals was found by Berezin in 1964, using a Grassmann variable formulation, as we have presented. And the key lies in forming the coherent state for Fermionic field with a resolution of the identity. However, for spin fields, the path integral is even more complicated, and the satisfactory formulation was not posted until 1980s.

9.3.1 Spinor Coherent State

The resolution of identity for spin path integral is also formed by its coherent state, namely the spin coherent state, which is the most classical angular momentum like state. To construct the state, let us firstly be reminded that spin-S is represented by the $2S+1$ dimensional irreducible representation of $SU(2)$ group, and the basis is given by $|S,M\rangle$ with $M=-S,\cdots,S$, which satisfy that

$$S^2\,|S,M\rangle = S(S+1)\,|S,M\rangle\,, \tag{9.77a}$$
$$S_z\,|S,M\rangle = M\,|S,M\rangle\,, \tag{9.77b}$$
$$\sum_{M=-S}^{S} |S,M\rangle\langle S,M| = \mathbb{I}_{2S+1}, \tag{9.77c}$$
$$S^{\pm}\,|S,M\rangle = \sqrt{(S\mp M)(S\pm M+1)}\,|S,M\pm 1\rangle\,, \tag{9.77d}$$

where the represented $\mathfrak{su}(2)$ Lie algebra satisfies that

$$[S^i,S^j] = i\epsilon^{ijk}S^k, \tag{9.78a}$$
$$[S^2,S^i] = 0, \tag{9.78b}$$
$$[S^z,S^{\pm}] = \pm S^{\pm}, \tag{9.78c}$$

with $S^{\pm}=S^x\pm iS^y$.

To define the coherent state of spin, we let $|0\rangle = |S,S\rangle$, $\vec{n}_0=\hat{e}_z$ and assign

$$|\vec{n}\rangle = D^S(\vec{n})\,|0\rangle = e^{i\theta(\widehat{\vec{n}_0\times\vec{n}})\cdot\hat{S}}\,|0\rangle = \sum_M D^S_M(\vec{n})\,|0\rangle\,, \tag{9.79}$$

where $\vec{n}$ is an arbitrary directional vector, and the big hat overhead means that we take the unit vector of the cross product. Notice that $D^S(\vec{n})$ represents the rotation operator in the $2S+1$ dimension through exponential of the angular momentum operator, with the unit vector as the rotational axis, and protecting the unitarity of the representation. Also notice that by inserting a rotation around $\vec{n}_0$ between D^S and $|0\rangle$ will not change the physical state, and in this sense we realize that the physical states live in the coset space $SU(2)/U(1)$. Indeed, if one actually knows a little about the algebraic topology, one sees a fibration

$$S^1 \overset{\iota}{\hookrightarrow} S^3 \overset{p}{\longrightarrow} S^2, \tag{9.80}$$

as the famous Hopf fibration as $U(1)$ bundle over S^2, by recognizing isomorphisms $U(1)\cong S^1$, $SU(2)\cong S^3$ and coset space $SU(2)/U(1)$ topologically as a 2-sphere, with ι the inclusion map and p the projection map. The fibration induces a long exact sequence

$$\cdots\to\pi_3(S^1)\to\pi_3(S^3)\to\pi_3(S^2)\to\pi_2(S^1)\to\cdots\,, \tag{9.81}$$

and since $\pi_{k\geq 2}(S^1)\cong\pi_k(\mathbb{R}/\mathbb{Z})\cong\pi_{k-1\geq 1}(\mathbb{Z})=0$ given by another trivial fibration $\mathbb{Z}\overset{\iota}{\hookrightarrow}\mathbb{R}\overset{p}{\longrightarrow}S^1$, we have

$$\pi_3(S^2)\cong\pi_3(S^3)\cong\mathbb{Z} \tag{9.82}$$

as the famous Hopf map. This should be the origin of topological charge quantization. Finally, of course, as a coherent state, $|n\rangle$ is a superposition of all M states.

There exists a geometric cocycle relation for the rotation operator, i.e.,

$$D^S(\vec{n}_1)D^S(\vec{n}_2) = D^S(\vec{n}_3)e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_3)S_3}, \tag{9.83}$$

with the appeared S_3 hinting that this is a pure phase, equal to the geometric angle formed by the three vectors in some way. For an important instance, by considering a loop in the coset space from $|0\rangle$ to $|0\rangle$, we can choose the accumulated phase as the solid angle formed by the loop, but there is an ambiguity since the orientation is not given. Thus following the same spirit of Wu-Yang Monopole, one has to say that the phase difference between the choices, which is given by $e^{i4\pi S}$ with 4π solid angle of S^2, does not matter, i.e., we should admit $4\pi S = 2\pi n$ so that $S = n/2$ with $n \in \mathbb{Z}$ is recognized. This is another way to get generally the half-quantized spin index. Now we want to show the cocycle phase, and we can do this by considering an inner product

$$\begin{aligned}\langle \vec{n}_1|\vec{n}_2\rangle &= \langle 0|\, D^S(\vec{n}_1)D^S(\vec{n}_2)\,|0\rangle = \langle 0|\, D^S(\vec{n}_0)e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_0)S_3}\,|0\rangle \\ &= \left(\frac{1+\vec{n}_1\cdot\vec{n}_2}{2}\right)^S e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_0)S}.\end{aligned} \tag{9.84}$$

Proof. Consider that $\vec{n} = \cos\theta\hat{z} + \sin\theta\cos\varphi\hat{x} + \sin\theta\sin\varphi\hat{y}$, we have $\widehat{\hat{n}_0\times\hat{n}} = \cos\varphi\hat{y} - \sin\varphi\hat{x}$, which leads to

$$D^S(\vec{n}) = \exp(i\theta(\cos\varphi S_y - \sin\varphi S_x)) = \exp\left(\frac{\theta}{2}(S_+e^{-i\varphi} - S_-e^{i\varphi})\right). \tag{9.85}$$

Now we claim that $D^S(\vec{n}) = e^{-\mu S_-}e^{-\ln\left(1+|\mu|^2\right)S_z}e^{\mu^* S_+}$, where $\mu = e^{i\phi}\tan\left(\frac{\theta}{2}\right)$. This can be down by considering generally that

$$e^{\omega_+S_+ + \omega_-S_- + \omega_zS_z} = \sum_{n=0}^{+\infty}\frac{1}{n!}A^n,$$

with

$$A = \omega_+S_+ + \omega_-S_- + \omega_zS_z = \begin{pmatrix}\omega_z/2 & \omega_+ \\ \omega_- & -\omega_z/2\end{pmatrix}$$

by considering the spin-1/2 representation where $S = \sigma/2$. Since A is diagonalizable with

$$\begin{cases}\lambda = \pm K = \pm\sqrt{\omega_+\omega_- + \omega_z^2/4} \\ v_\pm \propto \begin{pmatrix}\omega_z/2 \pm K \\ \omega_-\end{pmatrix}\end{cases},$$

we may let $A = VDV^{-1}$ with $D = \mathrm{diag}(\lambda_+, \lambda_-)$ and $V = (v_+, v_-)$ (notice that the unnormalized eigenvector does not affect the decomposition), and then we have

$$A^n = VD^nV^{-1} = \begin{pmatrix}\frac{K^n+(-K)^n}{2} + \frac{\omega_z}{2}\frac{K^n-(-K)^2}{2K} & \omega_+\frac{K^n-(-K)^n}{2K} \\ \omega_-\frac{K^n-(-K)^n}{2K} & \frac{K^n+(-K)^n}{2} - \frac{\omega_z}{2}\frac{K^n-(-K)^2}{2K}\end{pmatrix},$$

and

$$e^{\omega_+S_+ + \omega_-S_- + \omega_zS_z} = \begin{pmatrix}\cosh(K) + \frac{\omega_z}{2K}\sinh(K) & \omega_+\frac{\sinh(K)}{K} \\ \omega_-\frac{\sinh(K)}{K} & \cosh(K) - \frac{\omega_z}{2K}\sinh(K)\end{pmatrix}. \tag{9.86}$$

On the other hand, we have

$$\begin{aligned}e^{x_-S_-}e^{\ln x_zS_z}e^{x_+S_+} &= \begin{pmatrix}1 & \\ x_- & 1\end{pmatrix}\begin{pmatrix}x_z^{1/2} & \\ & x_z^{-1/2}\end{pmatrix}\begin{pmatrix}1 & x_+ \\ & 1\end{pmatrix} \\ &= \begin{pmatrix}x_z^{1/2} & x_z^{-1/2}x_+ \\ x_-x_z^{1/2} & x_-x_z^{-1/2}x_+ + 1\end{pmatrix}.\end{aligned} \tag{9.87}$$

Now, we can firstly let $\omega_\pm = \pm\theta e^{\mp i\varphi}/2$, $\omega_z = 0$, and then by assigning $x_+ = \mu^*$, $x_- = -\mu$, $x_z = \cosh^2\frac{\theta}{2} = (1+|\mu|^2)^{-1}$ so that

$$D^S(\vec{n}) = \begin{pmatrix} \cosh\frac{\theta}{2} & \sinh\frac{\theta}{2}e^{-i\varphi} \\ -\sinh\frac{\theta}{2}e^{i\varphi} & \cosh\frac{\theta}{2} \end{pmatrix} = e^{-\mu S_-}e^{-\ln\left(1+|\mu|^2\right)S_z}e^{\mu^* S_+}. \tag{9.88}$$

Notice that although the proof above relies on the 2D/spin-$\frac{1}{2}$ representation very much, it should hold generally for the $SU(2)$ group as an algebraic property.

Since $S_+\left|0\right\rangle = 0$, $S_z\left|0\right\rangle = S\left|0\right\rangle$, we have

$$\left|\vec{n}\right\rangle = D^S(\vec{n})\left|0\right\rangle = e^{-\mu S_-}e^{-\ln\left(1+|\mu|^2\right)S_z}e^{\mu^* S_+}\left|0\right\rangle = (1+|\mu|^2)^{-S}e^{-\mu S_-}\left|0\right\rangle, \tag{9.89}$$

where $\mu = e^{i\varphi}\tan\frac{\theta}{2}$ is determined by $\vec{n}$. Now we claim that

$$\begin{aligned} \left\langle\vec{n}_1|\vec{n}_2\right\rangle &= \left[(1+|\mu_1|^2)(1+|\mu_2|^2)\right]^{-S}\left\langle 0\right|e^{-\mu_1^* S_+}e^{-\mu_2 S_-}\left|0\right\rangle \\ &= \left[(1+|\mu_1|^2)(1+|\mu_2|^2)\right]^{-S}(1+\mu_1^*\mu_2)^{2S}. \end{aligned} \tag{9.90}$$

This can be done by considering that

$$e^{-\mu S_-}\left|0\right\rangle = \sum_{n=0}^{2S}\frac{(-\mu)^n}{n!}S_-^n\left|0\right\rangle, \tag{9.91}$$

which only contains a finite order of expansion due to the finite dimension of the representation, and

$$S_-^n\left|0\right\rangle = \sqrt{(S-n)(S-n+1)\cdots(S-1)S}\left|S,S-n\right\rangle = \sqrt{\frac{(2S)!n!}{(2S-n)!}}\left|S,S-n\right\rangle, \tag{9.92}$$

so that

$$e^{-\mu S_-}\left|0\right\rangle = \sum_{n=0}^{2S}\frac{(-\mu)^n}{n!}\sqrt{\frac{(2S)!n!}{(2S-n)!}}\left|S,S-n\right\rangle = \sum_{n=0}^{2S}(-\mu)^n\sqrt{C_{2S}^n}\left|S,S-n\right\rangle, \tag{9.93}$$

where $C_{2S}^n = \frac{(2S)!}{n!(2S-n)!}$ is the binomial coefficient. and

$$\begin{aligned} \left\langle\vec{n}_1|\vec{n}_2\right\rangle &\propto \sum_{nm}(-\mu_1^*)^n(-\mu_2)^m\sqrt{C_{2S}^n C_{2S}^m}\left\langle S,S-n|S,S-m\right\rangle \\ &= \sum_{n=0}^{2S}(\mu_1^*\mu_2)^n C_{2S}^n = (1+\mu_1^*\mu_2)^{2S}. \end{aligned} \tag{9.94}$$

Now we can consider the overlap of two coherent states, i.e.,

$$\left\langle\vec{n}_1|\vec{n}_2\right\rangle = \frac{(1+\mu_1^*\mu_2)^{2S}}{\left[(1+|\mu_1|^2)(1+|\mu_2|^2)\right]^S} = \left(\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} + e^{-i(\varphi_1-\varphi_2)}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\right)^{2S}, \tag{9.95}$$

and for the overlapping amplitude, we have

$$\left|\left\langle\vec{n}_1|\vec{n}_2\right\rangle\right| = \cdots = \left(\frac{1+\vec{n}_1\cdot\vec{n}_2}{2}\right)^S, \tag{9.96}$$

through a direct calculation. The cocycle is then given by

$$e^{i\Phi(\vec{n}_1,\vec{n}_2,\vec{n}_0)S} = \frac{\langle\vec{n}_1|\vec{n}_2\rangle}{|\langle\vec{n}_1|\vec{n}_2\rangle|}, \tag{9.97}$$

which is clearly not equal to one generally. □

Now we also claim the resolution of identity for spin coherent state, i.e.,

$$\mathbb{I}_{2S+1} = \int \mathrm{d}\mu(\vec{n})\,|\vec{n}\rangle\langle\vec{n}|\,,\ \mathrm{d}\mu(\vec{n}) = \frac{2S+1}{4\pi}\delta(n^2-1)\mathrm{d}^3n. \tag{9.98}$$

Proof. Notice that

$$|n\rangle\langle n| = (1+|\mu|)^{-2S}\sum_{m,n}^{2S}(-\mu)^m(-\mu^*)^n\sqrt{C_{2S}^m C_{2S}^n}\,|S,S-m\rangle\langle S,S-n|\,, \tag{9.99}$$

where the $\vec{n}$ dependent part is

$$(1+|\mu|)^{-2}(-\mu)^m(-\mu^*)^n = \cos^{4S}\frac{\theta}{2}(-1)^{m+n}\tan^{m+n}\frac{\theta}{2}e^{i\varphi(m-n)}, \tag{9.100}$$

which tells that upon integral measure $\mathrm{d}\mu(\vec{n}) = N\mathrm{d}^2S = N\sin\theta\mathrm{d}\theta\mathrm{d}\varphi$ where N is a to-be-defined norm, we must accept $m=n$ so that the integral does not vanish. This leads to

$$\int \mathrm{d}\mu(\vec{n})\,|n\rangle\langle n| = 2\pi N\sum_{n=0}^{2S}\int_0^\pi \sin\theta\mathrm{d}\theta\cos^{4S}\frac{\theta}{2}\tan^{2n}\frac{\theta}{2}C_{2S}^n\,|S,S-n\rangle\langle S,S-n|\,, \tag{9.101}$$

where the integral

$$\begin{aligned} I(S,n) &= \int_0^\pi \sin\theta\mathrm{d}\theta\cos^{4S}\frac{\theta}{2}\tan^{2n}\frac{\theta}{2} \\ &\overset{\phi=\theta/2}{=} 4\int_0^{\pi/2}\mathrm{d}\phi\cos^{4S-2n+1}\phi\sin^{2n+1}\phi \\ &= 2\mathrm{B}(2S-n+1,n+1) = 2\frac{\Gamma(2S-n+1)\Gamma(n+1)}{\Gamma(2S+2)} = \frac{2(2S-n)!(n)!}{(2S+1)!}, \end{aligned} \tag{9.102}$$

is given by the Beta function. Then we have

$$\int \mathrm{d}\mu(\vec{n})\,|n\rangle\langle n| = 2\pi N\sum_{n=0}^{2S} I(S,n)C_{2S}^n\,|S,S-n\rangle\langle S,S-n| = \frac{4\pi N}{2S+1}\mathbb{I}_{2S+1}, \tag{9.103}$$

which gives the norm $N = \dfrac{2S+1}{4\pi}$ and completes the proof. □

9.3.2 Partition Function for Spin

Let us consider the partition function for spin, which is given by

$$\begin{aligned} Z &= \operatorname{Tr} e^{-\beta\hat{H}} = \lim_{N\to\infty} \operatorname{Tr}\left(e^{-\delta\tau H}\right)^N,\ \delta\tau = \frac{\beta}{N}, \\ &\overset{\mathbb{Id}}{=} \lim_{N\to\infty} \left[\int \prod_{j=1}^{N} \mathrm{d}\mu(\vec{n}_j)\right] \left[\prod_{j=0}^{N-1} \langle \vec{n}(\tau_{j+1})|\, e^{-\delta\tau H}\, |\vec{n}_{\tau_j}\rangle\right] \\ &= \lim_{N\to\infty} \left[\int \prod_{j=1}^{N} \mathrm{d}\mu(\vec{n}_j)\right] \prod_{j=0}^{N-1} \left[\langle \vec{n}(\tau_{j+1})|\vec{n}(\tau_j)\rangle - \delta\tau\, \langle \vec{n}(\tau_{j+1})|H|\vec{n}(\tau_j)\rangle\right] \\ &= \int \mathrm{D}\vec{n} e^{-S_E[\vec{n}]}, \end{aligned} \tag{9.104}$$

where the path integral measure

$$\mathrm{D}\vec{n} = \lim_{N\to\infty} \prod_{j=1}^{N} \mathrm{d}\mu(\vec{n}_j), \tag{9.105}$$

with $\vec{n}_j = \vec{n}(\tau_j)$ identified. Let us now consider the simplest Hamiltonian as a Zeeman coupling $H = \mu \boldsymbol{B}\cdot\boldsymbol{S}$, where $\boldsymbol{B}$ is the magnetic field and μ is the magnetic moment, and $\boldsymbol{S} = S\vec{n}$. Then we have

$$\langle \vec{n}(\tau_{j+1})|\vec{n}(\tau_j)\rangle = \left(\frac{1+\vec{n}_{j+1}\cdot\vec{n}_j}{2}\right)^S \exp(i\Phi(\vec{n}_{j+1},\vec{n}_j,\vec{n}_0)S), \tag{9.106a}$$

$$\langle \vec{n}(\tau_{j+1})|H|\vec{n}(\tau_j)\rangle / \langle \vec{n}(\tau_{j+1})|\vec{n}(\tau_j)\rangle \cong \mu S\boldsymbol{B}\cdot\vec{n}_j. \tag{9.106b}$$

Then the Euclidean action reads

$$-S_E[\vec{n}] = \lim_{N\to\infty} iS \sum_{j=0}^{N-1} \Phi(\vec{n}_{j+1},\vec{n}_j,\vec{n}_0) + S\sum_{j=0}^{N-1} \ln\left(\frac{1+\vec{n}_{j+1}\cdot\vec{n}_j}{2}\right) - \delta\tau\mu S\sum_{j=0}^{N-1} \boldsymbol{B}\cdot\vec{n}_j, \tag{9.107}$$

where the cocycle phase given by the first term can be defined as $\sum_{j=0}^{N-1}\Phi(\vec{n}_{j+1},\vec{n}_j,\vec{n}_0) = A[\Omega^+] = 4\pi - A[\Omega^-]$, which can be written elegantly either by Berry phase as a one-form integral over trajectory as boundary given by $\partial\Omega^+$, or by the Wess-Zumino-Witten (WZW) term as a two-form integral over the solid angle area Ω^+, i.e.,

$$\begin{aligned} A[\Omega^+] &= \oint_{\partial\Omega^+} \mathrm{d}\tau\, \langle \vec{n}(\tau)|\partial_\tau\vec{n}(\tau)\rangle \\ &= \int_0^1 \mathrm{d}s \int_0^\beta \mathrm{d}\tau \vec{n}(\tau,s)\cdot\partial_\tau\vec{n}(\tau,s)\times\partial_s\vec{n}(\tau,s), \end{aligned} \tag{9.108}$$

where we have parametrized $\vec{n}(\tau,s)$ with $0\le s\le 1$ to be an extension between $\vec{n}(\tau)$ and $\vec{n}_0$, i.e.,

$$\vec{n}(\tau,0) = \vec{n}(\tau),\ \vec{n}(\tau,1) = \vec{n}_0,\ \vec{n}(\tau+\beta,0) = \vec{n}(\tau,0). \tag{9.109}$$

And one sees that as a real vector, $\vec{n}$ satisfies periodic boundary condition like a Boson. Also notice that we write the Berry phase using ket state rather than vector $\vec{n}$, since

$$0 = \partial_\tau(\vec{n}\cdot\vec{n}) = 2\vec{n}\cdot\partial_\tau\vec{n}, \tag{9.110}$$

which again reveals the phase nature of the cocycle term, which is purely geometric/topological. For a spin-1/2 system, the cocycle phase can be given through the adjoint representation of $SU(2)$ group, i.e., we relate $\vec{n} = z^\dagger \vec{\sigma} z$, such that $\langle \vec{n}(\tau) | \partial_\tau \vec{n}(\tau) \rangle = z^\dagger \partial_\tau z$ stands. The third Zeeman term is rather straightforward, and for the second which accounts for the magnitude change thus a regular kinetic term, it has to be one order higher thus negligible, since

$$\begin{aligned} \ln\left(\frac{1+\vec{n}_{j+1}\cdot\vec{n}_j}{2}\right) &= \ln\left(\frac{1+\vec{n}_j\cdot[\vec{n}_j+\delta\tau\partial_\tau\vec{n}_j+\frac{1}{2}\delta\tau^2\partial_\tau^2\vec{n}_j+O(\delta\tau^3)]}{2}\right) \\ &= \ln\left(1+\frac{1}{4}\delta\tau^2\partial_\tau\vec{n}_j\cdot\partial_\tau\vec{n}_j+O(\delta\tau^3)\right) \\ &\cong \frac{1}{4}\delta\tau^2\partial_\tau\vec{n}_j\cdot\partial_\tau\vec{n}_j = -\frac{1}{4}(\partial_\tau\vec{n}_j)(\delta\tau)^2. \end{aligned} \tag{9.111}$$

Finally, we obtain the Euclidean action as

$$\begin{aligned} S_E[\vec{n}] &= -iSS_{WZW}[\vec{n}] + \frac{S}{4}\delta\tau\int_0^\beta \mathrm{d}\tau(\partial_\tau\vec{n})^2 + \mu S\int_0^\beta \mathrm{d}\tau \boldsymbol{B}\cdot\vec{n} \\ &\cong -iSS_{WZW}[\vec{n}] + \mu S\int_0^\beta \mathrm{d}\tau\boldsymbol{B}\cdot\vec{n}. \end{aligned} \tag{9.112}$$

One may notice that the WZW term here is the same one as we have utilized for calculating Hall conductance from the two-band model, and in this sense we can similarly introduce an effective magnetic field like the Berry curvature by defining

$$S_{WZW} = \int_{\Omega^+} \mathrm{d}\boldsymbol{S}\cdot\nabla_{\vec{n}}\times\boldsymbol{A}_{\text{eff}}(\vec{n}), \tag{9.113}$$

where $\boldsymbol{A}_{\text{eff}}(\vec{n})$ is the effective gauge field, and

$$\boldsymbol{B}_{\text{eff}} = \nabla_{\vec{n}}\times\boldsymbol{A}_{\text{eff}}(\vec{n}) = S\vec{n}. \tag{9.114}$$

The total flux of the effective magnetic field is then given by $4\pi S = 2\pi N$, with $N = 2S$ as integer magnetic charge, and 2π comes from dimension consideration $h/e = 2\pi$ under $\hbar = 1 = e$. This again reflects the half-integer nature of angular momentum index.

Semi-Classical Approximation

In real time, the action for the spin path integral is given by

$$S = -\int_0^T \mathrm{d}t\mu S\vec{n}\cdot\boldsymbol{B} + S\int_0^1 \mathrm{d}s\int_0^T \mathrm{d}t\vec{n}(t,s)\cdot\partial_t\vec{n}(t,s)\times\partial_s\vec{n}(t,s), \tag{9.115}$$

and for the semi-classical equation of motion, we consider the variation of the action with respect to $\vec{n}$ to find its saddle point, where we need to take care about the WZW term. Imagine that we have an Ω^+ for integration, and a small variation of the trajectory, i.e. of $\partial\Omega^+$ amounts to say that

$$\delta S_{WZW} = \int_{\partial\Omega^+} \mathrm{d}\tau - \vec{n}\cdot\partial_\tau\vec{n}\times\delta\vec{n}\Big|_{s=0} = \int_0^T \mathrm{d}t\delta\vec{n}(t)\cdot\vec{n}\times\partial_t\vec{n}, \tag{9.116}$$

then the variational principle tells that the semi-classical equation of motion is given by

$$\vec{n}(t) \times \partial_t \vec{n} = \mu \boldsymbol{B}, \tag{9.117}$$

which, by left acting $\vec{n}\times$, reduces to the famous Bloch precession motion

$$\partial_t \vec{n} = \mu \vec{n} \times \boldsymbol{B}, \tag{9.118}$$

which is rather similar to the Larmor precession of a magnetic moment in a magnetic field, but with a half-integer spin index.

9.4 Observables from Path Integral

A physical observable does not come from a static field configuration, and we need to instead stimulate the system and see how it responds, which will finally leads to the (retarded) correlation function as we have shown in the linear response theory. In this sense, we consider the time ordering $T \ll t < t' \ll T'$, with a source J turned on/off adiabatically at t/t'.

Theorem 9.4.1 (Gell-Mann and Low Theorem)**.**

Chapter 10

Open-System Dynamics

"Decoherence leads to existence."

10.1 Environment-Induced Decoherence

In the double-slit experiment, the interference pattern is weakened or even destroyed when a first measurement is performed immediately after the particle passes through the slits, depending on the 'strength' of the measurement. Nevertheless, we are not talking about invoking collapse or Born's rule here. Instead, as we have learnt in the AB effect, an external vector field can leads to different superposition patterns of beams of electrons, while if we promote this external field to another system referred to as the environment, clearly similar effects can be expected under suitable setups. Especially, in the double-slit experiment, the environment can couple with the beams of electrons in an entangled way, such that the interference between the two beams is weakened.

Under the framework of Von-Neumann measurement theory, to perform a measurement over a quantum system S, one needs to firstly couple it with a measuring apparatus, which for our purpose can be regarded as the environment E. The joint evolution of the system and the environment is then described by the Schrödinger equation under a full Hermitian Hamiltonian H. The possible measurement happens with Born's rule applied on the environment, from which we can deduce information about the system.

Assuming that at $t = 0$, the system and the environment begin to couple together with joint density matrix $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$, with $\rho_E(0) = \sum_i p_i \left|E_i\right\rangle\!\left\langle E_i\right|$ mixed states ensemble of the environment, satisfying $\sum_i p_i = 1$ with $|E_i\rangle$ orthonormal basis of the environment E. The system and environment are then entangled under the joint evolution $U(t) = e^{-iHt/\hbar}$, as $\rho_{SE}(t) = U(t)\rho_{SE}(0)U^\dagger(t)$. Especially, to see how the system S evolves, we introduce the reduced density matrix

$$\begin{aligned}\rho_S(t) \equiv \mathrm{Tr}_E(\rho_{SE}(t)) &= \sum_i p_i \,\mathrm{Tr}_E(U(t)\left[\rho_S(0) \otimes \left|E_i\right\rangle\!\left\langle E_i\right|\right] U^\dagger(t)) \\ &= \sum_{ij} p_i \left\langle E_j|U(t)|E_i\right\rangle \rho_S(0) \left\langle E_i|U^\dagger(t)|E_j\right\rangle \\ &= \sum_k W_k \rho_S(0) W_k^\dagger,\end{aligned} \tag{10.1}$$

where we have introduced the Kraus operator

$$W_k = \sqrt{p_{i_k}} \left\langle E_{j_k}|U(t)|E_{i_k}\right\rangle, \tag{10.2}$$

with k labeling the pair indices (i_k, j_k). Clearly, for S to be N_S dimensional and E to be N_E dimensional, the represented matrix of a Kraus operator will be a $N_S \times N_S$ one, while the number of Kraus operators is N_E^2. The unitarity of the evolution is represented as the completeness constraint

$$\begin{aligned}\sum_k W_k^\dagger W_k &= \sum_k p_{i_k} \langle E_{i_k}|U^\dagger(t)|E_{j_k}\rangle \langle E_{j_k}|U(t)|E_{i_k}\rangle \\ &= \sum_i p_i \langle E_i| U^\dagger(t) \left(\sum_j |E_j\rangle\langle E_j|\right) U(t) |E_i\rangle \\ &= \sum_i p_i = \mathbb{I}_S,\end{aligned} \tag{10.3}$$

such that $\mathrm{Tr}(\rho_S(t)) = \mathrm{Tr}\Big[\Big(\sum_k W_k^\dagger W_k\Big)\rho_S(0)\Big] = \mathrm{Tr}(\rho_S(0)) = 1$. The map $\Phi : \rho_S(0) \to \rho_S(t)$ can be viewed as a positive map generated by the Kraus operators, and is called the quantum operation or quantum channel.

Now we introduce the measurement process, which is assumed to select a certain state of the environment under projection $P_\alpha = |\alpha\rangle\langle\alpha|$. Clearly, if only one system state is coupled with this selected probe state α, the system will be in this state after the measurement. This happens, for example in the Stern-Gerlach experiment, where it happens that the system is also the probe itself, and in a Bell state where the system and the environment are maximally entangled. However, generally, the system is not maximally entangled with the environment, and the measurement will not be perfect in the sense that the reduced density matrix of the system after the measurement is not that for a pure state, but also a mixed state, indicating that there are unresolved degrees of freedom through the environment. Basically, we can make an analogy between the reduced density matrix of the system after the measurement with the conditional probability distribution, which means that we need to firstly define a normalization probability

$$P(\alpha, t) = \mathrm{Tr}(P_\alpha \rho_E(t)) = \mathrm{Tr}\big(P_\alpha \,\mathrm{Tr}_S(U(t)\rho_S(0)\otimes\rho_E(0)U^\dagger(t))\big), \tag{10.4}$$

which is the probability of the environment being in state α at time t, based on the reduced density matrix of the environment. The reduced density matrix of the system, upon the measurement of the environment in state α, will then be

$$\begin{aligned}\rho_S^{(\alpha)}(t) &= \frac{\mathrm{Tr}_E\left[P_\alpha U(t)\rho_S(0)\otimes\rho_E(0)U^\dagger(t)\right]}{P(\alpha,t)} \\ &= \sum_i \frac{M_{\alpha,i}\rho_S(0)M_{\alpha,i}^\dagger}{P(\alpha,t)},\end{aligned} \tag{10.5}$$

where similarly, the quantum operation operator related to the measurement is

$$M_{\alpha,i} = \sqrt{p_i}\,\langle\alpha|U(t)|E_i\rangle\,, \tag{10.6}$$

which belongs to the set of Kraus operators with fixed trace element α, representing the selective projection of the measurement. The completeness constraint is then $\sum_{\alpha,i} M_{\alpha,i}^\dagger M_{\alpha,i} = \mathbb{I}_S$.

This quantum operation formalism, which couples the system with the environment, leads to decoherence in the system represented in the reduced density matrix. Especially,

when the interaction between the system and the environment is dominant, one expects prominent selection in states of the system. What does this selection leads to? Consider a system exposed to unavoidable environmental noise, then the states that are mostly entangled with the environment (imagine a Bell state) are expected to be mostly unobservable, since the noise will effectively average out such quantum states (notice that this is opposite to the measurement process, where we do want entanglement as we are assumed to control well-behavior of the environment or apparatus). It then becomes clear that only those states which are the most robust to the interaction with the environment, in the sense that they are least entangled with the environment, are immune to decoherence and thus stable and observable. Such environment-superselected preferred states (or observables) are sometimes also called pointer states (or pointer observables). To give a definitive description, we consider that $H = H_S + H_E + H_I \approx H_I$, with H_I the interaction Hamiltonian dominant, and H_S and H_E the Hamiltonians of the system and the environment, respectively. Specifically, $H_I = \sum_\alpha S_\alpha \otimes E_\alpha$. Assuming pointer state $|s_i\rangle$ to be coupled with the environment as $\rho_{s_i} \otimes \rho_E(0)$ at $t = 0$, with $\rho_{s_i} = |s_i\rangle\langle s_i|$, and least entanglement requirement leads to that $U(t)\rho_{s_i} \otimes \rho_E(0)U^\dagger(t) = \rho_{s_i}(t) \otimes \rho_E(t)$, and a sufficient condition for this to stand is that

$$S_\alpha |s_i\rangle = \lambda_i^\alpha |s_i\rangle , \ \forall\alpha, \tag{10.7}$$

since this leads to

$$U(t)|s_i\rangle = e^{-i\sum_\alpha S_\alpha \otimes E_\alpha t/\hbar}|s_i\rangle = |s_i\rangle\, e^{-i\sum_\alpha \lambda_i^\alpha E_\alpha t/\hbar}$$

through a direct expansion. Especially, when $\lambda_i^\alpha \equiv \lambda^\alpha$ for all i, we get a decoherence-free subspace spanned by $|s_i\rangle$. A point observable follows with a similar requirement that $[O, S_\alpha] = 0$.

10.2 Quantum Master Equation

In the realm of probability theory, the time-evolution of a probability distribution is referred to as a master equation, which is familiar to physicists in the context of Brownian motion or the diffusion equation. In quantum mechanics, the master equation is a differential equation that describes the time-evolution of the density matrix of a quantum system, i.e.,

$$\rho_S(t) = \mathscr{V}(t) \circ \rho_S(0), \tag{10.8}$$

with $\mathscr{V}(t)$ the superoperator, or the quantum operation, that maps the initial density matrix to the density matrix at time t. Apparently and exactly, we have $\mathscr{V}(t) \circ \rho_S(0) = \mathrm{Tr}\big(U(t)\rho_{SE}(0)U^\dagger(t)\big)$, which, however, is impractical to be directly computed, so does its derived form using Kraus operators. In this sense, we have to make some approximations to the superoperator such that we can determine the approximate time evolution of $\rho_S(t)$ even when it is impossible to calculate the exact global dynamics of $\rho_{SE}(t)$. In general, we expect the resulting master equation to be a first-order differential equation local in time as

$$\frac{\mathrm{d}\rho_S(t)}{\mathrm{d}t} = \mathscr{L}\rho_S(t) = -\frac{i}{\hbar}[H_S', \rho_S(t)] + \mathscr{D}(\rho_S(t)), \tag{10.9}$$

with the unitary part given by the usual Liouville-von Neumann commutator with the renormalized Hamiltonian H_S', and a non-unitary part represented by superoperator $\mathscr{D}$ counting on the decoherence effect. In the following, we shall derive the master equation with Born-Markov, and rotating-wave approximations step by step.

10.2.1 The Redfield Equation

The total Hamiltonian is again $H = H_S + H_E + H_{\text{int}}$, with $H_{\text{int}} = \sum_\alpha S_\alpha \otimes E_\alpha$. Under the current technique, where we are able to proceed to is a perturbative regime with a weak coupling (interaction) between the system and the environment, which allows us to work in the interaction picture with $H_0 = H_S + H_E$. The full evolution of ρ_{SE} is the governed by the interaction Liouville-von Neumann equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho^I_{SE}(t) = -\frac{i}{\hbar}[H^I_{\text{int}}(t), \rho^I_{SE}(t)], \tag{10.10}$$

with $\rho^I_{SE}(t) = U_I(t)\rho_{SE}(0)U_I^\dagger(t)$, and $O^I(t) = U_0^\dagger(t)OU_0(t)$, where $U_0(t) = e^{-iH_0t/\hbar}$, $U_I(t) = U_0^\dagger(t)U(t)$.

Now we expand the formal solution of $\rho^I_{SE}(t)$ into the equation, which is equivalent to a second-order expansion,

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho^I_{SE}(t) = -\frac{i}{\hbar}[H^I_{\text{int}}, \rho^I_{SE}(0)] - \frac{1}{\hbar^2}\int_0^t \mathrm{d}t'\ [H^I_{\text{int}}(t), [H^I_{\text{int}}(t'), \rho^I_{SE}(t')]]. \tag{10.11}$$

The philosophy behind may be understand as, in linear response theory, we expand to the first order of perturbation, since the external source, which usually appears as a background field, is assumed to be invariant under the feedback of the system; however, in the current case, the environment appears explicitly as a part of the total Hamiltonian, thus its response to the change shall be considered as well, which leads to a second-order expansion to include possible back-action effects.

Since we focus on the system, we partial trace out the environment and obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho^I_S(t) = -\frac{1}{\hbar^2}\int_0^t \mathrm{d}t'\ \mathrm{Tr}_E\left([H^I_{\text{int}}(t), [H^I_{\text{int}}(t'), \rho^I_{SE}(t')]]\right), \tag{10.12}$$

where we have assumed that $\mathrm{Tr}_E([H^I_{\text{int}}, \rho^I_{SE}(0)]) = 0$ under a thermal equilibrium environment, which means that the initial state does not generate any dynamics for the system or the environment, reasonable since we do not expect time non-local effects. Indeed, notice that due to the decoupled form of $\rho^I_{SE}(0) = \rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$ and $U_0(t) = e^{-iH_St/\hbar} \otimes e^{-iH_Et/\hbar}$,

$$\begin{aligned}[H^I_{\text{int}}, \rho^I_{SE}(0)] &= \sum_\alpha [S^I_\alpha \otimes E^I_\alpha, \rho_S(0) \otimes \rho_E(0)] \\ &= \sum_\alpha [S^I_\alpha, \rho_S(0)] \otimes E^I_\alpha\rho_E(0) - \rho_S(0)S^I_\alpha \otimes [E^I_\alpha, \rho_E(0)],\end{aligned}$$

with its non-vanishing partial trace represented by the first term. Especially, its trace reads that

$$\mathrm{Tr}_E([H^I_{\text{int}}, \rho^I_{SE}(0)]) = \sum_\alpha \mathrm{Tr}\big(E^I_\alpha\rho_E(0)\big)[S^I_\alpha, \rho_S(0)] = \sum_\alpha \mathrm{Tr}(E_\alpha\rho_E(0))[S^I_\alpha, \rho_S(0)]$$

where in the second equality we have used that the environment is in a thermal equilibrium state, and thus the expectation value does not vary with time. Then it is easy to verify that, by introducing $\Delta H_S = \sum_\alpha \mathrm{Tr}(E_\alpha\rho_E(0))S_\alpha$ in H_0 (thus subtract from H_{int}), the partial trace related to the initial state vanishes. The resulting master equation is exact,

but is certainly neither closed with $\rho_S(t)$ only, nor local in time, which is impractical for computation.

To proceed, we firstly mention that the thermal equilibrium assumption over the environment is quite close to the approximations we shall made, known as the Born-Markov approximation. The Born approximation utilizes the weak-coupling nature of the system and the environment, thus assuming not only invariant product form of density matrix, but also a constant density matrix for the environment (in Schrödinger picture)

$$\rho_{SE}(t) \approx \rho_S(t) \otimes \rho_E, \tag{10.13}$$

which is readily to be pushed forward into the interaction picture due to the decoupled form of H_0. The master equation accordingly reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{1}{\hbar^2}\int_0^t \mathrm{d}t' \ \mathrm{Tr}_E\left([H_{\mathrm{int}}^I(t), [H_{\mathrm{int}}^I(t'), \rho_S^I(t)\otimes\rho_E]]\right), \tag{10.14}$$

which leads to the closed form.

A closer look over the master equation reads

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) &= -\frac{1}{\hbar^2}\int_0^t \mathrm{d}t' \ \mathrm{Tr}_E\left([H_{\mathrm{int}}^I(t), [H_{\mathrm{int}}^I(t'), \rho_S^I(t)\otimes\rho_E]]\right) \\
&= -\frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^t \mathrm{d}t' \ \mathrm{Tr}_E\left([S_\alpha^I(t)\otimes E_\alpha^I(t), [S_\beta^I(t')\otimes E_\beta^I(t'), \rho_S^I(t)\otimes\rho_E]]\right) \\
&= -\frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^t \mathrm{d}t' \left[C_{\alpha\beta}(t-t')\left(S_\alpha^I(t)S_\beta^I(t')\rho_S^I(t') - S_\beta^I(t')\rho_S^I(t')S_\alpha^I(t)\right)\right. \\
&\quad \left. + C_{\beta\alpha}(t'-t)\left(\rho_S^I(t')S_\beta^I(t')S_\alpha^I(t) - S_\alpha^I(t)\rho_S^I(t')S_\beta^I(t')\right)\right],
\end{aligned}$$

where we have utilized that $H_{\mathrm{int}}^I(t) = \sum_\alpha S_\alpha^I(t)\otimes E_\alpha^I(t)$ due to the decoupling form of H_0. The defined environment self-correlation functions read

$$C_{\alpha\beta}(t,t') = \langle E_\alpha^I(t)E_\beta^I(t')\rangle_E = \mathrm{Tr}\big(E_\alpha^I(t-t')E_\beta^I(0)\rho_E\big) \equiv C_{\alpha\beta}(t-t'), \tag{10.15}$$

where we have utilized the thermal equilibrium assumption such that $[H_E, \rho_E] = 0$ to make the correlation functions time translational invariant. Notice that

$$C_{\beta\alpha}^*(-t) = \langle E_\beta^I(-t)E_\alpha^I(0)\rangle_E^* = \langle E_\alpha^I(t)E_\beta^I(0)\rangle_E = C_{\alpha\beta}(t).$$

The Markov approximation, which utilizes the fact that given the environment a large thermal equilibrium bath, the correlation functions decay rapidly at high temperatures, compared with the characteristic timescale of the system during which ρ_S changes appreciably, assumes that $C_{\alpha\beta}(t)$ is peaked only at $t = 0$. Such an assumption immediately drags $\rho_S(t') \to \rho_S(t)$ (as bears Markovian) and leads to the Redfield equation

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^t \mathrm{d}t' &\left[C_{\alpha\beta}(t-t')\left(S_\alpha^I(t)S_\beta^I(t')\rho_S^I(t) - S_\beta^I(t')\rho_S^I(t)S_\alpha^I(t)\right)\right. \\
&\left. + C_{\beta\alpha}(t'-t)\left(\rho_S^I(t)S_\beta^I(t')S_\alpha^I(t) - S_\alpha^I(t)\rho_S^I(t)S_\beta^I(t')\right)\right],
\end{aligned} \tag{10.16}$$

which fulfills the requirement of a closed and local-in-time master equation.

Indeed, due to the peaked nature, we can extend the integral to $-\infty$, and by replacing $\tau = t - t'$, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^{+\infty}\mathrm{d}\tau\left[C_{\alpha\beta}(\tau)\left(S_\alpha^I(t)S_\beta^I(t-\tau)\rho_S^I(t) - S_\beta^I(t-\tau)\rho_S^I(t)S_\alpha^I(t)\right)\right.$$
$$\left. + C_{\beta\alpha}(-\tau)\left(\rho_S^I(t)S_\beta^I(t-\tau)S_\alpha^I(t) - S_\alpha^I(t)\rho_S^I(t)S_\beta^I(t-\tau)\right)\right].$$

We can further transform the equation into the Schrödinger picture, with replacement $\rho_S^I(t) = e^{iH_St/\hbar}\rho_S(t)e^{-iH_St/\hbar}$, $S_\alpha^I(t) = e^{iH_St/\hbar}S_\alpha(t)e^{-iH_St/\hbar}$, which leads to

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\rho_S(t) &= -\frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^{+\infty}\mathrm{d}\tau\left[C_{\alpha\beta}(\tau)\left(S_\alpha S_\beta(-\tau)\rho_S(t) - S_\beta(-\tau)\rho_S(t)S_\alpha\right)\right.\\ &\left. + C_{\beta\alpha}(-\tau)\left(\rho_S(t)S_\beta(-\tau)S_\alpha - S_\alpha\rho_S(t)S_\beta(-\tau)\right)\right] - \frac{i}{\hbar}[H_S,\rho_S(t)]\\ &= -\frac{i}{\hbar}[H_S,\rho_S(t)] - \frac{1}{\hbar^2}\sum_{\alpha\beta}\int_0^{+\infty}\mathrm{d}\tau\left(C_{\alpha\beta}(\tau)[S_\alpha, S_\beta(-\tau)\rho_S(t)]\right.\\ &\left. + C_{\beta\alpha}(-\tau)[\rho_S(t)S_\beta(-\tau), S_\alpha]\right),\end{aligned}$$

then by further introducing

$$B_\alpha = \sum_\beta\int_0^{+\infty}\mathrm{d}\tau C_{\alpha\beta}(\tau)S_\beta(-\tau), \tag{10.17a}$$

$$C_\alpha = \sum_\beta\int_0^{+\infty}\mathrm{d}\tau C_{\beta\alpha}(-\tau)S_\beta(-\tau), \tag{10.17b}$$

we have the Born-Markov master equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S(t) = -\frac{i}{\hbar}[H_S,\rho_S(t)] - \frac{1}{\hbar^2}\sum_\alpha\left([S_\alpha, B_\alpha\rho_S(t)] + [\rho_S(t)C_\alpha, S_\alpha]\right). \tag{10.18}$$

10.2.2 The Lindblad Equation

The Redfield equation is a general form of the master equation, and is not guaranteed to be positive defined as $\langle\psi|\rho_S(t)|\psi\rangle \geq 0$ for any t and $|\psi\rangle$. Imposing such a requirement over the master equation, together with the Markov approximations will lead to the Lindblad form master equations, which can be viewed as a special case of the Redfield equation. Nevertheless, such a derivation based on the semigroup property of the superoperator does not encode the physical meaning of the master equation, thus we shall derive the Lindblad equation from the Redfield equation with a further rotating wave approximation.

The rotating wave approximation, or more appropriately the secular approximation, is a common technique in quantum optics, which is used to simplify the interaction Hamiltonian by neglecting the fast oscillating terms by long period average. Thus by denoting $H_S|\varepsilon\rangle = \varepsilon|\varepsilon\rangle$, we project the interaction into energy space by

$$S_\alpha(\omega) = \sum_{\varepsilon'-\varepsilon=\hbar\omega}P_\varepsilon S_\alpha P_{\varepsilon'}, \tag{10.19}$$

where P_ε is the projector onto the energy eigenstate $|\varepsilon\rangle$. It is ready to verify the sum rule

$$\sum_\omega S_\alpha(\omega) = \sum_\omega \sum_{\varepsilon'-\varepsilon=\hbar\omega} P_\varepsilon S_\alpha P_{\varepsilon'} = \sum_\varepsilon \sum_{\varepsilon'} P_\varepsilon S_\alpha P_{\varepsilon'} = S_\alpha,$$

and by assuming Hermitian S_α, we have $S_\alpha(\omega) = S_\alpha^\dagger(-\omega)$. Essentially, by noticing that

$$e^{iH_S t/\hbar} S_\alpha(\omega) e^{-iH_S t/\hbar} = e^{i\varepsilon t/\hbar} S_\alpha(\omega) e^{-i\varepsilon' t/\hbar} = e^{-i\omega t} S_\alpha(\omega), \tag{10.20}$$

we have $S_\alpha^I(t) = \sum_\omega e^{-i\omega t} S_\alpha(\omega)$, and

$$S_\alpha^I(t) S_\beta^I(t-\tau) = \sum_{\omega\omega'} S_\alpha^\dagger(\omega') S_\beta(\omega) e^{i(\omega'-\omega)t} e^{i\omega\tau},$$

which leads to the variable-replaced Redfield equation

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{1}{\hbar^2} \sum_{\omega\omega'} \sum_{\alpha\beta} \int_0^{+\infty} \mathrm{d}\tau \Big[& C_{\alpha\beta}(\tau) e^{i(\omega'-\omega)t} e^{i\omega\tau} \left(S_\alpha^\dagger(\omega') S_\beta(\omega) \rho_S^I(t) - S_\beta(\omega) \rho_S^I(t) S_\alpha^\dagger(\omega') \right) \\
& + C_{\beta\alpha}(-\tau) e^{i(\omega-\omega')t} e^{-i\omega'\tau} \left(\rho_S^I(t) S_\beta^\dagger(\omega') S_\alpha(\omega) - S_\alpha(\omega) \rho_S^I(t) S_\beta^\dagger(\omega') \right) \Big] \\
= -\frac{1}{\hbar^2} \sum_{\omega\omega'} \sum_{\alpha\beta} \Big(& \Gamma_{\alpha\beta}(\omega) e^{i(\omega'-\omega)t} [S_\alpha^\dagger(\omega'), S_\beta(\omega)\rho_S^I(t)] + \Gamma_{\beta\alpha}^*(\omega') e^{i(\omega-\omega')t} [\rho_S^I(t) S_\alpha^\dagger(\omega'), S_\beta(\omega)] \Big),
\end{aligned} \tag{10.21}$$

where we have interchanged $\alpha\beta$ for the second term in the second line, and the coefficient

$$\Gamma_{\alpha\beta}(\omega) = \int_0^{+\infty} \mathrm{d}\tau C_{\alpha\beta}(\tau) e^{i\omega\tau}, \tag{10.22}$$

is time-independent.

Due to the existence of double summation ω, ω' with oscillating factor $e^{i(\omega'-\omega)t}$, it is further assumed, as the rotating wave approximation, that the contributions not coming from $\omega = \omega'$ vanishes, thus we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{1}{\hbar^2} \sum_\omega \sum_{\alpha\beta} \Gamma_{\alpha\beta}(\omega) [S_\alpha^\dagger(\omega), S_\beta(\omega)\rho_S^I(t)] + \Gamma_{\beta\alpha}^*(\omega) [\rho_S^I(t) S_\alpha^\dagger(\omega), S_\beta(\omega)]. \tag{10.23}$$

Now we split $\Gamma(\omega) = \frac{1}{2}\gamma(\omega) + i\pi(\omega)$, where the Hermitian part

$$\begin{aligned}
\gamma_{\alpha\beta} = \Gamma_{\alpha\beta} + \Gamma_{\beta\alpha}^* &= \int_0^{+\infty} \mathrm{d}\tau C_{\alpha\beta}(\tau) e^{i\omega\tau} + \int_0^{+\infty} \mathrm{d}\tau C_{\beta\alpha}^*(\tau) e^{-i\omega\tau} \\
&= \int_0^{+\infty} \mathrm{d}\tau C_{\alpha\beta}(\tau) e^{i\omega\tau} + \int_{-\infty}^0 \mathrm{d}\tau C_{\beta\alpha}^*(-\tau) e^{i\omega\tau} = \int_{-\infty}^{+\infty} \mathrm{d}\tau C_{\alpha\beta}(\tau) e^{i\omega\tau}
\end{aligned} \tag{10.24}$$

defines a positive definite matrix. From the expression, γ is the Fourier transform of the correlation function, which is expected to be nearly constant with ω due to the peaked nature of C. Notice that π is also a Hermitian matrix, with element $\pi_{\alpha\beta} = (\Gamma_{\alpha\beta} - \Gamma_{\beta\alpha}^*)/2i$, while coefficient i makes $i\pi$ anti-Hermitian. Under this splitting, we have the Lindblad

equation in the interaction picture

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) &= -\frac{1}{2\hbar^2}\sum_{\omega}\sum_{\alpha\beta}\gamma_{\alpha\beta}\left([S_\alpha^\dagger(\omega), S_\beta(\omega)\rho_S^I(t)] + [\rho_S^I(t)S_\alpha^\dagger(\omega), S_\beta(\omega)]\right)\\ &\quad - \frac{i}{\hbar^2}\sum_{\omega}\sum_{\alpha\beta}\pi_{\alpha\beta}\left([S_\alpha^\dagger(\omega), S_\beta(\omega)\rho_S^I(t)] - [\rho_S^I(t)S_\alpha^\dagger(\omega), S_\beta(\omega)]\right)\\ &= -\frac{i}{\hbar}[H_{\mathrm{LS}}, \rho_S^I(t)] + \frac{1}{\hbar^2}\sum_{\omega}\sum_{\alpha\beta}\gamma_{\alpha\beta}\left(S_\beta(\omega)\rho_S^I(t)S_\alpha^\dagger(\omega) - \frac{1}{2}\left\{S_\alpha^\dagger(\omega)S_\beta(\omega), \rho_S^I(t)\right\}\right),\end{aligned} \tag{10.25}$$

where the Lamb-shift Hamiltonian reads

$$H_{\mathrm{LS}} = \frac{1}{\hbar}\sum_{\alpha\beta}\pi_{\alpha\beta}(\omega)S_\alpha^\dagger(\omega)S_\beta(\omega). \tag{10.26}$$

Notice that since γ and π are Hermitian, we can diagonalize then with a $U_\gamma^\dagger\gamma U_\gamma = d_\gamma$, $U_\pi^\dagger\pi U_\pi = d_\pi$, with $d_{\gamma/\pi}$ diagonal matrices whose diagonal elements are the eigenvalues of $d_\mu^{\gamma/\pi}$. Thus we can rewrite

$$\begin{aligned}\hbar H_{\mathrm{LS}} &= \boldsymbol{S}^\dagger\cdot\pi\cdot\boldsymbol{S} = (\boldsymbol{S}^\dagger U_\pi)\cdot d_\pi\cdot(U_\pi^\dagger\boldsymbol{S})\\ &= \sum_\mu d_\mu^\pi\tilde{S}_\mu^\dagger\tilde{S}_\mu,\end{aligned}$$

with $\tilde{S}_\mu = \sum_\alpha (U_\pi^\dagger)_{\mu\alpha}S_\alpha$. And similarly, the dissipative part reads

$$\begin{aligned}&\sum_{\alpha\beta}\gamma_{\alpha\beta}\left(S_\beta(\omega)\rho_S^I(t)S_\alpha^\dagger(\omega) - \frac{1}{2}\left\{S_\alpha^\dagger(\omega)S_\beta(\omega), \rho_S^I(t)\right\}\right)\\ =&\sum_\mu d_\mu^\gamma\left(L_\mu(\omega)\rho_S^I(t)L_\mu^\dagger(\omega) - \frac{1}{2}\left\{L_\mu^\dagger(\omega)L_\mu(\omega), \rho_S^I(t)\right\}\right),\end{aligned}$$

such that

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S^I(t) = -\frac{i}{\hbar}[H_{\mathrm{LS}}, \rho_S^I(t)] + \frac{1}{\hbar^2}\sum_{\omega,\mu}d_\mu^\gamma\left(L_\mu(\omega)\rho_S^I(t)L_\mu^\dagger(\omega) - \frac{1}{2}\left\{L_\mu^\dagger(\omega)L_\mu(\omega), \rho_S^I(t)\right\}\right). \tag{10.27}$$

Finally, we get the celebrated Lindblad-Gorini-Kossakowski-Sudarshan master equation by transforming back to the Schrödinger picture as

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\rho_S(t) &= \mathscr{L}\circ\rho_S\\ &= -\frac{i}{\hbar}[H_S', \rho_S(t)] + \frac{1}{\hbar^2}\sum_{\omega,\mu}d_\mu^\gamma\left(L_\mu(\omega)\rho_S(t)L_\mu^\dagger(\omega) - \frac{1}{2}\left\{L_\mu^\dagger(\omega)L_\mu(\omega), \rho_S(t)\right\}\right),\end{aligned} \tag{10.28}$$

with $H_S' = H_S + H_{\mathrm{LS}}$ the renormalized system Hamiltonian. One notices that due to the bilinear form of $S(\omega)$ operators, the $e^{i\omega t}$ phase is cancelled out during the transformation. The Lindblad operators $L_\mu(\omega)$ are also referred to as the jump operators. Essentially, notice that the right hand side is traceless, which preserves the trace of the density matrix, and thus the normalization of the state. To physically interpret the Lindblad equation,

we may roughly consider L to be some annihilation operator, thus the first term in the dissipative part describes eliminating particle from the system, while the second term claims that the eliminating rate is promotional to the particle number. Also notice that if we take L to be dimensionless, then the constant d^γ will include a dimension of inverse time, thus directly quantifying the decaying rate of the system.

10.2.3 Application: Spontaneous Emission

Spontaneous emission generally happens when an atom in an excited state decays to a lower energy state by emitting a photon, which is a typical example of the Lindblad equation. For simplicity, we may consider a two-level system with $H_S = -\frac{\omega}{2}\sigma_z$, whose $|0\rangle = (1,0)^T$ state has energy $\frac{\omega}{2}$ and is thus the stable ground state, while its $|1\rangle = (0,1)^T$ state has energy $-\frac{\omega}{2}$ and is the excited state. We shall take $\hbar = 1$ for simplicity. Now assume that the emission is introduced by the Lindblad operator $L(\omega') = \delta_{\omega'\omega}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, which is the annihilation operator for the excited state, while its strength is $d(\omega) = \Gamma$. We ignore the Lamb shift for simplicity. Accordingly, we have the Lindblad equation

$$\dot{\rho}_S = -i[H_S, \rho_S] + \Gamma(L\rho_S L^\dagger - \frac{1}{2}\{L^\dagger L, \rho_S\}), \tag{10.29}$$

which in explicit form reads

$$\begin{pmatrix} \dot{\rho}_{00} & \dot{\rho}_{01} \\ \dot{\rho}_{10} & \dot{\rho}_{11} \end{pmatrix} = i\omega \begin{pmatrix} & \rho_{01} \\ -\rho_{10} & \end{pmatrix} + \Gamma \begin{pmatrix} \rho_{11} & -\rho_{01}/2 \\ -\rho_{10}/2 & -\rho_{11} \end{pmatrix}, \tag{10.30}$$

and leads to the component equations as

$$\begin{cases} \dot{\rho}_{00} = \Gamma\rho_{11}, \\ \dot{\rho}_{01} = i\omega\rho_{01} - \frac{\Gamma}{2}\rho_{01}, \\ \dot{\rho}_{10} = -i\omega\rho_{10} - \frac{\Gamma}{2}\rho_{10}, \\ \dot{\rho}_{11} = -\Gamma\rho_{11}. \end{cases}$$

Solving these equations, we have

$$\begin{cases} \rho_{00}(t) = \rho_{00}(0) + \rho_{11}(0)(1 - e^{-\Gamma t}) = 1 - \rho_{11}(0)e^{-\Gamma t}, \\ \rho_{01}(t) = \rho_{01}(0)e^{i\omega t - \Gamma t/2}, \\ \rho_{10}(t) = \rho_{10}(0)e^{-i\omega t - \Gamma t/2}, \\ \rho_{11}(t) = \rho_{11}(0)e^{-\Gamma t}, \end{cases} \tag{10.31}$$

and one can easily see that as $t \to +\infty$, the excited state population $\rho_{11}(t)$ decays to zero, while the ground state population $\rho_{00}(t)$ approaches to 1. Meanwhile, the off-diagonal terms $\rho_{01}(t)$ and $\rho_{10}(t)$ oscillate with frequency ω and decay with rate $\Gamma/2$, which also vanishes after a long time, indicating that the system is in a pure state. In the intermediate case, it is clear that both the ground and excited state populations are non-vanishing, and to see whether the system is in a mixed state or a superposition state, we can either calculate the purity of the state or the von Neumann entropy, as a measure of the mixedness of the state, while for the latter, it reads

$$S(t) = -\operatorname{Tr}(\rho_S(t)\log\rho_S(t)) = -\sum_i \lambda_i \log\lambda_i, \tag{10.32}$$

with λ_i the eigenvalues of $\rho_S(t)$ reading

$$\lambda_\pm(t) = \frac{1 \pm \sqrt{1 + 4\rho_{11}(0)e^{-\Gamma t}(\rho_{11}e^{-\Gamma t} - 1) - |\rho_{01}(0)|^2 e^{-\Gamma t}}}{2}.$$

For instance, if we start from a pure excite state with $\rho_{11}(0) = 1$ while other elements vanish, we have $\lambda(t) = e^{-\Gamma t}, 1 - e^{-\Gamma t}$, and

$$S(t) = -e^{-\Gamma t} \log e^{-\Gamma t} - (1 - e^{-\Gamma t}) \log\big(1 - e^{-\Gamma t}\big), \tag{10.33}$$

which starts from 0, increasing first, and finally decreases again to 0 as the system approaches to a pure state.

10.3 The Keldysh Formalism

Chapter 11

Renormalization Group

"Variant is invariant."

11.1 To be

Upon taking the partition function as an invariant, the variants and scaling laws show up.

Appendix A

Some Mathematics

A.1 Review of Classical Mechanics

Newton's 2nd law: $\boldsymbol{F} = m\ddot{\boldsymbol{r}} = \dot{\boldsymbol{p}}$.
Definition of angular momentum: $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$.
Velocity for pure rotation: $\boldsymbol{v} = \dot{\boldsymbol{r}} = \boldsymbol{\omega} \times \boldsymbol{r}$.
Torque:

$$\boldsymbol{\tau} = \frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \dot{\boldsymbol{r}} \times \boldsymbol{p} + \boldsymbol{r} \times \dot{\boldsymbol{p}} = \boldsymbol{r} \times \boldsymbol{F}.$$

Angular momentum for rigid body:

$$\begin{aligned} L_i &= \epsilon_{ijk} \int r_j \rho(\boldsymbol{r}) \dot{r}_k \mathrm{d}V = \epsilon_{ijk}\epsilon_{kmn} \int \rho(\boldsymbol{r}) r_j r_m \mathrm{d}V \omega_n \\ &= \int \rho(\boldsymbol{r})[r_j r_j \omega_i - r_i r_m \omega_m]\mathrm{d}V = \int \rho(\boldsymbol{r})[r^2\delta_{ij} - r_i r_j]\mathrm{d}V\omega_j \\ &= I_{ij}\omega_j, \\ \Longleftrightarrow \boldsymbol{L} &= \mathbf{I} \cdot \boldsymbol{\omega}. \end{aligned}$$

Moment of inertia: $I_{ij} = \int \rho(\boldsymbol{r})[r^2\delta_{ij} - r_i r_j]\mathrm{d}V$.
Kinetic energy for pure rotation:

$$\begin{aligned} T &= \frac{1}{2}\int \rho(\boldsymbol{r}) \dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} \mathrm{d}V = \frac{1}{2}\int \rho(\boldsymbol{r}) \dot{\boldsymbol{r}} \cdot (\boldsymbol{\omega} \times \boldsymbol{r}) \mathrm{d}V \\ &= \frac{1}{2}\boldsymbol{\omega} \cdot \int \rho(\boldsymbol{r})(\boldsymbol{r} \times \dot{\boldsymbol{r}})\mathrm{d}V = \frac{1}{2}\boldsymbol{\omega} \cdot \boldsymbol{L} = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{I} \cdot \boldsymbol{\omega}. \end{aligned}$$

At any point, the moment of inertia is a symmetric matrix and is diagonalizable, with eigenvectors as principle axes and eigenvalues $\mathbf{I} \longrightarrow \mathrm{diag}(I_1, I_2, I_3)$ as principle moments. The kinetic energy for pure rotation under principle axes coordinates (chosen orthogonal) becomes

$$T = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2),$$

with mapping analogy $m \leftrightarrow I$, $v \leftrightarrow \omega$.

Kinetic energy for general motion:

$$T = \frac{1}{2}M\dot{\boldsymbol{R}}^2 + \sum_{i=1}^{3}\frac{1}{2}I_i^{CM}\omega_i^2,$$

where $\boldsymbol{R}$ is coordinates for central of mass, and I_i^{CM} is principle moment in CM principle axes coordinates (the body frame). The introduced CM frame relative to lab frame is essential for decoupling of translation and rotation.

Euler's EoM: as a vector, $\boldsymbol{L} = L_i^{CM}\hat{e}_i^{CM}$ in CM principle axes coordinates, with $L_i^{CM} = I_i^{CM}\omega_i^{CM}$ (no summation).

$$\boldsymbol{\tau} = \frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \frac{\mathrm{d}L_i^{CM}}{\mathrm{d}t}\hat{e}_i^{CM} + L_i^{CM}\frac{\mathrm{d}\hat{e}_i^{CM}}{\mathrm{d}t} = I_i\dot{\omega}_i\hat{e}_i + L_i(\boldsymbol{\omega}\times\hat{e}_i),$$

where we drop CM. In component,

$$\tau_i = I_i\dot{\omega}_i + \sum_{mn} I_n\epsilon_{imn}\omega_m\omega_n, \text{ e.g. } \tau_1 = I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3.$$

Basically, the first term is the change of angular momentum measured in the body frame, and the second term is the rotation of the body frame measured in lab frame. This is just a result from general transformation of coordinates of non-inertial frame.

Precession of free symmetric top: symmetric means $I_1 = I_2 \equiv I$, which makes $\dot{\omega}_3 = 0$. Free means $\boldsymbol{\tau} = 0$, which makes the angular momentum **vector** conserved. Also as vectors,

$$\begin{cases} \boldsymbol{\omega} = \omega_1\hat{e}_1 + \omega_2\hat{e}_2 + \omega_3\hat{e}_3 \\ \boldsymbol{L} = I(\omega_1\hat{e}_1 + \omega_2\hat{e}_2) + I_3\omega_3\hat{e}_3 \end{cases} \implies \boldsymbol{L} = I(\boldsymbol{\omega} + \Omega\hat{e}_3),\ \Omega = \frac{I_3 - I}{I}.$$

The restriction makes the three vectors $\boldsymbol{L}, \boldsymbol{\omega}, \hat{e}_3$ coplanar. Similar to $\boldsymbol{L}$, the body frame axis $\hat{e}_3$ has EoM as

$$\frac{\mathrm{d}\hat{e}_3}{\mathrm{d}t} = \boldsymbol{\omega}\times\hat{e}_3 = \frac{1}{I}\boldsymbol{L}\times\hat{e}_3,$$

which reveals the precession of $\hat{e}_3$ around the invariant vector $\boldsymbol{L}$, with frequency L/I. By coplanarity, $\boldsymbol{\omega}$ also rotates around $\boldsymbol{L}$ with the same frequency, i.e., synchronized.

Lagrangian for the rigid body:

$$L = T - U = \frac{1}{2}M\dot{\boldsymbol{R}}^2 + \sum_{i=1}^{3}\frac{1}{2}I_i^{CM}\omega_i^2 - U,$$

and especially for rigid body, the basic six general coordinates are R_i, θ_i ($i = 1, 2, 3$, $\dot{\theta}_i = \omega_i$). However, θ_i are not good coordinates, and we prefer to change them to Eular angles, and one interpretation for them is they describe a two-step rotation of z-axis from lab frame to body frame by (ϕ, θ), just like a spherical coordinate, plus a self-rotation around z-axis in body frame by ψ. With the relations, we have

$$\begin{cases} \omega_1 = \dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi \\ \omega_2 = \dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi \\ \omega_3 = \dot{\psi} + \dot{\phi}\cos\theta \end{cases}.$$

Using these Euler angles as new general coordinates and derive the E-L EoM will also leads to the Euler equations.

Lagrangian mechanics: given Lagrangian $L(q, \dot{q}, t)$, we have least action principle

$$\begin{aligned}
S &= \int_{t_i}^{t_f} L(q, \dot{q}, t)\mathrm{d}t \\
\delta S &= \int_{t_i}^{t_f} \delta L(q, \dot{q}, t)\mathrm{d}t = \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial \dot{q}}\delta\dot{q}\right)\mathrm{d}t \\
&= \int_{t_i}^{t_f} \left[\frac{\partial L}{\partial q}\delta q + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\delta q\right) - \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right)\delta q\right]\mathrm{d}t \\
&= \frac{\partial L}{\partial \dot{q}}\delta q\Big|_{t_i}^{t_f} + \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right)\delta q\mathrm{d}t \\
&= \int_{t_i}^{t_f} \left(\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right)\delta q\mathrm{d}t,
\end{aligned}$$

which leads to the famous Euler-Lagrange equation

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}} = 0.$$

The difference between the least principle and the symmetry variance is that the former is a functional of the trajectory with fixed boundary, while in the latter, generally we change the boundary or change the measure.

The general coordinate q has its canonical momentum partner defined as $p = \partial L/\partial\dot{q}$, and the Hamiltonian is defined as the Legendre transformation of Lagrangian

$$H(q, p, t) = \dot{q}p - L(q, \dot{q}, t),\ \ \dot{q} = \dot{q}(q, p, t).$$

Especially notice that the argues are different. The first Hamiltonian equation follows as

$$\frac{\partial H}{\partial p} = \frac{\partial \dot{q}}{\partial p}p + \dot{q} - \frac{\partial L}{\partial \dot{q}}\frac{\partial \dot{q}}{\partial p} = \dot{q},$$

where the definition of p is used. The second Hamiltonian equation reads

$$\frac{\partial H}{\partial q} = \frac{\partial \dot{q}}{\partial q}p - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial \dot{q}}\frac{\partial \dot{q}}{\partial q} = -\frac{\partial L}{\partial q} = -\dot{p},$$

where the Euler-Lagrange equation is used. In the derivation, we observe that even if we ignore the dependence of $\dot{q}$ over p and q, we still have the same result, which is a feature of Legendre transformation. Notice that in Hamiltonian formalism, we obtain twice the number of equations, but lower the order the differential from 2 to 1, which is a good thing.

Imagine that we have a general function $O(q, p, t)$, and we want to find its time evolution, we have

$$\begin{aligned}
\frac{\mathrm{d}O}{\mathrm{d}t} &= \frac{\partial O}{\partial q}\dot{q} + \frac{\partial O}{\partial p}\dot{p} + \frac{\partial O}{\partial t} \\
&= \frac{\partial O}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial O}{\partial p}\frac{\partial H}{\partial q} + \frac{\partial O}{\partial t} \\
&\equiv \{O, H\} + \partial_t O,
\end{aligned}$$

where the Poisson bracket is defined as

$$\{A, B\} = \frac{\partial A}{\partial q}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial q}.$$

In Hamiltonian mechanics, a quantity is conserved if the RHS of the above formula is zero, and especially, a time-independent Hamiltonian is naturally conserved, i.e., energy conservation. Interestingly, Dirac is inspired by the Poisson bracket form and generalized it to quantum mechanics, where the Poisson bracket is replaced by the operator commutation relation

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},$$

and the Hamiltonian is replaced by the Hamiltonian operator. Indeed, the evolution form here is much close to the Heisenberg equation of motion for operator time evolution, which is given by

$$\frac{\mathrm{d}\hat{O}}{\mathrm{d}t} = \frac{\mathrm{i}}{i\hbar}[\hat{O}(t), \hat{H}(t)] + \partial_t \hat{O}(t),$$

where all operators are in Heisenberg picture, with i being the imaginary unit and $\hbar$ being the reduced Planck constant. The similarity is not just a coincidence, as the time evolution is basically generated by Lie group with Lie algebra, which is the Poisson bracket in classical mechanics and the commutation relation in quantum mechanics.

Let us come back. We know sometimes for simplicity we choose not-so-evident generalized coordinates, and we can even make transformation between different generalized coordinates, which is called *canonical transformation*

$$\begin{cases} P = P(q, p, t) \\ Q = Q(q, p, t) \end{cases}.$$

The requirement is that the equation of motion, a physical observable, should be invariant under such transformation, which comes from the least action principle:

$$\delta \int_{t_i}^{t_f} (p\dot{q} - H)\mathrm{d}t = 0 \Rightarrow \delta \int_{t_i}^{t_f} (P\dot{Q} - H')\mathrm{d}t = 0,$$

where H' is the Hamiltonian in the new coordinate system.

Let us recall that for $L' = L + \frac{\mathrm{d}U}{\mathrm{d}t}$, we have the same EoM, which is a consequence of the least action principle, since $U(t_f) - U(t_i)$ term is a boundary term and does not affect the changing of the action. Then we have the following relation

$$\begin{aligned} (p\dot{q} - H) - (P\dot{Q} - H') &= \frac{\mathrm{d}U}{\mathrm{d}t}, \\ \Rightarrow p\mathrm{d}q - P\mathrm{d}Q + (H' - H)\mathrm{d}t &= \mathrm{d}U, \end{aligned}$$

where U is called *generating function* of the canonical transformation. We have four useful types of generating function, which are related to each other by Legendre transformation

- $U_1 = U_1(q, Q, t)$, which is just the definition above, and shares

$$\begin{cases} p = \partial_q U_1 \\ P = -\partial_Q U_1 \\ H' = H + \partial_t U_1 \end{cases}.$$

- $U_2 = U_2(p, Q, t)$, where one Legendre transformation over q is performed as

$$U_2(p, Q, t) = U_1(q, Q, t) - (\partial_q U_1)q = U_1 - pq.$$

Basically, we have

$$-q\mathrm{d}p - P\mathrm{d}Q + (H' - H)\mathrm{d}t = \mathrm{d}U_2,$$

and

$$\begin{cases} q = -\partial_p U_2 \\ P = -\partial_Q U_2 \\ H' = H + \partial_t U_2 \end{cases}.$$

- $U_3 = U_3(q, P, t)$, where one Legendre transformation over Q is performed as

$$\begin{gathered} U_3(q, P, t) = U_1(q, Q, t) - (\partial_Q U_1)Q = U_1 + PQ, \\ p\mathrm{d}q + Q\mathrm{d}P + (H' - H)\mathrm{d}t = \mathrm{d}U_3, \end{gathered},$$

and

$$\begin{cases} p = \partial_q U_3 \\ Q = \partial_P U_3 \\ H' = H + \partial_t U_3 \end{cases}.$$

- $U_4 = U_4(p, P, t)$, where one Legendre transformation over q and Q is performed as

$$\begin{gathered} U_4(p, P, t) = U_1(q, Q, t) - (\partial_q U_1)q - (\partial_Q U_1)Q = U_1 - pq + PQ, \\ -q\mathrm{d}p + Q\mathrm{d}P + (H' - H)\mathrm{d}t = \mathrm{d}U_4, \end{gathered}$$

and

$$\begin{cases} q = -\partial_q U_4 \\ Q = \partial_P U_4 \\ H' = H + \partial_t U_4 \end{cases}.$$

People familiar with thermodynamics or differential forms know what we are doing here.

Especially, the Poisson bracket is invariant under canonical transformation, i.e.,

$$\{O, H\}_{pq} = \{O, H\}_{PQ},$$

which should be the way since the time-evolution of a physical observable should not depend on the choice of coordinates.

Now we wish to find a special canonical transformation such that

$$\begin{cases} \dot{P} = -\partial_Q H = 0 \\ \dot{Q} = \partial_P H = 0 \end{cases},$$

i.e., the Hamiltonian is constant in phase space. Let us take $U_3 = U_3(q, O, t)$, with

$$p\mathrm{d}q + Q\mathrm{d}P + (H' - H)\mathrm{d}t = \mathrm{d}U_3, \quad \begin{cases} p = \partial_q U_3 \\ Q = \partial_P U_3 \\ H' = H + \partial_t U_3 \end{cases}.$$

Let us choose the simplest case, $H' = 0$, then we obtain the *Hamilton-Jacobi equation*

$$H\left(q, \frac{\partial U_3}{\partial q}, t\right) + \frac{\partial U_3}{\partial t} = 0,$$

where the generating function U_3 is called *action function* or *Hamilton's principal function*, and the solution U_3 is called *action integral* or *Hamilton's principal integral*. Let us denote $U_3(q,P,t) = S(q,P,t)$, such that a formal re-write gives

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0.$$

Since once knowing the solution S, we immediately get invariant generalized coordinates P and Q, and a inverse canonical transformation gives $q(t)$ and $p(t)$, we say that the Hamilton-Jacobi equation is equivalent to the Hamiltonian equations of motion. This is the third form of theoretical mechanics, apart from Lagrangian and Hamiltonian mechanics.

Especially notice that

$$\begin{aligned}\frac{\mathrm{d}S}{\mathrm{d}t} &= \frac{\partial S}{\partial t} + \dot{q}\frac{\partial S}{\partial q} + \dot{P}\frac{\partial S}{\partial P} \\ &= \frac{\partial S}{\partial t} + p\dot{q} = p\dot{q} - H = L,\end{aligned}$$

we find that $S = \int \mathrm{d}tL$ is just the action, i,e, action is the Hamilton's principal function.

If we now let $S = -i\hbar \ln U$, we obtain

$$i\hbar\partial_t U = HU,$$

which resembles the Schrödinger equation for time evolution operator. This similarity can find its way in path integral, as generally

$$\langle x_f|U(t_f,t_i)|x_i\rangle = \int \mathscr{D}q \exp\left(\frac{i}{\hbar}S[q(t)]\right),$$

and if we ignore the basis, then roughly we have

$$U \sim e^{iS/\hbar}.$$

A.2 Coordinate Transformation

A tensor is invariant under coordinate transformation. We use Einstein summation convention, where the repeated indices (dummy or bound indices, contrary to the free index that appears only once) are summed over, if there is no specific explanation. For instance, the scalar product of two vectors $\boldsymbol{A} = (A_1, A_2, \cdots, A_n)$ and $\boldsymbol{B} = (B_1, B_2, \cdots, B_n)$ is written as

$$\boldsymbol{A}\cdot\boldsymbol{B} = A_iB_i = A_1B_1 + A_2B_2 + \cdots + A_nB_n.$$

Given basis transformation

$$\hat{e}'_i = \lambda_{ij}\hat{e}_j,\ \ \hat{e}_i = (\lambda^T)_{ij}\hat{e}'_j,$$

where the transformation matrix λ with component $\lambda_{ij} = \hat{e}'_i\cdot\hat{e}_j$ and transverse $(\lambda^T)_{ij} = \lambda_{ji}$ satisfies that

$$\lambda^T\lambda = \lambda\lambda^T = \mathbb{I},$$

i.e., an orthogonal matrix. We can simply denote

$$\hat{e}' = \lambda \hat{e},\ \hat{e} = \lambda^T \hat{e}'.$$

Now, the position vector, which is recognized as a rank-1 tensor and is invariant under basis transformation, is written as

$$\boldsymbol{r} = r_i \hat{e}_i = r'_i \hat{e}'_i = r'_i \lambda_{ij} \hat{e}_j,$$

then one has

$$r_j = r'_i \lambda_{ij} = (\lambda^T)_{ji} r'_i,$$

i.e., when represented in certain basis, the position vector components transform as

$$r' = \lambda r,\ r = \lambda^T r',$$

with $r = (r_1, r_2, \cdots)^T$ recognized.

Scalar as rank-0 tensor. A scalar $\boldsymbol{S}$ has single component representation under basis as $\boldsymbol{S} = S$. The invariance of the scalar means $S' = S$, and with position dependence we have $S'(r') = S(r') = S(r)$.

Vector as rank-1 tensor. A vector $\boldsymbol{V}$ shares similar representation form as position vector as $\boldsymbol{V} = V_i \hat{e}_i$, its invariance under basis transformation is then similar to the position vector, $V'_i = \lambda_{ij} V_j$, or

$$V' = \lambda V,\ V = \lambda^T V',$$

with $V = (V_1, V_2, \cdots)$. Further counting on the position dependence leads to $V'(r') = \lambda V(r)$. It is not strange here that the represented vector V shares the same transformation form with the basis, since by the representation form we write, we have already chosen both basis and vector to be arranged in the column form, which leads to the co-variant transformation form. A more formal writing which defines $r = (r_1, r_2, \cdots)$ will lead to the contra-variant transformation $r' = r\lambda^T$.

Rank-2 tensor. A rank-2 tensor is represented as $\boldsymbol{T} = T_{ij} \hat{e}_i \hat{e}_j$, with the operation between $\hat{e}_i$ and $\hat{e}_j$ as tensor product to be explained. Its invariance under basis transformation is given by

$$\boldsymbol{T} = T_{ij} \hat{e}_i \hat{e}_j = \boldsymbol{T} = T'_{ij} \hat{e}'_i \hat{e}'_j = T'_{ij} \lambda_{im} \hat{e}_m \lambda_{jn} \hat{e}_n,$$

which leads to

$$T_{ij} = T'_{ij} \lambda_{im} \lambda_{jn} = (\lambda^T)_{mi} (\lambda^T)_{nj} T'_{ij},$$

and inversely,

$$T'_{ij} = \lambda_{im} \lambda_{jn} T_{mn}.$$

Sometimes, to get a better visual effect and relate the transformation to the similar transformation in linear algebra that people are more familiar with, one writes

$$T'_{ij} = \lambda_{im} T_{mn} (\lambda^T)_{nj} \Longleftrightarrow T' = \lambda T \lambda^T,$$

with the represented matrix form T of the rank-2 tensor $\boldsymbol{T}$ recognized.

General rank-N tensor. Generalize procedure above, one has that for a rank-N tensor $\boldsymbol{T} = T_{i_1 i_2 \cdots i_N} \hat{e}_{i_1} \hat{e}_{i_2} \cdots \hat{e}_{i_N}$, with N indices,

$$T'_{i_1 i_2 \cdots i_N} = \lambda_{i_1 j_1} \lambda_{i_2 j_2} \cdots \lambda_{i_N j_N} T_{j_1 j_2 \cdots j_N}.$$

Just like scalars and vectors, a general tensor shares common linear laws, such as

- Addition: $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B} \Leftrightarrow C_{i_1 \cdots i_n} = A_{i_1 \cdots i_n} + B_{i_1 \cdots i_n}$.
- Subtraction: $\boldsymbol{C} = \boldsymbol{A} - \boldsymbol{B} \Leftrightarrow C_{i_1 \cdots i_n} = A_{i_1 \cdots i_n} - B_{i_1 \cdots i_n}$.
- Scalar multiplication: $\boldsymbol{C} = a\boldsymbol{A} \Leftrightarrow C_{i_1 \cdots i_n} = aA_{i_1 \cdots i_n}$.

These operations are all element-wise and require that the tensors to be added or multiplied have the same rank and the same dimension.

Tensor product. A rank-p tensor does tensor product with a rank-q tensor results in a rank-$p+q$ tensor. Denote $\boldsymbol{R} \otimes \boldsymbol{S} = \boldsymbol{T}$, we have

$$T_{i_1 \cdots i_p j_1 \cdots j_q} = R_{i_1 \cdots i_p} S_{j_1 \cdots j_q}.$$

For instance, as we claimed $\hat{e}_i \otimes \hat{e}_j$ should be understand as tensor product. In their own representation, $(\hat{e}_i)_m = \delta_{im}$ with d the dimension and we have

$$(\hat{e}_i \otimes \hat{e}_j)_{mn} = (\hat{e}_i)_m (\hat{e}_j)_n = \delta_{im}\delta_{jn},$$

which gives a matrix with (i, j) element 1 and others zero. Also, we recognize $\hat{e}_i \otimes \hat{e}_i = \mathbb{I}$, which is just the Kronecker tensor of rank-2 since $\mathbb{I} = \delta_{ij}\hat{e}_i \otimes \hat{e}_j$. The definition is not very formal here.

Another example is tensor between rank-2 tensors, and since they can be represented as matrices, one can use Kronecker product to replace the tensor product. Say $\boldsymbol{A}$ and $\boldsymbol{B}$ are two rank-2 tensors with representation matrices A and B, the Kronecker product is defined as

$$A \otimes_K B = \begin{pmatrix} A_{11}B & A_{12}B & \\ A_{21}B & A_{22}B & \\ & & \ddots \end{pmatrix}.$$

Contraction. Given a tensor $\boldsymbol{T}$ of rank-p, we can construct a new tensor $\boldsymbol{S}$ of rank-$(p-2)$ by contracting on two indices using δ_{ij} as

$$S_{k_1 \cdots k_{p-2}} = \delta_{ij} T_{ijk_1 \cdots k_{p-2}} = T_{iik_1 \cdots k_{p-2}}.$$

The contraction of a rank-2 tensor is called trace,

$$\mathrm{Tr}(T) = T_{ii}.$$

The inner product between two vectors can be recognized as contraction after $(1, 1)$ tensor product,

$$V \cdot L = (V \otimes L)_{ii} = V_i L_i.$$

In this way, we find the representation of vector under basis natural,

$$\boldsymbol{V} = \boldsymbol{V} \cdot \mathbb{I} = \boldsymbol{V} \cdot \hat{e}_i \otimes \hat{e}_i = (\boldsymbol{V} \cdot \hat{e}_i)\hat{e}_i = V_i \hat{e}_i,$$

with $V_i = \boldsymbol{V} \cdot \hat{e}_i$ defined as inner product between vectors. This can be generalized to general tensors.

Symmetric and anti-symmetric tensors. This definition only applies for tensors with rank $n \geq 2$. A rank-2 tensor (a matrix) $\boldsymbol{T}$ is symmetric if

$$T_{ij} = T_{ji},$$

and is anti-symmetric or skew-symmetric if

$$T_{ij} = -T_{ji}.$$

They both have direct applications in physics, e.g., the energy-momentum tensor is (at many times) symmetric, while the electromagnetic tensor is anti-symmetric.

For a rank-n tensor, we generally only talk that it is symmetric or anti-symmetric under the exchange of any two indices, i.e., symmetric in two indices if

$$T_{\cdots i\cdots j\cdots} = T_{\cdots j\cdots i\cdots},$$

and anti-symmetric in two indices if

$$T_{\cdots i\cdots j\cdots} = -T_{\cdots j\cdots i\cdots}.$$

Tensor can be symmetrized or anti-symmetrized. For rank-2 tensor, we can directly decompose it into a symmetric and anti-symmetric part as

$$T_{ij} = T_{(ij)} + T_{[ij]},\ T_{(ij)} = \frac{1}{2}(T_{ij} + T_{ji}),\ T_{[ij]} = \frac{1}{2}(T_{ij} - T_{ji}).$$

For general rank-n tensor, we can symmetrize or anti-symmetrize it as

$$\begin{aligned} T_{(i_1\cdots i_n)} &= \frac{1}{n!}\sum_{\sigma\in S_n} T_{i_{\sigma(1)}\cdots i_{\sigma(n)}}, \\ T_{[i_1\cdots i_n]} &= \frac{1}{n!}\sum_{\sigma\in S_n} \mathrm{sgn}(\sigma) T_{i_{\sigma(1)}\cdots i_{\sigma(n)}}, \end{aligned}$$

where S_n is the permutation group of n elements, and $\mathrm{sgn}(\sigma)$ is the signature of the permutation σ, meaning that $\mathrm{sgn}(\sigma) = 1$ if σ is an even permutation, and $\mathrm{sgn}(\sigma) = -1$ if σ is an odd permutation. Notice that sometimes we also symmetrize or anti-symmetrize a tensor in several indices, for instance,

$$\begin{aligned} T_{(ij)k} &= \frac{1}{2}(T_{ijk} + T_{jik}), \\ T_{[ij]k} &= \frac{1}{2}(T_{ijk} - T_{jik}). \end{aligned}$$

At many times, we do not perform symmetrize or anti-symmetrize process, as some tensors are naturally symmetric or anti-symmetric.

Consider that we have two tensors, and we want to contract them. Such a contraction will be zero if the contracted indices are symmetric for one tensor while anti-symmetric for the other tensor. For instance, consider contraction of a $A_{(ij)}$ and $B_{[ij]}$ leads to

$$\begin{aligned} A_{(ij)}B_{[ij]} &= \frac{1}{2}(A_{ij} + A_{ji})\frac{1}{2}(B_{ij} - B_{ji}) \\ &= \frac{1}{4}(A_{ij}B_{ij} - A_{ij}B_{ji} + A_{ji}B_{ij} - A_{ji}B_{ji}) \\ &= \frac{1}{4}(A_{ij}B_{ij} - A_{ij}B_{ij} + A_{ij}B_{ij} - A_{ij}B_{ij}) = 0. \end{aligned}$$

A.3 Some Linear Algebra

Schur complement

The Schur complement arises when performing a block Gaussian elimination on the matrix $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. Consider that A is invertible, then the block Gaussian elimination gives

$$\begin{aligned} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \mathbb{I}_A & -A^{-1}B \\ 0 & \mathbb{I}_D \end{pmatrix} &= \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix}, \\ \begin{pmatrix} \mathbb{I}_A & 0 \\ -CA^{-1} & \mathbb{I}_D \end{pmatrix} \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix} &= \begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}. \end{aligned} \tag{A.1}$$

Consequently, the Schur complement of A in M is defined as $S = D - CA^{-1}B$. Meanwhile, we obtain a determinant formula for M as

$$\det(M) = \det(A) \det\big(D - CA^{-1}B\big), \tag{A.2}$$

utilizing $\det(AB) = \det(A)\det(B)$ and $\det \begin{pmatrix} A & \\ & B \end{pmatrix} = \det(A)\det(B)$.

Baker-Campbell-Hausdorff formula

A.4 de Rham's Theorem

The integration over a manifold is about integrating differential forms over chains as $I(c_p, \alpha^p) = \int_{c_p} \alpha^p$, such an integration can be viewed as a functional over either the vector space of p-forms $I(c_p, :)$ or the vector space of p-chains $I(:, \alpha^p)$, by allowing the attached Abelian group to be the real space. Different orders of forms and chains are related by the exterior derivative d (coboundary operator) and the boundary operator ∂, respectively as

$$\begin{cases} \alpha^{p+1} = \mathrm{d}\alpha^p \\ c_{p-1} = \partial c_p \end{cases} . \tag{A.3}$$

Essentially, due to the antisymmetric property over differential forms and the intuition (which can also be proved directly) that a boundary has no boundary, the following property goes as

$$d^2 = 0,\ \partial^2 = 0, \tag{A.4}$$

which leads to the non-trivial topological property.

To proceed, we first define the pth chain group $C_p(M, G) = \{c_p | c_p = \sum_{g_i \in G} g_i \sigma_p^i\}$ over manifold M with coefficients in group Abelian group G, where σ_p^i are p-simplices. We define circle as chain without boundary thus writing down the pth circle group $Z_p(M, G) = \{c_p | \partial c_p = 0\}$. Since $\partial^2 = 0$, we see immediately that the pth boundary group $B_p(M, G) = \{c_p | c_p = \partial c_{p+1}, c_{p+1} \in C_{p+1}\}$ is a subgroup of $Z_p(M, G)$. Due to the Abelian nature of these group with group product as addition, we see B_p is a normal subgroup of Z_p, and we correspondingly have the quotient group $H_p(M, G) = Z_p(M, G)/B_p(M, G)$, which is the pth homology group. Such a group is always Abelian, and can be calculated using exact sequences. For $G = \mathbb{R}$, the dimension of the pth homology group is the pth Betti number

b_p, which is a topological invariant. For $\dim M = D$, we always have $H_0(M, G) = G^{n_{\text{part}}}$, with n_{part} the number of connected components of M, and $H_D(M, G) = G$ if M is orientable, and $H_D(M, G) = 0$ if M is non-orientable (defined by no coherent simplex basis, i.e., non-vanishing boundary of M under the basis).

Similar things happen for differential forms, with the pth form group $A^p(M, G) = \{\alpha^p | \alpha^p = \sum_{g_i \in G} g_i \omega_p^i\}$ over manifold M with coefficients in group Abelian group G, where ω_p^i are p-forms. We define closed form with vanishing exterior derivative thus the pth closed form group $F^p(M, G) = \{\alpha^p | \mathrm{d}\alpha^p = 0\}$, together with exact form as some exterior derivative of another form, the pth exact form group $E^p(M, G) = \{\alpha^p | \alpha^p = \mathrm{d}\alpha^{p-1},\ \alpha^{p-1} \in A^{p-1}\}$. Again due to that $\mathrm{d}^2 = 0$ and the Abelian nature, we have the pth cohomology group $H^p(M, G) = F^p(M, G)/E^p(M, G)$.

We may now come back to $I(c_p, \alpha^p)$ by focusing on integrating closed forms over cycles. We can not directly apply Stokes' theorem to such kind, however, we can utilize it to show the equivalence classes of such kind, as

$$I(c_p, \alpha^p + \mathrm{d}\beta^{p-1}) = \int_{c_p} \alpha^p + \int_{c_p} \mathrm{d}\beta^{p-1} = \int_{c_p} \alpha^p + \int_{\partial c_p} \beta^{p-1} = \int_{c_p} \alpha^p = I(c_p, \alpha^p), \quad \text{(A.5a)}$$

$$I(c_p + \partial c_{p+1}, \alpha^p) = \int_{c_p} \alpha^p + \int_{\partial c_{p+1}} \alpha^p = \int_{c_p} \alpha^p + \int_{c_{p+1}} \mathrm{d}\alpha^p = \int_{c_p} \alpha^p = I(c_p, \alpha^p), \quad \text{(A.5b)}$$

i.e., the integration depends only on the equivalence class of the form and the cycle, i.e., on the (co)homology group $H^p(M, G)$ and $H_p(M, G)$. Let $G = \mathbb{R}$, and by viewing the integration as a functional over the vector space of homology group, we have the de Rham's theorem that this dual space is isomorphic to the cohomology group, i.e., $H^p(M, \mathbb{R}) \cong H_p^*(M, \mathbb{R}) \cong H_p(M, \mathbb{R})$, i.e., the pth cohomology group is isomorphic to the pth homology group.